

# Status of Yelloweye Rockfish off the U.S. West Coast in 2025



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Please cite this publication as:

Johnston\*, M. A., Rosemond\*, R. C., Whitman, A., Perl, E., Barros, M., Champagnat, J., Schamp, A., Schiano, S., Prior Caltabellotta, F., Gertseva, V., Taylor, I., Oken, K. and Berger, A. (2025) Status of Yelloweye rockfish off the U.S. West Coast in 2025. Pacific Fishery Management Council. 147 pp.

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## AT-A-GLANCE

# U.S. West Coast Yelloweye Rockfish in 2025



### Data and Stock Assessment Model

This update assessment reports the status of the Yelloweye Rockfish (*Sebastes ruberrimus*) resource off the coast of the United States (U.S.) using data through 2024.

View all groundfish  
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Landings &  
Discards



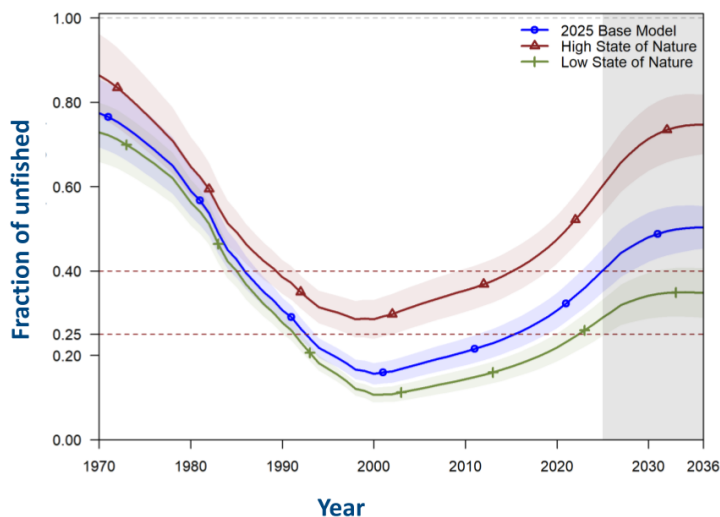
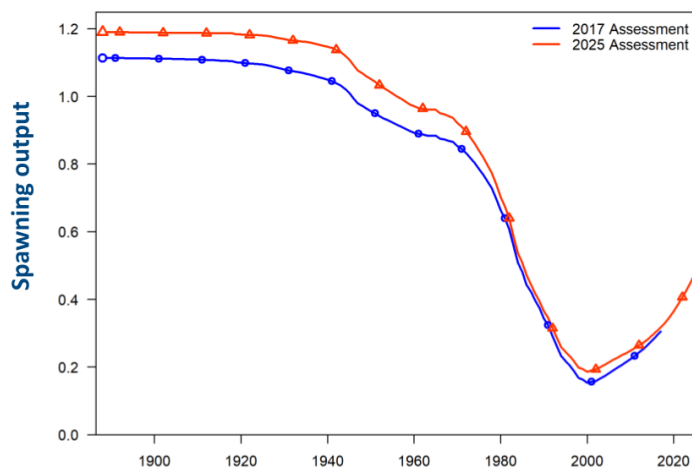
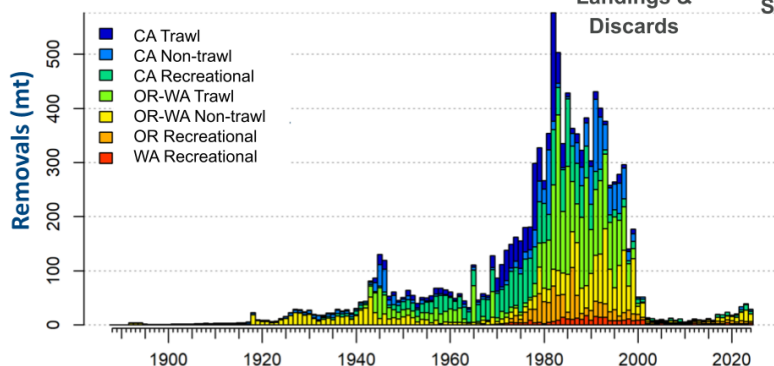
Survey Indices



Age  
Composition



Length  
Composition



### Key Takeaways

- At the start of 2025, the estimated stock status is 40.1%, **at the target of 40%** of its unfished condition.
- Stock trend is **consistent with 2017 benchmark assessment** and rebuilding analysis.
- Assessment model is **sensitive to natural mortality**, with alternative states of nature estimating 2025 fraction unfished between 29% and 60%, for low and high state respectively.

### Base Model Removals & Stock Projections

| Year | Removals (mt)      | Spawning Output (billions of eggs) | Stock Status (target = 40%) |
|------|--------------------|------------------------------------|-----------------------------|
| 2025 | 45.7 <sup>a</sup>  | 478                                | 40%                         |
| 2026 | 46.4 <sup>a</sup>  | 502                                | 42%                         |
| 2027 | 112.9 <sup>b</sup> | 527                                | 44%                         |
| 2028 | 113.1 <sup>b</sup> | 544                                | 46%                         |
| 2029 | 113.0 <sup>b</sup> | 558                                | 47%                         |
| 2030 | 112.6 <sup>b</sup> | 571                                | 48%                         |
| 2031 | 111.8 <sup>b</sup> | 581                                | 49%                         |
| 2032 | 110.9 <sup>b</sup> | 588                                | 49%                         |

<sup>a</sup> Based on Groundfish Management Team Recommendations for Forecasting

<sup>b</sup> Annual Catch Limit (ACL) from the default harvest control rule with  $P^* = 0.4$ ,  $\sigma = 0.5$

Johnston, M. et al. 2025. Status of Yelloweye rockfish off the U.S. West Coast in 2025

## Executive Summary

### Stock

This update assessment reports the status of Yelloweye Rockfish (*Sebastes ruberrimus*) off the U.S. West Coast using data through 2024. Yelloweye Rockfish are found from the Gulf of Alaska to northern Baja California in Mexico across the northeastern Pacific Ocean. Their core distribution is from southeast Alaska to central California on the west coast of the United States. Yelloweye Rockfish are strongly associated with rocky bottom habitat and adults are considered to be solitary and sedentary after settlement. Given the general perception of the sedentary nature of Yelloweye Rockfish adults and the moderate amount of mixing that occurs during the pelagic larval stage, the previous Yelloweye Rockfish assessment, conducted in 2017, modeled the West coast population as a two-area assessment (California and a combined Oregon-Washington area) with a common stock recruitment relationship. This update assessment necessarily maintains this same structure.

### Catches

Catches for Yelloweye Rockfish have averaged over 20 mt in recent years (Figure i, Table i). The Yelloweye Rockfish stock was declared overfished in 2002 and remains under a rebuilding plan that substantially limits catch. However, as other rockfish stocks have rebuilt and Yelloweye Rockfish has progressed under its rebuilding plan, catches have slowly increased in recent years, primarily in the Oregon-Washington non-trawl fleet and the recreational fleets.

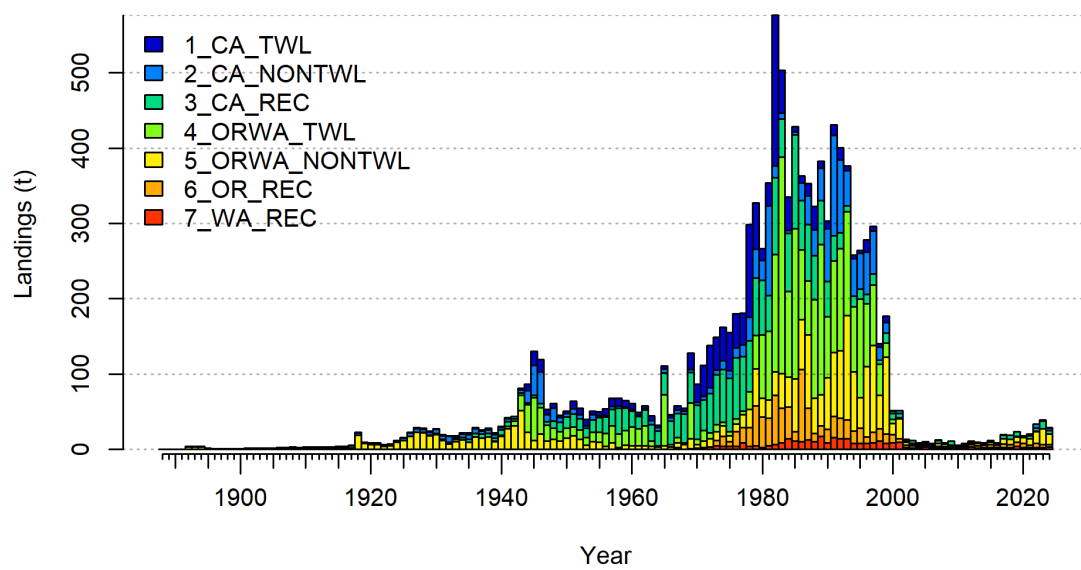


Figure i: Yelloweye Rockfish landing history in metric tons (mt) between 1889 and 2024 for each fleet.

Table i: Recent catches by fleet and total catch (mt) summed across fleets (total catch includes WA REC catch converted to mt).

| Year | CA<br>TWL<br>(mt) | CA<br>NON-<br>TWL<br>(mt) | CA<br>REC<br>(mt) | ORWA<br>TWL<br>(mt) | ORWA<br>NON-<br>TWL<br>(mt) | OR<br>REC<br>(mt) | WA<br>REC<br>(mt) | Catch<br>(mt) |
|------|-------------------|---------------------------|-------------------|---------------------|-----------------------------|-------------------|-------------------|---------------|
| 2015 | 0.00              | 0.40                      | 2.00              | 0.03                | 3.15                        | 4.26              | 2.10              | 11.94         |
| 2016 | 0.00              | 0.00                      | 1.00              | 0.07                | 2.59                        | 2.84              | 2.46              | 8.96          |
| 2017 | 0.01              | 1.23                      | 4.52              | 0.24                | 6.97                        | 4.27              | 2.49              | 19.73         |
| 2018 | 0.00              | 0.00                      | 4.99              | 0.54                | 6.38                        | 4.01              | 2.57              | 18.49         |
| 2019 | 0.04              | 0.00                      | 6.16              | 0.59                | 7.43                        | 5.04              | 4.23              | 23.49         |
| 2020 | 0.13              | 0.00                      | 1.95              | 0.32                | 7.52                        | 6.00              | 2.24              | 18.16         |
| 2021 | 0.12              | 2.43                      | 3.96              | 0.39                | 7.97                        | 3.34              | 2.54              | 20.75         |
| 2022 | 0.10              | 5.60                      | 3.80              | 0.76                | 15.55                       | 5.20              | 2.65              | 33.66         |
| 2023 | 0.09              | 1.83                      | 9.59              | 0.40                | 20.64                       | 3.84              | 2.88              | 39.27         |
| 2024 | 0.19              | 3.27                      | 4.65              | 0.49                | 13.51                       | 3.66              | 2.92              | 28.69         |



## Data and Assessment

The last assessment for Yelloweye Rockfish occurred in 2017. This update assessment extends the data used in the 2017 assessment through 2024. This assessment uses the stock assessment framework Stock Synthesis (SS3 Version 3.30.23.2) by Methot and Wetzel (2013). Data includes catch, length and age data from seven fishery fleets and multiple indices of abundance in California and Oregon/Washington. Two new historical catch reconstructions from Oregon and Washington were incorporated. Four indices of abundance were updated for this assessment, including two recreational fishery indices in Oregon, the West Coast Groundfish Bottom Trawl Survey (WCG BTS), and the International Pacific Halibut Commission (IPHC) longline survey. In addition, sample sizes and assignment of aging error were corrected in the compositional data. No new data streams were considered in this update assessment.

## Stock Spawning Output and Dynamics

The Yelloweye Rockfish assessment uses estimates of fecundity (eggs-at-length) from the Dick et al. (2017) method, and spawning output is reported in billions of eggs. The unexploited level of spawning stock output is estimated to be 1190 billion eggs (95% confidence interval: 1,048.1 - 1,331.9 billion eggs) (Figure ii). At the beginning of 2025, the spawning stock output is estimated to be 477.63 billion eggs (95% confidence interval: 384 – 571 billion eggs), which represents 40.1% of the unfished spawning output level.

Estimated relative spawning output was below the minimum stock size threshold in the late 1990s and was lowest in the early 2000s before increasing over the last 20 years. The 2025 estimated relative spawning output follows an increasing trajectory and is slightly above the management target threshold (Figure ii, Figure iii). Though Yelloweye Rockfish are considered a single stock due to their population's even genetic and spatial structure throughout their range, this assessment is modeled with two areas (California and Oregon-Washington). Current population status differs by area which may be valuable information for making management and allocation decisions (Figure iv).

Table ii: Estimated recent trend in spawning output (billions of eggs) and the fraction of unfished spawning output and the 95 percent confidence intervals.

| Year | Spawning output | Lower Interval. | Upper Interval. | Fraction Unfished | Lower Interval | Upper Interval |
|------|-----------------|-----------------|-----------------|-------------------|----------------|----------------|
| 2015 | 293.07          | 233.20          | 352.95          | 0.246             | 0.211          | 0.282          |
| 2016 | 304.71          | 242.84          | 366.58          | 0.256             | 0.220          | 0.293          |
| 2017 | 317.87          | 253.80          | 381.94          | 0.267             | 0.230          | 0.305          |
| 2018 | 331.52          | 265.00          | 398.05          | 0.279             | 0.240          | 0.317          |
| 2019 | 347.17          | 277.86          | 416.48          | 0.292             | 0.252          | 0.332          |
| 2020 | 364.48          | 292.01          | 436.95          | 0.306             | 0.265          | 0.348          |
| 2021 | 384.48          | 308.44          | 460.51          | 0.323             | 0.280          | 0.366          |
| 2022 | 406.33          | 326.37          | 486.28          | 0.341             | 0.296          | 0.387          |
| 2023 | 428.74          | 344.53          | 512.94          | 0.360             | 0.313          | 0.408          |
| 2024 | 452.09          | 363.37          | 540.81          | 0.380             | 0.330          | 0.430          |
| 2025 | 477.63          | 384.18          | 571.08          | 0.401             | 0.349          | 0.453          |

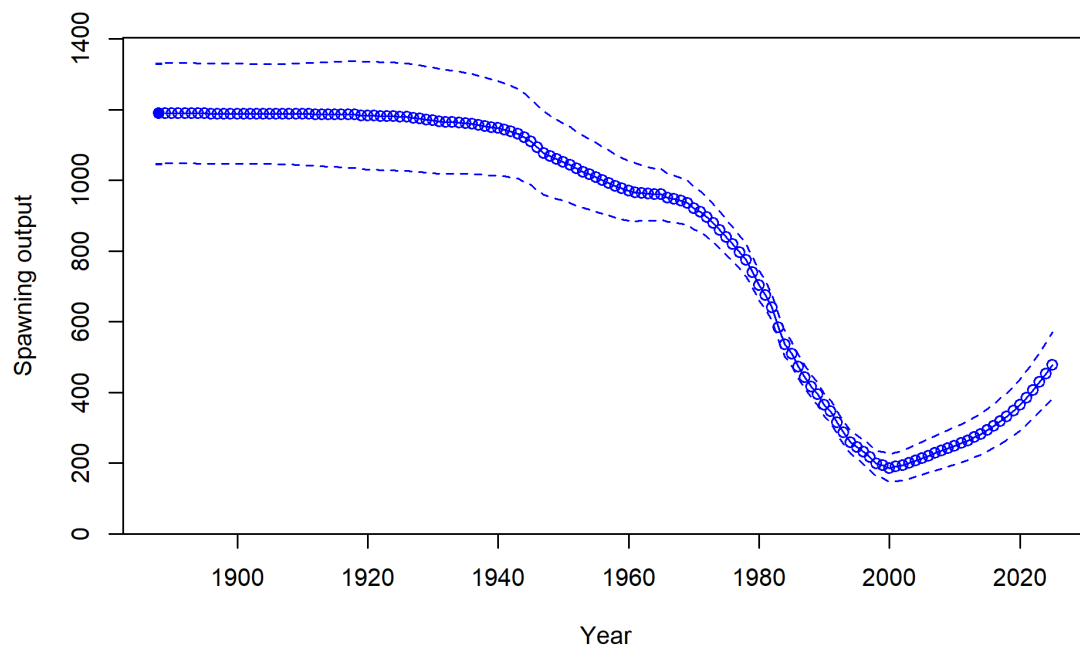


Figure ii: Time series of estimated spawning output (billions of eggs) for the base model (circles) with ~ 95% interval (dashed lines).

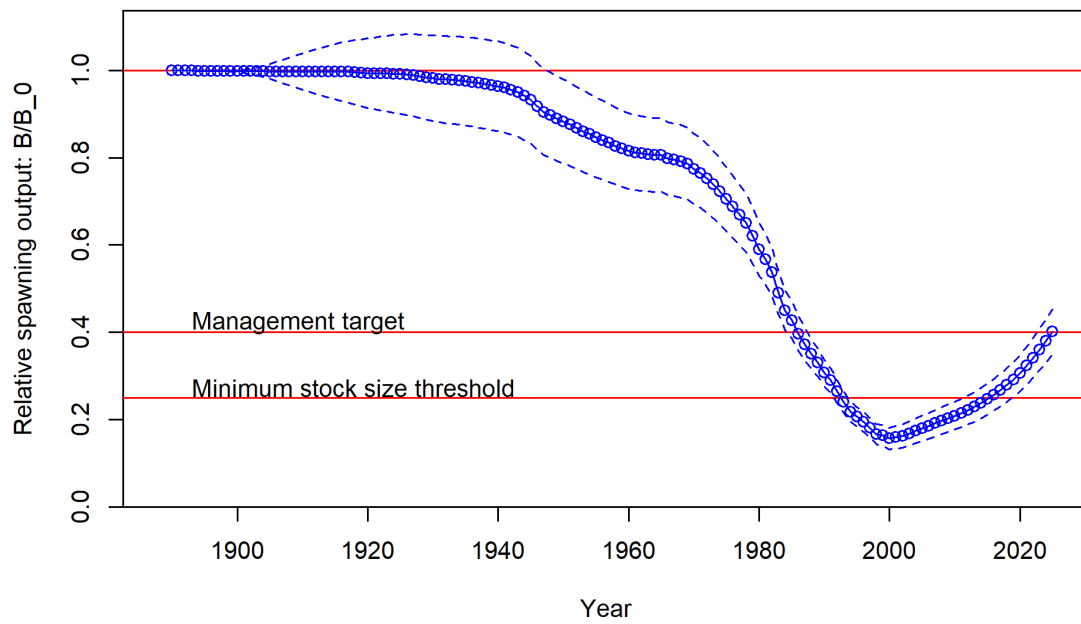


Figure iii: Time series of estimated relative spawning output for the base model.

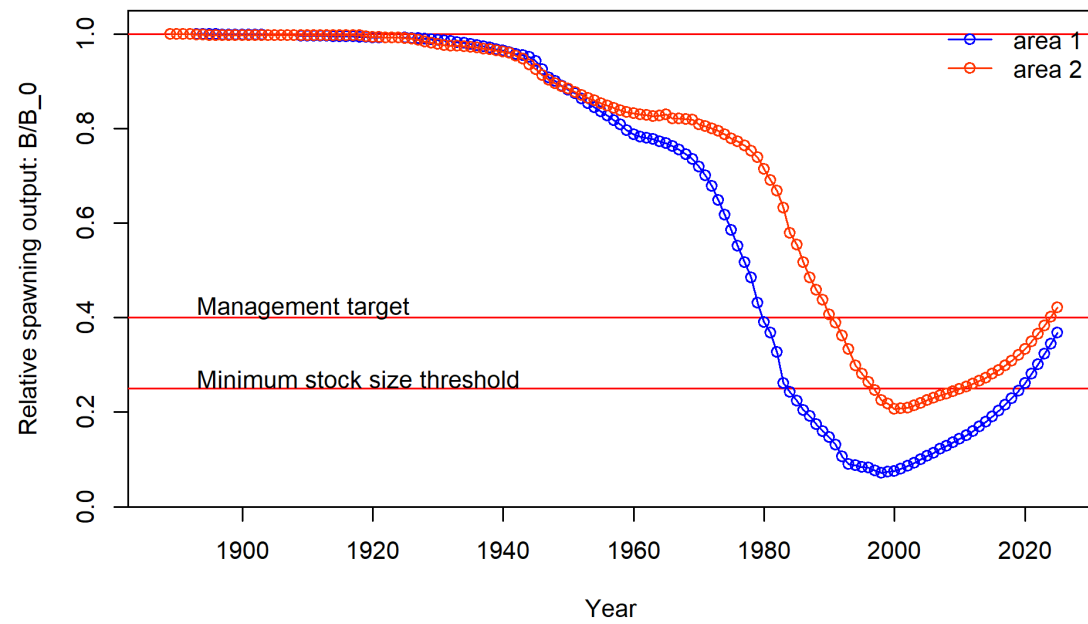


Figure iv: Time series of relative spawning output estimated by area (area 1= California, area 2 = Oregon and Washington).

## Recruitment

The largest estimated recruitment events were in 1971, followed by more recently, in 2013 and 2008 (Figure v, Figure vi, Table iii). Trends in recruitment are largely consistent with the previous assessment, apart from the most recent elevated time period that is more informed with additional length and age composition data. Recruits for this assessment appear to have extended this more recent time period starting in 2005, with peaks in 2008 and 2013, and lower recruitment in 2017.

Table iii: Estimated recent trend in recruitment (1,000s) and recruitment deviations and the 95 percent confidence intervals.

| Year | Recruit-<br>ment<br>(1,000s) | Lower<br>Interval<br>(1,000s) | Upper<br>Interval<br>(1,000s) | Recruit-<br>ment<br>Deviations | Lower<br>Interval | Upper<br>Interval |
|------|------------------------------|-------------------------------|-------------------------------|--------------------------------|-------------------|-------------------|
| 2015 | 359                          | 200                           | 643                           | 0.729                          | 0.153             | 1.306             |
| 2016 | 242                          | 126                           | 467                           | 0.315                          | -0.347            | 0.977             |
| 2017 | 121                          | 56                            | 259                           | -0.404                         | -1.193            | 0.386             |
| 2018 | 115                          | 53                            | 250                           | -0.472                         | -1.276            | 0.331             |
| 2019 | 118                          | 54                            | 262                           | -0.467                         | -1.291            | 0.357             |
| 2020 | 117                          | 51                            | 267                           | -0.501                         | -1.364            | 0.362             |
| 2021 | 153                          | 64                            | 364                           | -0.259                         | -1.172            | 0.655             |
| 2022 | 174                          | 70                            | 429                           | -0.154                         | -1.107            | 0.800             |
| 2023 | 179                          | 72                            | 445                           | -0.141                         | -1.102            | 0.819             |
| 2024 | 209                          | 82                            | 531                           | 0.000                          | -0.980            | 0.980             |

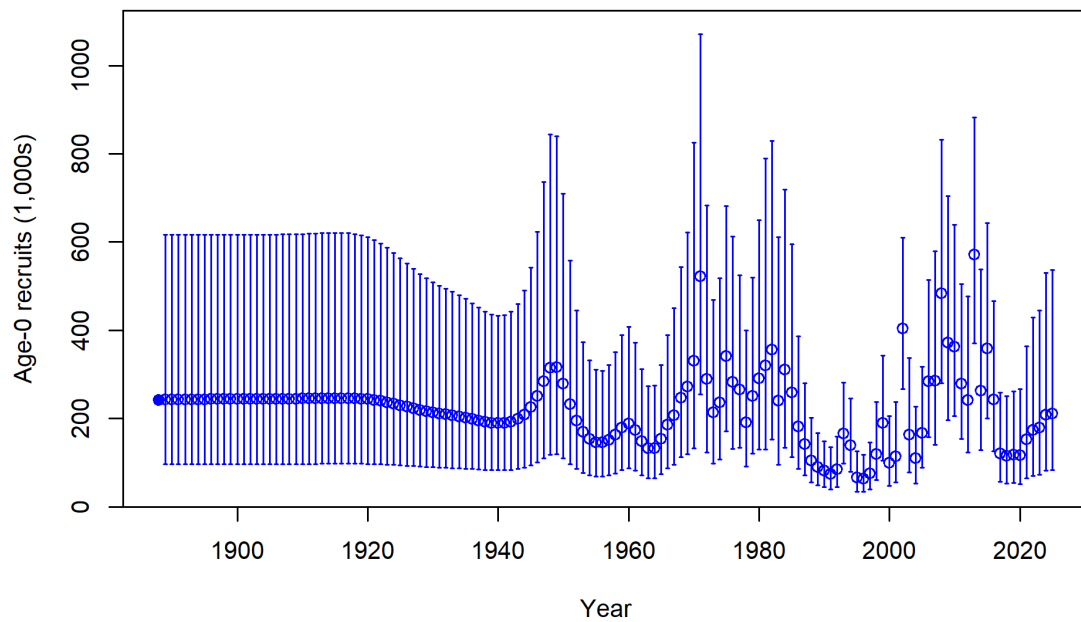


Figure v: Time series of estimated yelloweye rockfish recruitments for the base model (circles) with approximate 95% intervals (vertical lines).

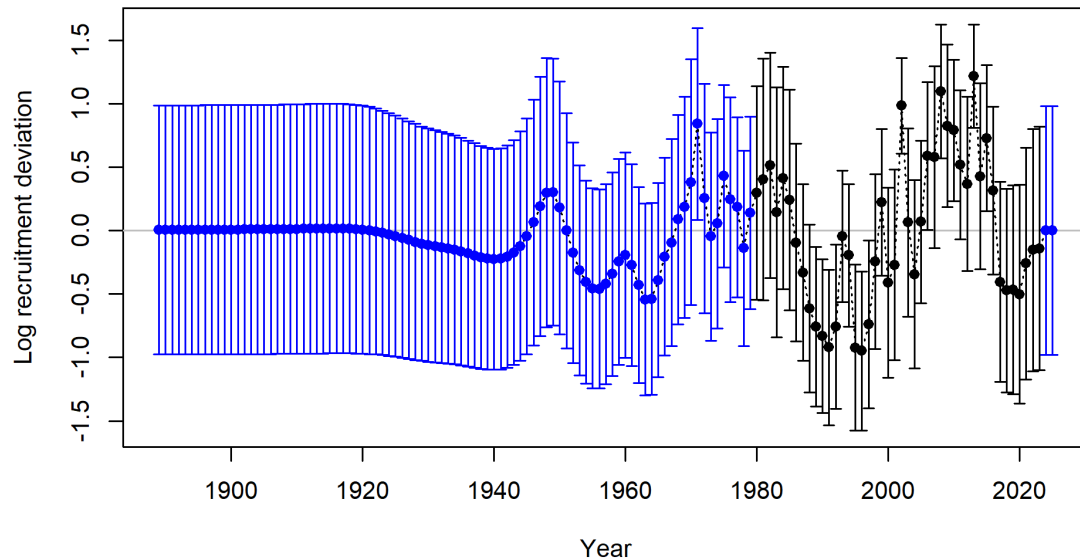


Figure vi: Estimated recruitment deviations with 95% intervals.

### Exploitation Status

This assessment estimates that the stock of Yelloweye Rockfish off the continental U.S. Pacific Coast is currently at 40.1% of its unexploited level. This is above the overfished threshold of  $SO_{25\%}$ , and slightly above the management target  $SO_{40\%}$  of unfished spawning output. Fishing intensity increased throughout the 1900s as the stock was fished down, until stabilizing at peak intensity between the mid-1980s and late 1990s and substantially decreasing in the late 1990s and early 2000s, around the time the stock was declared overfished. Fishing intensity has since been relatively stable (Figure [vii](#), Table [iv](#)).

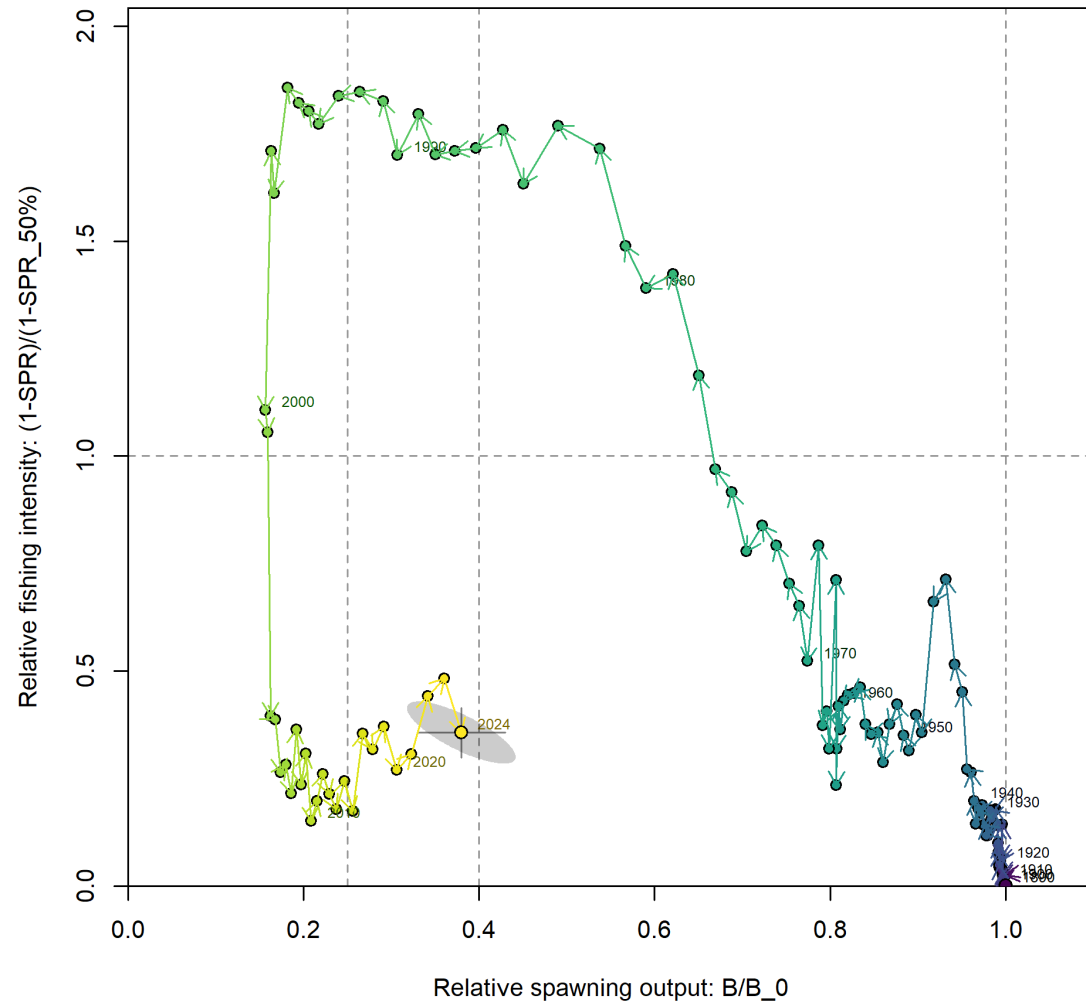


Figure vii: Phase plot of fishing intensity versus fraction unfished. Each point represents the biomass ratio at the start of the year and the relative fishing intensity in that same year. Lines through the final point show 95% intervals based on the asymptotic uncertainty for each dimension. The shaded ellipse is a 95% region which accounts for the estimated correlation between the two quantities.



Table iv: Estimated recent trend in relative fishing intensity  $(1-SPR)/(1-SPR_{50\%})$ , where SPR is the spawning potential ratio, and the exploitation rate, along with the 95 percent confidence intervals for both quantities.

| Year | $(1-SPR)/(1-SPR_{50\%})$ | Lower Interval (SPR) | Upper Interval (SPR) | Exploitation Rate | Lower Interval (Rate) | Upper Interval (Rate) |
|------|--------------------------|----------------------|----------------------|-------------------|-----------------------|-----------------------|
| 2015 | 0.244                    | 0.201                | 0.286                | 0.004             | 0.003                 | 0.005                 |
| 2016 | 0.174                    | 0.143                | 0.204                | 0.003             | 0.002                 | 0.003                 |
| 2017 | 0.355                    | 0.296                | 0.414                | 0.006             | 0.005                 | 0.007                 |
| 2018 | 0.317                    | 0.264                | 0.370                | 0.005             | 0.004                 | 0.006                 |
| 2019 | 0.371                    | 0.310                | 0.431                | 0.006             | 0.005                 | 0.007                 |
| 2020 | 0.270                    | 0.225                | 0.314                | 0.004             | 0.004                 | 0.005                 |
| 2021 | 0.306                    | 0.255                | 0.358                | 0.005             | 0.004                 | 0.006                 |
| 2022 | 0.441                    | 0.372                | 0.509                | 0.007             | 0.006                 | 0.009                 |
| 2023 | 0.482                    | 0.408                | 0.555                | 0.008             | 0.007                 | 0.010                 |
| 2024 | 0.357                    | 0.299                | 0.414                | 0.006             | 0.005                 | 0.007                 |

### Ecosystem Considerations

No ecosystem or environmental data was used in the previous Yelloweye Rockfish assessment and no new data were considered for this update assessment.

### Reference Points

A list of estimates of the current state of the population, as well as reference points based on 1) a target unfished spawning output of 40%, 2) a spawning potential ratio of 0.5, and 3) the model estimate of maximum sustainable yield, are all listed in Table v. Unfished spawning stock output for Yelloweye Rockfish was estimated to be 1190 billion eggs (95% confidence interval: 1,048.1 - 1,331.9 billion eggs). The management target for Yelloweye Rockfish is defined as 40% of the unfished spawning output ( $SO_{40\%}$ ), which is estimated by the model to be 476 billion eggs (95% confidence interval: 419 - 533), which corresponds to an exploitation rate of 0.026. This harvest rate provides an equilibrium yield of 122 mt at  $SO_{40\%}$  (95% confidence interval: 108 - 137 mt).

Table v: Summary of reference points and management quantities, including estimates of the 95 percent confidence intervals. SO is spawning output, SPR is the spawning potential ratio, and MSY is maximum sustainable yield.

| Reference Point                                | Estimate | Lower Interval | Upper Interval |
|------------------------------------------------|----------|----------------|----------------|
| Unfished Spawning output                       | 1,190.0  | 1,048.1        | 1,331.9        |
| Unfished Age 8+ Biomass (mt)                   | 10,331   | 9,101          | 11,561         |
| Unfished Recruitment (R0)                      | 242      | 213            | 271            |
| 2025 Spawning output                           | 478      | 384            | 571            |
| 2025 Fraction Unfished                         | 0.401    | 0.349          | 0.453          |
| Reference Points Based SO40%                   | —        | —              | —              |
| Proxy Spawning output SO40%                    | 476      | 419            | 533            |
| SPR Resulting in SO40%                         | 0.458    | 0.458          | 0.458          |
| Exploitation Rate Resulting in SO40%           | 0.026    | 0.026          | 0.027          |
| Yield with SPR Based On SO40% (mt)             | 122      | 108            | 137            |
| Reference Points Based on SPR Proxy for MSY    | —        | —              | —              |
| Proxy Spawning output (SPR50)                  | 531      | 468            | 594            |
| SPR50                                          | 0.500    | —              | —              |
| Exploitation Rate Corresponding to SPR50       | 0.023    | 0.023          | 0.023          |
| Yield with SPR50 at SO SPR (mt)                | 117      | 103            | 131            |
| Reference Points Based on Estimated MSY Values | —        | —              | —              |
| Spawning output at MSY (SO MSY)                | 343      | 303            | 384            |
| SPR MSY                                        | 0.358    | 0.356          | 0.359          |
| Exploitation Rate Corresponding to SPR MSY     | 0.037    | 0.036          | 0.037          |
| MSY (mt)                                       | 128      | 113            | 143            |

## Management Performance

Recent trends in total catch relative to management guidelines is available in Table vi and shows that total catch of Yelloweye Rockfish has remained below both the overfishing limit (OFL) and annual catch limit (ACL) in each year since the previous assessment. Catch in Table vi combines the two areas in this model as catch limits for Yelloweye Rockfish are managed as a single coast wide unit and includes both landings and estimated discard mortality.

Table vi: Recent trend in the overfishing limits (OFL), the acceptable biological catches (ABCs), the annual catch limits (ACLs), and the total dead catch (landings + discards) all in metric tons (mt).

| Year | OFL (mt) | ABC (mt) | ACL (mt) | Catch (mt) |
|------|----------|----------|----------|------------|
| 2015 | 52       | 43       | 18       | 12         |
| 2016 | 52       | 43       | 19       | 9          |
| 2017 | 57       | 47       | 20       | 20         |
| 2018 | 58       | 48       | 20       | 19         |
| 2019 | 82       | 74       | 48       | 23         |
| 2020 | 84       | 77       | 49       | 18         |
| 2021 | 97       | 83       | 50       | 21         |
| 2022 | 98       | 83       | 51       | 34         |
| 2023 | 123      | 103      | 66       | 39         |
| 2024 | 123      | 103      | 66       | 29         |

## Harvest Projections

This section will be updated after SSC GFSC review.

Table vii: Potential OFLs (mt), ABCs (mt), ACLs (mt), the buffer between the OFL and ABC, estimated spawning output, and fraction of unfished spawning output with adopted OFLs and ACLs and assumed catch for the first two years of the projection period.

| Year | Adopted<br>OFL (mt) | Adopted<br>ACL (mt) | Assumed<br>Catch (mt) | OFL (mt) | Buffer | ABC (mt) | ACL (mt) | Spawning<br>output | Fraction<br>Unfished |
|------|---------------------|---------------------|-----------------------|----------|--------|----------|----------|--------------------|----------------------|
| 2025 | 105.8               | 55.8                | 45.7                  | —        | —      | —        | —        | 477.630            | 0.401                |
| 2026 | 108.3               | 56.6                | 46.4                  | —        | —      | —        | —        | 502.268            | 0.422                |
| 2027 | —                   | —                   | —                     | 129.3    | 0.873  | 112.9    | 112.9    | 526.772            | 0.443                |
| 2028 | —                   | —                   | —                     | 130.9    | 0.864  | 113.1    | 113.1    | 543.555            | 0.457                |
| 2029 | —                   | —                   | —                     | 132.0    | 0.856  | 113.0    | 113.0    | 558.406            | 0.469                |
| 2030 | —                   | —                   | —                     | 132.7    | 0.848  | 112.6    | 112.6    | 570.839            | 0.480                |
| 2031 | —                   | —                   | —                     | 133.1    | 0.840  | 111.8    | 111.8    | 580.654            | 0.488                |
| 2032 | —                   | —                   | —                     | 133.3    | 0.832  | 110.9    | 110.9    | 587.932            | 0.494                |
| 2033 | —                   | —                   | —                     | 133.2    | 0.824  | 109.8    | 109.8    | 592.973            | 0.498                |
| 2034 | —                   | —                   | —                     | 133.1    | 0.817  | 108.8    | 108.8    | 596.202            | 0.501                |
| 2035 | —                   | —                   | —                     | 132.9    | 0.809  | 107.6    | 107.6    | 598.069            | 0.503                |
| 2036 | —                   | —                   | —                     | 132.8    | 0.801  | 106.3    | 106.3    | 599.025            | 0.503                |

## Decision Table

This section will be updated after SSC GFSC review.

Table viii: Decision table with 12-year projections. ‘Mgmt’ refers to the management scenario (A) with the default harvest control rule  $P = 0.40$ . In each case the 2025 and 2026 catches are fixed at the estimates provided by the GMT. The catch for the Washington recreational fleet is input in numbers so the GMT estimate was converted from 3.22 metric tons to 1.53 thousands of fish based on a mean weight of 2.105 kg estimated by SS3 for this fleet in 2024. The alternative states of nature (‘Low’, ‘Base’, and ‘High’ as discussed in the text) are provided in the columns, with Spawning Output (‘Spawn’, in billions of eggs) and Fraction of unfished spawning output (‘Frac’) provided for each state.

| Mgmt | Year | Catch | Low<br>Spawn | Low<br>Frac | Base<br>Spawn | Base<br>Frac | High<br>Spawn | High<br>Frac |
|------|------|-------|--------------|-------------|---------------|--------------|---------------|--------------|
| A    | 2025 | 46    | 330.67       | 0.289       | 477.63        | 0.401        | 839.65        | 0.603        |
|      | 2026 | 46    | 347.96       | 0.305       | 502.27        | 0.422        | 878.58        | 0.630        |
|      | 2027 | 113   | 365.20       | 0.320       | 526.77        | 0.443        | 916.84        | 0.658        |
|      | 2028 | 113   | 374.94       | 0.328       | 543.56        | 0.457        | 946.40        | 0.679        |
|      | 2029 | 113   | 383.27       | 0.335       | 558.41        | 0.469        | 972.47        | 0.698        |
|      | 2030 | 113   | 389.86       | 0.341       | 570.84        | 0.480        | 994.20        | 0.713        |
|      | 2031 | 112   | 394.60       | 0.345       | 580.65        | 0.488        | 1011.25       | 0.726        |
|      | 2032 | 111   | 397.56       | 0.348       | 587.93        | 0.494        | 1023.78       | 0.735        |
|      | 2033 | 110   | 398.98       | 0.349       | 592.97        | 0.498        | 1032.35       | 0.741        |
|      | 2034 | 109   | 399.19       | 0.349       | 596.20        | 0.501        | 1037.71       | 0.745        |
|      | 2035 | 108   | 398.51       | 0.349       | 598.07        | 0.503        | 1040.63       | 0.747        |
|      | 2036 | 106   | 397.30       | 0.348       | 599.03        | 0.503        | 1041.85       | 0.748        |

## Scientific Uncertainty

The model estimate of the log-scale standard deviation of the 2025 spawning output is 0.0996. The model estimate of the log-scale standard deviation of the 2025 OFL is 0.095. Each of these are likely underestimates of overall uncertainty due to the necessity to fix several key population dynamics parameters (e.g. steepness and recruitment variance) and also because there is no explicit incorporation of model structural uncertainty (although see the decision table for alternative states of nature).

## Research and Data Needs

Please refer to the 2017 benchmark assessment for a detailed list of research and data needs for Yelloweye Rockfish (Gertseva and Cope (2017)). In addition to those, the following research and recommendations could improve the ability of future stock assessments to determine the status and productivity of the Yelloweye Rockfish population:

- Continue refining the ORFS index analysis and ultimately use either the ORBS or ORFS index to describe the CPUE trends in the Oregon recreational fishery after 2000.
- Expand the IPHC age composition bins to an older maximum age for the IPHC age composition data to spread out the distribution of length data in the oldest age bins for conditional age-at-length.

- Explore potential indices of abundance in untrawlable areas.

### Rebuilding Projections

This section will be updated after SSC GFSC review.

## 1 Introduction

Yelloweye Rockfish (*Sebastes ruberrimus*) are found from the Gulf of Alaska to northern Baja California in Mexico across the northeastern Pacific Ocean (Hart 1973; Love, Yoklavich, and Thorsteinson 2002). Their core distribution is from southeast Alaska to central California on the west coast of the United States (Love, Yoklavich, and Thorsteinson 2002). Yelloweye Rockfish in Puget Sound are considered isolated from the coastal waters population (I. J. Stewart, Wallace, and McGilliard 2009) and have been listed as threatened under the Endangered Species Act since 2010 (Drake et al. 2010).

Yelloweye Rockfish are strongly associated with rocky bottom habitat, particularly areas of high relief (Love, Yoklavich, and Thorsteinson 2002), and adults are considered to be solitary and sedentary after settlement (Coombs 1979; DeMott 1983). However, new tagging studies suggest that adult Yelloweye Rockfish exhibit larger scale movement patterns more commonly than previously considered (Hannah and Rankin 2011; Rasmuson et al. 2025).

There has been little advancement on information pertaining to the stock structure of Yelloweye Rockfish since the previous benchmark assessment. As noted in Gertseva and Cope (2017), there is evidence of genetic differences between Canadian waters (Strait of Georgia) and West coast coastal populations of Yelloweye Rockfish, but no evidence of differentiation across coastal populations (Siegle et al. 2013). Gao et al. (2010) found that there was complete mixing of offspring from Oregon and Washington waters using otolith isotope analyses, indicating a single spawning stock in this portion of the Yelloweye Rockfish stock. Given the general perception of the sedentary nature of Yelloweye Rockfish adults and the moderate amount of mixing that occurs during the pelagic larval stage, the previous Yelloweye Rockfish assessment modeled the West coast population as a two-area assessment (California and a combined Oregon-Washington area) with a common stock recruitment relationship (Gertseva and Cope 2017). This update assessment maintains this basic structure.

### 1.1 Life History

This section is not required for an update assessment; please refer to the most recent full assessment (Gertseva and Cope 2017) for additional information.

### 1.2 Ecosystem Considerations

This section is not required for an update assessment; please refer to the most recent full assessment (Gertseva and Cope 2017) for additional information.

### 1.3 Fishery Description

This section is not required for an update assessment; please refer to the most recent full assessment (Gertseva and Cope 2017) for additional information.

### 1.4 Management History

Since the 2017 assessment, catch restrictions for Yelloweye Rockfish have continued, though as the stock recovers, the rebuilding plan has allowed for small catch increases that slightly loosened the constraining impact of this species.

This section is not required for an update assessment; please refer to the most recent full assessment (Gertseva and Cope 2017) and Section 1.5 for additional information.

### 1.5 Management Performance

Yelloweye Rockfish removals have been substantially reduced since its designation as overfished in 2002 through a variety of management measures that eliminated retention in recreational fisheries, limited commercial retention, created broad spatial closures, and implemented new gear restrictions that reduced trawling in rocky habitats. Many of these restrictions remain in effect, though as Yelloweye Rockfish and other groundfish stocks have begun to rebuild, some management measures have been modified or removed in recent years. These include some additional allocations to recreational fisheries that remain constrained by Yelloweye Rockfish estimated discard mortality, the recent removal of the Yelloweye Rockfish Conservation Area (RCA) for the trawl sector off of California and Oregon, and eliminating some gear restrictions in the RCAs for the non-trawl sector.

Recent trends in total catch relative to management guidelines are available in Table 1 and show that total catch of Yelloweye Rockfish has remained below both the OFL and ACL in each year since the previous assessment. Catch in Table 1 combines the two areas in this model as catch limits for Yelloweye Rockfish are managed as a single coast wide unit and includes both landings and estimated discard mortality. As in the previous assessment, total catches for each fleet in this update include both landings and estimated dead discard mortality.

### 1.6 Fisheries off Canada and Alaska

This section is not required for an update assessment; please refer to the most recent full assessment (Gertseva and Cope 2017) for additional information.



## 2 Data

A summary of available data by type and fleet used in the Yelloweye Rockfish assessment is available in Figure 1. Data that have changed or been added since the previous 2017 assessment are summarized below. No new data sources were considered in this update assessment.

### Removals:

- Post-2016 landings and discards were added for all three states for the commercial and recreational fleets.
- A new Oregon historical recreational catch reconstruction was incorporated, which covered 1979 - 2000.
- A new Washington historical recreational catch reconstruction was provided by Washington Department of Fish and Wildlife (WDFW) and included changes to data from 1990 - 2016.

### Composition Data:

- Length and age composition data were added from 2017 - 2024 for all states for the commercial and recreational fleets.
- Length and age composition data were also extended for the WCGBTS and the IPHC Longline survey.
- Some length and age composition data from the 2017 assessment had minor errors in how sample numbers were calculated, ageing error assignment, or doubled age samples and thus needed to be fixed. See Section 2.1.2 below.

### Indices of Abundance:

- Indices that were updated with more recent data and/or updated methodology include:
  - Oregon Onboard Observer (2001 - 2024)
  - Oregon Oregon Department of Fish and Wildlife Oregon Recreational Boat Survey (ORBS) Dockside (release only) (2004-2024)
  - WCGBTS (2003 - 2024)
  - IPHC Longline Survey (2002 - 2024)

### Biological Data:

- Length-weight relationship parameters were updated to include all the recent (2017 - 2024) WCGBTS data.
- Ageing error matrices were unchanged but some Oregon recreation ages were assigned the wrong ageing error in the 2017 assessment and were corrected based on ODFW recommendations.

### 2.1 Fishery-Dependent Data

Updated fishery-dependent data, including removals, length and age compositions, and indices of abundance are detailed below.

### 2.1.1 Landings

A summary of total removals are provided in Table 2 and Figure 2.

Recent commercial landings (2017 - 2024) were obtained from [Pacific Fisheries Information Network \(PacFIN\)](#) for California, Oregon and Washington. For the period from 2016 through 2023, updated West Coast Groundfish Observer Program (WCGOP) discard estimates were added to PacFIN landings by adding the annual dead discard mortality rate for the commercial sector in the Groundfish Expanded Mortality Multi-Year (GEMM) recorded discards to obtain the total catch of Yelloweye Rockfish within commercial fleets. At this time no WCGOP data were available for 2024, so we used the average total discard for 2021 - 2023 to approximate total 2024 discards for each commercial fleet.

Bycatch for the At-Sea Pacific Hake fleet (A-SHOP) was updated from 2017 through 2024.

Recreational removals from [Recreational Fishery Information Network \(RecFIN\)](#) were updated for California, Oregon and Washington from 2017 - 2024. RecFIN removals include an estimate of discard mortality and represent total estimated removals. The Oregon Department of Fish and Wildlife (ODFW) provided updated historical recreational removals for Oregon from 1979 through 2000 (Whitman 2024). The Washington Department of Fish and Wildlife (WDFW) provided updated historical recreational removals (1967 - 1989) and WDFW Ocean Sampling Program (OSP) estimates (1990 - 2001). The historical recreational removals for 1971, 1974, and 1979 were not available and were filled in as the average of the two preceding and two following years. Historical data were filtered to marine catch areas 1-4. For OSP catch estimates, data included marine catch areas 1-4, up to the Bonilla-Tatoosh line. WDFW also provided updated catch estimates for 2002 - 2004, which did not include discard mortality. To adjust for this, we multiplied the average discard mortality rate from the following five years (2005 - 2009) by the total discards to calculate total mortality for those years.

### 2.1.2 Fishery-Dependent Length and Age Compositions

Updated length composition data for commercial catches (trawl and non-trawl) were available from PacFIN (extracted April 4, 2025) and from WCGOP for all three states. These include the years 2017 - 2024 for PacFIN data and 2017 - 2023 for WCGOP data. Updated recreational length composition data were available from RecFIN (extracted April 4, 2025) for all three states, and include years 2017 - 2024. Additionally, updated length compositions from the California On-Board CPFV Observer Sampling Program and from the Ocean Recreational Fishery Survey (ORFS, previously the Oregon onboard recreational observer program), both of which measure fish discarded at sea, were also available up through 2024 on RecFIN.

New commercial age composition data from PacFIN and WCGOP for 2017 - 2024 was included for Oregon and Washington. No new commercial age data were available from California. New recreational age composition data was available from RecFIN from 2017 - 2024 for Washington only (extracted May 13, 2025). These data were collected in the OSP. There were also some historical updates to Oregon and Washington recreational age data provided by the state representatives.

In addition to extending the length data time series, we also fixed minor data errors found in the 2017 assessment. For length composition data, years with small samples sizes ( $N = 1$ ) were excluded. There

were no changes in how commercial length sample numbers were calculated. However, for all recent recreational fleet length data, the total number of trips information used to calculate the number of samples was not available. Using data from the 2017 assessment, we built fleet-specific linear regressions to approximate the relationship of samples to the number of fish. Then, we applied that regression to the total number of fish for data between 2017 and 2024 to estimate the number of samples. A future benchmark assessment should investigate how to get the number of sampled trips from RecFIN to calculate the number of samples using the Stewart and Hamel (2014) method.

We also found that conditional age-at-length (CAAL) data from the 2017 assessment had all sample sizes and relative proportions doubled, potentially from when Yelloweye Rockfish was changed from a two-sex to single-sex model. For most fleets this was not a problem because the proportions of age-at-length counts were the same, however, some of the commercial fleets included discard age proportions that were not doubled, leading to small differences in proportions. To fix the CAAL data so it accurately represented the number of fish in each age class, we either rebuilt the entire fleet's CAAL data frame using the most recently pulled information from PacFIN and RecFIN, or divided the number of samples or relative proportions in each length bin by two. How these problems were treated for each fleet specifically is detailed below, including other minor data changes. Otherwise, length and age composition data are unchanged from the previous assessment; please refer to the most recent benchmark assessment (Gertseva and Cope 2017) for additional information.

#### 2.1.2.1 Fleet Specific Changes in the Compositional Data

##### Fleet 2. California Non-Trawl:

- For ages, all the CAAL and marginal ages (used to explore fits only, not included in the likelihood) data were recalculated using the most recent age data pulled from PacFIN and WCGOP, to account for age doubling in 2017.

##### Fleet 3. California Recreational:

- CAAL data for 1979-1984 were doubled, so the number of samples and age-at-length proportions were divided by two.
- CAAL data for 2009-2016 were doubled, but the raw data we received from RecFIN were correct, without doubled ages, so this time series was replaced with newly pulled data.
- We then re-built the marginal age data from the updated CAAL for both time periods because there were errors in previous data entry and sample number calculations.

##### Fleet 4 & 5. Oregon/Washington Trawl & Non-Trawl:

- Both the OR/WA commercial fleets had all CAAL and marginal age (not included in the likelihood) data recalculated using the most recent age data pulled from PacFIN and WCGOP, to account for age doubling in 2017.

##### Fleet 6. Oregon Recreational:

- CAAL data sample sizes and proportions were doubled so numbers from 1979 - 2017 were divided by two.
- We included 2015 unsexed ages.
- We also reassigned the aging error for this fleet for the correct years. The ODFW data representative confirmed that all fish from 1979 - 2002 were aged by WDFW (ageing error 1), and fish from 2009 - 2016 were aged by the NWFSC (ageing error 2). No new ages since 2016 were provided. Marginal data were then recalculated from the updated CAAL so that the ageing error labels and number of samples matched.

#### Fleet 7. Washington Recreational:

- All age data from 1998 to 2024 were replaced with the most recent data provided in RecFIN, following the recommendation of the WDFW representative. CAAL and the marginal age data were calculated using this data.

#### 2.1.3 Indices of Abundance

Two fishery-dependent indices of abundance were updated with new data and up-to-date methodology. These are detailed below. Otherwise, indices of abundances are unchanged from the previous assessment; please refer to the most recent benchmark assessment (Gertseva and Cope 2017) for additional information.

##### 2.1.3.1 Oregon Onboard Observer CPUE, 2001 -- 2024

The Oregon Onboard Observer (now Ocean Recreational Fisheries Survey, or ORFS) index was updated from the previous Yelloweye Rockfish assessment, and updated drift-level catch-per-unit-effort data were obtained from ODFW through the end of 2024. The database contains information on catch by species (number of retained and released fish), effort (angler hours), sample depth, and bag limits and other relevant regulations (Monk et al. 2013).

The unfiltered data set contained 18,410 drifts. Multiple standardized filters are applied to remove outliers and data unsuitable for an index. These filters are very similar to filters applied in 2017 and include removing drifts without data needed for CPUE information, long drifts (above 95th percentile), drifts in deeper waters (more than 64fm, 99th percentile), drifts that were targeting primarily mid-water species, and drifts outside of the legal fishing depth (with a five fathom buffer). Additionally, years with extremely low sample sizes (< 50) were excluded. Finally, drifts on charters from Port Orford were removed due to small sample sizes. Years that were excluded due to small sample sizes include 2003, 2016, and 2019. The final filtered data set included 6,839 trips with a 6.1% encounter rate for Yelloweye Rockfish (Table 3).

Covariates evaluated included year, month, port, the open depths to fishing (all depths or inside 20/30/40fm), and a five fm-binned depth of drift covariate. This is in contrast to the 2017 index, which was only able to evaluate a year covariate. The covariates listed above are standard to evaluate for this index in other assessments. Negative binomial models were fit using [Species Distribution Models with Template Model Builder](#) (`sdmTMB`) version 0.6.0 (Anderson et al. 2024) to the drift-level data (catch with a log offset for adjusted angler hours). A model without the open fishing depths or month was selected as the best fit model by AIC (Table 4). Acceptable diagnostics for the model were achieved, as

evidenced by passing the sanity function in `sdmTMB` (Figure 3). A comparison to the ORFS index used in the previous assessment indicates that despite the change in modeling approach and the covariates included, most years overlap between the two indices and similar trends are observed (Figure 4). For the two years that differed significantly, 2001 was a pilot study year and 2013 had high variance. The 2017 assessment stated that the previous modeling approach struggled with convergence, whereas this `sdmTMB` model from 2024 has multiple checks and passed all internal package diagnostics. Thus, the updated index has reduced within-year variance and a lack of extreme swings in the standardized index value relative to the index from 2017.

#### 2.1.3.2 Oregon ORBS Dockside (release only) CPUE, 2004-2024

The ORBS dockside index for Oregon was updated for this assessment. CPUE, expressed in terms of fish per angler-hour, was calculated by multiplying the number of anglers and the total trip time, minus the boat type-specific travel time. The database contains information on released fish by species (number of angler-reported released fish), effort (angler hours), sample location (port where data were collected), date, bag limits and other relevant regulations, boat type (charter or private), and trip type (e.g., bottom associated fish).

The unfiltered data set contained 504,128 trips from 2001 - 2024. Since the previous Yelloweye assessment, multiple data filters have been standardized, which are very similar to the 2017 assessment, and are applied to ORBS trip-level data to remove outliers and data unsuitable for an index. For this index, the time period was restricted to years when retention of Yelloweye Rockfish was prohibited, which began on January 1, 2004. There were two differences in the filtering in this updated index. First, the previous index began in 2005, which was determined to be an error in the timing of the implementation of prohibited status for Yelloweye. Given that prohibition was in effect on January 1, the year 2004 is included in this updated index. The second difference in filtering is the elimination of the Stephens-MacCall filter in the updated index. This filter has not been used for several assessment cycles, based on a recommendation from NWFSC staff (pers. comm. A. Whitman, ODFW). The final dataset included 133,039 trips from 2004 – 2024 with an overall encounter rate of 7.4% (Table 5).

Covariates evaluated included year, month, port, the open depths to fishing (all depths or inside 20/30/40 fm), and boat type. These are the same covariates evaluated in the 2017 ORBS index, apart from the open depths of the fishery. The final model in 2017 included boat type, port, and year. Negative binomial models were fit in `sdmTMB` (Version 0.6.0) to the trip-level data (catch with a log offset for adjusted angler hours). The final model selected includes year, month, port, boat type and open fishery depths, which was the best fit model by AIC in this series (Table 6). Acceptable diagnostics for the model were achieved, as evidenced by passing the sanity function in `sdmTMB` (Figure 5). The index of abundance for the Oregon recreational fleet, including both Marine Recreational Fisheries Statistics Survey (MRFS) (1980 - 1999) and ORBS (2004 - 2024) indices is shown in Figure 6. ODFW no longer maintains the deltaGLM code that was used to develop the 2017 index and so the index was updated to use the currently accepted modeling approach for Pacific Fishery Management Council (PFMC) groundfish assessments (`sdmTMB`, version 0.6.0) (Anderson et al. 2024). To bridge this change, the 2017 model index structure was applied to the current data set using `sdmTMB` and compared to the deltaGLM index used in the 2017 assessment and the current recommended updated index in Figure 7. There are some differences observed in 2005 – 2009 between the deltaGLM index and the two `sdmTMB` indices; however, this appears to be largely driven by the updated modeling approach.

## 2.2 Fishery-Independent Data

Two sources of fishery-independent data were updated: the WCGBTS and the IPHC Longline survey.

### 2.2.1 West Coast Groundfish Bottom Trawl Survey (WCGBTS)

The WCGBTS survey methods are most recently described in detail in Keller, Wallace, and Methot (2017). Geostatistical models of biomass density were fit to survey data from the WCGBTS using [Template Model Builder \(TMB\)](#) (Kristensen et al. 2016) via the `sdmTMB` R package (Anderson et al. 2024) as configured within the `indexwc` R package (Johnson et al. 2025). These models can account for latent spatial factors with a constant spatial Gaussian random field and spatiotemporal deviations to evolve as a random walk Gaussian random field (Thorson et al. 2015). Delta-gamma and delta-lognormal distributions were investigated. Results are only shown for the model that led to the best model diagnostics, defined as similar distributions of theoretical normal quantiles and model quantiles (Figure 8), high precision, lack of extreme predictions, and low Akaike information criterion (AIC). Estimates of biomass from this best model were predicted using a grid based on available survey locations.

The final model used a delta model with a lognormal distribution for the catch-rate component. A logit-link was used for encounter probability and a log-link for positive catch rates. The response variable was catch (mt) with an offset of area swept ( $\text{km}^2$ ) to account for differences in effort. Fixed effects were estimated for each year and pass. The index was estimated for the area north of 42 degrees North (Oregon and Washington) to be consistent with the previous assessment. The data were truncated to depths shallower than 325 m prior to modeling given that there were zero positive encounters in depths deeper than 325 m. The prediction grid was also truncated to only include available survey locations in depths between 55-325 m to limit extrapolating beyond the data and edge effects. Spatial variation was included in the encounter probability and the positive catch rate model. Spatial variation was approximated using 200 knots, where more knots led to non-estimable standard errors because the positive encounters are too sparse to support the dense spatial structure. Anisotropy was not estimated.

The biomass estimates produced for this assessment using `sdmTMB` are comparable to the biomass estimates produced in the previous benchmark assessment (Figure 9). The index is relatively flat with a peak in 2014, but variation is high throughout the time series.

### 2.2.2 IPHC Setline Survey

The IPHC has conducted an annual longline survey for Pacific halibut off the coast of Oregon and Washington (IPHC area “2A”) since 1997 (no surveys were performed in 1998 or 2000). Beginning in 1999, this has been a fixed station design, with roughly 1,800 hooks deployed at each of 84 locations. Before 1999, station locations were not fixed, and thus, those years are not used in the index. Rockfish bycatch, primarily Yelloweye, were recorded during this survey, although values for 1999 and 2001 are estimates based on sub-sampling the first 20 hooks of each 100-hook skate. The gear used to conduct this survey, while designed to sample Pacific Halibut, is similar to gear previously used in line fisheries targeting adult Yelloweye Rockfish. Some variability in sampling location is unavoidable due to wind and currents affecting gear deployment. This can result in different habitats accessed at each fixed location between years. The number of skates used can also differ from year to year; skates hauled (i.e., 100 hooks/skate) are thus used as the unit of effort for all years. This has been the standard effort used in past Yelloweye Rockfish stock assessments.

New to this assessment is the consideration of eight additional survey stations (1527 to 1534) conducted in a collaborative effort between IPHC and WDFW from 2007 - 2009, 2013 - 2019 and 2021 - 2023. These stations are set around IPHC station 1082 due to high capture probability of Yelloweye. Only summer months are considered to match the time of year sampled by the IPHC survey. Survey sets at the WDFW stations used three skates with 100 hooks each for most years, except for 2021 - 2023, where a total of four skates were used. Like the IPHC survey, effort was standardized to 100 hooks/skate. These stations were integrated into the IPHC stations when calculating the index of abundance. The full survey used in this assessment combined all stations in Oregon and Washington into a single index. Data were first filtered to remove all depths with few or no encounters, and then we excluded stations that rarely encountered Yelloweye Rockfish (averaging less than one encounter a year). This left a total of 11 stations for analysis. Both filtering levels increased the percentage of encounters from an initial 11% to 80%.

A log-normal generalized linear model with a log link using `sdmTMB` (Anderson et al. 2024) was used to standardize the CPUE. Model selection using AICc was conducted to select which variables were included in the model. The final model included year, station, and depth as explanatory variables. Acceptable diagnostics for the model were achieved, as evidenced by passing the sanity function in `sdmTMB`. We compared the updated 2025 index to the 2017 index and found no change in trends (Figure 10).

### 2.2.3 Fishery-Independent Length and Age Compositions

Updated length and age composition data were available for the two updated fishery-independent surveys. Composition data from 2017 through 2024 were updated for WCG BTS and were obtained using functions from the `nwfscSurvey` R package (Wetzel, Johnson, and Hicks 2025). The IPHC survey compositional data were provided by WDFW.

A summary of sampling efforts (number of hauls and number of individual fish) in both surveys is provided in Table 7 and Table 8. Updated year-specific length frequency distributions generated for each survey are shown in Figure 11 and Figure 12, respectively. Updated year-specific CAAL frequencies for each survey are shown in Figure 13 and Figure 14 for the WCG BTS and Figure 15 and Figure 16 for the IPHC.

## 2.3 Biological Parameters and Data

Several biological parameters used in the assessment were estimated outside the model or obtained from literature. Their values were treated in the model as fixed, and therefore uncertainty reported for the stock assessment results does not include any uncertainty in these quantities (however, some were investigated via sensitivity analyses described later in this report). These parameters include length-weight relationship parameters, maturity and fecundity parameters, natural mortality, and ageing error. Ageing error matrices were unchanged. The methods used to derive these parameters in the assessment are described below.

### 2.3.1 Length-Weight Relationships

The parameters for the length-weight relationship were updated to include the most recent WCG BTS data from 2017 - 2024. Length-weight curves were fitted using this equation:

$$W = 7.18331 \cdot 10^{-6} L^{3.2448}$$

Where  $W$  is individual weight (kg) and  $L$  is total natural length (cm) (Figure 17).

### 2.3.2 Maturity

The length at maturity relationship remained unchanged from the 2017 assessment. They used a functional maturity approach to assess individual maturity and account for possible false spawning events (Gertseva and Cope 2017). Figure 18 shows the logistic curve applied in this assessment.

### 2.3.3 Fecundity

The fecundity-at-length relationship also remained unchanged from the 2017 assessment and was developed by Dick et al. (2017) using a hierarchical Bayesian modeling framework. The fecundity was assumed to be related to female body size:

$$F = aL^b$$

Where  $F$  is fecundity (number of eggs),  $L$  is fish length (cm), and  $a$  and  $b$  are constant coefficients (Figure 19). For Yelloweye Rockfish, Dick et al. (2017) estimated  $a = 7.21847 \times 10^{-8}$  and  $b = 4.043$ .

### 2.3.4 Natural Mortality

The 2025 base model used the 2017 assessment's initial value of natural Mortality ( $M = 0.044\text{y}^{-1}$ ) which was estimated based on Hamel (2015).

## 2.4 Environmental and ecosystem data

No environmental or ecosystem data were used in the 2017 assessment (Gertseva and Cope 2017) and no new data sources were considered for this update assessment.



### 3 Assessment Model

#### 3.1 History of Modeling Approaches

This section is not required for an update assessment; please refer to the most recent full assessment (Gertseva and Cope 2017) for additional information.

#### 3.2 Responses to SSC Groundfish Subcommittee Requests

The 2017 assessment was a benchmark assessment reviewed by a STAR panel and the PFMC SSC, and therefore, the Groundfish Subcommittee of the PFMC SSC did not review the 2017 assessment. No Yelloweye Rockfish benchmark or update assessments have been conducted since.

Responses to the 2025 GFSC review of this update assessment will be added following their review.

#### 3.3 Model Structure and Assumptions

##### 3.3.1 Description of New Modeling Approaches

This section is not required for an update assessment.

##### 3.3.2 Modeling Platform and Structure

The assessment was updated to use the most recent version of Stock Synthesis 3 (Version 3.30.23.2 - available [online](#)). Bridging between SS versions is discussed in Section 3.3.3.

Briefly, the Yelloweye Rockfish model is a coastwide, single-sex, two-area model. California is Area 1, and Oregon and Washington are combined into Area 2; the areas are separated because of differences in potential exploitation rates by area over time. Yelloweye Rockfish composition data are primarily reported as both sexes combined, and therefore, the assessment used a single sex model to facilitate the use of all available data. Growth is assumed to be the same in both areas, though future benchmark assessments may want to re-evaluate this assumption if more spatially-explicit data become available. Both growth and initial recruitment parameters were estimated internally, while all other biological parameters were fixed. The modeling period starts in the first year of available catches from historical reconstructions (1889) and the stock is assumed to be at an unfished equilibrium prior to that time. No changes were made to the fleet structure of the model. Fishery removals were divided among seven area- and sector-specific fleets. Estimated discard mortality was added to landings and included in the model as fleet-specific total removals. Length compositions for discarded and retained fish were combined as well. Data weighting was done using the Francis method (Francis 2011). More detailed information on the model structure and justification is available in Gertseva and Cope (2017) and summarized in Table 9.

##### 3.3.3 Model Changes from the Last Assessment

A list of changes that were made to the model compared to the previous assessment (Gertseva and Cope 2017) are listed below.

- Data:

- Detailed information on specific updates and changes to the data included in the model are described in Section 2 but are summarized below.
- The removals time series were corrected and updated through the end of 2024 for California, Oregon and Washington. Overall, there was little change in the model results when updating and extending the catch time series, even after changes in the historical catch (Figure 20 and Figure 21). Though, the age 8+ biomass increased more for the most recent years than the forecast estimates from 2017 (Figure 22).
- Indices of abundance were updated with recent data, where available, and re-analyzed using more up-to-date methods. Updating indices lead to a slight decrease in  $\ln(R_0)$ , reducing the spawning output by very little (Figure 23).
- Length and age compositions from all fishery removal and index fleets were updated and tuned through 2024. The addition of composition data slightly increased the spawning output for 1920 - 1980 (Figure 24). This new data also changed the pattern of the recruitment deviations from 1990 - 2024 (Figure 25). From 1990 - 2005 recruitment was lower than predicted in the last assessment, then from 2006 - 2016 recruitment was higher than the last assessment, and finally recruitment dropped lower for 2017-2021. Due to the estimated recruitment in recent years, the recovery curve slowed (Figure 24).
- Fleet structure:
  - No changes were made to the fleet structure.
- Biology:
  - No changes were made to the biological parameterization of the model; however, the length-weight relationship was updated to include the most recent data from the WCGTBS and did not change the model fit. The impact of updating the length-weight relationship is evaluated as a sensitivity.
- Recruitment:
  - The control file settings for the bias adjustment were updated to improve the fit of the bias adjustment relationship for recruitment deviations (Methot and Taylor 2011).
- Selectivity and Catchability:
  - The end year for all time blocks was extended to 2024.
  - All final updates in the control, starter, and forecast files showed little to no additional change in the model, with the exception of the updated parameterization of the catchability offset for the Oregon recreational index, which increased spawning output and final stock status (Figure 26 and Figure 27).
- Software and Workflow:
  - Updating to SS3 3.30.23.2 and to the most recent version of the SS3 executable had no discernable impact on model results (Figure 28).
  - We used the most up-to-date R packages to process input and output files for the assessment, including *nwfscDiag*, *r4ss*, and *pacfintools*.
  - A public github repository for Yelloweye Rockfish (“sebastes\_ruberrimus\_2025”) is available to provide a transparent and reproducible system for processing the data and creating the model and assessment document ([available online](#)).

### 3.3.4 Key Assumptions and Structural Choices

This section is not required for an update assessment; please refer to the most recent full assessment (Gertseva and Cope 2017) for additional information.

### 3.3.5 Priors

The natural mortality prior standard deviation was updated to be consistent with methods described in Hamel and Cope (2022). No other updates to the priors or prior standard deviations were made. Please refer to the most recent full assessment (Gertseva and Cope 2017) for additional information on the priors used in this assessment.

### 3.3.6 Data Weighting

Length data from the WCGBTS and fishery discards, conditional age-at-length compositions from the WCGBTS, and marginal age compositions from the fishery fleets and other surveys were fit and appropriately weighted using an iterative approach. The Francis (2011) method was used twice with two iterations to tune the length and age data simultaneously (Table 10). For conditional-age-at-length data, it was assumed that each age was a random sample within the length bin, and thus, the model started with a sample size equal to the number of fish in that length bin. A sensitivity is included to examine differences in parameter estimates when data weighting was implemented using the McAllister and Ianelli (1997) method, which is based on the harmonic mean.

Additional variance was estimated and added to the input variance for all indices with the exception of the WCGBTS.

### 3.3.7 Model Parameters

The base model had 189 estimated parameters (tallied by type in Table 11). A single-sex growth curve was estimated (Figure 29). Natural mortality was fixed at 0.044, as in the 2017 assessment. Unfished recruitment and the distribution of recruits between areas are estimated. Steepness of the stock-recruit relationship was fixed at 0.72, updated from the 2017 assessment which was fixed at 0.718. Estimating steepness was evaluated as a sensitivity. As is current practice, recruitment deviations during the “main” period (from 1980 to 2023) were forced to sum to zero and the bias adjustment ramp was updated (Figure 30).

We extended the time blocks on catchability parameters to 2024 to encompass additional data. To best fit the index for the Oregon recreational fleet, an offset for catchability between the MRFFS and ORBS sampling time periods was estimated in the 2017 assessment. In order to estimate this parameter under current best practices, the float was set to 0 and the phase to 1. See Section 3.4.3 for more information about the model fit to the index.

All selectivities were assumed to be length-based and used a double-normal functional form. Selectivities for all fleets were estimated to be asymptotic (Figure 31), though selectivity for the California Onboard Observer CPUE was mirrored to the California recreational fleet. Selectivities were constant through time. Dome-shaped selectivity and various time blocks for specific fleets were explored in Gertseva and Cope (2017) but not re-evaluated in this update assessment.

Aging error matrices were estimated outside the assessment model and were unchanged from the previous assessment, with the exception of designating the correct error matrix in some years for the Oregon recreational ages.

### 3.4 Base Model Results

#### 3.4.1 Base Model Selection

As a supplement to the model results figures included in this report and described below, a full set of diagnostic plots created by the *r4ss* package (Taylor et al. 2021) is available [online](#) along with the Stock Synthesis input files.

#### 3.4.2 Parameter Estimates

Estimated and fixed parameter values are shown in Table 12. Unfished recruitment was estimated at 5.488 thousands of fish. The von Bertalanffy growth function (Bertalanffy 1938) was used to model the relationship between length and age in Yelloweye Rockfish. Conditional age-at-length data is the main source of information to estimate growth. Yelloweye length-at-age 70 (the second reference age) equals 61.4 and an L-infinity of 61.7. Figure 29 shows the estimated growth curve. Spawning output-at-length (cm) is shown in Figure 32. Spawning output in the assessment is expressed in billions of eggs.

Estimated stock-recruit function for the assessment model is shown in Figure 33. Estimated recruitment deviations are shown Figure 34. Recruitment of Yelloweye Rockfish was estimated to be variable over time, with the most recent years experiencing slightly lower than average recruitment after periods of low and high recruitment during the late 1980s to the early 2000s, and late 2000s to late 2010s, respectively. Reflecting these variable recruitment patterns, the estimated stock-recruit function predicts a relatively wide range of cohort sizes over the observed range of spawning biomass. The model output recruitment variance ( $RMSE = 0.48$ ) is consistent with the fixed input recruitment variance ( $R = 0.5$ ) (Methot and Taylor 2011).

Length-based selectivity curves estimated in the assessment are shown for all fleets together in Figure 31. Estimated selectivity curves for the fishing fleets indicate that the recreational fleets access somewhat smaller fish than the commercial fisheries. All fleets for which curves were allowed to be dome-shaped (commercial trawl and non-trawl fleets) were estimated to be asymptotic. Estimated selectivity curves for the IPHC survey indicate a selection of the largest Yelloweye available, and select the least amount of smaller Yelloweye Rockfish. The WCGBTS selected smaller Yelloweye among all the fishery-independent surveys.

#### 3.4.3 Fits to the Data

Model fits to the fishery CPUE and survey indices are presented in Figure 35 through Figure 42, and log scale plots were used to see finer changes. Model fits to the indices generally fall within the uncertainty intervals but do not capture the temporal dynamics of each index, most notably in the WCGBTS and in the IPHC survey.

The model fitted length data aggregated across years reasonably well for all fleets (Figure 43). Pearson residuals for the fits by fleet and year are shown in Figure 44 and Figure 45. The length data are very sparse in many years and model fit varies among years and fleets, reflecting the differences in the quantity of the data. For example, lengths for the IPHC survey, the California recreational, and OR/WA non-trawl fleets, which also have the highest input sample sizes, are fit by the model relatively well.

The fits to the mean age by fleet are generally acceptable and are responsive to short-term temporal changes (Figure 46 - Figure 53), with the exception of the Oregon recreational fleet and the IPHC survey, where the model predicts a higher and a lower mean age, respectively, than the data suggest. This is consistent with the 2017 assessment. Pearson residuals by year and fleet show strong residual patterns but are difficult to interpret given the general lack of age data. See the [GitHub repository](#) for individual Pearson residual plots.

#### 3.4.4 Population Trajectory

The estimated time series of spawning output for the entire stock and by area are shown in Figure 54 and Figure 55, respectively. Spawning output relative to unfished spawning output for the entire stock and by area are shown in Figure 56 and Figure 57. Total biomass, summary biomass and recruitment are shown in Figure 58, Figure 59 and Figure 60, respectively. Trends in total and summary biomass, absolute and relative spawning output track one another very closely. The spawning output of Yelloweye Rockfish started to decline in the 1940s during World War II, but are estimated to have been lightly exploited until the mid-1970s when catches increased and a rapid decline in biomass and spawning output began. The combined relative spawning output reached a minimum of 15.6 of unexploited levels in 2000 (Figure 54). Yelloweye Rockfish spawning output and relative status is estimated to have been gradually increasing since that time, in response to large reductions in harvest and spatial area closures. The trend from the 2025 update model is very similar that from the 2017 assessment, however, the 2017 assessment estimated a slightly quicker recovery in biomass since 2002 than the 2025 model (Figure 61). Relative spawning output has differed between the two areas modeled in the assessment, with the California resource estimated to have a lower unfished equilibrium spawning output and estimated to be more depleted in 2025 than the Oregon and Washington resource (Figure 57).

Recruitment has been relatively dynamic over time, with several large peaks and troughs estimated in the age-0 recruits (Figure 60). From 1900 to 2006, the 2025 base model recruitment deviations closely track the 2017 model recruitment deviations, without any significant differences (Figure 62). From 2007 to 2016 we begin to see an increase in 2025 model recruitment deviations and a departure from lower 2017 model recruitment deviations due to composition data now being available to inform year class strength for that time period. This extended recent period of higher recruitment is likely driving the overall estimated population increase. Recruitment deviations for 2017 to 2024 are based on the assumed shape of the spawner-recruit curve for years without compositional data to inform year class strength.

### 3.5 Model Diagnostics

#### 3.5.1 Convergence

Model convergence was evaluated by starting the minimization process from dispersed values of the maximum likelihood estimates to determine if the model found a better minimum. Starting parameters were jittered using the jitter function built into Stock Synthesis, using a jitter input of 0.10. This was repeated 100 times with 82% of runs returning to the base model likelihood. A better, lower negative log-likelihood, model fit was not found. The spread of this search indicates that the jitter was sufficient to search a large portion of the likelihood surface, and that the base model is at a global minimum (Figure 63). Through the jittering and the likelihood profiles, we are confident that the base model, as presented, represents the best fit to the data given the assumptions made. There were no difficulties in inverting the Hessian to obtain estimates of variability. The final gradient was 0.00215. The maximum

gradient component was slightly higher than the convergence criterion, but the “hess\_step” option was used successfully to confirm that the gradient could be improved without changing the model results, providing strong evidence of convergence to a mode with quadratic log-likelihood surface.

### 3.5.2 Sensitivity Analyses

#### 3.5.2.1 Sensitivity to assumptions about model structure

Sensitivity analyses to examine the impact of different assumptions about model structure on management quantities included a model with an estimated natural mortality rate ( $M$ ), one with estimated steepness ( $h$ ) of the stock-recruit relationship, and one using the 2017 length-weight relationship. Summaries of model results for these sensitivities are presented in Table 13.

The estimated natural mortality was slightly higher than that of the base model ( $\sim 0.053$  compared to  $\sim 0.044$  from the base model). Steepness was estimated much higher at the maximum of 1.0, slightly higher than the sensitivity of the 2017 assessment estimated, which would otherwise indicate recruitment is less dependent on the spawning stock biomass, especially at small stock sizes. However, steepness values close to 1 are implausible for slow growing rockfish, supporting the decision to fix steepness at 0.72.

Model results are sensitive to whether natural mortality and steepness are estimated or fixed, with the alternative models estimating a higher spawning output and a higher relative spawning output (status) when compared to the base model (Figure 64 and Figure 65). In contrast, outputs of the model using the 2017 length-weight relationship showed the base model is not sensitive to this update, with similar results between the base and alternative model.

#### 3.5.2.2 Sensitivity to dataset choice and weighting schemes

Sensitivity analyses to dataset choices to examine the impact of including different data streams were conducted by selectively removing each data source using emphasis factors, as well as by including different weighting schemes for composition data. Summaries of model results for these sensitivities are presented in Table 14 to Table 18. Among these, the base model appears to be most sensitive to removing all length compositions, which show a significantly different biomass trajectory and a more optimistic estimate of stock status at the end of the time series (Figure 66 through Figure 69).

Removing all age compositions, IPHC age compositions only or applying the McAllister & Ianelli weighting scheme also result in slightly more optimistic estimates of stock status at the end of the time series compared to the base model (Figure 70 to Figure 73). Furthermore, models with all abundance and IPHC indices removed resulted in slightly less optimistic estimates of spawning output and stock status at the end of the time series (Figure 74 and Figure 75). Models removing the remaining indices consecutively did not show large differences in the final stock status estimate. A summary of the relative changes in management quantities from all sensitivity models is shown in Figure 76, and without the removal of all length composition data in Figure 77.

### 3.5.3 Retrospective Analysis

A retrospective analysis was conducted by running the base model with data removed for the past 5 years. Comparisons of the time series of absolute and relative spawning output and recruitment deviations time series for the runs are shown in Figure 78, Figure 79, and Figure 80, respectively. Recent recruitment deviations, which were negative in the base model, regressed towards the spawner-recruit curve, indicating closer to average recruitments in those years, with higher uncertainty, as data were removed. However, the change is not large, indicating that the new data are consistent with previous values or the sample sizes are too small to have any impact.

General trends in relative depletion (or spawning output) have been relatively stable across assessments (Figure 81), with a decline throughout the later half of the 1900's as the stock was fished down, followed by a reversal in the overall status after substantial catch restrictions were implemented in 2002. Across the most recent assessments, the 2017 assessment (Gertseva and Cope 2017) appears to have the most pessimistic depletion across the stock decline in the 1900s but otherwise, the relative depletion trend appears to be similar across the 2011, 2017 and the 2025 assessment as the stock has rebounded from its lowest status.

### 3.5.4 Likelihood Profiles

Likelihood profiles were conducted for natural mortality ( $M$ ), stock-recruit steepness ( $h$ ), and equilibrium recruitment ( $\ln(R_0)$ ). These likelihood profiles were conducted by fixing the parameter of interest at specific values and estimating the remaining parameters based on the fixed parameter value.

In the assessment,  $M$  was fixed at the value of 0.044, based on Hamel's prior. The profile analysis over  $M$  showed that the negative log-likelihood was minimized with a value around 0.050 (Figure 82), which is close to what was assumed in the assessment. The time series of absolute and relative spawning output associated with different values of  $M$  ranging from 0.03 to 0.06 are shown in Figure 83.

In the base model,  $h$  is fixed at the mean of the meta-analytic steepness prior, 0.72; much higher values (e.g. 0.9) are considered implausible given the life-history of slow growing rockfish. The likelihood profile for  $h$  shows that the negative log-likelihood for the base model declines with increasing  $h$  to a value around 0.8, notably different from the sensitivity analysis (Figure 84). Time series of relative unfished biomass associated with different values of  $h$  ranging from 0.25 to 1.0 are shown in Figure 85.

A likelihood profile analysis for  $\ln(R_0)$  shows a strongly informed initial recruitment value in the base model (Figure 86). Most of the information for this parameter is coming from the recruitment estimates. Within the length composition likelihood component, all sources of length compositions are equally informative. The index and age data are relatively uninformative. Changes in  $\ln(R_0)$  results in relatively small changes in the scale of the population (Figure 87 and Figure 88) compared to the 2017 assessment model.

## 3.6 Unresolved Problems and Major Uncertainties

Main life history parameters, such as natural mortality and stock-recruit curve steepness, generally contribute significant uncertainty to stock assessments. These values were fixed in this assessment, as

they were in the benchmark assessment because the benchmark model was unable to reliably estimate these quantities (Gertseva and Cope 2017). These quantities are essential for understanding the dynamics of the stock and determining projected rebuilding. Alternative values of these parameters were explored through both sensitivity and likelihood profile analyses. Maturity parameters were fixed in this assessment, as they were in benchmark assessment (Gertseva and Cope 2017). Maturity schedules are generally estimated outside of the assessment, and the maturity parameter estimates were not updated in the time since the benchmark assessment. These parameters, once updated, may influence estimates of spawning stock output.

Although significant progress has been made in reconstructing historical landings, early catches of Yelloweye Rockfish continue to be uncertain.

The model fits to the indices generally fall within uncertainty intervals but do not capture the temporal dynamics of each index, thus the indices may not be as informative as they would be if the temporal dynamics were fit well. In addition, the fishery-independent indices available (e.g., the WCGBTS and IPHC indices) do not target Yelloweye Rockfish specifically and, thus, may not be the most appropriate indicator of abundance dynamics.



## 4 Management

### 4.1 Reference Points

This assessment estimates that the stock of Yelloweye Rockfish off the continental U.S. Pacific Coast is currently at 40.1% of its unexploited level. This is above the overfished threshold of  $SO_{25\%}$ , and slightly above the management target of  $SO_{40\%}$  of unfished spawning output. Both areas (California and Oregon-Washington) are above the overfished level of 25%. The assessment estimates that the coastwide spawning output of Yelloweye Rockfish dropped below the  $SO_{40\%}$  target for the first time in 1986 and below the overfished  $SO_{25\%}$  threshold in 1993, as a result of intense fishing by commercial and recreational fleets. It continued to decline and reached 15.6% of its unfished output in 2000 (Table 19). The stock was declared overfished in 2002. Since then, the spawning output has slowly increased due to management regulations implemented to foster stock rebuilding.

Reference points for the base model are summarized in Table 20. Unfished spawning stock output for Yelloweye Rockfish was estimated to be 1190 billion eggs (95% confidence interval: 1,048.1 - 1,331.9 billion eggs). The stock is declared overfished if the current spawning output is estimated to be below the minimum stock size threshold (MSST) of 25% of unfished level ( $SO_{25\%}$ ). The management target for Yelloweye Rockfish is defined as 40% of the unfished spawning output ( $SO_{40\%}$ ), which is estimated by the model to be 476 billion eggs (95% confidence interval: 419 - 533 billion eggs), which corresponds to an exploitation rate of 0.026. This harvest rate provides an equilibrium yield of 122 mt at  $SO_{40\%}$  (95% confidence interval: 108 - 137 mt). The model estimate of maximum sustainable yield (MSY) is 128 mt (95% confidence interval: 113 - 143 mt). The estimated spawning stock output at MSY is 343 billion eggs (95% confidence interval: 303-384 billion eggs). The exploitation rate corresponding to the estimated SPRMSY is 0.037.

This assessment estimates that the 2024 SPR is 82.15% (Figure 89). The SPR used for setting the OFL is 50%, while the SPR-based management fishing mortality target specified in the current rebuilding plan and used to determine the ACL is 76% (when the SPR is greater than this value, the exploitation is below the target). Relative exploitation rates (calculated as catch/biomass of age-8 and older fish) are estimated to have been below 1% during the last decade. This assessment estimates that Yelloweye Rockfish was fished beyond the relative SPR ratio (calculated as  $1-SPR/1-SPR_{Target} = 0.5$ ) between 1977 and 2000. The equilibrium yield curve is shown in Figure 90.

### 4.2 Harvest Projections and Decision Tables

The base model estimate for 2025 spawning depletion is 40.1% (Table 21). The primary axis of uncertainty about this estimate used in the decision table was based on natural mortality. Natural mortality in the assessment model is fixed at the median of the Hamel prior ( $0.044\ y^{-1}$ ), estimated using the maximum age of 123 years. The natural mortality value for the high state of nature was calculated to correspond to 97 years of age, which was the 99th percentile of the age data available for the 2017 assessment (Gertseva and Cope (2017)); this value was  $0.056\ y^{-1}$ . The natural mortality value for low state of nature was calculated to correspond to 147 years of age, which is the maximum age reported for the Yelloweye Rockfish at the time of the 2017 assessment (Gertseva and Cope (2017)); this value was  $0.037\ y^{-1}$ .

Twelve-year forecasts for each state of nature were calculated (Table 22).

### 4.3 Evaluation of Scientific Uncertainty

The model estimate of the log-scale standard deviation of the 2025 spawning output is 0.0996. The model estimate of the log-scale standard deviation of the 2025 OFL is 0.095. Each of these are likely underestimates of overall uncertainty due to the necessity to fix several key population dynamics parameters (e.g. steepness and recruitment variance) and also because there is no explicit incorporation of model structural uncertainty (although see the decision table for alternative states of nature).

### 4.4 Regional management considerations

Yelloweye Rockfish is modeled in two areas (California and Oregon-Washington) in this assessment. Current population status does differ by area and may be valuable information for making management and allocation decisions (Figure 57).

### 4.5 Research and Data Needs

Please refer to the 2017 benchmark assessment for a detailed list of research and data needs for Yelloweye Rockfish (Gertseva and Cope (2017)). In addition to those, the following research and recommendations could improve the ability of future stock assessments to determine the status and productivity of the Yelloweye Rockfish population:

- Continue refining the ORFS index analysis and ultimately use either the ORBS or ORFS index to describe the CPUE trends in the Oregon recreational fishery after 2000.
- Expand the IPHC age composition bins to an older maximum age for the IPHC age composition data to spread out the distribution of length data in the oldest age bins for conditional age-at-length.

## 5 Acknowledgements

The authors would like to thank the other members of the University of Washington FSH 577 class, particularly those on the Yelloweye Rockfish team (Emily Branam, Julia Coates, Kimberly Fitzpatrick, Madison Sandquist, Olivia Boisen, and Rachel Brooks). The authors also thank all data providers.

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## 7 Tables

Table 1: Recent trend in the overfishing limits (OFL), the acceptable biological catches (ABCs), the annual catch limits (ACLs), and the total dead catch (landings + discards) in metric tons (mt).

| Year | OFL (mt) | ABC (mt) | ACL (mt) | Catch (mt) |
|------|----------|----------|----------|------------|
| 2015 | 52       | 43       | 18       | 12         |
| 2016 | 52       | 43       | 19       | 9          |
| 2017 | 57       | 47       | 20       | 20         |
| 2018 | 58       | 48       | 20       | 19         |
| 2019 | 82       | 74       | 48       | 23         |
| 2020 | 84       | 77       | 49       | 18         |
| 2021 | 97       | 83       | 50       | 21         |
| 2022 | 98       | 83       | 51       | 34         |
| 2023 | 123      | 103      | 66       | 39         |
| 2024 | 123      | 103      | 66       | 29         |

Table 2: Time series of Yelloweye Rockfish catches by fleet and total catch (mt) summed across fleets (total catch includes WA REC catch converted to mt). Trawl fleets include Yelloweye Rockfish bycatch in foreign POP and in at-sea Pacific hake fisheries.

| Year | CA TWL<br>(mt) | CA<br>NONTWL<br>(mt) | CA REC<br>(mt) | ORWA<br>TWL (mt) | ORWA<br>NONTWL<br>(mt) | OR REC<br>(mt) | WA REC<br>(mt) | Catch (mt) |
|------|----------------|----------------------|----------------|------------------|------------------------|----------------|----------------|------------|
| 1889 | 0.00           | 0.00                 | 0.00           | 0.00             | 0.04                   | 0.00           | 0.00           | 0.04       |
| 1890 | 0.02           | 0.07                 | 0.00           | 0.00             | 0.04                   | 0.00           | 0.00           | 0.13       |
| 1891 | 0.03           | 0.13                 | 0.00           | 0.00             | 0.07                   | 0.00           | 0.00           | 0.23       |
| 1892 | 0.05           | 0.20                 | 0.00           | 0.00             | 3.64                   | 0.00           | 0.00           | 3.89       |
| 1893 | 0.06           | 0.26                 | 0.00           | 0.00             | 3.55                   | 0.00           | 0.00           | 3.87       |
| 1894 | 0.08           | 0.33                 | 0.00           | 0.00             | 3.55                   | 0.00           | 0.00           | 3.96       |
| 1895 | 0.09           | 0.39                 | 0.00           | 0.00             | 0.92                   | 0.00           | 0.00           | 1.40       |
| 1896 | 0.11           | 0.46                 | 0.00           | 0.00             | 0.22                   | 0.00           | 0.00           | 0.79       |
| 1897 | 0.12           | 0.52                 | 0.00           | 0.00             | 0.22                   | 0.00           | 0.00           | 0.86       |
| 1898 | 0.14           | 0.59                 | 0.00           | 0.00             | 0.13                   | 0.00           | 0.00           | 0.86       |
| 1899 | 0.16           | 0.66                 | 0.00           | 0.00             | 0.23                   | 0.00           | 0.00           | 1.05       |
| 1900 | 0.17           | 0.72                 | 0.00           | 0.00             | 0.30                   | 0.00           | 0.00           | 1.19       |
| 1901 | 0.19           | 0.79                 | 0.00           | 0.00             | 0.39                   | 0.00           | 0.00           | 1.37       |
| 1902 | 0.20           | 0.85                 | 0.00           | 0.00             | 0.48                   | 0.00           | 0.00           | 1.53       |
| 1903 | 0.22           | 0.92                 | 0.00           | 0.00             | 0.56                   | 0.00           | 0.00           | 1.70       |
| 1904 | 0.23           | 0.98                 | 0.00           | 0.00             | 0.73                   | 0.00           | 0.00           | 1.94       |
| 1905 | 0.25           | 1.05                 | 0.00           | 0.00             | 0.74                   | 0.00           | 0.00           | 2.04       |
| 1906 | 0.26           | 1.11                 | 0.00           | 0.00             | 0.83                   | 0.00           | 0.00           | 2.20       |
| 1907 | 0.28           | 1.18                 | 0.00           | 0.00             | 0.91                   | 0.00           | 0.00           | 2.37       |
| 1908 | 0.30           | 1.25                 | 0.00           | 0.00             | 1.95                   | 0.00           | 0.00           | 3.50       |
| 1909 | 0.31           | 1.31                 | 0.00           | 0.00             | 1.09                   | 0.00           | 0.00           | 2.71       |
| 1910 | 0.33           | 1.38                 | 0.00           | 0.00             | 1.18                   | 0.00           | 0.00           | 2.89       |
| 1911 | 0.34           | 1.44                 | 0.00           | 0.00             | 1.26                   | 0.00           | 0.00           | 3.04       |
| 1912 | 0.36           | 1.51                 | 0.00           | 0.00             | 1.35                   | 0.00           | 0.00           | 3.22       |
| 1913 | 0.37           | 1.57                 | 0.00           | 0.00             | 1.44                   | 0.00           | 0.00           | 3.38       |
| 1914 | 0.39           | 1.64                 | 0.00           | 0.00             | 1.53                   | 0.00           | 0.00           | 3.56       |
| 1915 | 0.40           | 1.70                 | 0.00           | 0.00             | 2.23                   | 0.00           | 0.00           | 4.33       |
| 1916 | 0.42           | 1.77                 | 0.00           | 0.00             | 1.70                   | 0.00           | 0.00           | 3.89       |
| 1917 | 0.66           | 2.96                 | 0.00           | 0.00             | 1.79                   | 0.00           | 0.00           | 5.41       |
| 1918 | 0.77           | 3.48                 | 0.00           | 0.00             | 18.54                  | 0.00           | 0.00           | 22.79      |
| 1919 | 0.54           | 1.62                 | 0.00           | 0.00             | 7.61                   | 0.00           | 0.00           | 9.77       |
| 1920 | 0.55           | 1.84                 | 0.00           | 0.00             | 6.57                   | 0.00           | 0.00           | 8.96       |
| 1921 | 0.45           | 1.85                 | 0.00           | 0.00             | 6.33                   | 0.00           | 0.00           | 8.63       |

|      |       |       |       |       |       |      |      |        |
|------|-------|-------|-------|-------|-------|------|------|--------|
| 1922 | 0.39  | 1.68  | 0.00  | 0.00  | 4.38  | 0.00 | 0.00 | 6.45   |
| 1923 | 0.42  | 1.79  | 0.00  | 0.00  | 5.10  | 0.00 | 0.00 | 7.31   |
| 1924 | 0.24  | 2.58  | 0.00  | 0.00  | 9.29  | 0.00 | 0.00 | 12.11  |
| 1925 | 0.17  | 3.69  | 0.00  | 0.00  | 11.48 | 0.00 | 0.00 | 15.34  |
| 1926 | 0.62  | 4.25  | 0.00  | 0.00  | 17.48 | 0.00 | 0.00 | 22.35  |
| 1927 | 1.05  | 4.87  | 0.00  | 0.00  | 22.79 | 0.00 | 0.00 | 28.71  |
| 1928 | 1.34  | 4.18  | 0.64  | 0.00  | 22.09 | 0.00 | 0.00 | 28.25  |
| 1929 | 1.58  | 4.07  | 1.29  | 0.00  | 17.73 | 0.00 | 0.00 | 24.67  |
| 1930 | 1.47  | 5.30  | 1.48  | 0.00  | 19.50 | 0.00 | 0.00 | 27.75  |
| 1931 | 0.88  | 4.74  | 1.97  | 0.00  | 11.69 | 0.00 | 0.00 | 19.28  |
| 1932 | 1.05  | 7.08  | 2.47  | 0.02  | 7.33  | 0.00 | 0.00 | 17.95  |
| 1933 | 1.63  | 2.81  | 2.96  | 0.01  | 10.30 | 0.00 | 0.00 | 17.71  |
| 1934 | 1.61  | 4.17  | 3.45  | 0.00  | 12.66 | 0.00 | 0.00 | 21.89  |
| 1935 | 1.68  | 6.31  | 3.95  | 0.01  | 9.69  | 0.00 | 0.00 | 21.64  |
| 1936 | 1.49  | 6.60  | 4.44  | 0.03  | 16.65 | 0.00 | 0.00 | 29.21  |
| 1937 | 1.77  | 4.31  | 5.27  | 0.06  | 14.82 | 0.00 | 0.00 | 26.23  |
| 1938 | 1.67  | 4.69  | 5.18  | 0.00  | 16.35 | 0.00 | 0.00 | 27.89  |
| 1939 | 1.73  | 4.71  | 4.53  | 0.09  | 10.63 | 0.00 | 0.00 | 21.69  |
| 1940 | 1.60  | 2.97  | 6.51  | 2.06  | 17.14 | 0.00 | 0.00 | 30.28  |
| 1941 | 1.16  | 4.19  | 6.02  | 3.17  | 27.38 | 0.00 | 0.00 | 41.92  |
| 1942 | 0.27  | 3.10  | 3.20  | 5.95  | 31.38 | 0.00 | 0.00 | 43.90  |
| 1943 | 2.05  | 3.84  | 3.06  | 20.81 | 51.22 | 0.00 | 0.00 | 80.98  |
| 1944 | 8.36  | 16.52 | 2.51  | 36.51 | 22.60 | 0.00 | 0.00 | 86.50  |
| 1945 | 18.54 | 40.02 | 3.35  | 56.89 | 11.52 | 0.00 | 0.00 | 130.32 |
| 1946 | 16.33 | 41.42 | 5.76  | 34.85 | 20.68 | 0.00 | 0.00 | 119.04 |
| 1947 | 7.09  | 9.19  | 4.59  | 21.42 | 10.95 | 0.00 | 0.00 | 53.24  |
| 1948 | 6.49  | 16.81 | 9.18  | 15.14 | 13.38 | 0.00 | 0.00 | 61.00  |
| 1949 | 3.72  | 6.17  | 11.88 | 12.64 | 11.21 | 0.00 | 0.00 | 45.62  |
| 1950 | 3.42  | 4.61  | 14.49 | 13.69 | 14.78 | 0.00 | 0.00 | 50.99  |
| 1951 | 9.91  | 7.07  | 17.16 | 12.02 | 17.96 | 0.00 | 0.00 | 64.12  |
| 1952 | 8.70  | 5.44  | 15.00 | 12.79 | 13.06 | 0.00 | 0.00 | 54.99  |
| 1953 | 8.57  | 3.19  | 12.85 | 9.96  | 5.61  | 0.00 | 0.00 | 40.18  |
| 1954 | 4.99  | 6.78  | 16.17 | 12.81 | 10.25 | 0.00 | 0.00 | 51.00  |
| 1955 | 5.61  | 1.83  | 19.51 | 13.13 | 9.71  | 0.00 | 0.00 | 49.79  |
| 1956 | 8.58  | 1.81  | 21.90 | 16.99 | 4.34  | 0.00 | 0.00 | 53.62  |
| 1957 | 10.49 | 4.07  | 21.71 | 22.96 | 8.51  | 0.00 | 0.00 | 67.74  |
| 1958 | 10.34 | 3.05  | 33.84 | 18.38 | 2.39  | 0.00 | 0.00 | 68.00  |
| 1959 | 8.61  | 1.64  | 29.23 | 19.94 | 5.41  | 0.00 | 0.00 | 64.83  |
| 1960 | 7.48  | 2.24  | 20.86 | 25.20 | 4.92  | 0.00 | 0.00 | 60.70  |



|      |        |        |        |        |        |       |       |        |
|------|--------|--------|--------|--------|--------|-------|-------|--------|
| 1961 | 3.56   | 1.69   | 16.35  | 22.72  | 4.91   | 0.00  | 0.00  | 49.23  |
| 1962 | 3.68   | 1.75   | 20.81  | 26.40  | 5.16   | 0.00  | 0.00  | 57.80  |
| 1963 | 6.02   | 5.61   | 21.80  | 7.17   | 4.10   | 0.00  | 0.00  | 44.70  |
| 1964 | 3.12   | 4.56   | 18.96  | 1.95   | 3.11   | 0.00  | 0.00  | 31.70  |
| 1965 | 3.86   | 5.51   | 29.11  | 67.88  | 4.68   | 0.00  | 0.00  | 111.04 |
| 1966 | 3.62   | 4.45   | 31.60  | 3.03   | 3.24   | 0.00  | 0.00  | 45.94  |
| 1967 | 6.17   | 4.38   | 31.89  | 6.82   | 6.60   | 0.00  | 1.64  | 57.50  |
| 1968 | 3.78   | 3.89   | 37.66  | 2.97   | 5.66   | 0.00  | 0.32  | 54.28  |
| 1969 | 21.80  | 3.91   | 40.62  | 47.76  | 13.08  | 0.00  | 0.78  | 127.95 |
| 1970 | 24.22  | 3.47   | 45.79  | 7.05   | 4.31   | 0.00  | 1.20  | 86.04  |
| 1971 | 41.77  | 4.73   | 40.72  | 13.65  | 8.34   | 0.00  | 1.90  | 111.11 |
| 1972 | 56.22  | 7.44   | 52.36  | 7.35   | 10.86  | 0.00  | 2.48  | 136.71 |
| 1973 | 43.62  | 5.89   | 66.48  | 9.52   | 11.46  | 7.40  | 3.13  | 147.50 |
| 1974 | 44.80  | 11.59  | 70.15  | 4.41   | 14.46  | 12.78 | 2.90  | 161.09 |
| 1975 | 50.31  | 9.93   | 71.13  | 5.36   | 7.65   | 6.24  | 2.93  | 153.55 |
| 1976 | 45.27  | 13.39  | 80.63  | 6.91   | 10.15  | 19.38 | 3.06  | 178.79 |
| 1977 | 42.51  | 14.95  | 72.78  | 4.97   | 17.02  | 19.91 | 6.30  | 178.44 |
| 1978 | 123.44 | 30.76  | 67.89  | 23.64  | 24.10  | 24.52 | 3.12  | 297.47 |
| 1979 | 61.02  | 38.31  | 76.31  | 44.58  | 49.10  | 52.62 | 3.67  | 325.61 |
| 1980 | 15.48  | 26.58  | 72.51  | 83.95  | 24.96  | 40.43 | 1.84  | 265.75 |
| 1981 | 30.20  | 119.50 | 47.00  | 91.34  | 23.95  | 37.20 | 3.42  | 352.61 |
| 1982 | 199.93 | 15.59  | 102.00 | 156.08 | 31.45  | 65.06 | 4.91  | 575.02 |
| 1983 | 56.65  | 7.68   | 51.00  | 287.29 | 45.95  | 46.08 | 6.75  | 501.40 |
| 1984 | 44.03  | 4.42   | 77.00  | 113.98 | 39.39  | 41.86 | 11.44 | 332.12 |
| 1985 | 7.42   | 4.23   | 124.00 | 200.04 | 69.72  | 12.72 | 8.70  | 426.83 |
| 1986 | 9.89   | 23.43  | 65.00  | 92.92  | 66.15  | 95.62 | 8.46  | 361.47 |
| 1987 | 16.84  | 38.00  | 75.00  | 71.75  | 97.08  | 41.64 | 10.63 | 350.94 |
| 1988 | 30.57  | 34.95  | 58.00  | 130.64 | 47.45  | 10.78 | 8.33  | 320.72 |
| 1989 | 9.38   | 42.37  | 59.00  | 199.34 | 41.40  | 13.48 | 14.69 | 379.66 |
| 1990 | 10.08  | 70.26  | 46.25  | 81.07  | 68.95  | 17.57 | 7.47  | 301.65 |
| 1991 | 13.98  | 133.07 | 33.50  | 121.38 | 85.62  | 27.81 | 13.60 | 428.96 |
| 1992 | 15.83  | 96.85  | 20.75  | 135.66 | 89.87  | 27.55 | 12.27 | 398.78 |
| 1993 | 6.18   | 46.59  | 8.00   | 137.96 | 138.25 | 25.52 | 12.46 | 374.96 |
| 1994 | 4.70   | 49.78  | 14.00  | 86.00  | 79.29  | 16.19 | 7.19  | 257.15 |
| 1995 | 3.69   | 47.68  | 13.00  | 131.32 | 40.43  | 20.49 | 7.04  | 263.65 |
| 1996 | 16.32  | 56.18  | 12.00  | 83.88  | 93.25  | 8.29  | 7.58  | 277.50 |
| 1997 | 6.20   | 57.06  | 15.00  | 80.13  | 115.54 | 14.18 | 7.74  | 295.85 |
| 1998 | 4.10   | 17.64  | 5.00   | 41.18  | 45.05  | 16.22 | 9.98  | 139.17 |
| 1999 | 8.66   | 13.73  | 13.00  | 18.94  | 102.00 | 12.25 | 7.43  | 176.01 |

|      |      |      |      |      |       |       |      |       |
|------|------|------|------|------|-------|-------|------|-------|
| 2000 | 0.73 | 3.31 | 8.00 | 5.07 | 15.04 | 10.69 | 8.05 | 50.89 |
| 2001 | 0.62 | 3.90 | 5.00 | 1.63 | 26.31 | 4.69  | 8.66 | 50.81 |
| 2002 | 0.36 | 0.03 | 2.00 | 1.59 | 4.15  | 3.11  | 1.90 | 13.14 |
| 2003 | 0.13 | 0.05 | 4.00 | 0.55 | 2.24  | 3.32  | 1.40 | 11.69 |
| 2004 | 0.02 | 0.75 | 1.00 | 0.50 | 2.38  | 1.54  | 2.32 | 8.51  |
| 2005 | 0.02 | 0.73 | 1.00 | 1.24 | 1.66  | 2.13  | 2.75 | 9.53  |
| 2006 | 0.00 | 0.20 | 1.00 | 1.42 | 2.16  | 1.72  | 1.04 | 7.54  |
| 2007 | 0.00 | 0.93 | 4.00 | 0.09 | 3.68  | 2.13  | 1.54 | 12.37 |
| 2008 | 0.02 | 0.64 | 1.00 | 0.16 | 3.43  | 2.12  | 1.31 | 8.68  |
| 2009 | 0.02 | 0.19 | 5.00 | 0.09 | 2.18  | 1.88  | 1.34 | 10.70 |
| 2010 | 0.06 | 0.04 | 1.00 | 0.08 | 0.86  | 1.95  | 1.66 | 5.65  |
| 2011 | 0.00 | 0.20 | 2.00 | 0.06 | 1.21  | 2.17  | 1.87 | 7.51  |
| 2012 | 0.00 | 0.88 | 2.00 | 0.06 | 1.91  | 3.19  | 2.69 | 10.73 |
| 2013 | 0.01 | 0.56 | 1.00 | 0.11 | 2.94  | 3.22  | 1.70 | 9.54  |
| 2014 | 0.06 | 0.02 | 1.00 | 0.03 | 2.16  | 2.73  | 2.29 | 8.29  |
| 2015 | 0.00 | 0.40 | 2.00 | 0.03 | 3.15  | 4.26  | 2.10 | 11.94 |
| 2016 | 0.00 | 0.00 | 1.00 | 0.07 | 2.59  | 2.84  | 2.46 | 8.96  |
| 2017 | 0.01 | 1.23 | 4.52 | 0.24 | 6.97  | 4.27  | 2.49 | 19.73 |
| 2018 | 0.00 | 0.00 | 4.99 | 0.54 | 6.38  | 4.01  | 2.57 | 18.49 |
| 2019 | 0.04 | 0.00 | 6.16 | 0.59 | 7.43  | 5.04  | 4.23 | 23.49 |
| 2020 | 0.13 | 0.00 | 1.95 | 0.32 | 7.52  | 6.00  | 2.24 | 18.16 |
| 2021 | 0.12 | 2.43 | 3.96 | 0.39 | 7.97  | 3.34  | 2.54 | 20.75 |
| 2022 | 0.10 | 5.60 | 3.80 | 0.76 | 15.55 | 5.20  | 2.65 | 33.66 |
| 2023 | 0.09 | 1.83 | 9.59 | 0.40 | 20.64 | 3.84  | 2.88 | 39.27 |
| 2024 | 0.19 | 3.27 | 4.65 | 0.49 | 13.51 | 3.66  | 2.92 | 28.69 |

Table 3: Summary of trips with and without Yelloweye Rockfish from ORFS index

| year | tripsWithTarget | tripsWOTarget | totalTrips | percentpos |
|------|-----------------|---------------|------------|------------|
| 2001 | 11              | 334           | 345        | 0.03       |
| 2004 | 12              | 334           | 346        | 0.03       |
| 2005 | 10              | 392           | 402        | 0.02       |
| 2006 | 24              | 385           | 409        | 0.06       |
| 2007 | 20              | 478           | 498        | 0.04       |
| 2008 | 29              | 449           | 478        | 0.06       |
| 2009 | 23              | 285           | 308        | 0.07       |
| 2010 | 12              | 324           | 336        | 0.04       |
| 2011 | 20              | 317           | 337        | 0.06       |
| 2012 | 46              | 519           | 565        | 0.08       |
| 2013 | 31              | 391           | 422        | 0.07       |
| 2014 | 29              | 367           | 396        | 0.07       |
| 2015 | 10              | 312           | 322        | 0.03       |
| 2017 | 23              | 388           | 411        | 0.06       |
| 2022 | 11              | 216           | 227        | 0.05       |
| 2023 | 25              | 418           | 443        | 0.06       |
| 2024 | 54              | 540           | 594        | 0.09       |

Table 4: Model selection for top model covariate combinations considered for the ORFS index

| Gf_opendepth | Lgdepthbin | Month | Port  | Year  | Effort.Offset | Df | Log.Likelihood | AICc   | Delta |
|--------------|------------|-------|-------|-------|---------------|----|----------------|--------|-------|
| -            | Incl.      | -     | Incl. | Incl. | Incl.         | 28 | -1401.1        | 2858.5 | 0.0   |
| -            | Incl.      | Incl. | Incl. | Incl. | Incl.         | 35 | -1394.6        | 2859.6 | 1.2   |
| Incl.        | Incl.      | -     | Incl. | Incl. | Incl.         | 31 | -1399.3        | 2861.0 | 2.5   |
| Incl.        | Incl.      | Incl. | Incl. | Incl. | Incl.         | 38 | -1394.3        | 2865.1 | 6.7   |
| -            | Incl.      | Incl. | -     | Incl. | Incl.         | 29 | -1464.0        | 2986.3 | 127.8 |
| -            | Incl.      | -     | -     | Incl. | Incl.         | 22 | -1473.2        | 2990.5 | 132.1 |
| Incl.        | Incl.      | Incl. | -     | Incl. | Incl.         | 32 | -1463.7        | 2991.7 | 133.2 |
| Incl.        | Incl.      | -     | -     | Incl. | Incl.         | 25 | -1472.2        | 2994.5 | 136.1 |
| -            | -          | Incl. | Incl. | Incl. | Incl.         | 31 | -1497.3        | 3056.9 | 198.5 |
| Incl.        | -          | -     | Incl. | Incl. | Incl.         | 27 | -1502.7        | 3059.6 | 201.1 |

Table 5: Summary of trips with and without Yelloweye Rockfish from ORBS index

| year | tripsWithTarget | tripsWOTarget | totalTrips | percentpos |
|------|-----------------|---------------|------------|------------|
| 2004 | 111             | 3399          | 3510       | 0.03       |
| 2005 | 281             | 6561          | 6842       | 0.04       |
| 2006 | 278             | 6729          | 7007       | 0.04       |
| 2007 | 262             | 4588          | 4850       | 0.05       |
| 2008 | 273             | 5342          | 5615       | 0.05       |
| 2009 | 219             | 5430          | 5649       | 0.04       |
| 2010 | 287             | 5948          | 6235       | 0.05       |
| 2011 | 337             | 5203          | 5540       | 0.06       |
| 2012 | 415             | 5067          | 5482       | 0.08       |
| 2013 | 602             | 6655          | 7257       | 0.08       |
| 2014 | 429             | 5426          | 5855       | 0.07       |
| 2015 | 483             | 7945          | 8428       | 0.06       |
| 2016 | 328             | 6608          | 6936       | 0.05       |
| 2017 | 642             | 6653          | 7295       | 0.09       |
| 2018 | 681             | 6530          | 7211       | 0.09       |
| 2019 | 693             | 5610          | 6303       | 0.11       |
| 2020 | 802             | 6369          | 7171       | 0.11       |
| 2021 | 582             | 5256          | 5838       | 0.10       |
| 2022 | 628             | 5896          | 6524       | 0.10       |
| 2023 | 840             | 5992          | 6832       | 0.12       |
| 2024 | 680             | 5979          | 6659       | 0.10       |

Table 6: Model selection for top model covariate combinations considered for the ORBS index

| Boattype | Gf_opendepth | Month | Port  | Tgt.bag | Year  | Effort.Offset | Df | Log.Likelihood | AICc    | Delta |
|----------|--------------|-------|-------|---------|-------|---------------|----|----------------|---------|-------|
| Incl.    | Incl.        | Incl. | Incl. | Incl.   | Incl. | Incl.         | 48 | -45351.5       | 90799.1 | 0.0   |
| Incl.    | Incl.        | Incl. | Incl. | -       | Incl. | Incl.         | 44 | -45369.4       | 90826.8 | 27.7  |
| Incl.    | -            | Incl. | Incl. | Incl.   | Incl. | Incl.         | 45 | -45389.1       | 90868.2 | 69.2  |
| Incl.    | Incl.        | -     | Incl. | Incl.   | Incl. | Incl.         | 37 | -45414.1       | 90902.2 | 103.1 |
| Incl.    | -            | Incl. | Incl. | -       | Incl. | Incl.         | 41 | -45413.6       | 90909.2 | 110.2 |
| Incl.    | Incl.        | -     | Incl. | -       | Incl. | Incl.         | 33 | -45427.8       | 90921.6 | 122.6 |
| Incl.    | -            | -     | Incl. | Incl.   | Incl. | Incl.         | 34 | -45554.1       | 91176.1 | 377.1 |
| Incl.    | -            | -     | Incl. | -       | Incl. | Incl.         | 30 | -45583.4       | 91226.9 | 427.8 |
| -        | Incl.        | Incl. | Incl. | Incl.   | Incl. | Incl.         | 47 | -45634.6       | 91363.3 | 564.3 |
| -        | Incl.        | Incl. | Incl. | -       | Incl. | Incl.         | 43 | -45650.9       | 91387.7 | 588.7 |

Table 7: Summary of sampling effort within triennial survey, with total and Yelloweye positive hauls summarized by area.

|      | CA              |                          | OR-WA           |                          |
|------|-----------------|--------------------------|-----------------|--------------------------|
|      | Number of hauls | Number of positive hauls | Number of hauls | Number of positive hauls |
| 1980 | 68              | 1                        | 263             | 13                       |
| 1983 | 96              | 1                        | 416             | 26                       |
| 1986 | 95              | 2                        | 389             | 27                       |
| 1989 | 147             | 7                        | 300             | 30                       |
| 1992 | 135             | 2                        | 310             | 25                       |
| 1995 | 123             | 1                        | 241             | 7                        |
| 1998 | 129             | 0                        | 260             | 14                       |
| 2001 | 129             | 0                        | 246             | 15                       |
| 2004 | 103             | 3                        | 185             | 9                        |

Table 8: Summary of sampling effort within the WCGBTS, with total and Yelloweye positive hauls summarized by area.

|      | CA              |                          | ORWA            |                          |
|------|-----------------|--------------------------|-----------------|--------------------------|
|      | Number of hauls | positive.Number of hauls | Number of hauls | positive.Number of hauls |
| 2003 | 268             | 2                        | 274             | 17                       |
| 2004 | 247             | 1                        | 223             | 7                        |
| 2005 | 345             | 2                        | 296             | 11                       |
| 2006 | 346             | 1                        | 293             | 12                       |
| 2007 | 355             | 3                        | 332             | 9                        |
| 2008 | 382             | 2                        | 298             | 13                       |
| 2009 | 389             | 5                        | 292             | 6                        |
| 2010 | 413             | 1                        | 300             | 14                       |
| 2011 | 381             | 3                        | 314             | 10                       |
| 2012 | 389             | 2                        | 306             | 12                       |
| 2013 | 248             | 3                        | 220             | 10                       |
| 2014 | 0               | 0                        | 311             | 19                       |
| 2015 | 383             | 2                        | 283             | 11                       |
| 2016 | 383             | 5                        | 309             | 20                       |
| 2017 | 385             | 3                        | 320             | 16                       |
| 2018 | 396             | 5                        | 305             | 19                       |
| 2019 | 0               | 0                        | 161             | 9                        |
| 2021 | 382             | 4                        | 302             | 16                       |
| 2022 | 359             | 3                        | 275             | 15                       |
| 2023 | 365             | 4                        | 296             | 10                       |
| 2024 | 348             | 3                        | 310             | 19                       |

Table 9: Specifications and structure of the model.

| Section                   | Configuration         |
|---------------------------|-----------------------|
| Maximum model age         | 100                   |
| Sexes                     | Sexes combined        |
| Population bins           | 8-88 cm by 2 cm bins  |
| Summary biomass (mt) age  | 8+                    |
| Number of areas           | 2                     |
| Number of seasons         | 1                     |
| Number of growth patterns | 1                     |
| Start year                | 1889                  |
| End year                  | 2024                  |
| Data length bins          | 10-74 cm by 2 cm bins |
| Data age bins             | 0-65 by 1 year        |

Table 10: Data weightings applied to compositions according to the **Francis** method. **Obs.** refers to the number of unique composition vectors included in the likelihood. **N input** and **N adj.** refer to the sample sizes of those vectors before and after being adjusted by the the weights. **CAAL** is conditional age-at-length data.

| Type   | Fleet         | Francis | Obs. | Mean N input | Mean N adj. | Sum N adj. |
|--------|---------------|---------|------|--------------|-------------|------------|
| Length | 1_CA_TWL      | 0.519   | 38   | 9.9          | 5.2         | 196.0      |
| Length | 2_CA_NONTWL   | 0.287   | 44   | 34.9         | 10.0        | 440.7      |
| Length | 3_CA_REC      | 0.524   | 42   | 45.8         | 24.0        | 1008.7     |
| Length | 4_ORWA_TWL    | 0.255   | 29   | 37.2         | 9.5         | 275.4      |
| Length | 5_ORWA_NONTWL | 0.374   | 31   | 79.3         | 29.6        | 917.9      |
| Length | 6_OR_REC      | 0.364   | 43   | 52.5         | 19.1        | 820.6      |
| Length | 7_WA_REC      | 1.000   | 26   | 7.3          | 7.3         | 188.6      |
| Length | 8_CACPFV      | 0.560   | 32   | 37.6         | 21.0        | 673.5      |
| Length | 9_OR_RECOB    | 0.541   | 20   | 25.6         | 13.9        | 277.1      |
| Length | 10_TRI_ORWA   | 0.455   | 7    | 11.2         | 5.1         | 35.9       |
| Length | 11_NWFSC_ORWA | 0.511   | 21   | 16.1         | 8.2         | 172.5      |
| Length | 12_IPHC_ORWA  | 0.892   | 21   | 28.7         | 25.6        | 538.2      |
| CAAL   | 2_CA_NONTWL   | 1.000   | 42   | 1.4          | 1.4         | 58.0       |
| CAAL   | 3_CA_REC      | 1.000   | 102  | 1.5          | 1.5         | 153.0      |
| CAAL   | 4_ORWA_TWL    | 1.000   | 353  | 4.2          | 4.2         | 1486.0     |
| CAAL   | 5_ORWA_NONTWL | 0.222   | 266  | 8.9          | 2.0         | 523.4      |
| CAAL   | 6_OR_REC      | 1.000   | 195  | 4.1          | 4.1         | 798.0      |
| CAAL   | 7_WA_REC      | 1.000   | 177  | 3.6          | 3.6         | 643.0      |
| CAAL   | 11_NWFSC_ORWA | 1.000   | 382  | 2.3          | 2.3         | 870.0      |
| CAAL   | 12_IPHC_ORWA  | 0.088   | 531  | 16.0         | 1.4         | 748.8      |



Table 11: Estimated parameters in the model.

| Type                  | Count |
|-----------------------|-------|
| Growth mean           | 3     |
| Growth variability    | 2     |
| Stock-recruit         | 1     |
| Rec. dev. time series | 136   |
| Rec. dev. forecast    | 12    |
| Index                 | 8     |
| Index time-variation  | 1     |
| Size selectivity      | 25    |

Table 12: Parameter estimates, estimation phase, parameter bounds, estimation status, estimated standard deviation (SD), prior information [distribution(mean, SD)] used in the base model.

| Label                 | Value    | Phase | Bounds       | Status | SD      | Prior              |
|-----------------------|----------|-------|--------------|--------|---------|--------------------|
| NatM_break_1_Fem_GP_1 | 0.0439   | -1    | (0.01, 0.15) | fixed  |         | none               |
| L_at_Amin_Fem_GP_1    | 1.55     | 2     | (0.01, 35)   | ok     | 0.582   | none               |
| L_at_Amax_Fem_GP_1    | 61.4     | 2     | (40, 120)    | ok     | 0.225   | none               |
| VonBert_K_Fem_GP_1    | 0.0759   | 1     | (0.01, 0.2)  | ok     | 0.00133 | none               |
| CV_young_Fem_GP_1     | 0.148    | 3     | (0.01, 0.5)  | ok     | 0.00676 | none               |
| CV_old_Fem_GP_1       | 0.0644   | 7     | (0.01, 0.5)  | ok     | 0.00194 | none               |
| Wtlen_1_Fem_GP_1      | 7.18e-06 | -50   | (-3, 3)      | fixed  |         | none               |
| Wtlen_2_Fem_GP_1      | 3.24     | -50   | (-3, 4)      | fixed  |         | none               |
| Mat50%_Fem_GP_1       | 42.1     | -50   | (38, 45)     | fixed  |         | none               |
| Mat_slope_Fem_GP_1    | -0.402   | -50   | (-3, 3)      | fixed  |         | none               |
| Eggs_scalar_Fem_GP_1  | 7.22e-08 | -6    | (-3, 3e+05)  | fixed  |         | none               |
| Eggs_exp_len_Fem_GP_1 | 4.04     | -6    | (-3, 39000)  | fixed  |         | none               |
| RecrDist_GP_1         | 1        | -50   | (0, 2)       | fixed  |         | none               |
| RecrDist_Area_1       | 0        | -50   | (-4, 4)      | fixed  |         | none               |
| RecrDist_Area_2       | 0.473    | 3     | (-4, 4)      | ok     | 0.0235  | none               |
| RecrDist_month_1      | 1        | -50   | (0, 2)       | fixed  |         | none               |
| CohortGrowDev         | 1        | -50   | (0, 2)       | fixed  |         | none               |
| FracFemale_GP_1       | 0.5      | -99   | (1e-06, 1)   | fixed  |         | none               |
| SR_LN(R0)             | 5.49     | 3     | (3, 15)      | ok     | 0.0608  | none               |
| SR_BH_steep           | 0.72     | -3    | (0.2, 1)     | fixed  |         | none               |
| SR_sigmaR             | 0.5      | -2    | (0, 5)       | fixed  |         | none               |
| SR_regime             | 0        | -50   | (-5, 5)      | fixed  |         | none               |
| SR_autocorr           | 0        | -50   | (-1, 2)      | fixed  |         | none               |
| Early_RecrDev_1889    | 0.00546  | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |
| Early_RecrDev_1890    | 0.00565  | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |
| Early_RecrDev_1891    | 0.00585  | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |
| Early_RecrDev_1892    | 0.00604  | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |
| Early_RecrDev_1893    | 0.00625  | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |
| Early_RecrDev_1894    | 0.00646  | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |
| Early_RecrDev_1895    | 0.00667  | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |
| Early_RecrDev_1896    | 0.0069   | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |
| Early_RecrDev_1897    | 0.00713  | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |
| Early_RecrDev_1898    | 0.00737  | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |
| Early_RecrDev_1899    | 0.00762  | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |
| Early_RecrDev_1900    | 0.0079   | 7     | (-5, 5)      | dev    | 0.501   | normal(0.00, 0.50) |

|                    |          |   |         |     |       |                    |
|--------------------|----------|---|---------|-----|-------|--------------------|
| Early_RecrDev_1901 | 0.00818  | 7 | (-5, 5) | dev | 0.501 | normal(0.00, 0.50) |
| Early_RecrDev_1902 | 0.0085   | 7 | (-5, 5) | dev | 0.501 | normal(0.00, 0.50) |
| Early_RecrDev_1903 | 0.00881  | 7 | (-5, 5) | dev | 0.501 | normal(0.00, 0.50) |
| Early_RecrDev_1904 | 0.00916  | 7 | (-5, 5) | dev | 0.501 | normal(0.00, 0.50) |
| Early_RecrDev_1905 | 0.00955  | 7 | (-5, 5) | dev | 0.501 | normal(0.00, 0.50) |
| Early_RecrDev_1906 | 0.00998  | 7 | (-5, 5) | dev | 0.501 | normal(0.00, 0.50) |
| Early_RecrDev_1907 | 0.0105   | 7 | (-5, 5) | dev | 0.501 | normal(0.00, 0.50) |
| Early_RecrDev_1908 | 0.0111   | 7 | (-5, 5) | dev | 0.501 | normal(0.00, 0.50) |
| Early_RecrDev_1909 | 0.0118   | 7 | (-5, 5) | dev | 0.502 | normal(0.00, 0.50) |
| Early_RecrDev_1910 | 0.0126   | 7 | (-5, 5) | dev | 0.502 | normal(0.00, 0.50) |
| Early_RecrDev_1911 | 0.0135   | 7 | (-5, 5) | dev | 0.502 | normal(0.00, 0.50) |
| Early_RecrDev_1912 | 0.0145   | 7 | (-5, 5) | dev | 0.502 | normal(0.00, 0.50) |
| Early_RecrDev_1913 | 0.0155   | 7 | (-5, 5) | dev | 0.502 | normal(0.00, 0.50) |
| Early_RecrDev_1914 | 0.0164   | 7 | (-5, 5) | dev | 0.502 | normal(0.00, 0.50) |
| Early_RecrDev_1915 | 0.0171   | 7 | (-5, 5) | dev | 0.502 | normal(0.00, 0.50) |
| Early_RecrDev_1916 | 0.0173   | 7 | (-5, 5) | dev | 0.502 | normal(0.00, 0.50) |
| Early_RecrDev_1917 | 0.0169   | 7 | (-5, 5) | dev | 0.502 | normal(0.00, 0.50) |
| Early_RecrDev_1918 | 0.0155   | 7 | (-5, 5) | dev | 0.501 | normal(0.00, 0.50) |
| Early_RecrDev_1919 | 0.0127   | 7 | (-5, 5) | dev | 0.5   | normal(0.00, 0.50) |
| Early_RecrDev_1920 | 0.00827  | 7 | (-5, 5) | dev | 0.499 | normal(0.00, 0.50) |
| Early_RecrDev_1921 | 0.00191  | 7 | (-5, 5) | dev | 0.497 | normal(0.00, 0.50) |
| Early_RecrDev_1922 | -0.00657 | 7 | (-5, 5) | dev | 0.495 | normal(0.00, 0.50) |
| Early_RecrDev_1923 | -0.0172  | 7 | (-5, 5) | dev | 0.492 | normal(0.00, 0.50) |
| Early_RecrDev_1924 | -0.0298  | 7 | (-5, 5) | dev | 0.489 | normal(0.00, 0.50) |
| Early_RecrDev_1925 | -0.0441  | 7 | (-5, 5) | dev | 0.485 | normal(0.00, 0.50) |
| Early_RecrDev_1926 | -0.0593  | 7 | (-5, 5) | dev | 0.482 | normal(0.00, 0.50) |
| Early_RecrDev_1927 | -0.0748  | 7 | (-5, 5) | dev | 0.478 | normal(0.00, 0.50) |
| Early_RecrDev_1928 | -0.0896  | 7 | (-5, 5) | dev | 0.475 | normal(0.00, 0.50) |
| Early_RecrDev_1929 | -0.103   | 7 | (-5, 5) | dev | 0.472 | normal(0.00, 0.50) |
| Early_RecrDev_1930 | -0.115   | 7 | (-5, 5) | dev | 0.469 | normal(0.00, 0.50) |
| Early_RecrDev_1931 | -0.124   | 7 | (-5, 5) | dev | 0.467 | normal(0.00, 0.50) |
| Early_RecrDev_1932 | -0.133   | 7 | (-5, 5) | dev | 0.464 | normal(0.00, 0.50) |
| Early_RecrDev_1933 | -0.142   | 7 | (-5, 5) | dev | 0.462 | normal(0.00, 0.50) |
| Early_RecrDev_1934 | -0.153   | 7 | (-5, 5) | dev | 0.46  | normal(0.00, 0.50) |
| Early_RecrDev_1935 | -0.165   | 7 | (-5, 5) | dev | 0.457 | normal(0.00, 0.50) |
| Early_RecrDev_1936 | -0.18    | 7 | (-5, 5) | dev | 0.454 | normal(0.00, 0.50) |
| Early_RecrDev_1937 | -0.196   | 7 | (-5, 5) | dev | 0.451 | normal(0.00, 0.50) |
| Early_RecrDev_1938 | -0.21    | 7 | (-5, 5) | dev | 0.448 | normal(0.00, 0.50) |
| Early_RecrDev_1939 | -0.221   | 7 | (-5, 5) | dev | 0.446 | normal(0.00, 0.50) |

|                    |           |   |         |     |       |                    |
|--------------------|-----------|---|---------|-----|-------|--------------------|
| Early_RecrDev_1940 | -0.225    | 7 | (-5, 5) | dev | 0.445 | normal(0.00, 0.50) |
| Early_RecrDev_1941 | -0.221    | 7 | (-5, 5) | dev | 0.445 | normal(0.00, 0.50) |
| Early_RecrDev_1942 | -0.205    | 7 | (-5, 5) | dev | 0.447 | normal(0.00, 0.50) |
| Early_RecrDev_1943 | -0.174    | 7 | (-5, 5) | dev | 0.452 | normal(0.00, 0.50) |
| Early_RecrDev_1944 | -0.122    | 7 | (-5, 5) | dev | 0.461 | normal(0.00, 0.50) |
| Early_RecrDev_1945 | -0.0435   | 7 | (-5, 5) | dev | 0.475 | normal(0.00, 0.50) |
| Early_RecrDev_1946 | 0.0644    | 7 | (-5, 5) | dev | 0.495 | normal(0.00, 0.50) |
| Early_RecrDev_1947 | 0.192     | 7 | (-5, 5) | dev | 0.521 | normal(0.00, 0.50) |
| Early_RecrDev_1948 | 0.298     | 7 | (-5, 5) | dev | 0.541 | normal(0.00, 0.50) |
| Early_RecrDev_1949 | 0.303     | 7 | (-5, 5) | dev | 0.537 | normal(0.00, 0.50) |
| Early_RecrDev_1950 | 0.18      | 7 | (-5, 5) | dev | 0.509 | normal(0.00, 0.50) |
| Early_RecrDev_1951 | -0.000643 | 7 | (-5, 5) | dev | 0.474 | normal(0.00, 0.50) |
| Early_RecrDev_1952 | -0.174    | 7 | (-5, 5) | dev | 0.444 | normal(0.00, 0.50) |
| Early_RecrDev_1953 | -0.312    | 7 | (-5, 5) | dev | 0.422 | normal(0.00, 0.50) |
| Early_RecrDev_1954 | -0.406    | 7 | (-5, 5) | dev | 0.409 | normal(0.00, 0.50) |
| Early_RecrDev_1955 | -0.456    | 7 | (-5, 5) | dev | 0.402 | normal(0.00, 0.50) |
| Early_RecrDev_1956 | -0.46     | 7 | (-5, 5) | dev | 0.4   | normal(0.00, 0.50) |
| Early_RecrDev_1957 | -0.421    | 7 | (-5, 5) | dev | 0.402 | normal(0.00, 0.50) |
| Early_RecrDev_1958 | -0.343    | 7 | (-5, 5) | dev | 0.409 | normal(0.00, 0.50) |
| Early_RecrDev_1959 | -0.245    | 7 | (-5, 5) | dev | 0.414 | normal(0.00, 0.50) |
| Early_RecrDev_1960 | -0.194    | 7 | (-5, 5) | dev | 0.413 | normal(0.00, 0.50) |
| Early_RecrDev_1961 | -0.273    | 7 | (-5, 5) | dev | 0.406 | normal(0.00, 0.50) |
| Early_RecrDev_1962 | -0.429    | 7 | (-5, 5) | dev | 0.395 | normal(0.00, 0.50) |
| Early_RecrDev_1963 | -0.544    | 7 | (-5, 5) | dev | 0.386 | normal(0.00, 0.50) |
| Early_RecrDev_1964 | -0.538    | 7 | (-5, 5) | dev | 0.385 | normal(0.00, 0.50) |
| Early_RecrDev_1965 | -0.39     | 7 | (-5, 5) | dev | 0.392 | normal(0.00, 0.50) |
| Early_RecrDev_1966 | -0.206    | 7 | (-5, 5) | dev | 0.398 | normal(0.00, 0.50) |
| Early_RecrDev_1967 | -0.0938   | 7 | (-5, 5) | dev | 0.416 | normal(0.00, 0.50) |
| Early_RecrDev_1968 | 0.0875    | 7 | (-5, 5) | dev | 0.422 | normal(0.00, 0.50) |
| Early_RecrDev_1969 | 0.184     | 7 | (-5, 5) | dev | 0.445 | normal(0.00, 0.50) |
| Early_RecrDev_1970 | 0.383     | 7 | (-5, 5) | dev | 0.494 | normal(0.00, 0.50) |
| Early_RecrDev_1971 | 0.841     | 7 | (-5, 5) | dev | 0.385 | normal(0.00, 0.50) |
| Early_RecrDev_1972 | 0.254     | 7 | (-5, 5) | dev | 0.461 | normal(0.00, 0.50) |
| Early_RecrDev_1973 | -0.0456   | 7 | (-5, 5) | dev | 0.419 | normal(0.00, 0.50) |
| Early_RecrDev_1974 | 0.055     | 7 | (-5, 5) | dev | 0.421 | normal(0.00, 0.50) |
| Early_RecrDev_1975 | 0.431     | 7 | (-5, 5) | dev | 0.367 | normal(0.00, 0.50) |
| Early_RecrDev_1976 | 0.245     | 7 | (-5, 5) | dev | 0.412 | normal(0.00, 0.50) |
| Early_RecrDev_1977 | 0.185     | 7 | (-5, 5) | dev | 0.362 | normal(0.00, 0.50) |
| Early_RecrDev_1978 | -0.14     | 7 | (-5, 5) | dev | 0.393 | normal(0.00, 0.50) |

|                    |         |   |         |     |       |                    |
|--------------------|---------|---|---------|-----|-------|--------------------|
| Early_RecrDev_1979 | 0.14    | 7 | (-5, 5) | dev | 0.388 | normal(0.00, 0.50) |
| Main_RecrDev_1980  | 0.299   | 7 | (-5, 5) | dev | 0.429 | normal(0.00, 0.50) |
| Main_RecrDev_1981  | 0.403   | 7 | (-5, 5) | dev | 0.487 | normal(0.00, 0.50) |
| Main_RecrDev_1982  | 0.516   | 7 | (-5, 5) | dev | 0.453 | normal(0.00, 0.50) |
| Main_RecrDev_1983  | 0.143   | 7 | (-5, 5) | dev | 0.503 | normal(0.00, 0.50) |
| Main_RecrDev_1984  | 0.415   | 7 | (-5, 5) | dev | 0.447 | normal(0.00, 0.50) |
| Main_RecrDev_1985  | 0.242   | 7 | (-5, 5) | dev | 0.444 | normal(0.00, 0.50) |
| Main_RecrDev_1986  | -0.094  | 7 | (-5, 5) | dev | 0.397 | normal(0.00, 0.50) |
| Main_RecrDev_1987  | -0.33   | 7 | (-5, 5) | dev | 0.356 | normal(0.00, 0.50) |
| Main_RecrDev_1988  | -0.614  | 7 | (-5, 5) | dev | 0.337 | normal(0.00, 0.50) |
| Main_RecrDev_1989  | -0.757  | 7 | (-5, 5) | dev | 0.32  | normal(0.00, 0.50) |
| Main_RecrDev_1990  | -0.833  | 7 | (-5, 5) | dev | 0.31  | normal(0.00, 0.50) |
| Main_RecrDev_1991  | -0.921  | 7 | (-5, 5) | dev | 0.313 | normal(0.00, 0.50) |
| Main_RecrDev_1992  | -0.757  | 7 | (-5, 5) | dev | 0.33  | normal(0.00, 0.50) |
| Main_RecrDev_1993  | -0.0467 | 7 | (-5, 5) | dev | 0.264 | normal(0.00, 0.50) |
| Main_RecrDev_1994  | -0.195  | 7 | (-5, 5) | dev | 0.287 | normal(0.00, 0.50) |
| Main_RecrDev_1995  | -0.922  | 7 | (-5, 5) | dev | 0.333 | normal(0.00, 0.50) |
| Main_RecrDev_1996  | -0.949  | 7 | (-5, 5) | dev | 0.319 | normal(0.00, 0.50) |
| Main_RecrDev_1997  | -0.739  | 7 | (-5, 5) | dev | 0.338 | normal(0.00, 0.50) |
| Main_RecrDev_1998  | -0.244  | 7 | (-5, 5) | dev | 0.352 | normal(0.00, 0.50) |
| Main_RecrDev_1999  | 0.224   | 7 | (-5, 5) | dev | 0.295 | normal(0.00, 0.50) |
| Main_RecrDev_2000  | -0.41   | 7 | (-5, 5) | dev | 0.383 | normal(0.00, 0.50) |
| Main_RecrDev_2001  | -0.27   | 7 | (-5, 5) | dev | 0.383 | normal(0.00, 0.50) |
| Main_RecrDev_2002  | 0.985   | 7 | (-5, 5) | dev | 0.192 | normal(0.00, 0.50) |
| Main_RecrDev_2003  | 0.0639  | 7 | (-5, 5) | dev | 0.378 | normal(0.00, 0.50) |
| Main_RecrDev_2004  | -0.344  | 7 | (-5, 5) | dev | 0.379 | normal(0.00, 0.50) |
| Main_RecrDev_2005  | 0.0686  | 7 | (-5, 5) | dev | 0.326 | normal(0.00, 0.50) |
| Main_RecrDev_2006  | 0.588   | 7 | (-5, 5) | dev | 0.297 | normal(0.00, 0.50) |
| Main_RecrDev_2007  | 0.58    | 7 | (-5, 5) | dev | 0.367 | normal(0.00, 0.50) |
| Main_RecrDev_2008  | 1.1     | 7 | (-5, 5) | dev | 0.269 | normal(0.00, 0.50) |
| Main_RecrDev_2009  | 0.826   | 7 | (-5, 5) | dev | 0.327 | normal(0.00, 0.50) |
| Main_RecrDev_2010  | 0.791   | 7 | (-5, 5) | dev | 0.285 | normal(0.00, 0.50) |
| Main_RecrDev_2011  | 0.52    | 7 | (-5, 5) | dev | 0.3   | normal(0.00, 0.50) |
| Main_RecrDev_2012  | 0.369   | 7 | (-5, 5) | dev | 0.35  | normal(0.00, 0.50) |
| Main_RecrDev_2013  | 1.22    | 7 | (-5, 5) | dev | 0.208 | normal(0.00, 0.50) |
| Main_RecrDev_2014  | 0.429   | 7 | (-5, 5) | dev | 0.375 | normal(0.00, 0.50) |
| Main_RecrDev_2015  | 0.729   | 7 | (-5, 5) | dev | 0.294 | normal(0.00, 0.50) |
| Main_RecrDev_2016  | 0.315   | 7 | (-5, 5) | dev | 0.338 | normal(0.00, 0.50) |
| Main_RecrDev_2017  | -0.404  | 7 | (-5, 5) | dev | 0.403 | normal(0.00, 0.50) |

|                                   |        |    |           |       |        |                    |
|-----------------------------------|--------|----|-----------|-------|--------|--------------------|
| Main_RecrDev_2018                 | -0.472 | 7  | (-5, 5)   | dev   | 0.41   | normal(0.00, 0.50) |
| Main_RecrDev_2019                 | -0.467 | 7  | (-5, 5)   | dev   | 0.42   | normal(0.00, 0.50) |
| Main_RecrDev_2020                 | -0.501 | 7  | (-5, 5)   | dev   | 0.44   | normal(0.00, 0.50) |
| Main_RecrDev_2021                 | -0.259 | 7  | (-5, 5)   | dev   | 0.466  | normal(0.00, 0.50) |
| Main_RecrDev_2022                 | -0.154 | 7  | (-5, 5)   | dev   | 0.486  | normal(0.00, 0.50) |
| Main_RecrDev_2023                 | -0.141 | 7  | (-5, 5)   | dev   | 0.49   | normal(0.00, 0.50) |
| Late_RecrDev_2024                 | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2025                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2026                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2027                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2028                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2029                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2030                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2031                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2032                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2033                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2034                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2035                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| ForeRecr_2036                     | 0      | 8  | (-5, 5)   | dev   | 0.5    | normal(0.00, 0.50) |
| LnQ_base_3_CA_REC(3)              | -9.2   | -1 | (-15, 15) | fixed |        | none               |
| Q_extraSD_3_CA_REC(3)             | 0.122  | 5  | (0, 5)    | ok    | 0.078  | none               |
| LnQ_base_6_OR_REC(6)              | -9.14  | 1  | (-15, 15) | ok    | 0.111  | none               |
| Q_extraSD_6_OR_REC(6)             | 0.0842 | 5  | (0, 5)    | ok    | 0.029  | none               |
| LnQ_base_7_WA_REC(7)              | -8.85  | -1 | (-20, 15) | fixed |        | none               |
| Q_extraSD_7_WA_REC(7)             | 0.383  | 5  | (0, 5)    | ok    | 0.0763 | none               |
| LnQ_base_8_CACPFV(8)              | -9.25  | -1 | (-15, 15) | fixed |        | none               |
| Q_extraSD_8_CACPFV(8)             | 0.0966 | 5  | (0, 5)    | ok    | 0.0746 | none               |
| LnQ_base_9_OR_RECOB(9)            | -11.4  | -1 | (-15, 15) | fixed |        | none               |
| Q_extraSD_9_OR_RECOB(9)           | 0.165  | 5  | (0, 5)    | ok    | 0.0792 | none               |
| LnQ_base_10_TRI_ORWA(10)          | -1.51  | -1 | (-15, 15) | fixed |        | none               |
| Q_extraSD_10_TRI_ORWA(10)         | 0.14   | 5  | (0, 5)    | ok    | 0.122  | none               |
| LnQ_base_11_NWFSC_ORWA(11)        | -0.957 | -1 | (-15, 15) | fixed |        | none               |
| Q_extraSD_11_NWFSC_ORWA(11)       | 0      | -5 | (0, 5)    | fixed |        | none               |
| LnQ_base_12_IPHC_ORWA(12)         | -0.648 | -1 | (-15, 15) | fixed |        | none               |
| Q_extraSD_12_IPHC_ORWA(12)        | 0.552  | 5  | (0, 5)    | ok    | 0.107  | none               |
| LnQ_base_6_OR_REC(6)_BLK2add_2004 | -2.67  | 1  | (-4, 4)   | ok    | 0.112  | none               |
| Size_DbIN_peak_1_CA_TWL(1)        | 44     | 4  | (20, 60)  | ok    | 3.31   | none               |
| Size_DbIN_top_logit_1_CA_TWL(1)   | -15    | -5 | (-15, 4)  | fixed |        | none               |
| Size_DbIN_ascend_se_1_CA_TWL(1)   | 5.13   | 4  | (-1, 9)   | ok    | 0.404  | none               |

|                                        |      |    |            |       |       |      |
|----------------------------------------|------|----|------------|-------|-------|------|
| Size_DbIN_descend_se_1_CA_TWL(1)       | 18.3 | 5  | (-1, 30)   | ok    | 152   | none |
| Size_DbIN_start_logit_1_CA_TWL(1)      | -999 | -4 | (-1000, 9) | fixed |       | none |
| Size_DbIN_end_logit_1_CA_TWL(1)        | -999 | -5 | (-1000, 9) | fixed |       | none |
| Size_DbIN_peak_2_CA_NONTWL(2)          | 44.7 | 4  | (20, 60)   | ok    | 2.49  | none |
| Size_DbIN_top_logit_2_CA_NONTWL(2)     | -15  | -5 | (-15, 4)   | fixed |       | none |
| Size_DbIN_ascend_se_2_CA_NONTWL(2)     | 5.2  | 4  | (-1, 9)    | ok    | 0.281 | none |
| Size_DbIN_descend_se_2_CA_NONTWL(2)    | 17.4 | 5  | (-1, 30)   | ok    | 172   | none |
| Size_DbIN_start_logit_2_CA_NONTWL(2)   | -999 | -4 | (-1000, 9) | fixed |       | none |
| Size_DbIN_end_logit_2_CA_NONTWL(2)     | -999 | -5 | (-1000, 9) | fixed |       | none |
| Size_DbIN_peak_3_CA_REC(3)             | 41.7 | 4  | (20, 60)   | ok    | 1.35  | none |
| Size_DbIN_top_logit_3_CA_REC(3)        | -15  | -5 | (-15, 4)   | fixed |       | none |
| Size_DbIN_ascend_se_3_CA_REC(3)        | 5.21 | 4  | (-1, 9)    | ok    | 0.144 | none |
| Size_DbIN_descend_se_3_CA_REC(3)       | 20   | -5 | (-1, 30)   | fixed |       | none |
| Size_DbIN_start_logit_3_CA_REC(3)      | -999 | -4 | (-1000, 9) | fixed |       | none |
| Size_DbIN_end_logit_3_CA_REC(3)        | -999 | -5 | (-1000, 9) | fixed |       | none |
| Size_DbIN_peak_4_ORWA_TWL(4)           | 41.9 | 4  | (20, 60)   | ok    | 3.04  | none |
| Size_DbIN_top_logit_4_ORWA_TWL(4)      | -15  | -5 | (-15, 4)   | fixed |       | none |
| Size_DbIN_ascend_se_4_ORWA_TWL(4)      | 5.49 | 4  | (-1, 9)    | ok    | 0.343 | none |
| Size_DbIN_descend_se_4_ORWA_TWL(4)     | 18.2 | 5  | (-1, 30)   | ok    | 151   | none |
| Size_DbIN_start_logit_4_ORWA_TWL(4)    | -999 | -4 | (-1000, 9) | fixed |       | none |
| Size_DbIN_end_logit_4_ORWA_TWL(4)      | -999 | -5 | (-1000, 9) | fixed |       | none |
| Size_DbIN_peak_5_ORWA_NONTWL(5)        | 50.9 | 4  | (20, 60)   | ok    | 1.48  | none |
| Size_DbIN_top_logit_5_ORWA_NONTWL(5)   | -15  | -5 | (-15, 4)   | fixed |       | none |
| Size_DbIN_ascend_se_5_ORWA_NONTWL(5)   | 5.44 | 4  | (-1, 9)    | ok    | 0.147 | none |
| Size_DbIN_descend_se_5_ORWA_NONTWL(5)  | 20   | -5 | (-1, 30)   | fixed |       | none |
| Size_DbIN_start_logit_5_ORWA_NONTWL(5) | -999 | -4 | (-1000, 9) | fixed |       | none |
| Size_DbIN_end_logit_5_ORWA_NONTWL(5)   | -999 | -5 | (-1000, 9) | fixed |       | none |
| Size_DbIN_peak_6_OR_REC(6)             | 36.7 | 4  | (20, 60)   | ok    | 1.27  | none |
| Size_DbIN_top_logit_6_OR_REC(6)        | -15  | -5 | (-15, 4)   | fixed |       | none |
| Size_DbIN_ascend_se_6_OR_REC(6)        | 4.14 | 4  | (-1, 9)    | ok    | 0.28  | none |
| Size_DbIN_descend_se_6_OR_REC(6)       | 12   | -5 | (-1, 30)   | fixed |       | none |
| Size_DbIN_start_logit_6_OR_REC(6)      | -999 | -4 | (-1000, 9) | fixed |       | none |
| Size_DbIN_end_logit_6_OR_REC(6)        | -999 | -5 | (-1000, 9) | fixed |       | none |
| Size_DbIN_peak_7_WA_REC(7)             | 42.8 | 6  | (20, 60)   | ok    | 2.75  | none |
| Size_DbIN_top_logit_7_WA_REC(7)        | -15  | -5 | (-15, 4)   | fixed |       | none |
| Size_DbIN_ascend_se_7_WA_REC(7)        | 4.32 | 6  | (-1, 9)    | ok    | 0.518 | none |
| Size_DbIN_descend_se_7_WA_REC(7)       | 20   | -5 | (-1, 30)   | fixed |       | none |
| Size_DbIN_start_logit_7_WA_REC(7)      | -999 | -4 | (-1000, 9) | fixed |       | none |
| Size_DbIN_end_logit_7_WA_REC(7)        | -999 | -5 | (-1000, 9) | fixed |       | none |

|                                         |      |    |            |       |       |      |
|-----------------------------------------|------|----|------------|-------|-------|------|
| Size_DbIN_peak_9_OR_RECOB(9)            | 35.1 | 4  | (20, 60)   | ok    | 1.63  | none |
| Size_DbIN_top_logit_9_OR_RECOB(9)       | -15  | -5 | (-15, 4)   | fixed |       | none |
| Size_DbIN_ascend_se_9_OR_RECOB(9)       | 4.61 | 4  | (-1, 9)    | ok    | 0.292 | none |
| Size_DbIN_descend_se_9_OR_RECOB(9)      | 20   | -5 | (-1, 30)   | fixed |       | none |
| Size_DbIN_start_logit_9_OR_RECOB(9)     | -999 | -4 | (-1000, 9) | fixed |       | none |
| Size_DbIN_end_logit_9_OR_RECOB(9)       | -999 | -5 | (-1000, 9) | fixed |       | none |
| Size_DbIN_peak_10_TRI_ORWA(10)          | 80   | 4  | (20, 80)   | HI    | 0.899 | none |
| Size_DbIN_top_logit_10_TRI_ORWA(10)     | -15  | -5 | (-15, 4)   | fixed |       | none |
| Size_DbIN_ascend_se_10_TRI_ORWA(10)     | 7.08 | 4  | (-1, 9)    | ok    | 0.264 | none |
| Size_DbIN_descend_se_10_TRI_ORWA(10)    | 12   | -5 | (-1, 30)   | fixed |       | none |
| Size_DbIN_start_logit_10_TRI_ORWA(10)   | -999 | -4 | (-1000, 9) | fixed |       | none |
| Size_DbIN_end_logit_10_TRI_ORWA(10)     | -999 | -5 | (-1000, 9) | fixed |       | none |
| Size_DbIN_peak_11_NWFSC_ORWA(11)        | 48.9 | 4  | (20, 60)   | ok    | 5.59  | none |
| Size_DbIN_top_logit_11_NWFSC_ORWA(11)   | -15  | -5 | (-15, 4)   | fixed |       | none |
| Size_DbIN_ascend_se_11_NWFSC_ORWA(11)   | 6.23 | 4  | (-1, 9)    | ok    | 0.386 | none |
| Size_DbIN_descend_se_11_NWFSC_ORWA(11)  | 20   | -5 | (-1, 30)   | fixed |       | none |
| Size_DbIN_start_logit_11_NWFSC_ORWA(11) | -999 | -4 | (-1000, 9) | fixed |       | none |
| Size_DbIN_end_logit_11_NWFSC_ORWA(11)   | -999 | -5 | (-1000, 9) | fixed |       | none |
| Size_DbIN_peak_12_IPHC_ORWA(12)         | 54   | 4  | (20, 60)   | ok    | 1.21  | none |
| Size_DbIN_top_logit_12_IPHC_ORWA(12)    | -15  | -5 | (-15, 4)   | fixed |       | none |
| Size_DbIN_ascend_se_12_IPHC_ORWA(12)    | 4.14 | 4  | (-1, 9)    | ok    | 0.233 | none |
| Size_DbIN_descend_se_12_IPHC_ORWA(12)   | 20   | -5 | (-1, 30)   | fixed |       | none |
| Size_DbIN_start_logit_12_IPHC_ORWA(12)  | -999 | -4 | (-1000, 9) | fixed |       | none |
| Size_DbIN_end_logit_12_IPHC_ORWA(12)    | -999 | -5 | (-1000, 9) | fixed |       | none |



Table 13: Base model sensitivity to model parameters and specifications.

| Label                                      | Base    | Est. M  | Est.<br>steepness | 2017 LW<br>relationship |
|--------------------------------------------|---------|---------|-------------------|-------------------------|
| <b>Diff. in likelihood from base model</b> |         |         |                   |                         |
| Total                                      | 0       | -5.64   | -3.33             | 0                       |
| Index                                      | 0       | -3.174  | -1.576            | 0                       |
| Length comp                                | 0       | -4.67   | -3.35             | 0                       |
| Age comp                                   | 0       | 3.93    | 2.65              | 0                       |
| Recruitment                                | 0       | -1.488  | -0.911            | 0                       |
| Parm priors                                | 0       | 0       | 0                 | 0                       |
| <b>Estimates of key parameters</b>         |         |         |                   |                         |
| Recruitment unfished thousands             | 241.887 | 392.064 | 240.801           | 241.887                 |
| log(R0)                                    | 5.488   | 5.971   | 5.484             | 5.488                   |
| M Female                                   | 0.044   | 0.053   | 0.044             | 0.044                   |
| L at Amax Female                           | 61.4    | 61.4    | 61.4              | 61.4                    |
| <b>Estimates of derived quantities</b>     |         |         |                   |                         |
| Unfished age 8+ bio 1000 mt                | 10.331  | 11.865  | 10.281            | 10.331                  |
| B0 billions of eggs                        | 1190    | 1316.36 | 1184.16           | 1190                    |
| B2025 billions of eggs                     | 477.63  | 724.664 | 639.998           | 477.63                  |
| Fraction unfished 2025                     | 0.401   | 0.551   | 0.54              | 0.401                   |
| Fishing intensity 2024                     | 0.357   | 0.215   | 0.276             | 0.357                   |

Table 14: Base model sensitivity to the removal of data sources (indices).

| Label                                      | Base    | - CA<br>REC | - OR<br>REC | - WA<br>REC | - CA<br>CPFV | - ORFS  | -<br>Triennial | - WCG-<br>BTS | - IPHC  | Yellow<br>eye<br>Rockfish<br>stock<br>assessment<br>2025<br>indices |
|--------------------------------------------|---------|-------------|-------------|-------------|--------------|---------|----------------|---------------|---------|---------------------------------------------------------------------|
| <b>Diff. in likelihood from base model</b> |         |             |             |             |              |         |                |               |         |                                                                     |
| Total                                      | 0       | 4.67        | 29.12       | 5.15        | -163.01      | -103.81 | -25.74         | -1460.39      | -957.26 | -2664.61                                                            |
| Index                                      | 0       | 4.893       | 29.269      | 5.761       | 5.574        | 4.043   | 0.897          | 4.76          | 0.932   | NA                                                                  |
| Length comp                                | 0       | 0.11        | 0.48        | -2.23       | -165.77      | -99.48  | -24.5          | -81.58        | -67.71  | -457.753                                                            |
| Age comp                                   | 0       | -0.02       | 0.09        | 2.29        | -5.05        | -8.32   | -2.3           | -1379.01      | -889.55 | -2876.84                                                            |
| Recruitment                                | 0       | -0.334      | -0.722      | -0.741      | 2.242        | -0.01   | 0.183          | -4.526        | -1.172  | -5.48                                                               |
| Parm priors                                | 0       | 0           | 0           | 0           | 0            | 0       | 0              | 0             | 0       | 0                                                                   |
| <b>Estimates of key parameters</b>         |         |             |             |             |              |         |                |               |         |                                                                     |
| Recruitment unfished thousands             | 241.887 | 237.747     | 238.639     | 231.896     | 246.229      | 242.162 | 243.372        | 244.802       | 222.331 | 209.223                                                             |
| log(R0)                                    | 5.488   | 5.471       | 5.475       | 5.446       | 5.506        | 5.49    | 5.495          | 5.5           | 5.404   | 5.343                                                               |
| M Female                                   | 0.044   | 0.044       | 0.044       | 0.044       | 0.044        | 0.044   | 0.044          | 0.044         | 0.044   | 0.044                                                               |
| L at Amax Female                           | 61.4    | 61.4        | 61.4        | 61.4        | 61.4         | 61.4    | 61.3           | 61.5          | 62.4    | 62.7                                                                |
| <b>Estimates of derived quantities</b>     |         |             |             |             |              |         |                |               |         |                                                                     |
| Unfished age 8+ bio 1000 mt                | 10.331  | 10.156      | 10.194      | 9.9         | 10.52        | 10.358  | 10.382         | 10.489        | 9.916   | 9.454                                                               |
| B0 billions of eggs                        | 1190    | 1169.99     | 1174.29     | 1140.22     | 1211.93      | 1193.95 | 1195.11        | 1205.33       | 1162.9  | 1111.28                                                             |
| B2025 billions of eggs                     | 477.63  | 459.638     | 464.736     | 434.657     | 499.77       | 488.3   | 485.989        | 484.887       | 391.733 | 339.272                                                             |
| Fraction unfished 2025                     | 0.401   | 0.393       | 0.396       | 0.381       | 0.412        | 0.409   | 0.407          | 0.402         | 0.337   | 0.305                                                               |
| Fishing intensity 2024                     | 0.357   | 0.368       | 0.365       | 0.385       | 0.343        | 0.35    | 0.351          | 0.357         | 0.428   | 0.484                                                               |

Table 15: Base model sensitivity to the removal of data sources (length compositional data - 1 of 2).

| Label                                      | Base    | - CA<br>TWL | - CA<br>NON-<br>TWL | - CA<br>REC | - ORWA<br>TWL | - ORWA<br>NON-<br>TWL | - OR<br>REC | - WA<br>REC |
|--------------------------------------------|---------|-------------|---------------------|-------------|---------------|-----------------------|-------------|-------------|
| <b>Diff. in likelihood from base model</b> |         |             |                     |             |               |                       |             |             |
| Total                                      | 0       | -139.43     | -109.74             | -271.06     | -93.92        | -95.57                | -194.67     | -119.41     |
| Index                                      | 0       | 0.235       | -1.775              | -0.893      | -1.404        | -0.586                | -2.52       | -0.162      |
| Length comp                                | 0       | -136.54     | -97.36              | -263.41     | -84.08        | -88.64                | -173.64     | -116.68     |
| Age comp                                   | 0       | -3.43       | -15.27              | -9.08       | -4.56         | -5.8                  | -18.88      | -2.76       |
| Recruitment                                | 0       | 0.305       | 4.435               | 2.435       | -3.845        | -0.547                | 0.428       | 0.193       |
| Parm priors                                | 0       | 0           | 0                   | 0           | 0             | 0                     | 0           | 0           |
| <b>Estimates of key parameters</b>         |         |             |                     |             |               |                       |             |             |
| Recruitment unfished thousands             | 241.887 | 240.231     | 243.282             | 243.821     | 241.236       | 244.452               | 247.886     | 242.171     |
| log(R0)                                    | 5.488   | 5.482       | 5.494               | 5.496       | 5.486         | 5.499                 | 5.513       | 5.49        |
| M Female                                   | 0.044   | 0.044       | 0.044               | 0.044       | 0.044         | 0.044                 | 0.044       | 0.044       |
| L at Amax Female                           | 61.4    | 61.3        | 61.3                | 61.3        | 61.2          | 61.4                  | 61.3        | 61.2        |
| <b>Estimates of derived quantities</b>     |         |             |                     |             |               |                       |             |             |
| Unfished age 8+ bio 1000 mt                | 10.331  | 10.24       | 10.38               | 10.41       | 10.278        | 10.468                | 10.661      | 10.304      |
| B0 billions of eggs                        | 1190    | 1178.36     | 1194.97             | 1198.67     | 1182.05       | 1207.27               | 1231.32     | 1184.65     |
| B2025 billions of eggs                     | 477.63  | 471.102     | 447.839             | 511.303     | 520.246       | 500.739               | 543.75      | 475.296     |
| Fraction unfished 2025                     | 0.401   | 0.4         | 0.375               | 0.427       | 0.44          | 0.415                 | 0.442       | 0.401       |
| Fishing intensity 2024                     | 0.357   | 0.36        | 0.371               | 0.341       | 0.333         | 0.344                 | 0.322       | 0.361       |

Table 16: Base model sensitivity to the removal of data sources (length compositional data - 2 of 2).

| Label                                      | Base    | - CA<br>CPFV | - ORFS  | -<br>Triennial | - WCG-<br>BTS | - IPHC  | No<br>length<br>comps |
|--------------------------------------------|---------|--------------|---------|----------------|---------------|---------|-----------------------|
| <b>Diff. in likelihood from base model</b> |         |              |         |                |               |         |                       |
| Total                                      | 0       | -169.32      | -108.27 | -26.84         | -66.36        | -55.02  | -1480.61              |
| Index                                      | 0       | -0.05        | -0.517  | 0.051          | -0.339        | -0.08   | -16.679               |
| Length comp                                | 0       | -165.81      | -99.39  | -25.21         | -65.86        | -52.19  | -1387.28              |
| Age comp                                   | 0       | -5.16        | -8.03   | -1.64          | -0.4          | -3.41   | -78.02                |
| Recruitment                                | 0       | 1.653        | -0.303  | -0.02          | 0.26          | 0.66    | 0.188                 |
| Parm priors                                | 0       | 0            | 0       | 0              | 0             | 0       | 0                     |
| <b>Estimates of key parameters</b>         |         |              |         |                |               |         |                       |
| Recruitment unfished thousands             | 241.887 | 240.462      | 242.106 | 241.355        | 242.363       | 241.152 | 383.142               |
| log(R0)                                    | 5.488   | 5.483        | 5.489   | 5.486          | 5.49          | 5.485   | 5.948                 |
| M Female                                   | 0.044   | 0.044        | 0.044   | 0.044          | 0.044         | 0.044   | 0.044                 |
| L at Amax Female                           | 61.4    | 61.4         | 61.4    | 61.3           | 61.4          | 61.3    | 61.4                  |
| <b>Estimates of derived quantities</b>     |         |              |         |                |               |         |                       |
| Unfished age 8+ bio 1000 mt                | 10.331  | 10.277       | 10.357  | 10.295         | 10.353        | 10.298  | 16.697                |
| B0 billions of eggs                        | 1190    | 1184.01      | 1193.73 | 1185.09        | 1192.46       | 1186.1  | 1935.01               |
| B2025 billions of eggs                     | 477.63  | 473.383      | 490.053 | 476.66         | 488.707       | 477.218 | 1450.16               |
| Fraction unfished 2025                     | 0.401   | 0.4          | 0.411   | 0.402          | 0.41          | 0.402   | 0.749                 |
| Fishing intensity 2024                     | 0.357   | 0.359        | 0.349   | 0.357          | 0.351         | 0.356   | 0.163                 |

Table 17: Base model sensitivity to the removal of data sources (age compositional data; 1 of 2).

| Label                                      | Base    | - CA<br>NON-<br>TWL | - CA<br>REC | - ORWA<br>TWL | - ORWA<br>NON-<br>TWL | - OR<br>REC | - WA<br>REC |
|--------------------------------------------|---------|---------------------|-------------|---------------|-----------------------|-------------|-------------|
| <b>Diff. in likelihood from base model</b> |         |                     |             |               |                       |             |             |
| Total                                      | 0       | -126.99             | -338.81     | -1631.1       | -435.31               | -1020.72    | -813.89     |
| Index                                      | 0       | 0.079               | 0.284       | -0.634        | 0.256                 | 0.912       | -0.952      |
| Length comp                                | 0       | -1.02               | -5.06       | -16.08        | -3.06                 | -8.19       | -5.31       |
| Age comp                                   | 0       | -126.67             | -333.22     | -1610.46      | -433.04               | -1010.16    | -806.67     |
| Recruitment                                | 0       | 0.635               | -0.785      | -3.912        | 0.533                 | -3.299      | -0.916      |
| Parm priors                                | 0       | 0                   | 0           | 0             | 0                     | 0           | 0           |
| <b>Estimates of key parameters</b>         |         |                     |             |               |                       |             |             |
| Recruitment unfished thousands             | 241.887 | 241.952             | 244.735     | 242.069       | 243.986               | 244.196     | 250.708     |
| log(R0)                                    | 5.488   | 5.489               | 5.5         | 5.489         | 5.497                 | 5.498       | 5.524       |
| M Female                                   | 0.044   | 0.044               | 0.044       | 0.044         | 0.044                 | 0.044       | 0.044       |
| L at Amax Female                           | 61.4    | 61.4                | 61.3        | 61.7          | 61.3                  | 61.3        | 61.1        |
| <b>Estimates of derived quantities</b>     |         |                     |             |               |                       |             |             |
| Unfished age 8+ bio 1000 mt                | 10.331  | 10.32               | 10.411      | 10.185        | 10.418                | 10.351      | 10.61       |
| B0 billions of eggs                        | 1190    | 1188.12             | 1197.32     | 1169.27       | 1200.16               | 1189.66     | 1217.61     |
| B2025 billions of eggs                     | 477.63  | 475.113             | 475.18      | 461.464       | 485.851               | 469.622     | 502.097     |
| Fraction unfished 2025                     | 0.401   | 0.4                 | 0.397       | 0.395         | 0.405                 | 0.395       | 0.412       |
| Fishing intensity 2024                     | 0.357   | 0.358               | 0.358       | 0.372         | 0.35                  | 0.359       | 0.341       |

Table 18: Base model sensitivity to the removal of data sources (age compositional data; 2 of 2).

| Label                                      | Base    | - WCG BTS | - IPHC  | No age comps | McAllister-lanelli |
|--------------------------------------------|---------|-----------|---------|--------------|--------------------|
| <b>Diff. in likelihood from base model</b> |         |           |         |              |                    |
| Total                                      | 0       | -1398.96  | -899.62 | -6628.94     | -368.18            |
| Index                                      | 0       | -0.684    | 4.295   | 1.068        | -2.963             |
| Length comp                                | 0       | -14.72    | -15.03  | -77.93       | 91.11              |
| Age comp                                   | 0       | -1378.25  | -887.3  | -6526.61     | -456.34            |
| Recruitment                                | 0       | -5.283    | -1.82   | -25.344      | 0.151              |
| Parm priors                                | 0       | 0         | 0       | 0            | 0                  |
| <b>Estimates of key parameters</b>         |         |           |         |              |                    |
| Recruitment unfished thousands             | 241.887 | 243.671   | 222.982 | 274.842      | 270.471            |
| log(R0)                                    | 5.488   | 5.496     | 5.407   | 5.616        | 5.6                |
| M Female                                   | 0.044   | 0.044     | 0.044   | 0.044        | 0.044              |
| L at Amax Female                           | 61.4    | 61.5      | 62.4    | 60.8         | 61.2               |
| <b>Estimates of derived quantities</b>     |         |           |         |              |                    |
| Unfished age 8+ bio 1000 mt                | 10.331  | 10.441    | 9.945   | 11.047       | 10.986             |
| B0 billions of eggs                        | 1190    | 1199.98   | 1166.44 | 1228.11      | 1241.2             |
| B2025 billions of eggs                     | 477.63  | 471.858   | 393.043 | 469.146      | 549.181            |
| Fraction unfished 2025                     | 0.401   | 0.393     | 0.337   | 0.382        | 0.442              |
| Fishing intensity 2024                     | 0.357   | 0.365     | 0.427   | 0.372        | 0.316              |

Table 19: Time series of population estimates from the base model, including projections from 2025 to 2036.

| Year | Total<br>Biomass<br>(mt) | Spawning<br>output | Total<br>Biomass<br>8+ (mt) | Fraction<br>Unfished | Age-0<br>Recruits<br>(1,000s) | Total<br>Mortality<br>(mt) | (1-<br>SPR)/(1-<br>SPR_-<br>50%) | Exploita-<br>tion Rate |
|------|--------------------------|--------------------|-----------------------------|----------------------|-------------------------------|----------------------------|----------------------------------|------------------------|
| 1889 | 10484                    | 1190.00            | 10331                       | 1.000                | 243                           | 0                          | 0.000                            | 0.000                  |
| 1890 | 10484                    | 1189.99            | 10331                       | 1.000                | 243                           | 0                          | 0.001                            | 0.000                  |
| 1891 | 10483                    | 1189.98            | 10331                       | 1.000                | 243                           | 0                          | 0.002                            | 0.000                  |
| 1892 | 10483                    | 1189.95            | 10331                       | 1.000                | 243                           | 4                          | 0.025                            | 0.000                  |
| 1893 | 10480                    | 1189.49            | 10327                       | 1.000                | 243                           | 4                          | 0.025                            | 0.000                  |
| 1894 | 10476                    | 1189.03            | 10323                       | 0.999                | 243                           | 4                          | 0.026                            | 0.000                  |
| 1895 | 10472                    | 1188.57            | 10319                       | 0.999                | 243                           | 1                          | 0.009                            | 0.000                  |
| 1896 | 10472                    | 1188.42            | 10318                       | 0.999                | 244                           | 1                          | 0.005                            | 0.000                  |
| 1897 | 10471                    | 1188.34            | 10318                       | 0.999                | 244                           | 1                          | 0.006                            | 0.000                  |
| 1898 | 10471                    | 1188.26            | 10318                       | 0.999                | 244                           | 1                          | 0.006                            | 0.000                  |
| 1899 | 10471                    | 1188.19            | 10318                       | 0.998                | 244                           | 1                          | 0.007                            | 0.000                  |
| 1900 | 10471                    | 1188.10            | 10318                       | 0.998                | 244                           | 1                          | 0.008                            | 0.000                  |
| 1901 | 10471                    | 1188.01            | 10318                       | 0.998                | 244                           | 1                          | 0.009                            | 0.000                  |
| 1902 | 10471                    | 1187.92            | 10318                       | 0.998                | 244                           | 2                          | 0.010                            | 0.000                  |
| 1903 | 10471                    | 1187.82            | 10318                       | 0.998                | 244                           | 2                          | 0.011                            | 0.000                  |
| 1904 | 10471                    | 1187.73            | 10318                       | 0.998                | 244                           | 2                          | 0.013                            | 0.000                  |
| 1905 | 10471                    | 1187.62            | 10317                       | 0.998                | 244                           | 2                          | 0.014                            | 0.000                  |
| 1906 | 10471                    | 1187.53            | 10317                       | 0.998                | 244                           | 2                          | 0.015                            | 0.000                  |
| 1907 | 10471                    | 1187.44            | 10317                       | 0.998                | 244                           | 2                          | 0.016                            | 0.000                  |
| 1908 | 10470                    | 1187.34            | 10317                       | 0.998                | 245                           | 4                          | 0.023                            | 0.000                  |
| 1909 | 10469                    | 1187.13            | 10315                       | 0.998                | 245                           | 3                          | 0.018                            | 0.000                  |
| 1910 | 10469                    | 1187.03            | 10315                       | 0.998                | 245                           | 3                          | 0.019                            | 0.000                  |
| 1911 | 10469                    | 1186.93            | 10315                       | 0.997                | 245                           | 3                          | 0.020                            | 0.000                  |
| 1912 | 10468                    | 1186.82            | 10314                       | 0.997                | 245                           | 3                          | 0.022                            | 0.000                  |
| 1913 | 10468                    | 1186.71            | 10314                       | 0.997                | 246                           | 3                          | 0.023                            | 0.000                  |
| 1914 | 10467                    | 1186.59            | 10313                       | 0.997                | 246                           | 4                          | 0.024                            | 0.000                  |
| 1915 | 10467                    | 1186.47            | 10313                       | 0.997                | 246                           | 4                          | 0.029                            | 0.000                  |
| 1916 | 10466                    | 1186.27            | 10312                       | 0.997                | 246                           | 4                          | 0.026                            | 0.000                  |
| 1917 | 10466                    | 1186.14            | 10311                       | 0.997                | 246                           | 5                          | 0.036                            | 0.001                  |
| 1918 | 10464                    | 1185.85            | 10309                       | 0.997                | 246                           | 23                         | 0.142                            | 0.002                  |
| 1919 | 10445                    | 1183.51            | 10290                       | 0.995                | 245                           | 10                         | 0.063                            | 0.001                  |
| 1920 | 10440                    | 1182.75            | 10285                       | 0.994                | 244                           | 9                          | 0.059                            | 0.001                  |
| 1921 | 10436                    | 1182.11            | 10281                       | 0.993                | 242                           | 9                          | 0.056                            | 0.001                  |
| 1922 | 10432                    | 1181.54            | 10277                       | 0.993                | 240                           | 6                          | 0.043                            | 0.001                  |
| 1923 | 10430                    | 1181.25            | 10276                       | 0.993                | 237                           | 7                          | 0.048                            | 0.001                  |
| 1924 | 10428                    | 1180.90            | 10274                       | 0.992                | 234                           | 12                         | 0.078                            | 0.001                  |
| 1925 | 10421                    | 1180.01            | 10268                       | 0.992                | 230                           | 15                         | 0.099                            | 0.001                  |
| 1926 | 10411                    | 1178.78            | 10258                       | 0.991                | 226                           | 22                         | 0.141                            | 0.002                  |
| 1927 | 10394                    | 1176.75            | 10242                       | 0.989                | 223                           | 29                         | 0.178                            | 0.003                  |
| 1928 | 10370                    | 1174.02            | 10220                       | 0.987                | 219                           | 28                         | 0.176                            | 0.003                  |
| 1929 | 10346                    | 1171.40            | 10198                       | 0.984                | 216                           | 25                         | 0.157                            | 0.002                  |
| 1930 | 10325                    | 1169.24            | 10179                       | 0.983                | 213                           | 28                         | 0.175                            | 0.003                  |
| 1931 | 10300                    | 1166.75            | 10156                       | 0.980                | 211                           | 19                         | 0.125                            | 0.002                  |
| 1932 | 10282                    | 1165.28            | 10140                       | 0.979                | 209                           | 18                         | 0.117                            | 0.002                  |
| 1933 | 10264                    | 1163.97            | 10124                       | 0.978                | 207                           | 18                         | 0.116                            | 0.002                  |
| 1934 | 10245                    | 1162.64            | 10107                       | 0.977                | 205                           | 22                         | 0.142                            | 0.002                  |

Table 19: Time series of population estimates from the base model, including projections from 2025 to 2036. (continued)

| Year | Total<br>Biomass<br>(mt) | Spawning<br>output | Total<br>Biomass<br>8+ (mt) | Fraction<br>Unfished | Age-0<br>Recruits<br>(1,000s) | Total<br>Mortality<br>(mt) | (1-<br>SPR)/(1-<br>SPR_<br>50%) | Exploita-<br>tion Rate |
|------|--------------------------|--------------------|-----------------------------|----------------------|-------------------------------|----------------------------|---------------------------------|------------------------|
| 1935 | 10219                    | 1160.75            | 10084                       | 0.975                | 202                           | 22                         | 0.141                           | 0.002                  |
| 1936 | 10193                    | 1158.80            | 10059                       | 0.974                | 199                           | 29                         | 0.187                           | 0.003                  |
| 1937 | 10156                    | 1155.81            | 10024                       | 0.971                | 195                           | 26                         | 0.170                           | 0.003                  |
| 1938 | 10121                    | 1153.01            | 9990                        | 0.969                | 192                           | 28                         | 0.181                           | 0.003                  |
| 1939 | 10082                    | 1149.81            | 9952                        | 0.966                | 190                           | 22                         | 0.143                           | 0.002                  |
| 1940 | 10047                    | 1147.12            | 9919                        | 0.964                | 189                           | 30                         | 0.197                           | 0.003                  |
| 1941 | 10001                    | 1143.17            | 9875                        | 0.961                | 190                           | 42                         | 0.264                           | 0.004                  |
| 1942 | 9942                     | 1137.58            | 9818                        | 0.956                | 193                           | 44                         | 0.271                           | 0.004                  |
| 1943 | 9879                     | 1131.49            | 9756                        | 0.951                | 198                           | 81                         | 0.450                           | 0.008                  |
| 1944 | 9778                     | 1120.78            | 9657                        | 0.942                | 209                           | 86                         | 0.514                           | 0.009                  |
| 1945 | 9671                     | 1109.25            | 9550                        | 0.932                | 225                           | 130                        | 0.712                           | 0.014                  |
| 1946 | 9519                     | 1092.38            | 9398                        | 0.918                | 250                           | 119                        | 0.661                           | 0.013                  |
| 1947 | 9379                     | 1076.61            | 9257                        | 0.905                | 284                           | 53                         | 0.357                           | 0.006                  |
| 1948 | 9306                     | 1068.42            | 9180                        | 0.898                | 315                           | 61                         | 0.397                           | 0.007                  |
| 1949 | 9227                     | 1059.12            | 9096                        | 0.890                | 316                           | 46                         | 0.315                           | 0.005                  |
| 1950 | 9166                     | 1051.47            | 9027                        | 0.884                | 279                           | 51                         | 0.350                           | 0.006                  |
| 1951 | 9104                     | 1043.03            | 8955                        | 0.876                | 232                           | 64                         | 0.422                           | 0.007                  |
| 1952 | 9036                     | 1032.92            | 8874                        | 0.868                | 195                           | 55                         | 0.376                           | 0.006                  |
| 1953 | 8983                     | 1023.86            | 8809                        | 0.860                | 170                           | 40                         | 0.287                           | 0.005                  |
| 1954 | 8950                     | 1016.60            | 8768                        | 0.854                | 154                           | 51                         | 0.357                           | 0.006                  |
| 1955 | 8912                     | 1008.23            | 8731                        | 0.847                | 146                           | 50                         | 0.352                           | 0.006                  |
| 1956 | 8878                     | 1000.35            | 8712                        | 0.841                | 146                           | 54                         | 0.375                           | 0.006                  |
| 1957 | 8842                     | 992.62             | 8699                        | 0.834                | 151                           | 68                         | 0.461                           | 0.008                  |
| 1958 | 8793                     | 984.06             | 8670                        | 0.827                | 163                           | 68                         | 0.448                           | 0.008                  |
| 1959 | 8742                     | 976.55             | 8634                        | 0.821                | 180                           | 65                         | 0.444                           | 0.008                  |
| 1960 | 8692                     | 970.54             | 8593                        | 0.816                | 189                           | 61                         | 0.431                           | 0.007                  |
| 1961 | 8644                     | 966.11             | 8548                        | 0.812                | 174                           | 49                         | 0.364                           | 0.006                  |
| 1962 | 8603                     | 963.88             | 8508                        | 0.810                | 149                           | 58                         | 0.418                           | 0.007                  |
| 1963 | 8551                     | 961.17             | 8453                        | 0.808                | 132                           | 45                         | 0.319                           | 0.005                  |
| 1964 | 8509                     | 960.06             | 8406                        | 0.807                | 133                           | 32                         | 0.234                           | 0.004                  |
| 1965 | 8477                     | 960.12             | 8370                        | 0.807                | 154                           | 111                        | 0.711                           | 0.013                  |
| 1966 | 8363                     | 950.38             | 8253                        | 0.799                | 185                           | 46                         | 0.319                           | 0.006                  |
| 1967 | 8309                     | 947.34             | 8203                        | 0.796                | 207                           | 58                         | 0.406                           | 0.007                  |
| 1968 | 8241                     | 941.98             | 8143                        | 0.792                | 247                           | 54                         | 0.373                           | 0.007                  |
| 1969 | 8174                     | 936.24             | 8081                        | 0.787                | 272                           | 128                        | 0.792                           | 0.016                  |
| 1970 | 8034                     | 921.14             | 7937                        | 0.774                | 331                           | 87                         | 0.523                           | 0.011                  |
| 1971 | 7937                     | 910.42             | 7829                        | 0.765                | 522                           | 112                        | 0.652                           | 0.014                  |
| 1972 | 7817                     | 896.31             | 7692                        | 0.753                | 290                           | 138                        | 0.703                           | 0.018                  |
| 1973 | 7677                     | 878.83             | 7534                        | 0.739                | 214                           | 149                        | 0.791                           | 0.020                  |
| 1974 | 7538                     | 859.79             | 7374                        | 0.723                | 236                           | 162                        | 0.838                           | 0.022                  |
| 1975 | 7398                     | 838.94             | 7209                        | 0.705                | 342                           | 155                        | 0.778                           | 0.021                  |
| 1976 | 7276                     | 818.93             | 7069                        | 0.688                | 283                           | 180                        | 0.915                           | 0.025                  |
| 1977 | 7141                     | 796.14             | 6918                        | 0.669                | 265                           | 181                        | 0.969                           | 0.026                  |
| 1978 | 7016                     | 773.72             | 6794                        | 0.650                | 191                           | 299                        | 1.187                           | 0.044                  |
| 1979 | 6787                     | 738.84             | 6622                        | 0.621                | 250                           | 327                        | 1.423                           | 0.049                  |
| 1980 | 6541                     | 702.27             | 6383                        | 0.590                | 291                           | 266                        | 1.390                           | 0.042                  |
| 1981 | 6364                     | 674.66             | 6192                        | 0.567                | 320                           | 354                        | 1.489                           | 0.057                  |



Table 19: Time series of population estimates from the base model, including projections from 2025 to 2036. (continued)

| Year | Total<br>Biomass<br>(mt) | Spawning<br>output | Total<br>Biomass<br>8+ (mt) | Fraction<br>Unfished | Age-0<br>Recruits<br>(1,000s) | Total<br>Mortality<br>(mt) | (1-<br>SPR)/(1-<br>SPR_<br>50%) | Exploita-<br>tion Rate |
|------|--------------------------|--------------------|-----------------------------|----------------------|-------------------------------|----------------------------|---------------------------------|------------------------|
| 1982 | 6108                     | 639.85             | 5927                        | 0.538                | 355                           | 577                        | 1.715                           | 0.097                  |
| 1983 | 5637                     | 583.50             | 5477                        | 0.490                | 241                           | 503                        | 1.768                           | 0.092                  |
| 1984 | 5246                     | 535.86             | 5091                        | 0.450                | 311                           | 335                        | 1.633                           | 0.066                  |
| 1985 | 5029                     | 508.45             | 4874                        | 0.427                | 259                           | 429                        | 1.759                           | 0.088                  |
| 1986 | 4724                     | 472.11             | 4549                        | 0.397                | 182                           | 363                        | 1.716                           | 0.080                  |
| 1987 | 4491                     | 443.26             | 4305                        | 0.372                | 141                           | 353                        | 1.710                           | 0.082                  |
| 1988 | 4275                     | 416.46             | 4088                        | 0.350                | 105                           | 322                        | 1.702                           | 0.079                  |
| 1989 | 4092                     | 393.78             | 3915                        | 0.331                | 90                            | 382                        | 1.795                           | 0.098                  |
| 1990 | 3849                     | 365.16             | 3700                        | 0.307                | 81                            | 303                        | 1.701                           | 0.082                  |
| 1991 | 3684                     | 345.58             | 3541                        | 0.290                | 73                            | 431                        | 1.826                           | 0.122                  |
| 1992 | 3388                     | 314.08             | 3276                        | 0.264                | 84                            | 400                        | 1.848                           | 0.122                  |
| 1993 | 3114                     | 285.72             | 3030                        | 0.240                | 166                           | 376                        | 1.837                           | 0.124                  |
| 1994 | 2855                     | 258.83             | 2786                        | 0.218                | 139                           | 258                        | 1.772                           | 0.093                  |
| 1995 | 2704                     | 245.18             | 2647                        | 0.206                | 66                            | 264                        | 1.802                           | 0.100                  |
| 1996 | 2540                     | 231.39             | 2485                        | 0.194                | 63                            | 278                        | 1.822                           | 0.112                  |
| 1997 | 2356                     | 216.06             | 2300                        | 0.182                | 75                            | 297                        | 1.857                           | 0.129                  |
| 1998 | 2147                     | 197.94             | 2085                        | 0.166                | 120                           | 140                        | 1.611                           | 0.067                  |
| 1999 | 2086                     | 194.15             | 2016                        | 0.163                | 189                           | 177                        | 1.709                           | 0.088                  |
| 2000 | 1985                     | 185.83             | 1909                        | 0.156                | 99                            | 52                         | 1.107                           | 0.027                  |
| 2001 | 2005                     | 189.60             | 1944                        | 0.159                | 114                           | 52                         | 1.056                           | 0.027                  |
| 2002 | 2028                     | 192.82             | 1975                        | 0.162                | 403                           | 13                         | 0.394                           | 0.007                  |
| 2003 | 2086                     | 199.71             | 2024                        | 0.168                | 163                           | 12                         | 0.387                           | 0.006                  |
| 2004 | 2148                     | 206.64             | 2072                        | 0.174                | 109                           | 9                          | 0.264                           | 0.004                  |
| 2005 | 2218                     | 213.77             | 2123                        | 0.180                | 167                           | 10                         | 0.281                           | 0.005                  |
| 2006 | 2293                     | 220.73             | 2184                        | 0.185                | 284                           | 8                          | 0.215                           | 0.004                  |
| 2007 | 2374                     | 227.86             | 2267                        | 0.191                | 285                           | 13                         | 0.364                           | 0.006                  |
| 2008 | 2457                     | 234.43             | 2324                        | 0.197                | 483                           | 9                          | 0.235                           | 0.004                  |
| 2009 | 2550                     | 241.33             | 2389                        | 0.203                | 372                           | 11                         | 0.308                           | 0.005                  |
| 2010 | 2649                     | 248.18             | 2535                        | 0.209                | 362                           | 6                          | 0.151                           | 0.002                  |
| 2011 | 2765                     | 255.84             | 2636                        | 0.215                | 279                           | 8                          | 0.197                           | 0.003                  |
| 2012 | 2892                     | 263.91             | 2723                        | 0.222                | 242                           | 11                         | 0.260                           | 0.004                  |
| 2013 | 3034                     | 272.49             | 2822                        | 0.229                | 571                           | 10                         | 0.213                           | 0.003                  |
| 2014 | 3187                     | 282.19             | 2958                        | 0.237                | 263                           | 9                          | 0.179                           | 0.003                  |
| 2015 | 3355                     | 293.08             | 3106                        | 0.246                | 359                           | 12                         | 0.244                           | 0.004                  |
| 2016 | 3535                     | 304.71             | 3317                        | 0.256                | 242                           | 9                          | 0.174                           | 0.003                  |
| 2017 | 3729                     | 317.87             | 3520                        | 0.267                | 121                           | 20                         | 0.355                           | 0.006                  |
| 2018 | 3925                     | 331.52             | 3724                        | 0.279                | 115                           | 19                         | 0.317                           | 0.005                  |
| 2019 | 4130                     | 347.17             | 3918                        | 0.292                | 118                           | 23                         | 0.371                           | 0.006                  |
| 2020 | 4335                     | 364.48             | 4100                        | 0.306                | 117                           | 18                         | 0.270                           | 0.004                  |
| 2021 | 4545                     | 384.48             | 4384                        | 0.323                | 153                           | 21                         | 0.306                           | 0.005                  |
| 2022 | 4749                     | 406.33             | 4599                        | 0.341                | 174                           | 34                         | 0.441                           | 0.007                  |
| 2023 | 4932                     | 428.74             | 4829                        | 0.360                | 179                           | 39                         | 0.482                           | 0.008                  |
| 2024 | 5102                     | 452.09             | 5024                        | 0.380                | 209                           | 29                         | 0.357                           | 0.006                  |
| 2025 | 5270                     | 477.63             | 5190                        | 0.401                | 211                           | 46                         | 0.515                           | 0.009                  |
| 2026 | 5411                     | 502.27             | 5324                        | 0.422                | 213                           | 46                         | 0.509                           | 0.009                  |
| 2027 | 5539                     | 526.77             | 5443                        | 0.443                | 216                           | 113                        | 0.921                           | 0.021                  |
| 2028 | 5590                     | 543.56             | 5480                        | 0.457                | 217                           | 113                        | 0.915                           | 0.021                  |

Table 19: Time series of population estimates from the base model, including projections from 2025 to 2036. (continued)

| Year | Total<br>Biomass<br>(mt) | Spawning<br>output | Total<br>Biomass<br>8+ (mt) | Fraction<br>Unfished | Age-0<br>Recruits<br>(1,000s) | Total<br>Mortality<br>(mt) | (1-<br>SPR)/(1-<br>SPR_<br>50%) | Exploita-<br>tion Rate |
|------|--------------------------|--------------------|-----------------------------|----------------------|-------------------------------|----------------------------|---------------------------------|------------------------|
| 2029 | 5631                     | 558.41             | 5512                        | 0.469                | 218                           | 113                        | 0.910                           | 0.021                  |
| 2030 | 5662                     | 570.84             | 5537                        | 0.480                | 219                           | 113                        | 0.905                           | 0.020                  |
| 2031 | 5686                     | 580.65             | 5553                        | 0.488                | 220                           | 112                        | 0.899                           | 0.020                  |
| 2032 | 5704                     | 587.93             | 5569                        | 0.494                | 220                           | 111                        | 0.894                           | 0.020                  |
| 2033 | 5717                     | 592.97             | 5581                        | 0.498                | 220                           | 110                        | 0.889                           | 0.020                  |
| 2034 | 5726                     | 596.20             | 5590                        | 0.501                | 221                           | 109                        | 0.884                           | 0.019                  |
| 2035 | 5733                     | 598.07             | 5596                        | 0.503                | 221                           | 108                        | 0.878                           | 0.019                  |
| 2036 | 5739                     | 599.03             | 5601                        | 0.503                | 221                           | 106                        | 0.873                           | 0.019                  |

Table 20: Summary of reference points and management quantities, including estimates of the 95 percent confidence intervals. SO is spawning output, SPR is the spawning potential ratio, and MSY is maximum sustainable yield.

| Reference Point                                | Estimate | Lower Interval | Upper Interval |
|------------------------------------------------|----------|----------------|----------------|
| Unfished Spawning output                       | 1,190.0  | 1,048.1        | 1,331.9        |
| Unfished Age 8+ Biomass (mt)                   | 10,331   | 9,101          | 11,561         |
| Unfished Recruitment (R0)                      | 242      | 213            | 271            |
| 2025 Spawning output                           | 478      | 384            | 571            |
| 2025 Fraction Unfished                         | 0.401    | 0.349          | 0.453          |
| Reference Points Based SO40%                   | —        | —              | —              |
| Proxy Spawning output SO40%                    | 476      | 419            | 533            |
| SPR Resulting in SO40%                         | 0.458    | 0.458          | 0.458          |
| Exploitation Rate Resulting in SO40%           | 0.026    | 0.026          | 0.027          |
| Yield with SPR Based On SO40% (mt)             | 122      | 108            | 137            |
| Reference Points Based on SPR Proxy for MSY    | —        | —              | —              |
| Proxy Spawning output (SPR50)                  | 531      | 468            | 594            |
| SPR50                                          | 0.500    | —              | —              |
| Exploitation Rate Corresponding to SPR50       | 0.023    | 0.023          | 0.023          |
| Yield with SPR50 at SO SPR (mt)                | 117      | 103            | 131            |
| Reference Points Based on Estimated MSY Values | —        | —              | —              |
| Spawning output at MSY (SO MSY)                | 343      | 303            | 384            |
| SPR MSY                                        | 0.358    | 0.356          | 0.359          |
| Exploitation Rate Corresponding to SPR MSY     | 0.037    | 0.036          | 0.037          |
| MSY (mt)                                       | 128      | 113            | 143            |

Table 21: Potential OFLs (mt), ABCs (mt), ACLs (mt), the buffer between the OFL and ABC, estimated spawning output, and fraction of unfished spawning output with adopted OFLs and ACLs and assumed catch for the first two years of the projection period.

| Year | Adopted<br>OFL (mt) | Adopted<br>ACL (mt) | Assumed<br>Catch (mt) | OFL (mt) | Buffer | ABC (mt) | ACL (mt) | Spawning<br>output | Fraction<br>Unfished |
|------|---------------------|---------------------|-----------------------|----------|--------|----------|----------|--------------------|----------------------|
| 2025 | 105.8               | 55.8                | 45.7                  | —        | —      | —        | —        | 477.6              | 0.401                |
| 2026 | 108.3               | 56.6                | 46.4                  | —        | —      | —        | —        | 502.3              | 0.422                |
| 2027 | —                   | —                   | —                     | 129.3    | 0.873  | 112.9    | 112.9    | 526.8              | 0.443                |
| 2028 | —                   | —                   | —                     | 130.9    | 0.864  | 113.1    | 113.1    | 543.6              | 0.457                |
| 2029 | —                   | —                   | —                     | 132.0    | 0.856  | 113.0    | 113.0    | 558.4              | 0.469                |
| 2030 | —                   | —                   | —                     | 132.7    | 0.848  | 112.6    | 112.6    | 570.8              | 0.480                |
| 2031 | —                   | —                   | —                     | 133.1    | 0.840  | 111.8    | 111.8    | 580.7              | 0.488                |
| 2032 | —                   | —                   | —                     | 133.3    | 0.832  | 110.9    | 110.9    | 587.9              | 0.494                |
| 2033 | —                   | —                   | —                     | 133.2    | 0.824  | 109.8    | 109.8    | 593.0              | 0.498                |
| 2034 | —                   | —                   | —                     | 133.1    | 0.817  | 108.8    | 108.8    | 596.2              | 0.501                |
| 2035 | —                   | —                   | —                     | 132.9    | 0.809  | 107.6    | 107.6    | 598.1              | 0.503                |
| 2036 | —                   | —                   | —                     | 132.8    | 0.801  | 106.3    | 106.3    | 599.0              | 0.503                |

Table 22: Decision table with 10-year projections. ‘Mgmt’ refers to the three management scenarios (A) the default harvest control rule  $P = 0.40$ . In each case the 2025 and 2026 catches are fixed at the estimates provided by the GMT. The catch for the Washington recreational fleet is input in numbers so the GMT estimate was converted from 3.22 metric tons to 1.53 thousands of fish based on a mean weight of 2.105 kg estimated by SS3 for this fleet in 2024. The alternative states of nature (‘Low’, ‘Base’, and ‘High’ as discussed in the text) are provided in the columns, with Spawning Output (‘Spawn’, in billions of eggs) and Fraction of unfished spawning output (‘Frac’) provided for each state.

| Mgmt | Year | Catch | Low<br>Spawn | Low<br>Frac | Base<br>Spawn | Base<br>Frac | High<br>Spawn | High<br>Frac |
|------|------|-------|--------------|-------------|---------------|--------------|---------------|--------------|
| A    | 2025 | 46    | 330.67       | 0.289       | 477.63        | 0.401        | 839.65        | 0.603        |
|      | 2026 | 46    | 347.96       | 0.305       | 502.27        | 0.422        | 878.58        | 0.630        |
|      | 2027 | 113   | 365.20       | 0.320       | 526.77        | 0.443        | 916.84        | 0.658        |
|      | 2028 | 113   | 374.94       | 0.328       | 543.56        | 0.457        | 946.40        | 0.679        |
|      | 2029 | 113   | 383.27       | 0.335       | 558.41        | 0.469        | 972.47        | 0.698        |
|      | 2030 | 113   | 389.86       | 0.341       | 570.84        | 0.480        | 994.20        | 0.713        |
|      | 2031 | 112   | 394.60       | 0.345       | 580.65        | 0.488        | 1011.25       | 0.726        |
|      | 2032 | 111   | 397.56       | 0.348       | 587.93        | 0.494        | 1023.78       | 0.735        |
|      | 2033 | 110   | 398.98       | 0.349       | 592.97        | 0.498        | 1032.35       | 0.741        |
|      | 2034 | 109   | 399.19       | 0.349       | 596.20        | 0.501        | 1037.71       | 0.745        |
|      | 2035 | 108   | 398.51       | 0.349       | 598.07        | 0.503        | 1040.63       | 0.747        |
|      | 2036 | 106   | 397.30       | 0.348       | 599.03        | 0.503        | 1041.85       | 0.748        |

## 8 Figures

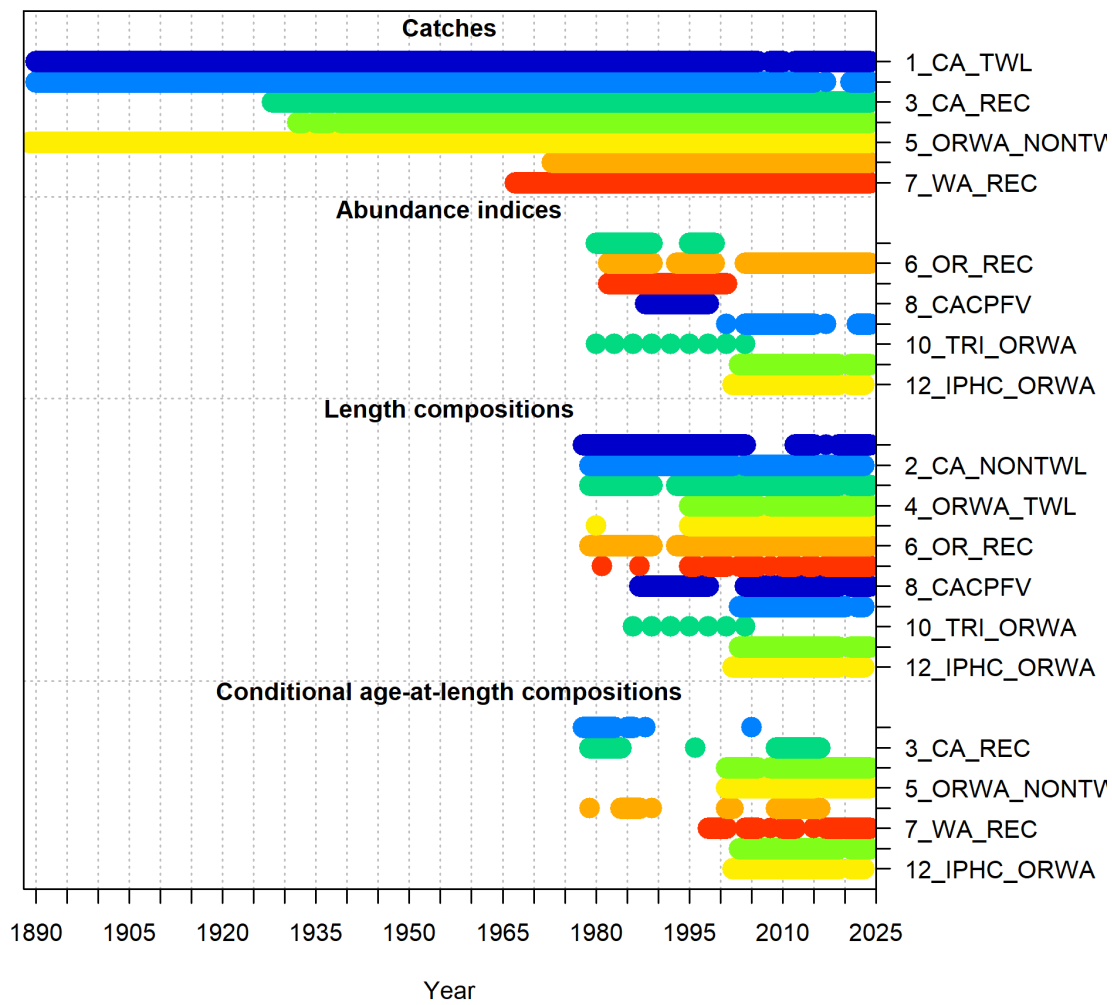


Figure 1: Summary of data sources used in the base model.

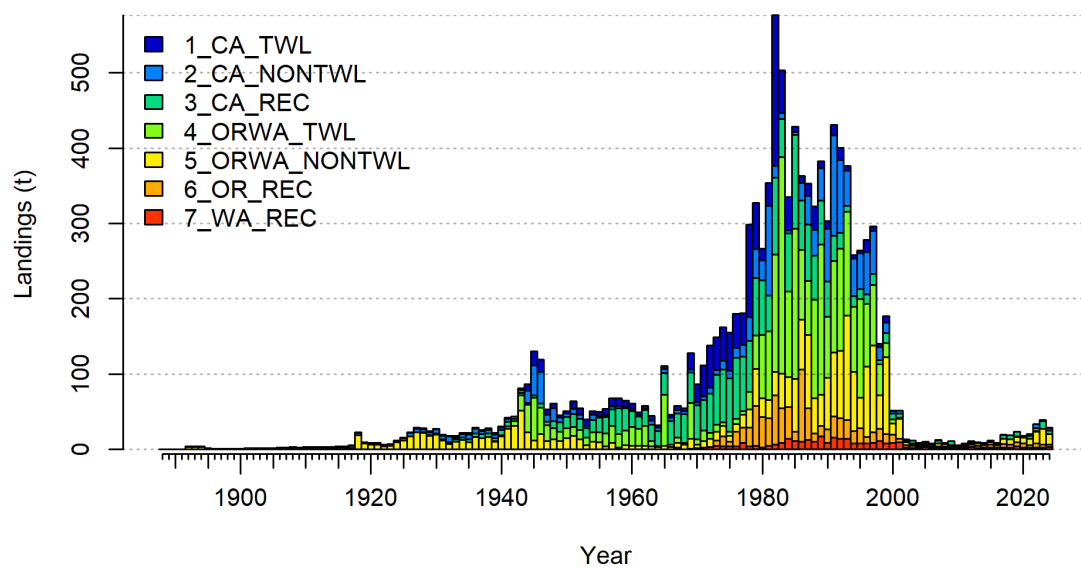


Figure 2: Yelloweye Rockfish landing history in metric tons (mt) between 1889 and 2024 for each fleet.



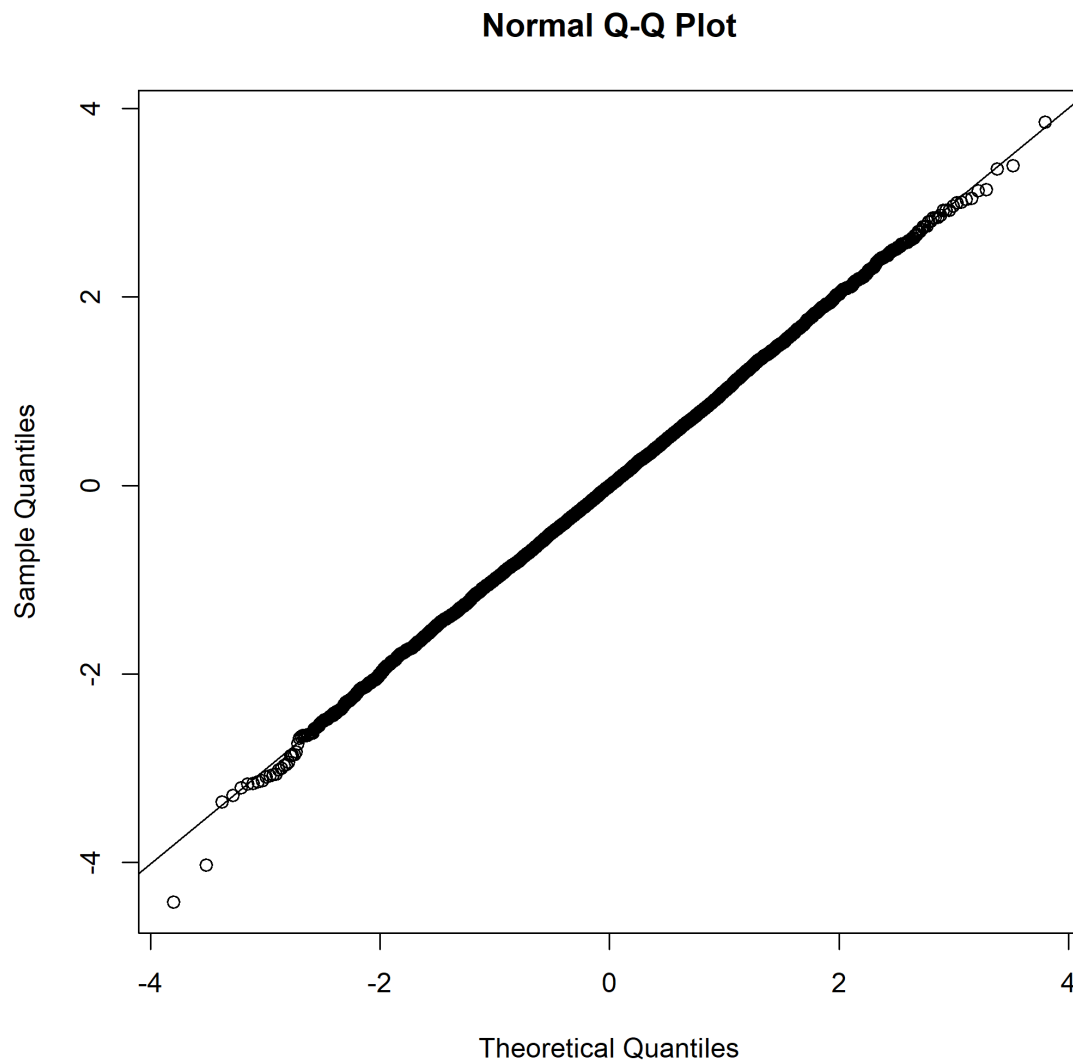


Figure 3: Quantile-quantile plot for the sdmTMB model fit for the Oregon Onboard Observer (ORFS) index.

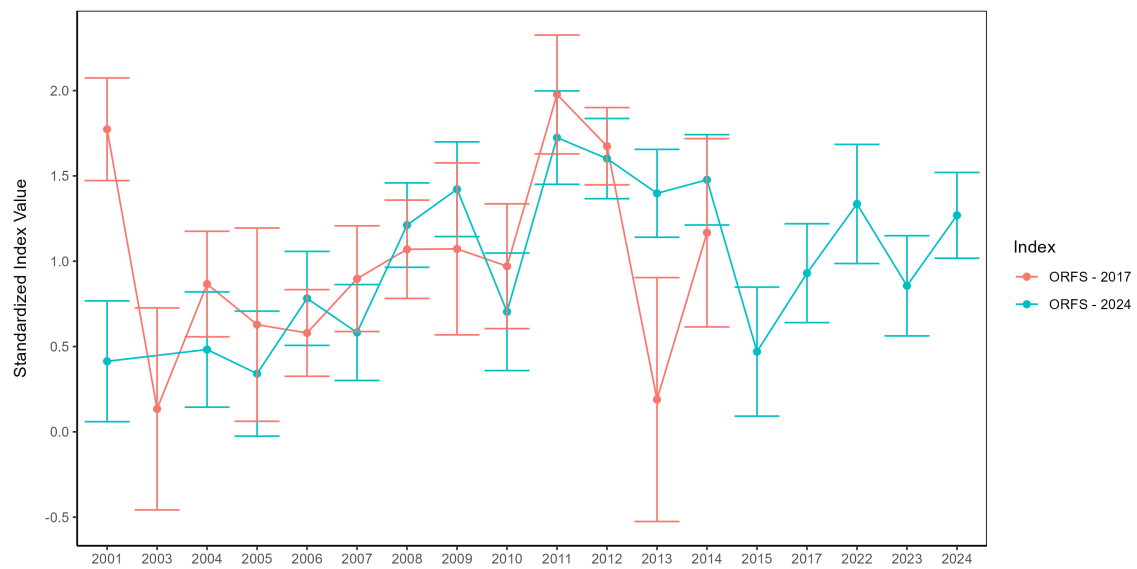


Figure 4: Comparison of Oregon Onboard Observer indices from the 2017 and the current assessment.

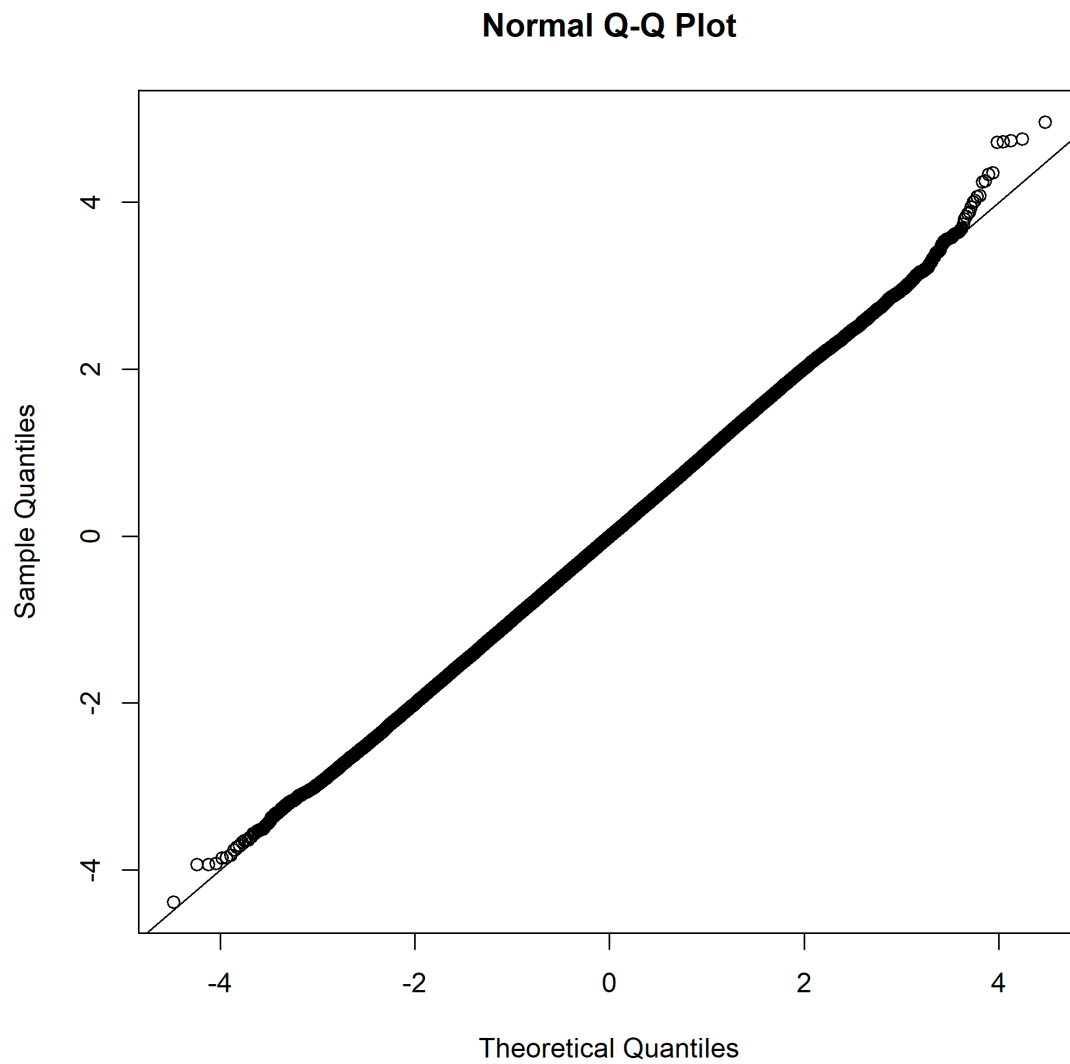


Figure 5: Quantile-quantile plot for the sdmTMB model fit for the updated portion of the Oregon recreational (ORBS) index.

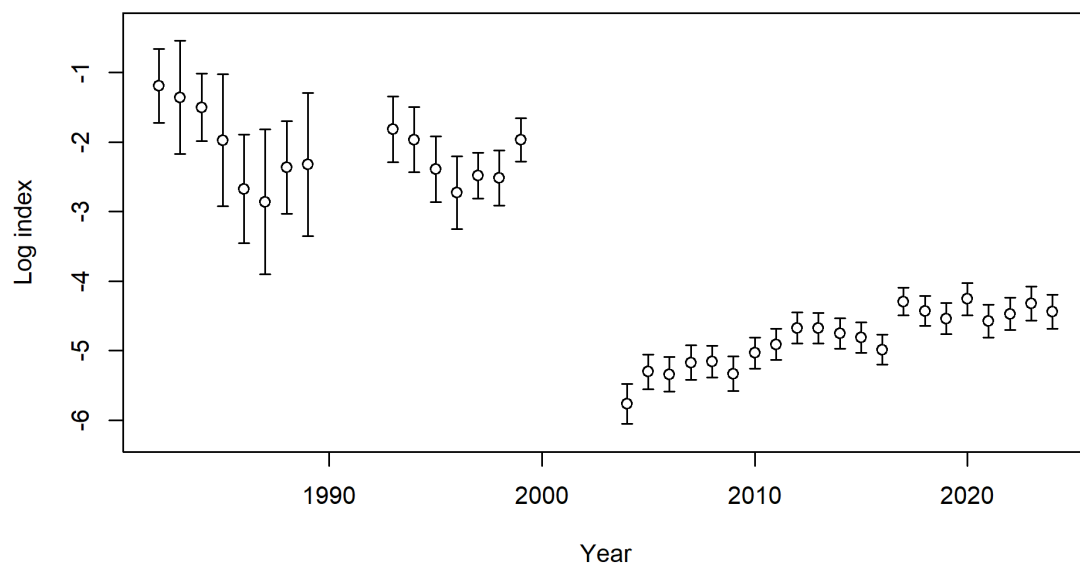


Figure 6: Annual relative log-scale index of abundance for the Oregon recreational index, including both MRFSS (1980 - 1999) and ORBS (2004 - 2024) indices.

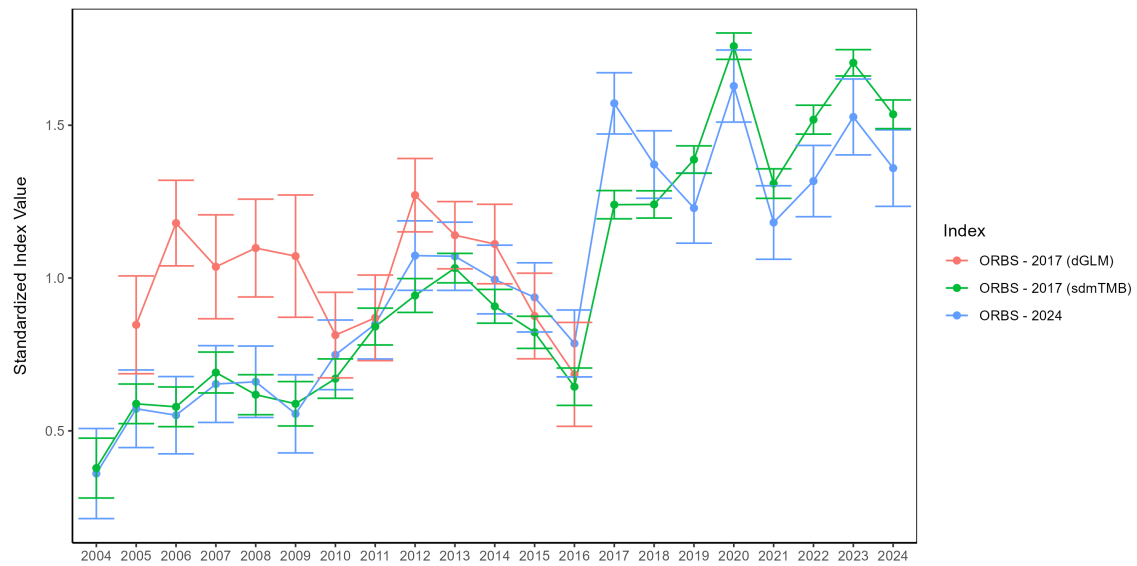


Figure 7: Comparison of the 2017 ORBS index (delta-GLM), the 2017 ORBS model structure (with the current dataset and implemented in sdmTMB), and the current ORBS index (sdmTMB).

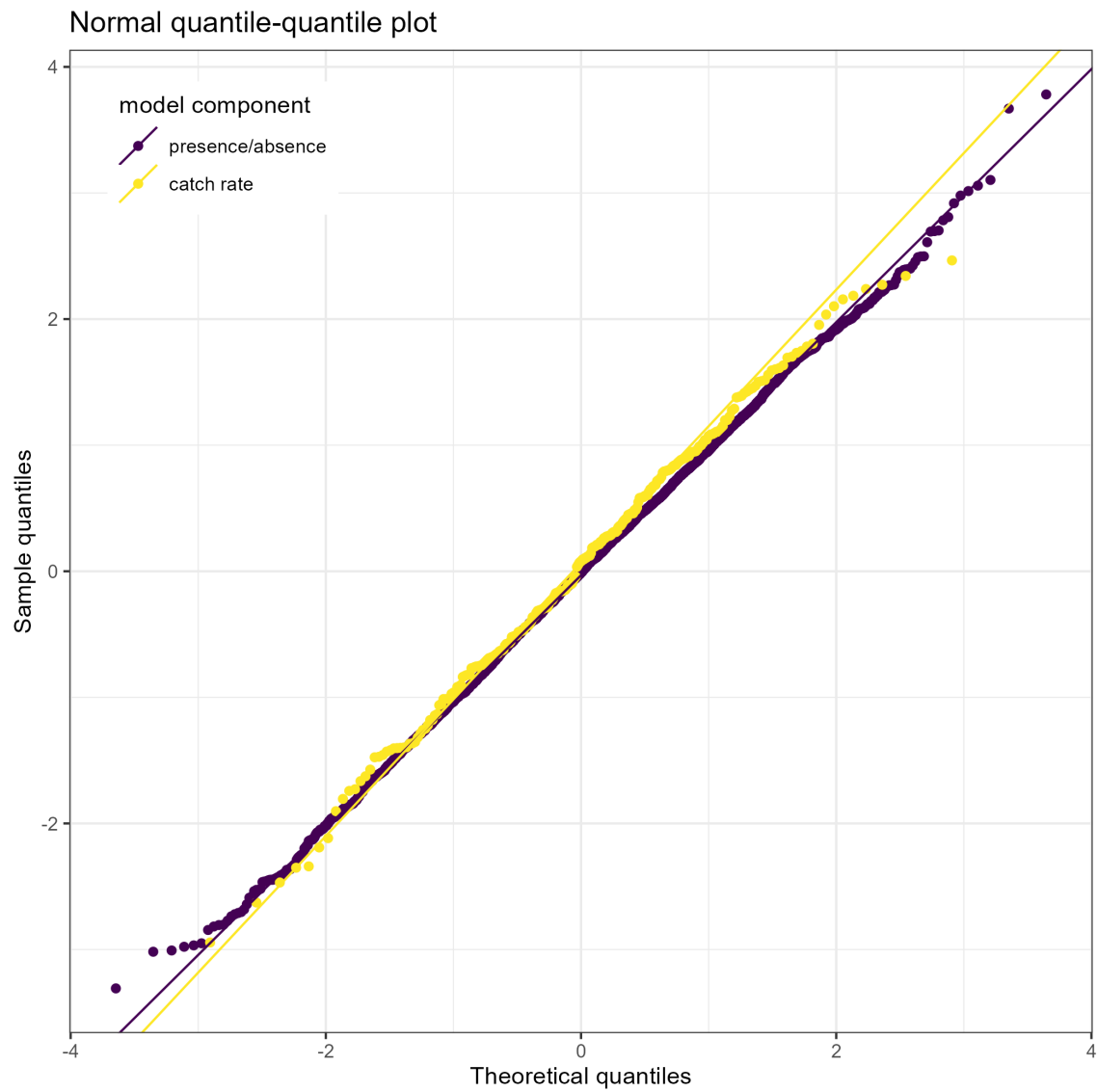


Figure 8: Quantile-quantile plot for the sdmTMB model fit for the West Coast Groundfish Bottom Trawl Survey (WCGBTS) index.

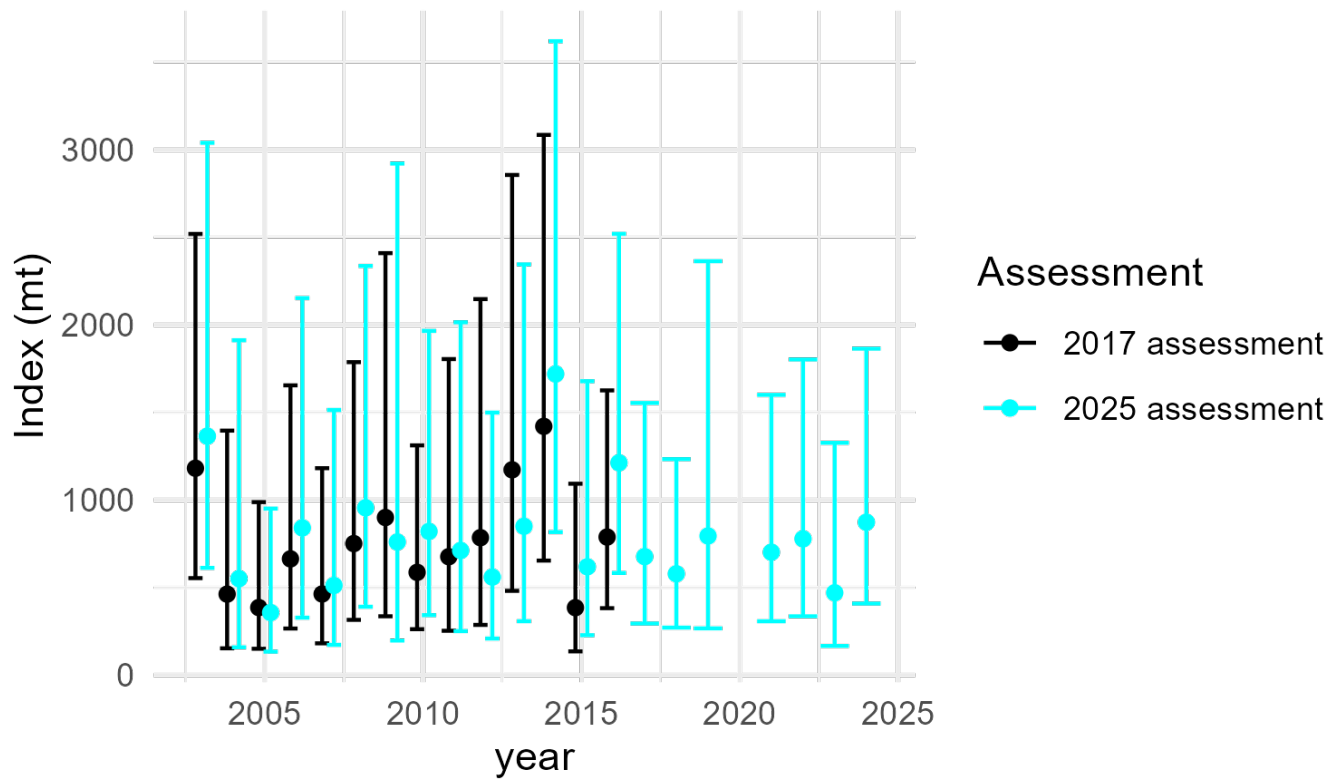


Figure 9: Comparison of the 2017 West Coast Groundfish Bottom Trawl Survey (WCGBTS) and the current WCGBTS index of abundance.

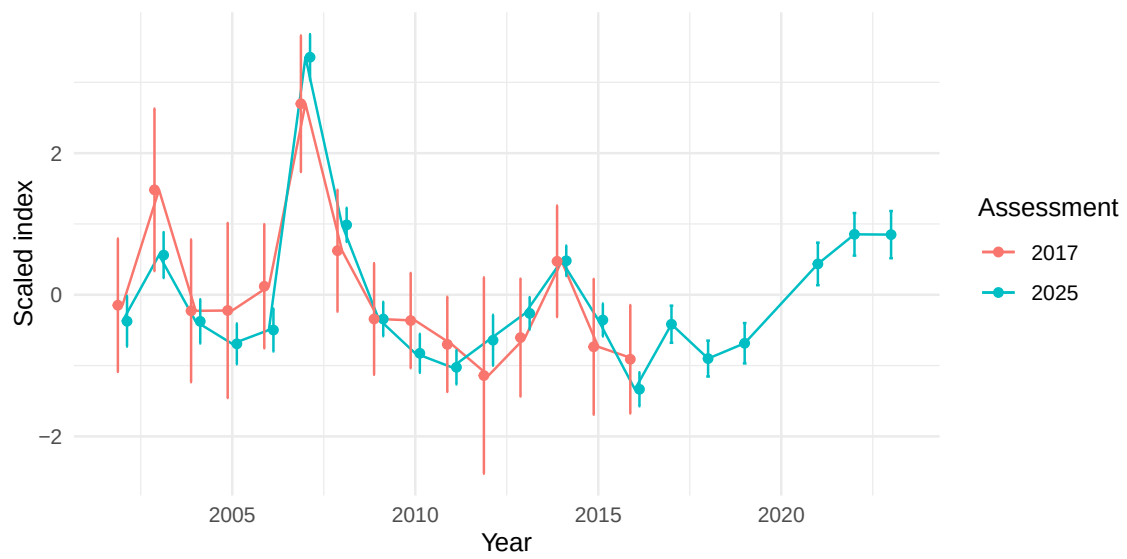


Figure 10: Comparison of the 2017 and the current IPHC index of abundance.



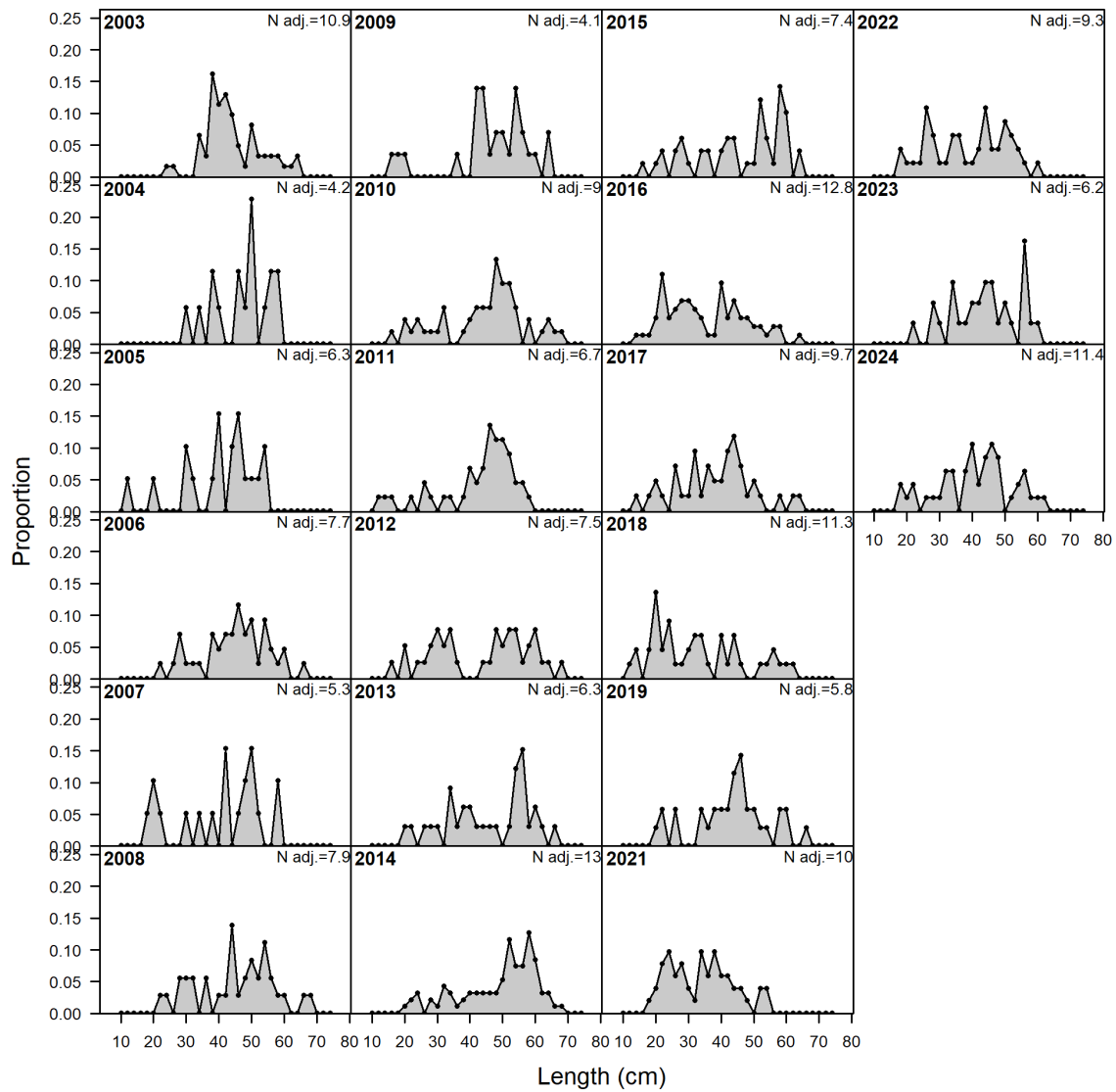


Figure 11: Annual length composition data for the WCBTS.

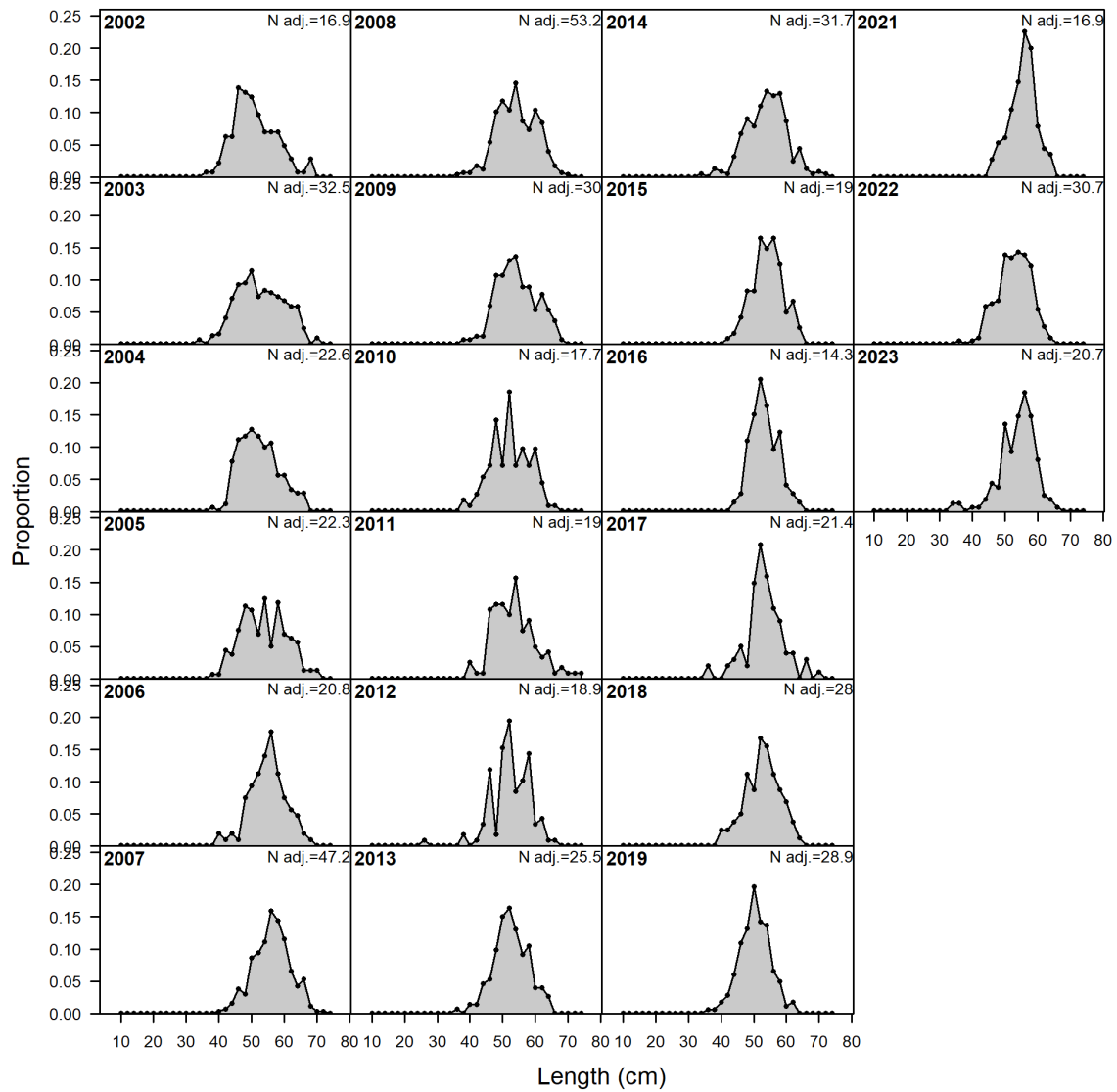


Figure 12: Annual length composition data from the IPHC longline survey.

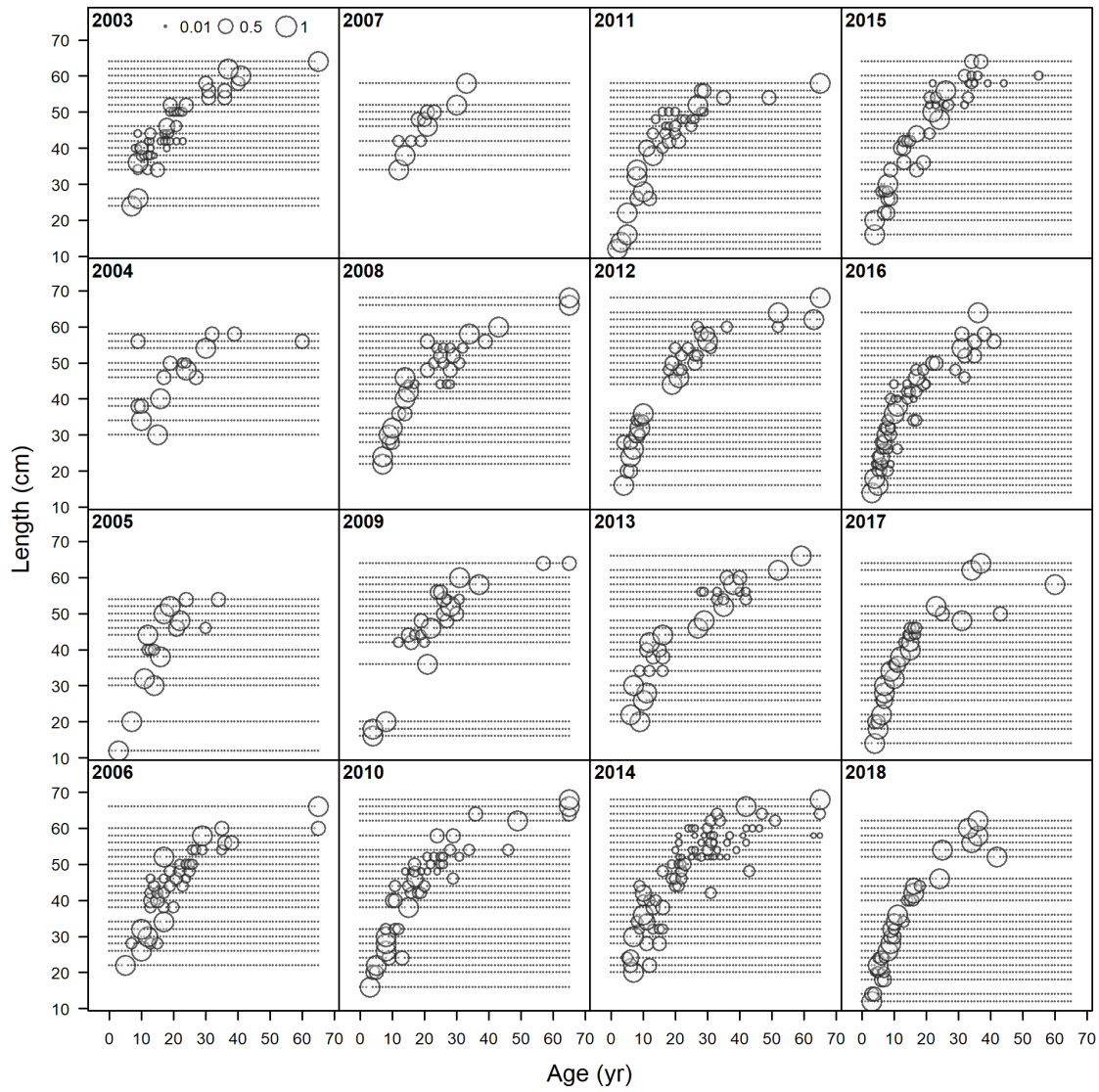


Figure 13: Annual unsexed conditional age-at-length data for the WCBTS (1 of 2).

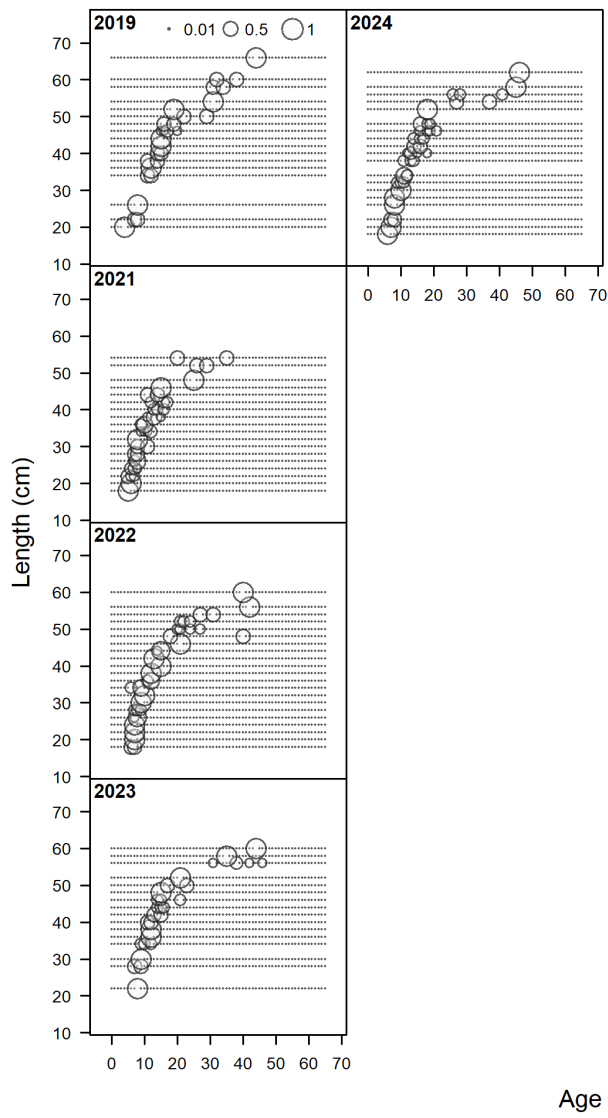


Figure 14: Annual unsexed conditional age-at-length data for the WCBTS (2 of 2).

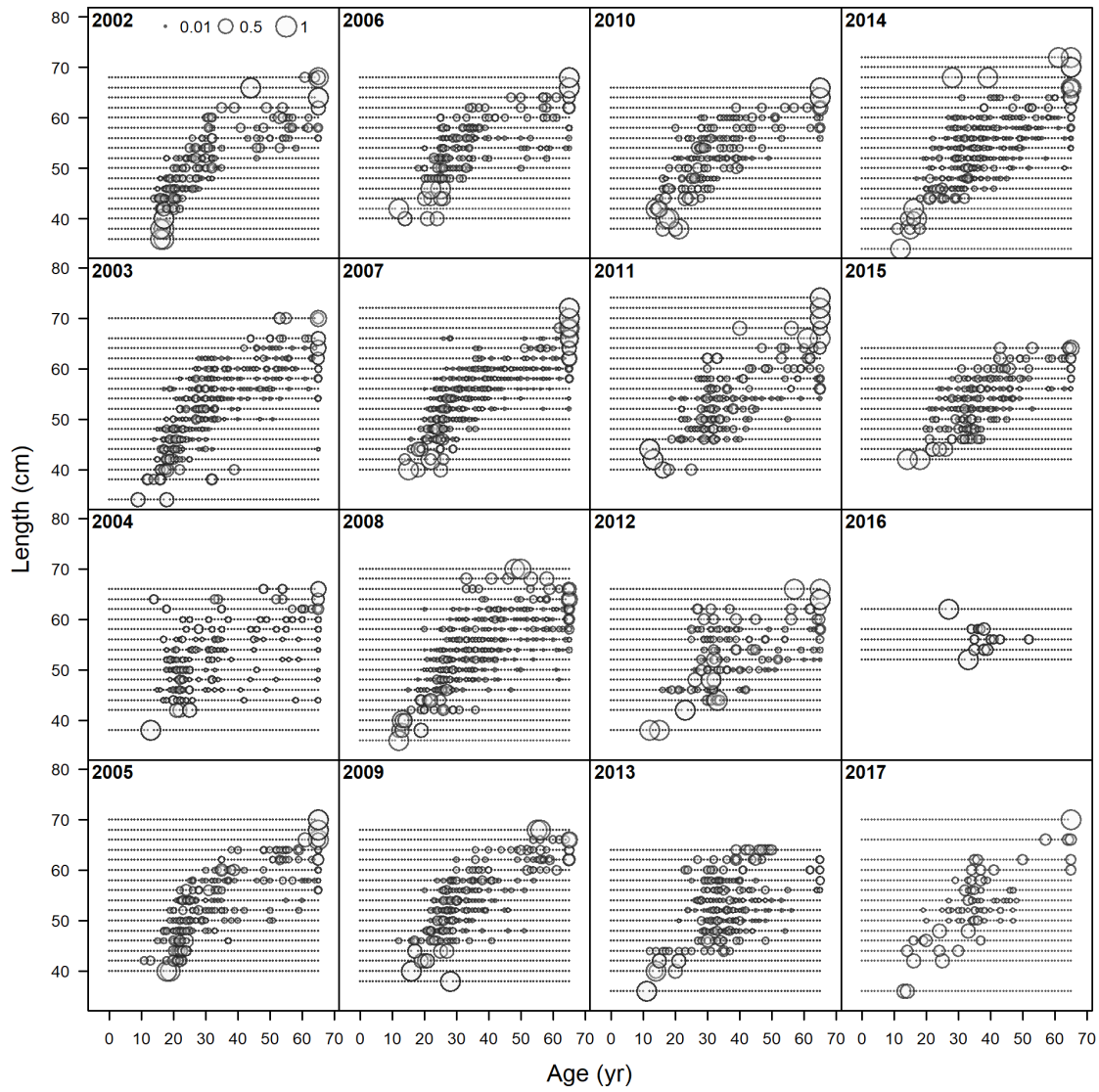


Figure 15: Annual unsexed conditional age-at-length data for the IPHC (1 of 2).

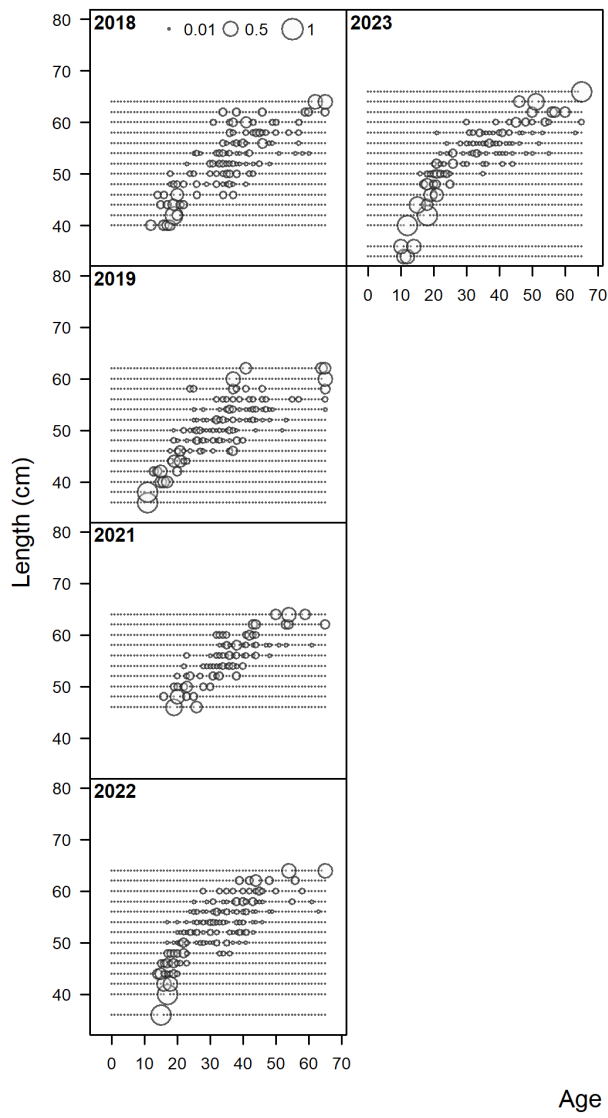


Figure 16: Annual unsexed conditional age-at-length data for the IPHC (2 of 2).

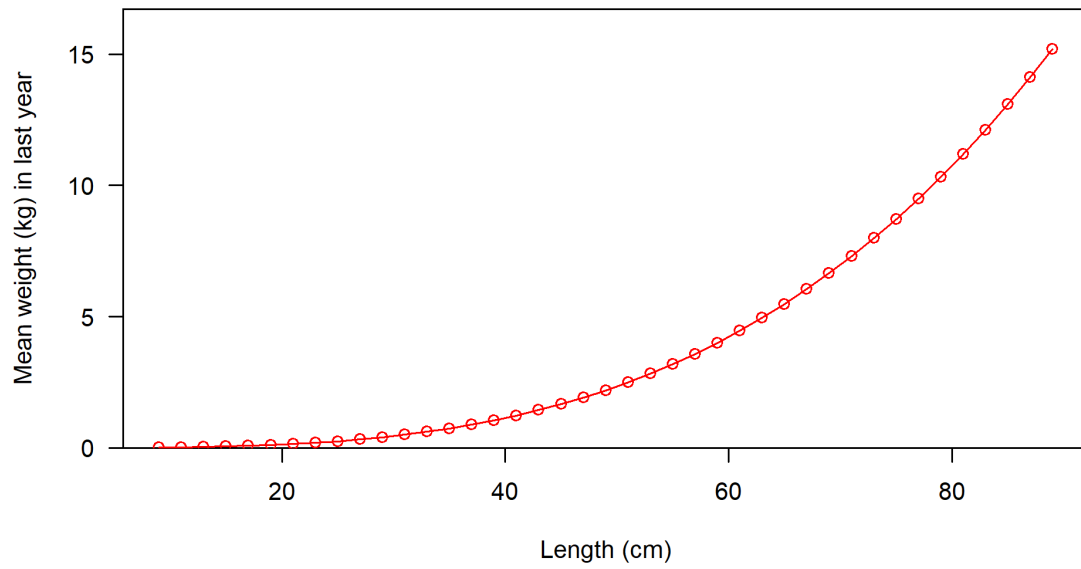


Figure 17: Updated weight-at-length relationship.

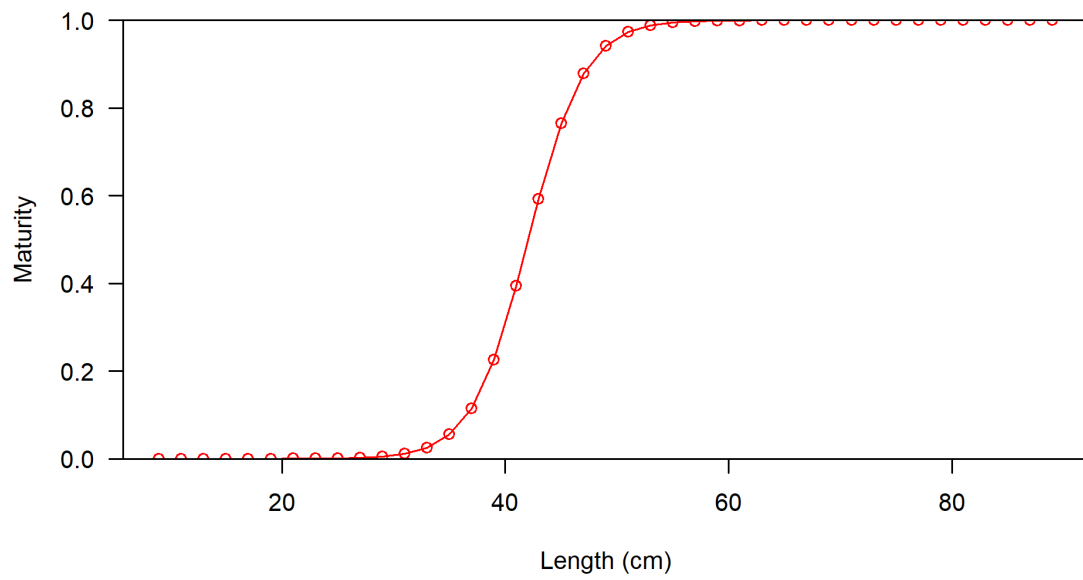


Figure 18: Maturity at length relationship used in the base model for Yelloweye Rockfish.



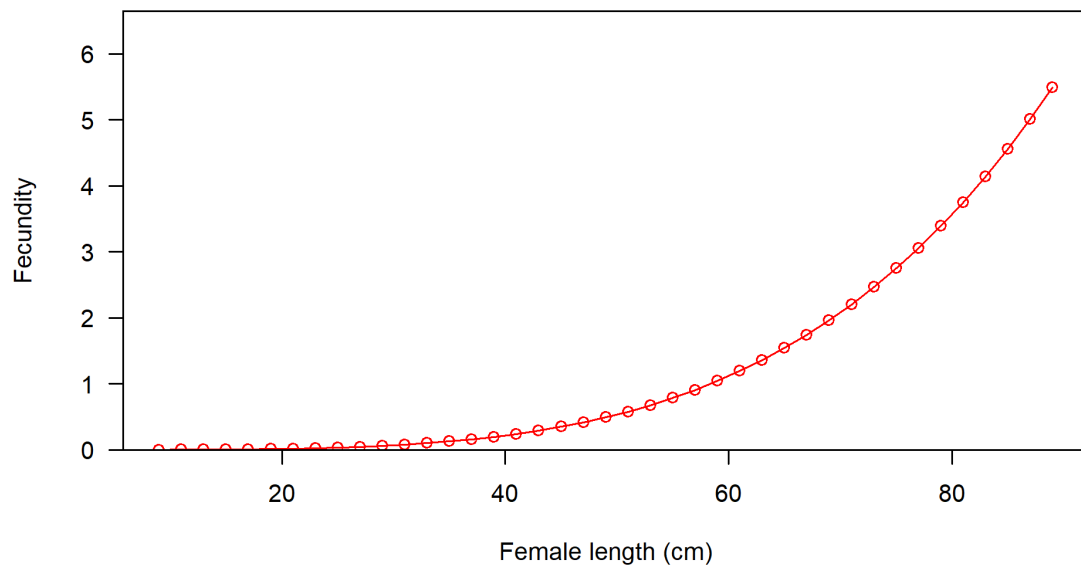


Figure 19: Fecundity at length relationship used in the base model for Yelloweye Rockfish.

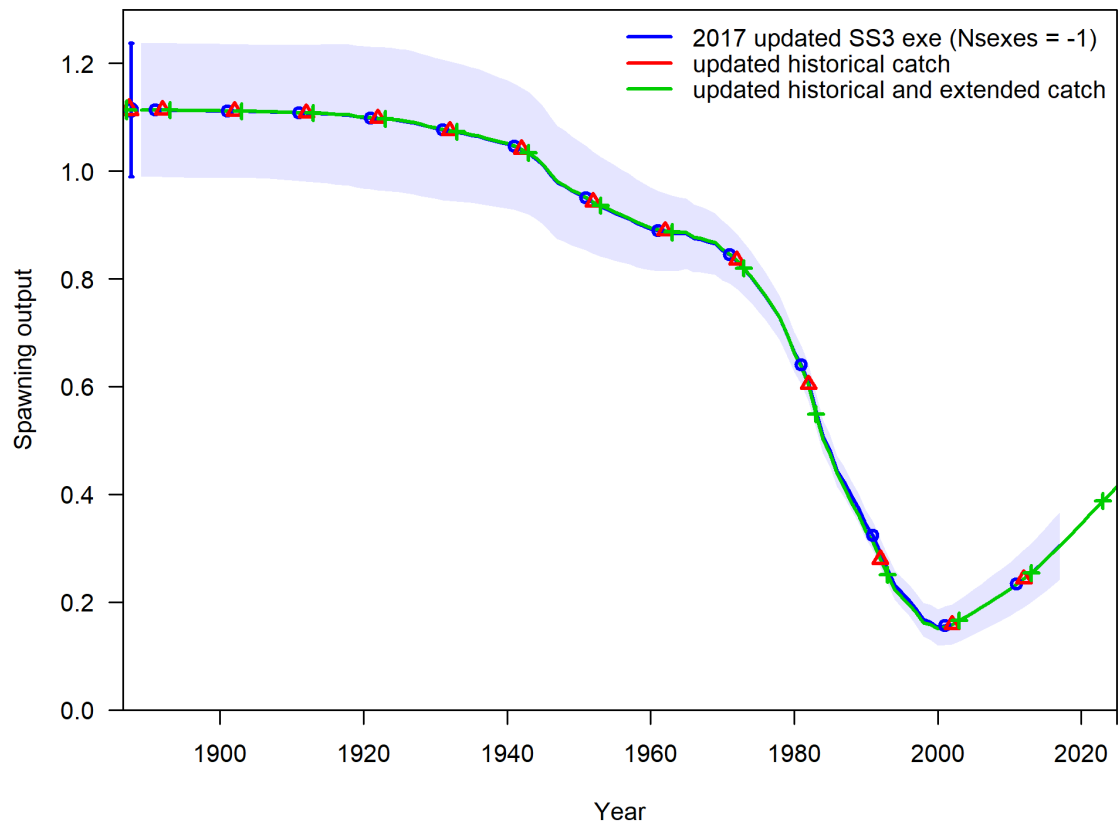


Figure 20: Comparison of the spawning output (billions of eggs) of the 2017 model with an updated SS3 executable (blue), updated historical catch data (red), and catch extended to 2024 (green).

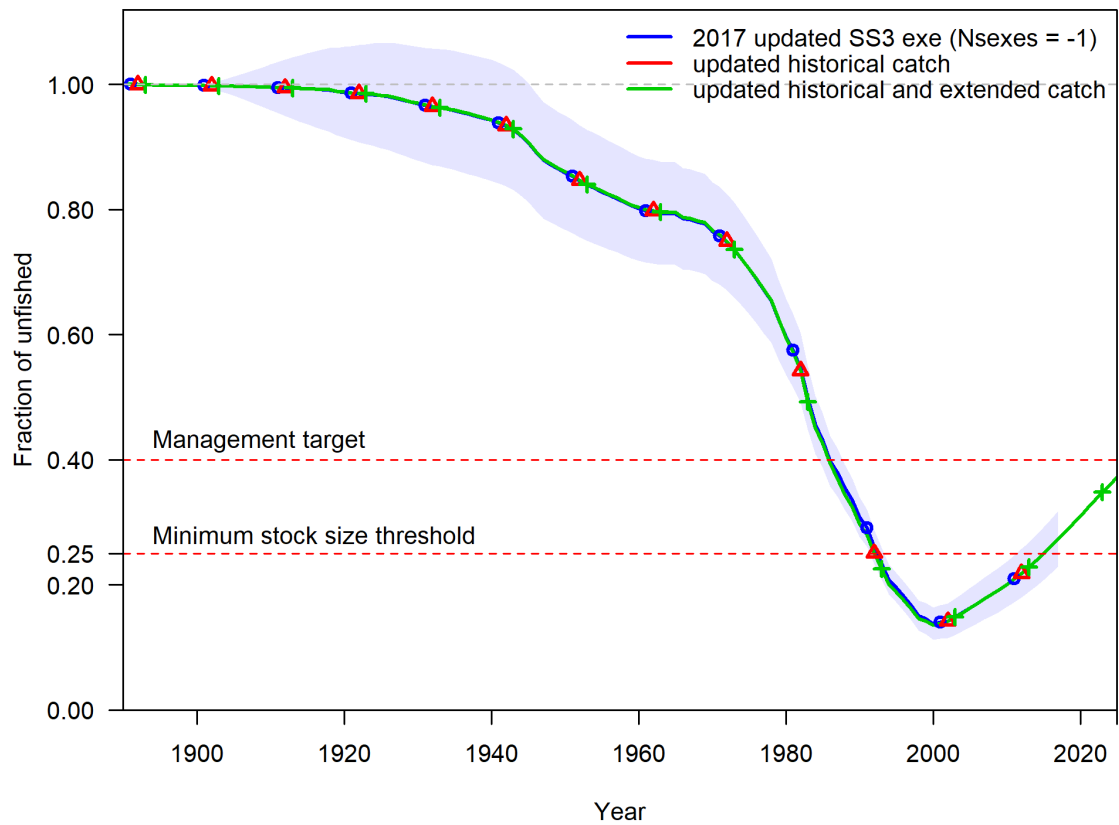


Figure 21: Comparison of the stock status of the 2017 model with an updated SS3 executable (blue), updated historical catch data (red), and catch extended to 2024 (green) relative to the management target and minimum stock size threshold.

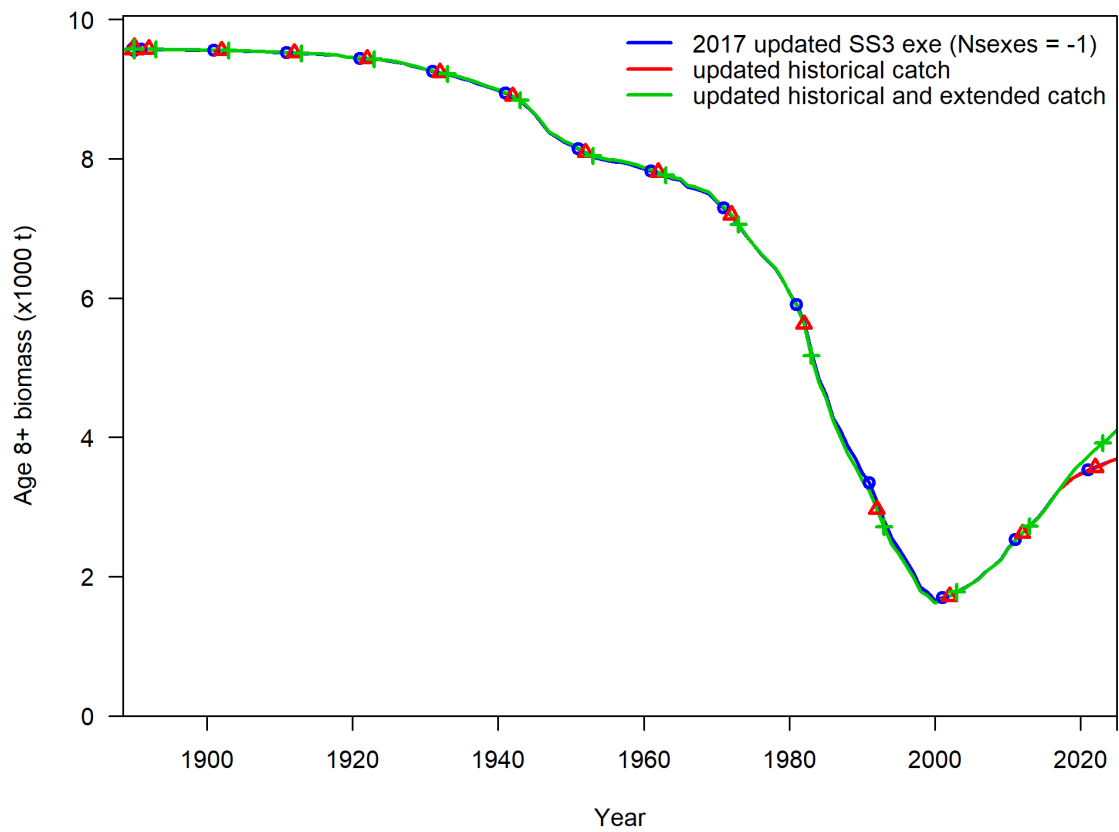


Figure 22: Comparison of adult Yelloweye Rockfish biomass of the 2017 model with an updated SS3 executable (blue), updated historical catch data (red), and catch extended to 2024 (green).

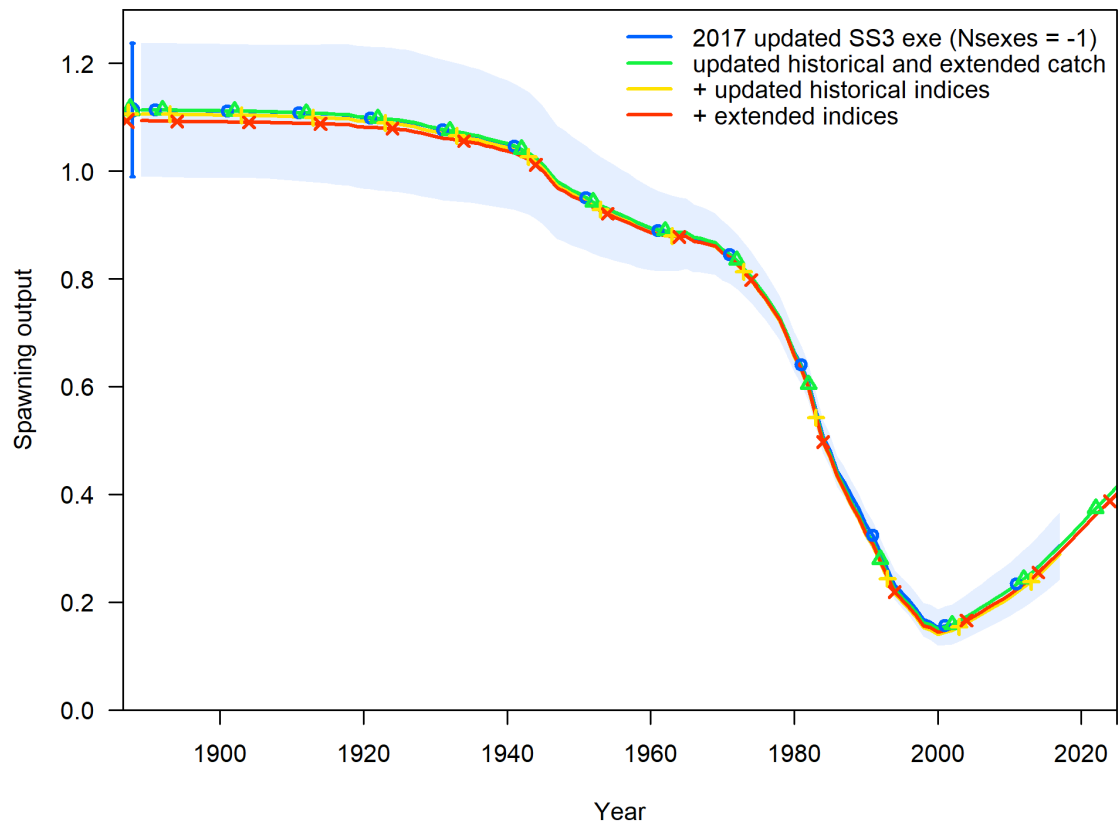


Figure 23: Comparison of the spawning output (billions of eggs) of the 2017 model with an updated SS3 executable (blue), updated and extended historical catch data (green), updated historical indices (yellow), and indices extended to 2024 (red).

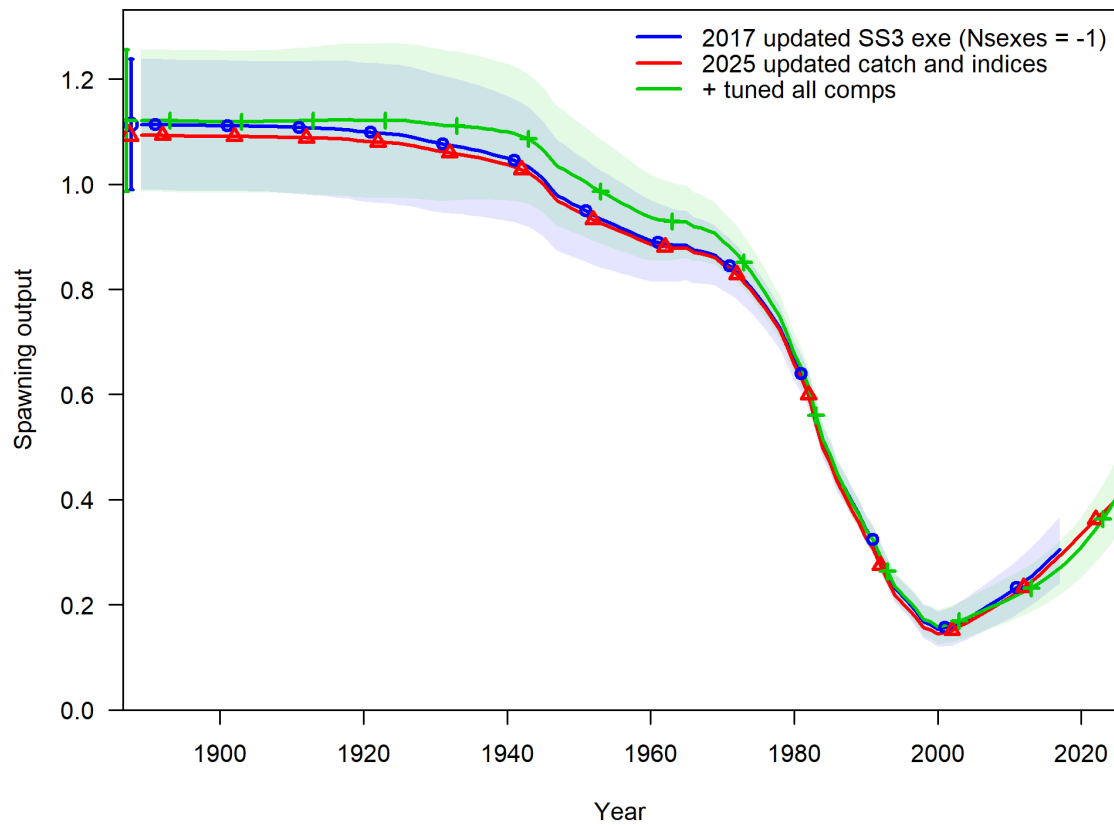


Figure 24: Comparison of the spawning output (billions of eggs) of the 2017 model with an updated SS3 executable (blue), updated and extended catch and indices (red), and all tuned length and age composition data (green).

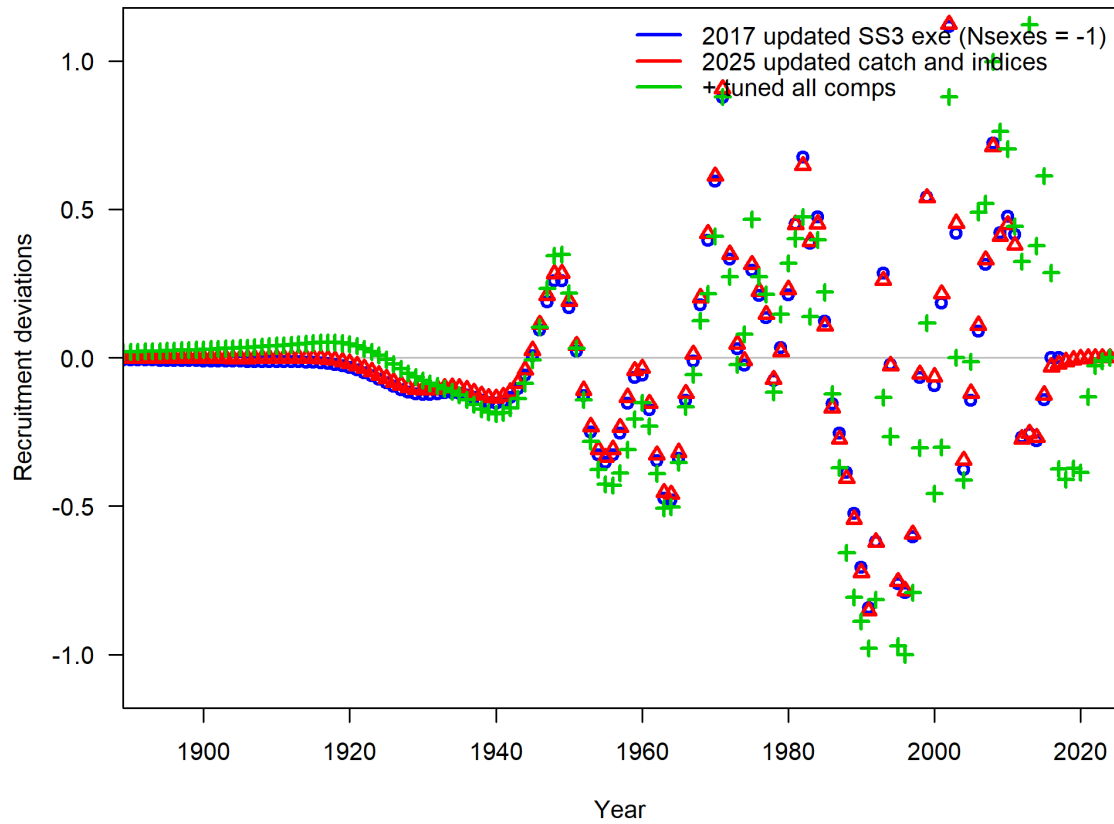


Figure 25: Recruitment deviation time-series comparing an updated SS3 executable (blue), updated and extended catch and indices (red), and all tuned length and age composition data (green).

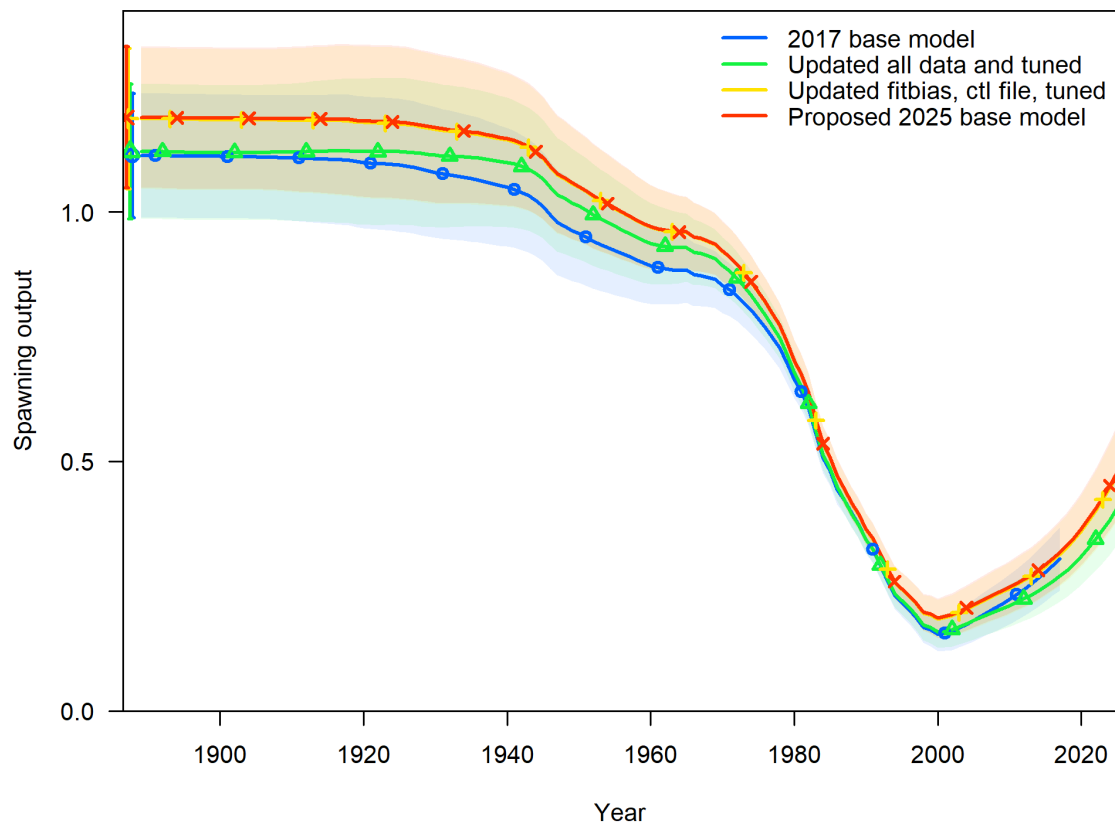


Figure 26: Comparison of the spawning output (billions of eggs) of the 2017 model with an updated SS3 executable (blue), updated and extended and tuned data (green), the updated SS input file changes with tuning (yellow), and the proposed 2025 base model after all final bridging steps (red), all with 95% confidence intervals.



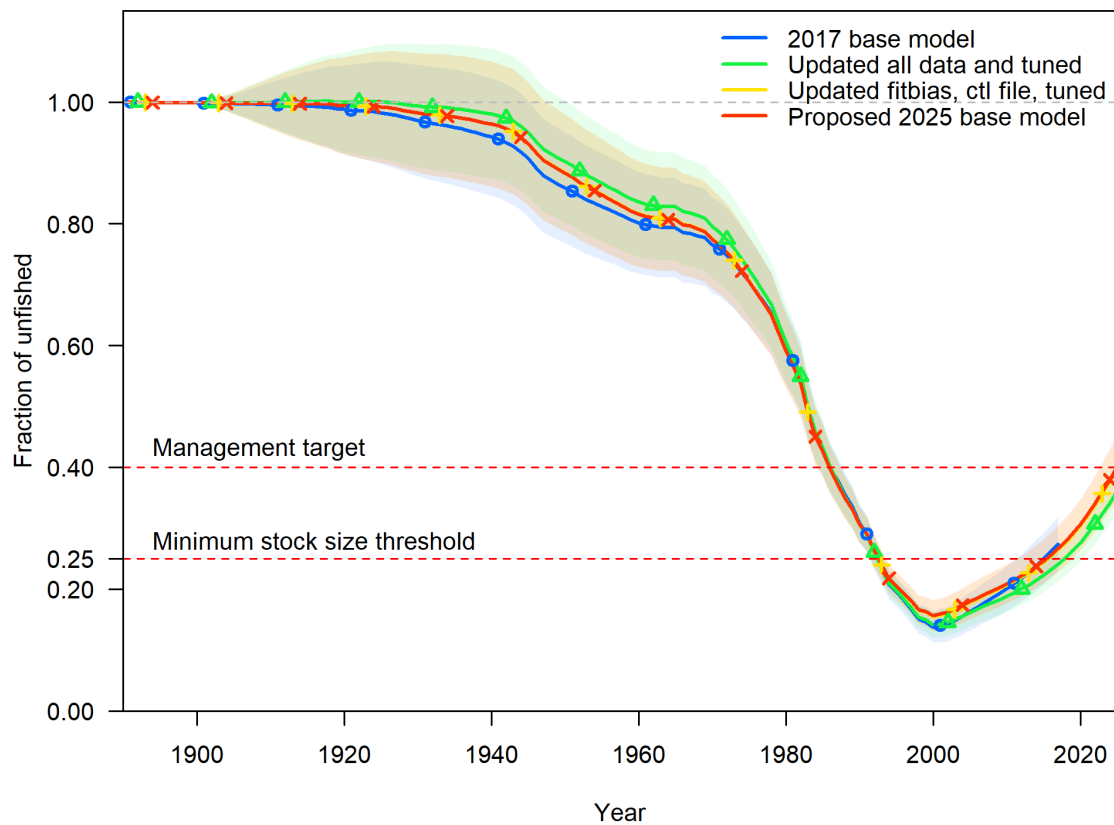


Figure 27: Comparison of the stock status output of the 2017 model with an updated SS3 executable (blue), updated and extended and tuned data (green), the updated SS input file changes with tuning (yellow), and the proposed 2025 base model after all final bridging steps (red) relative to the management target and minimum stock size threshold, including 95% confidence intervals.

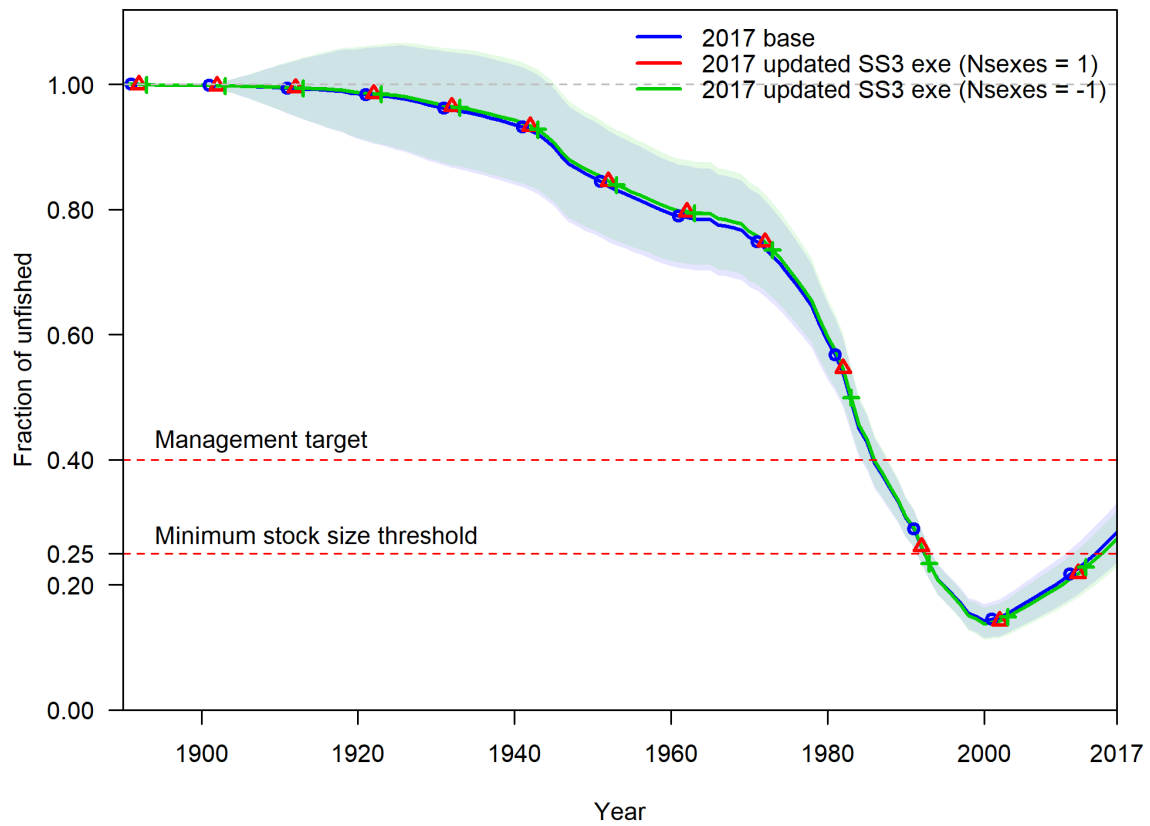


Figure 28: Comparison of the stock status for the 2017 model with the updated SS3 executable and a single-sex model.

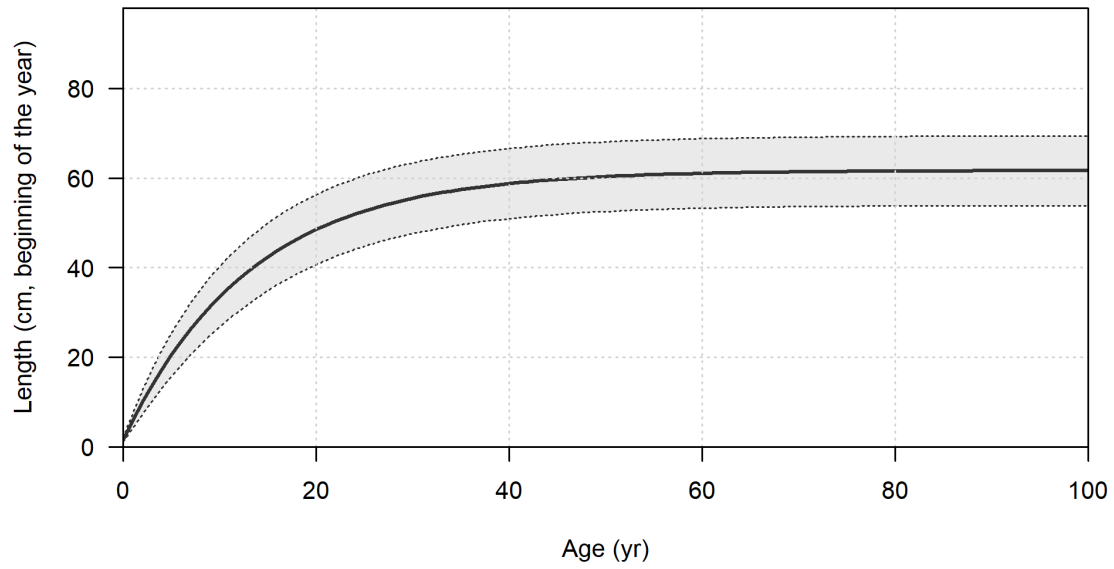


Figure 29: Length at age in the beginning of the year in the ending year of the model.

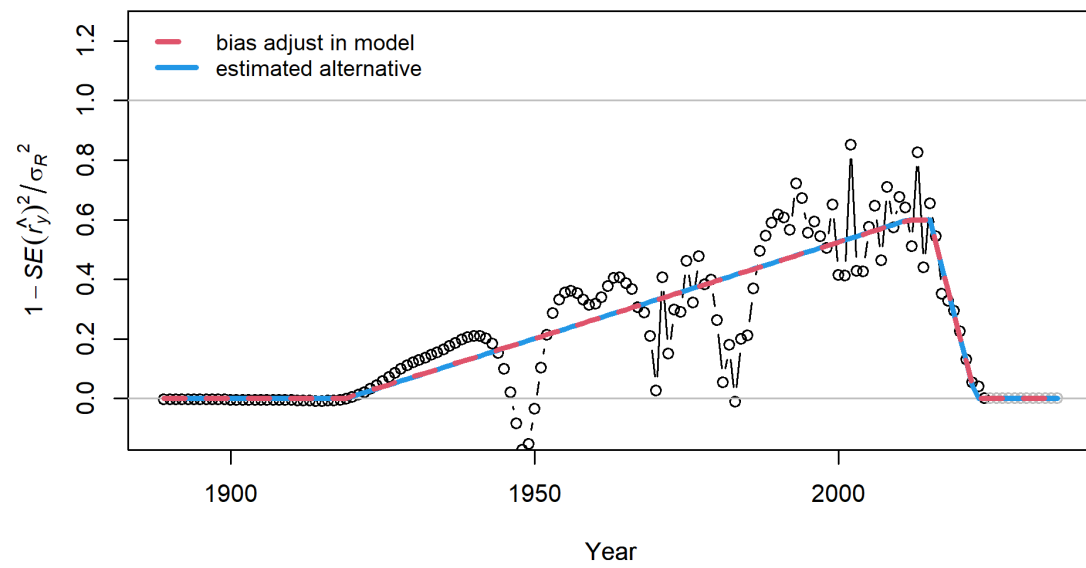


Figure 30: Points are transformed variances. Red line shows current settings for bias adjustment specified in the control file. Blue line shows least squares estimate of alternative bias adjustment relationship for recruitment deviations.

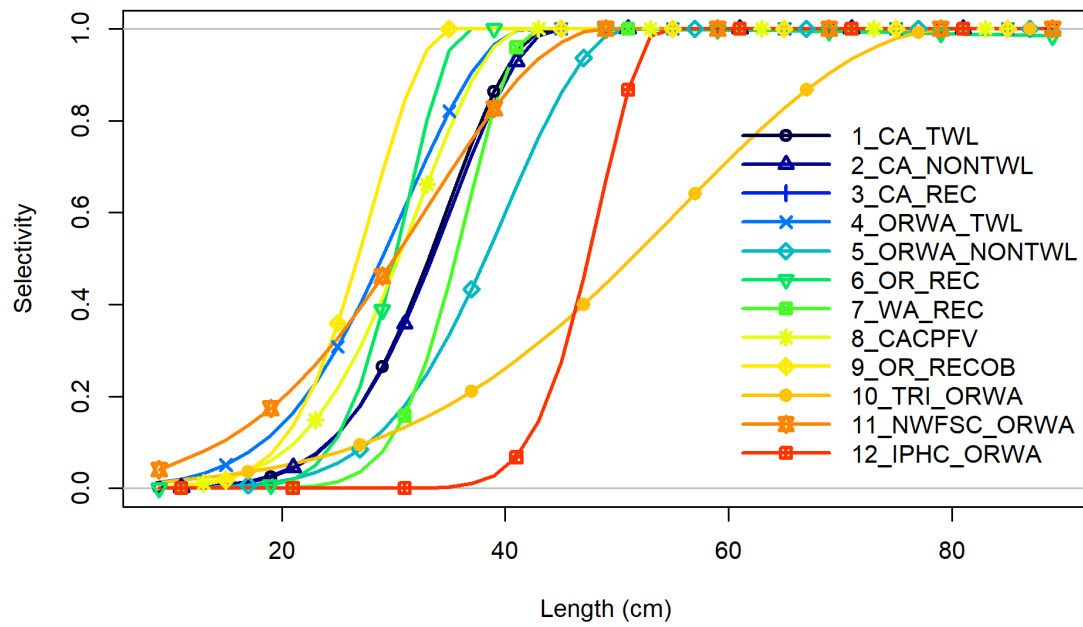


Figure 31: Estimated selectivity at length for all fleets.

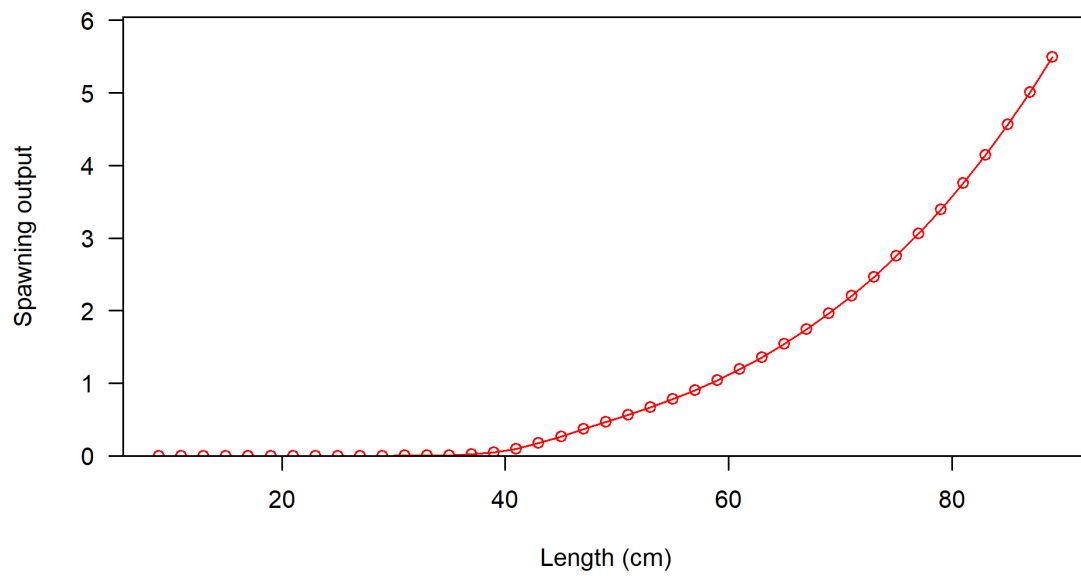


Figure 32: Spawning output (billions of eggs) at length. Yelloweye length-at-age 70 (the second reference age) equals 61.4 and an L-infinity of 61.7.

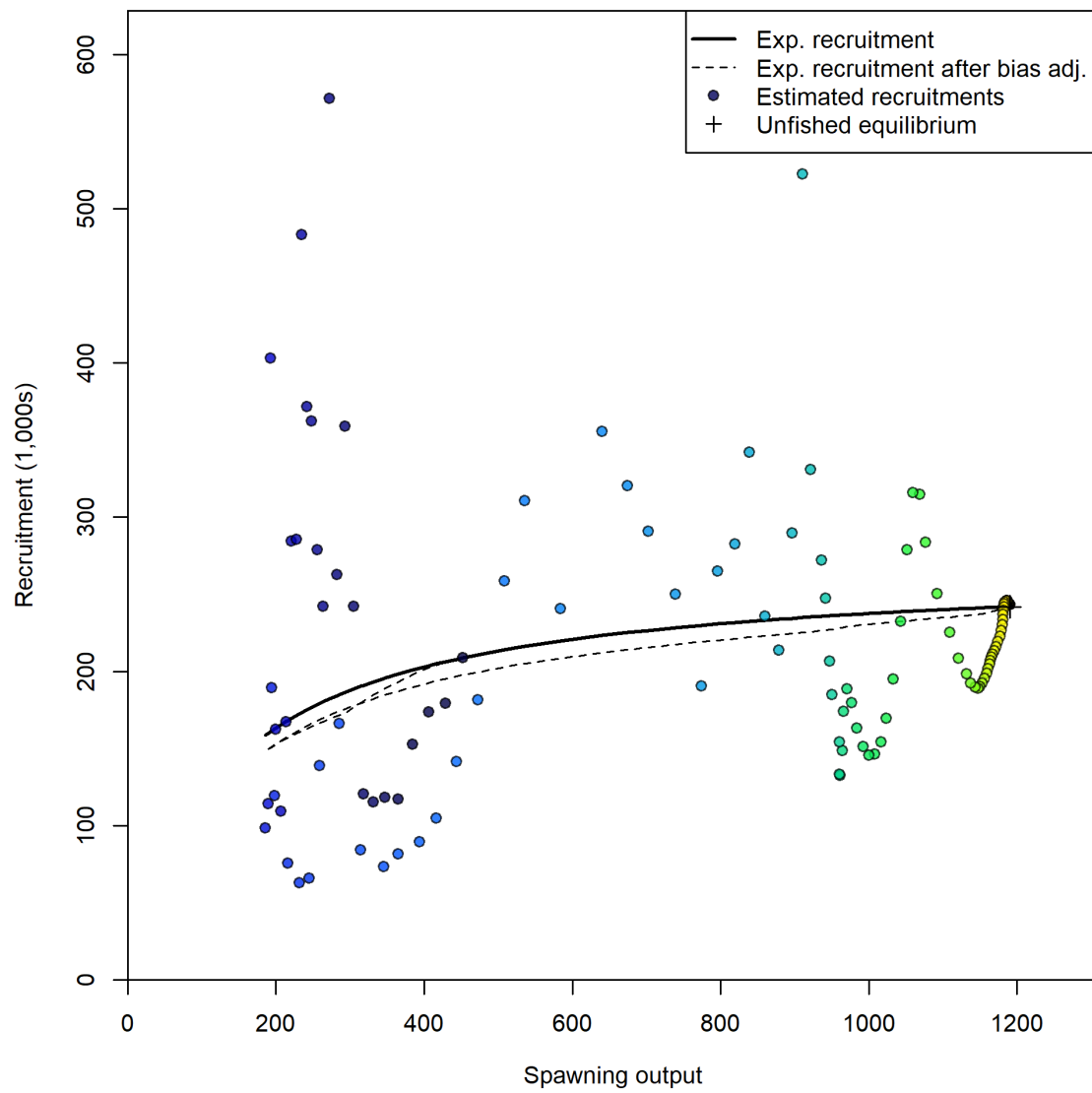


Figure 33: Stock-recruit curve. Point colors indicate year, with warmer colors indicating earlier years and cooler colors in showing later years.

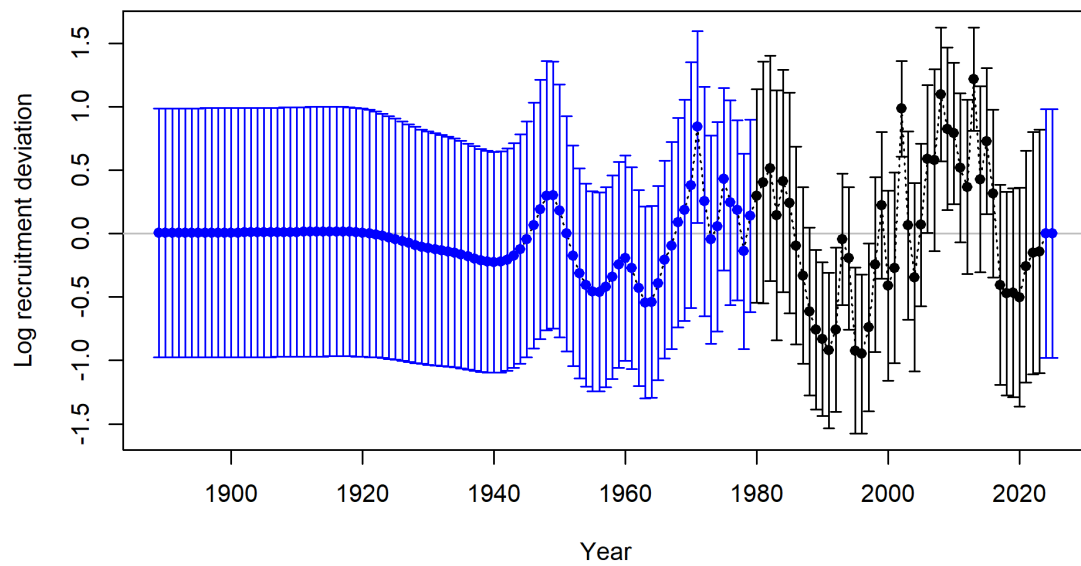


Figure 34: Estimated recruitment deviations with 95% intervals.



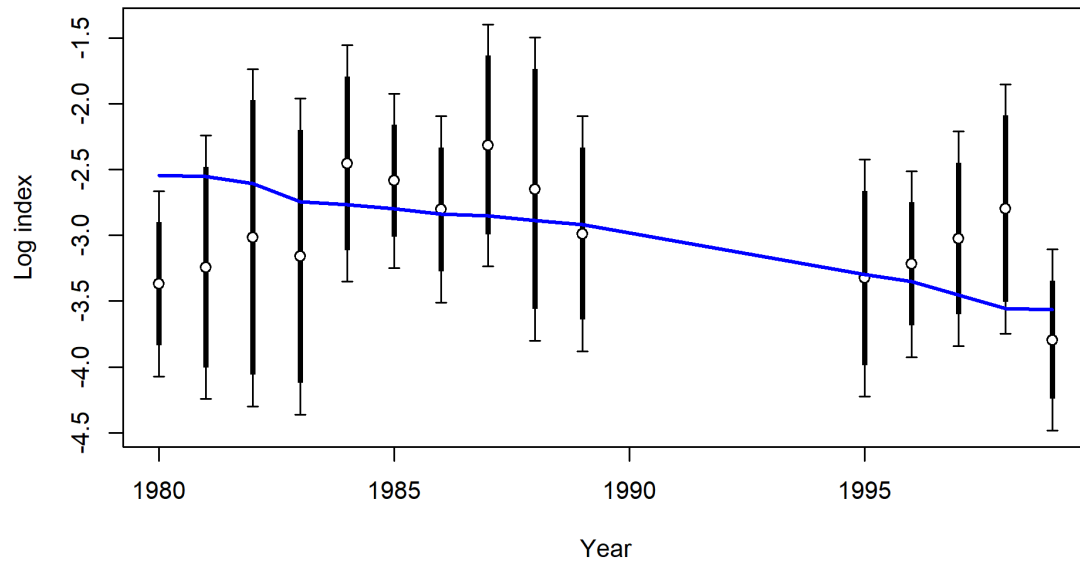


Figure 35: Fit to the log-scale California MRFSS recreational index.

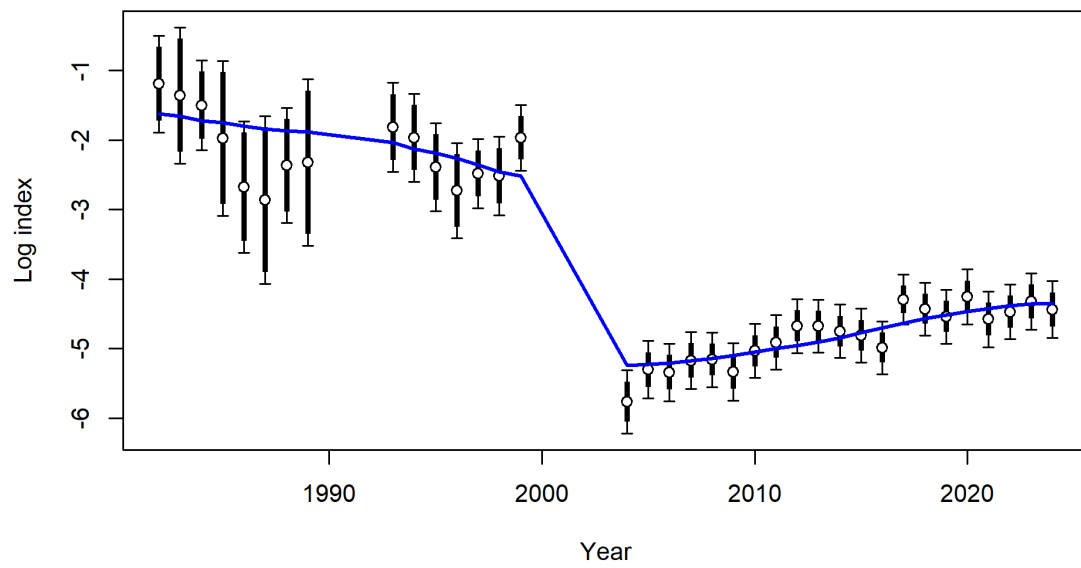


Figure 36: Fit to the log-scale Oregon recreational index.

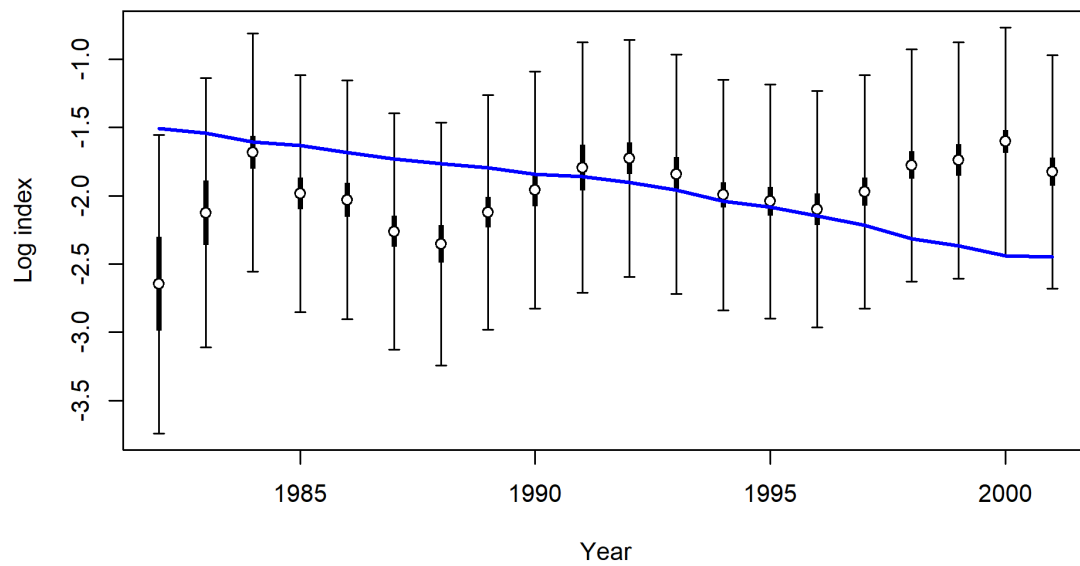


Figure 37: Fit to the log-scale Washington recreational index.

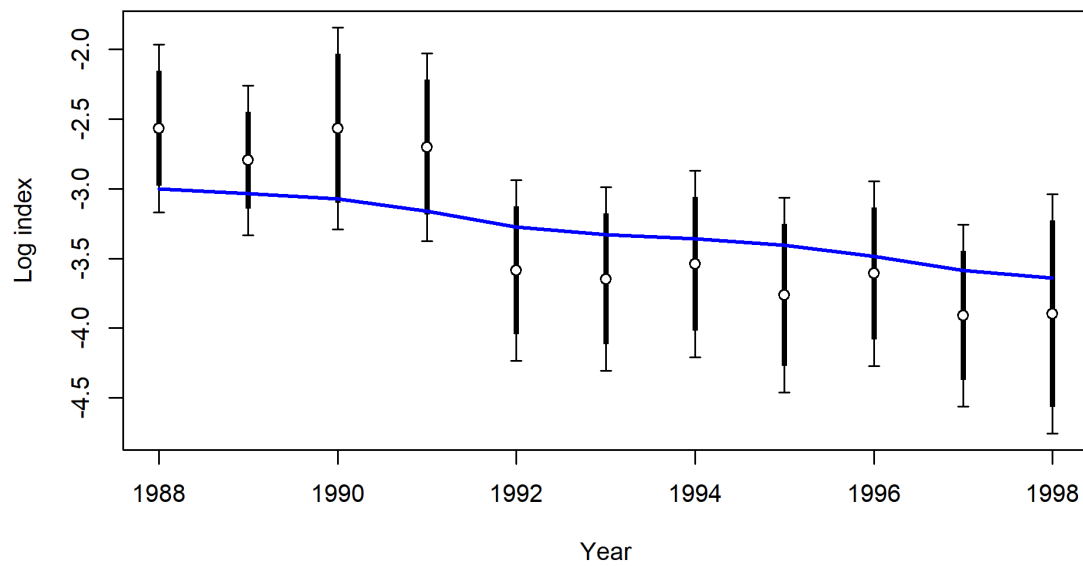


Figure 38: Fit to the log-scale California CPFV observer index.

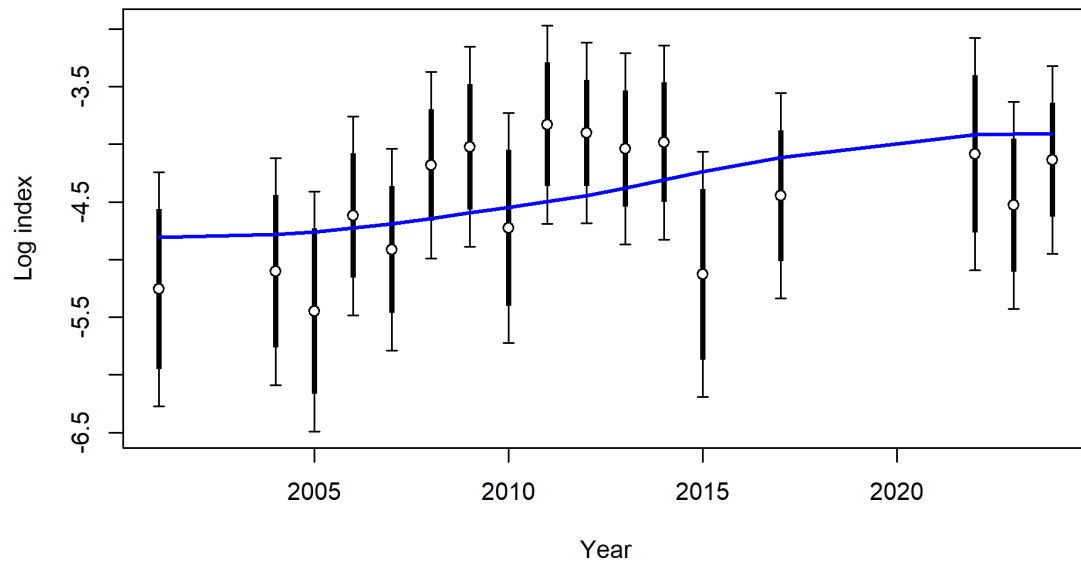


Figure 39: Fit to the log-scale Oregon onboard observer (ORFS) index.

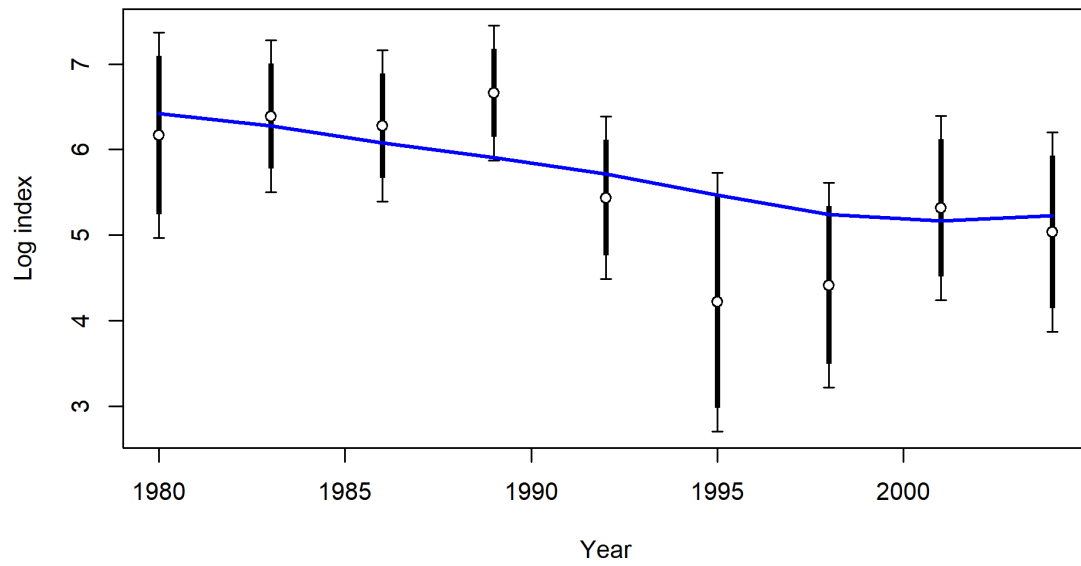


Figure 40: Fit to the log-scale Triennial survey index.

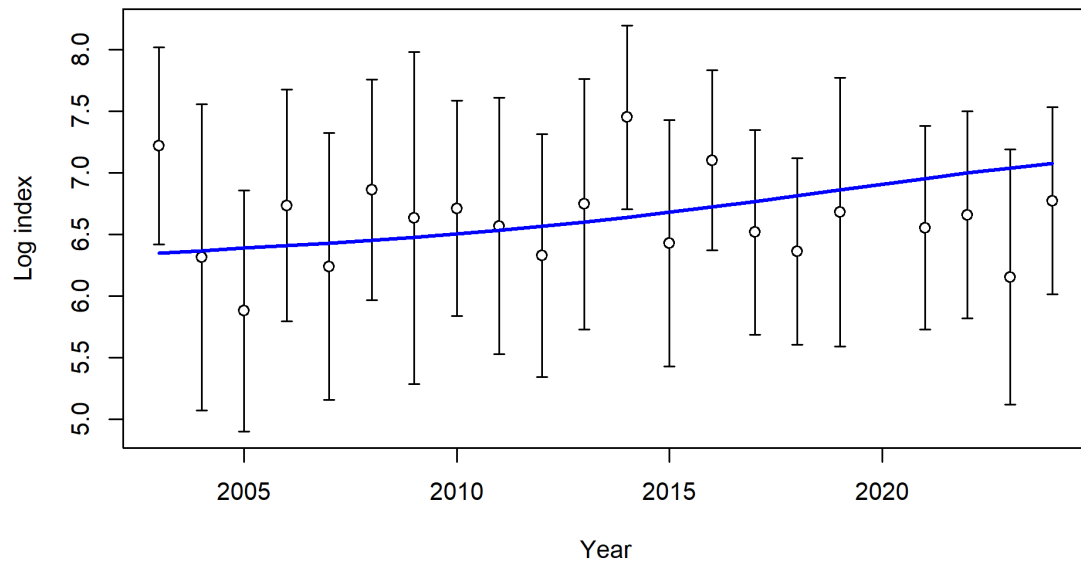


Figure 41: Fit to the log-scale WCGBTS index.

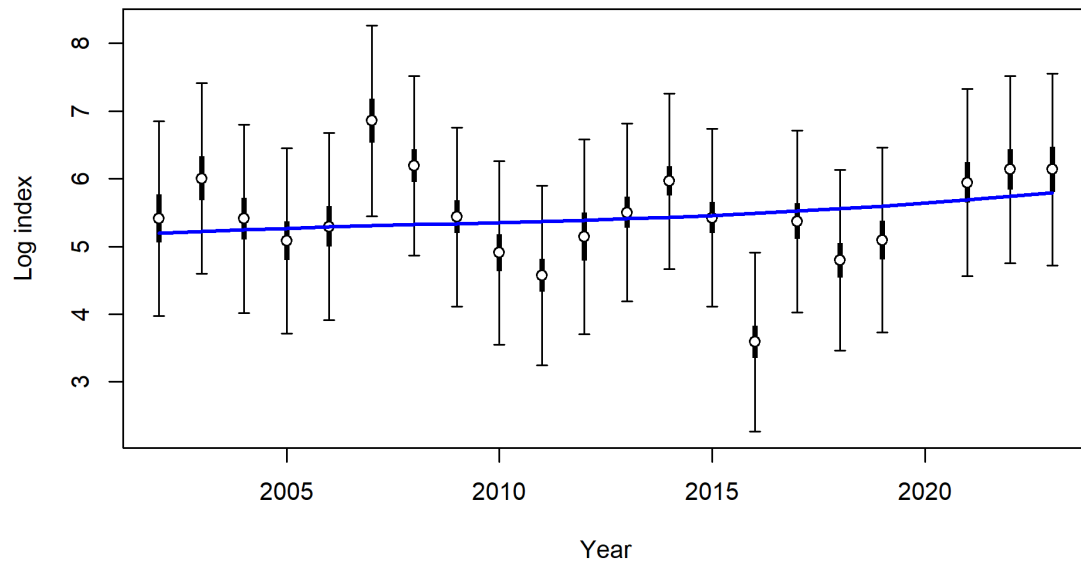


Figure 42: Fit to the log-scale IPHC survey index.



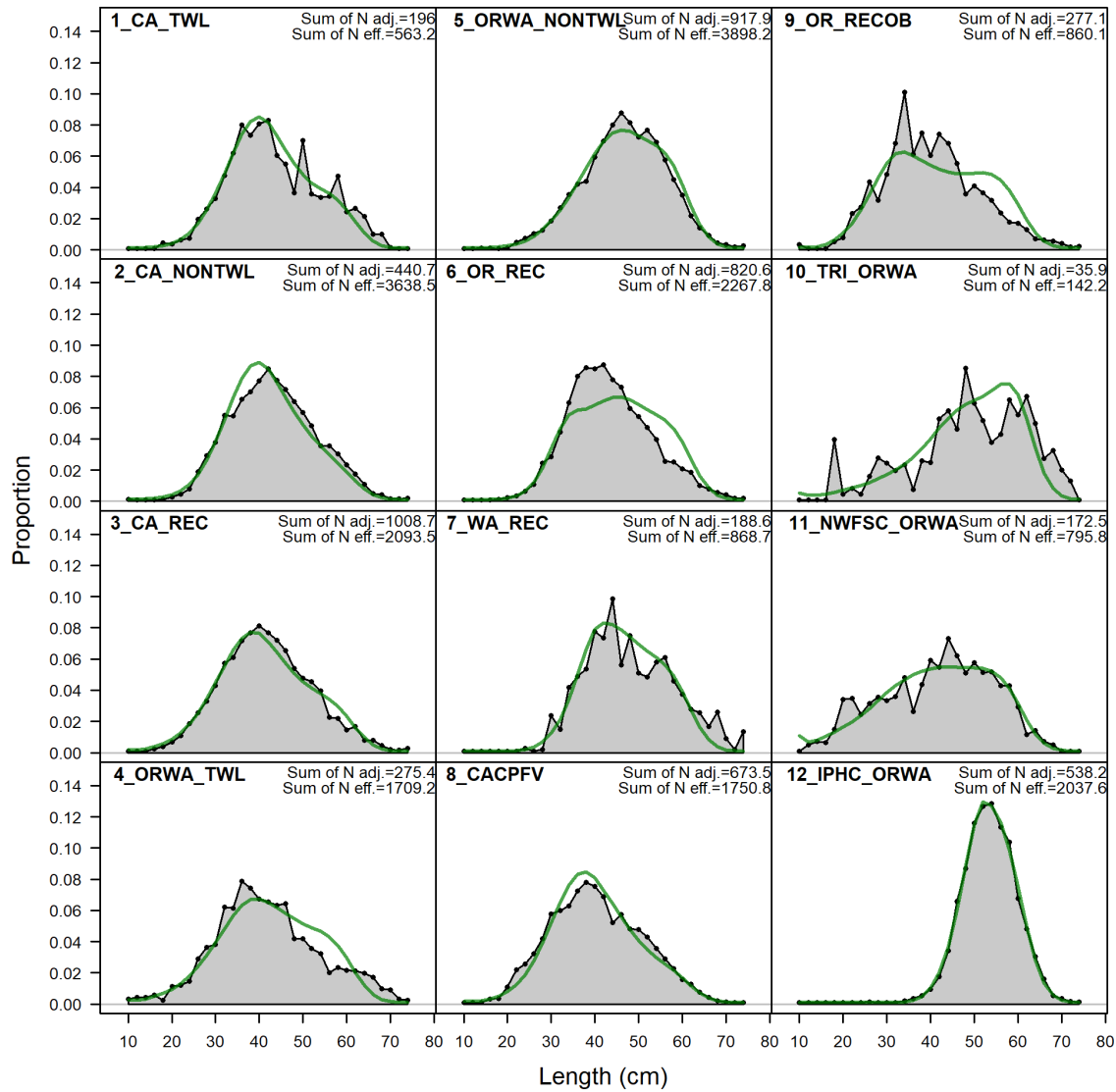


Figure 43: Fit to length composition data, aggregated across time by fleet.

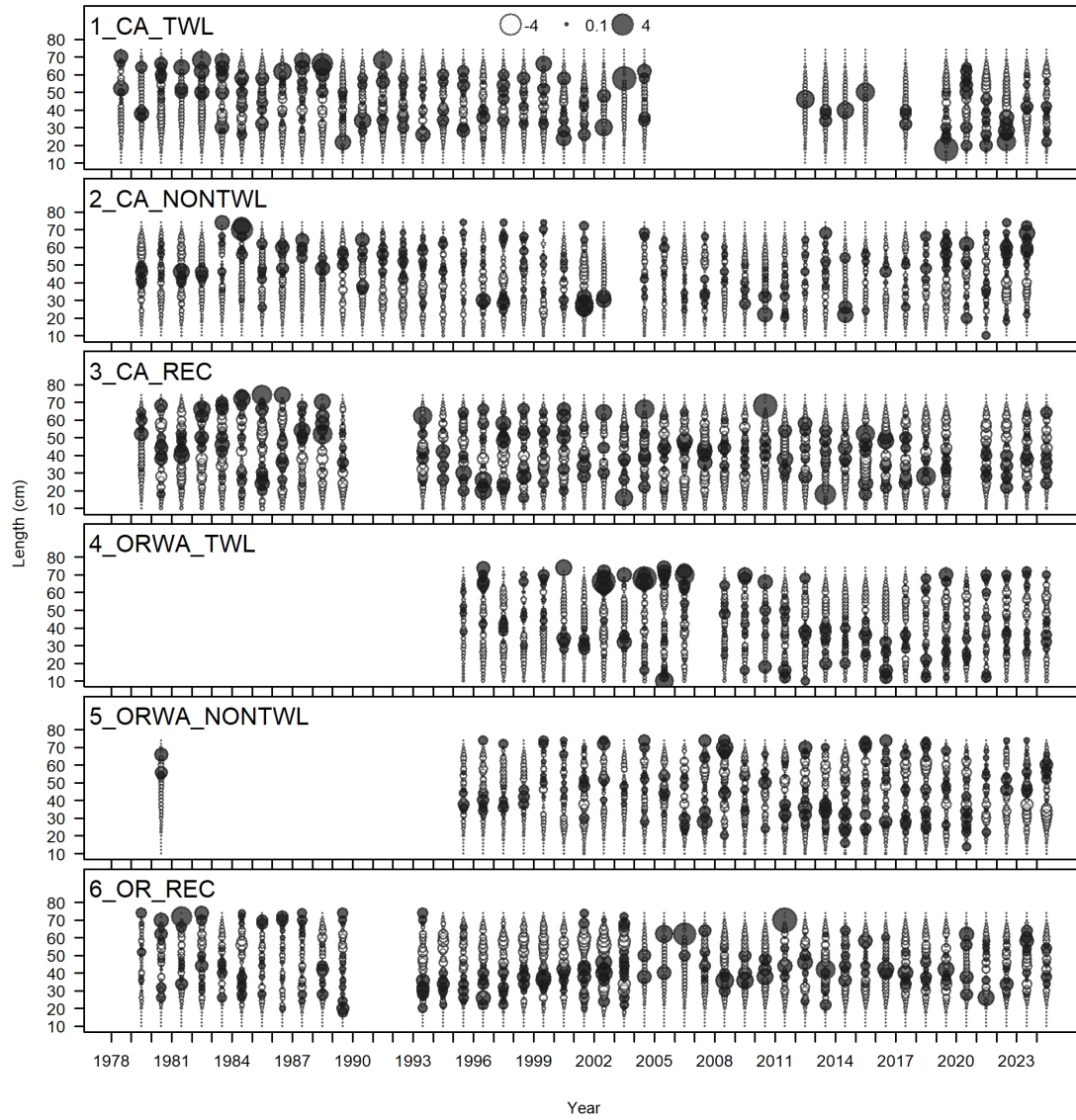


Figure 44: Pearson residuals, comparing across fleets, for length composition data (1 of 2). Closed bubbles are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

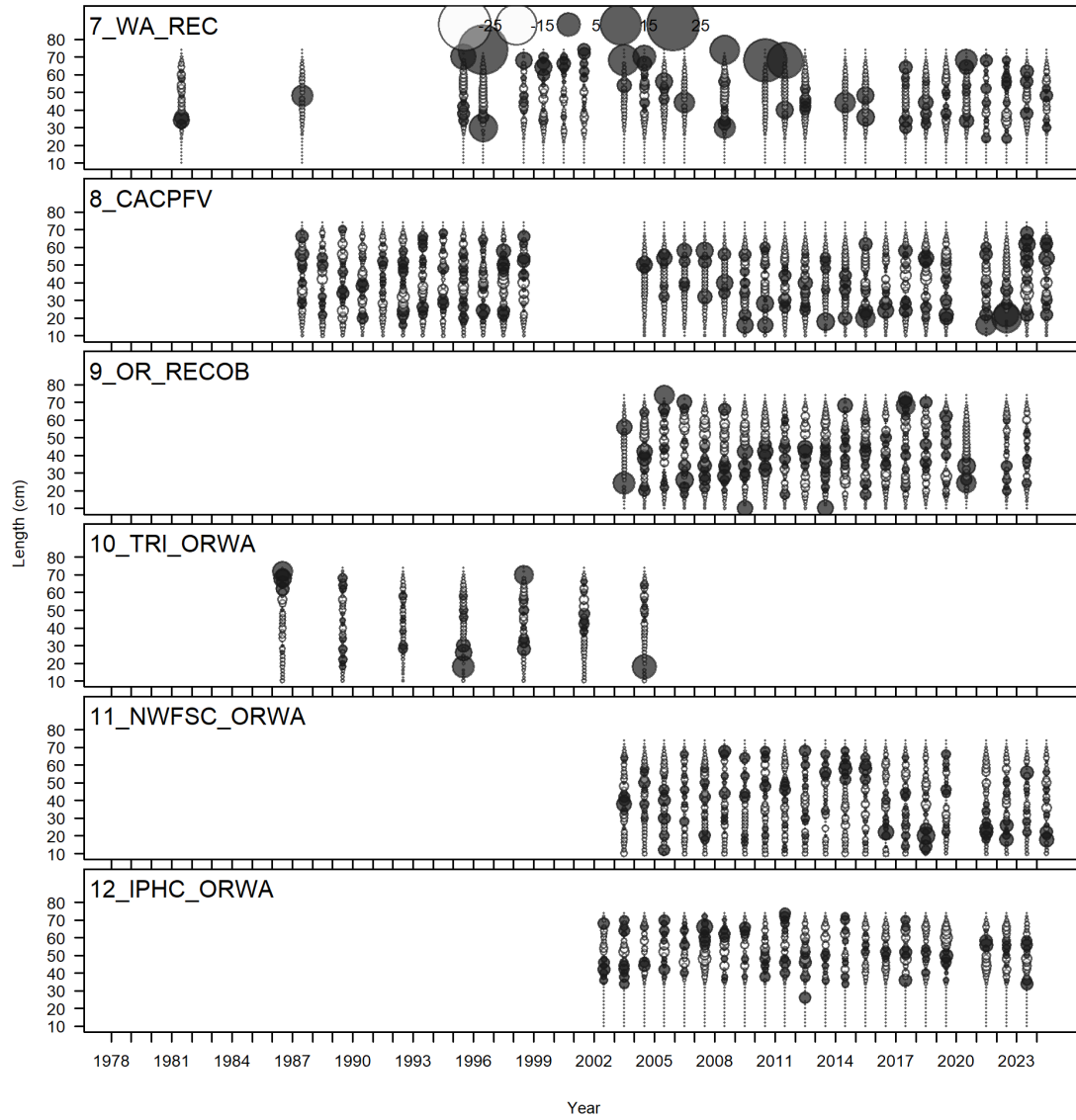


Figure 45: Pearson residuals, comparing across fleets, for length composition data (2 of 2). Closed bubbles are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

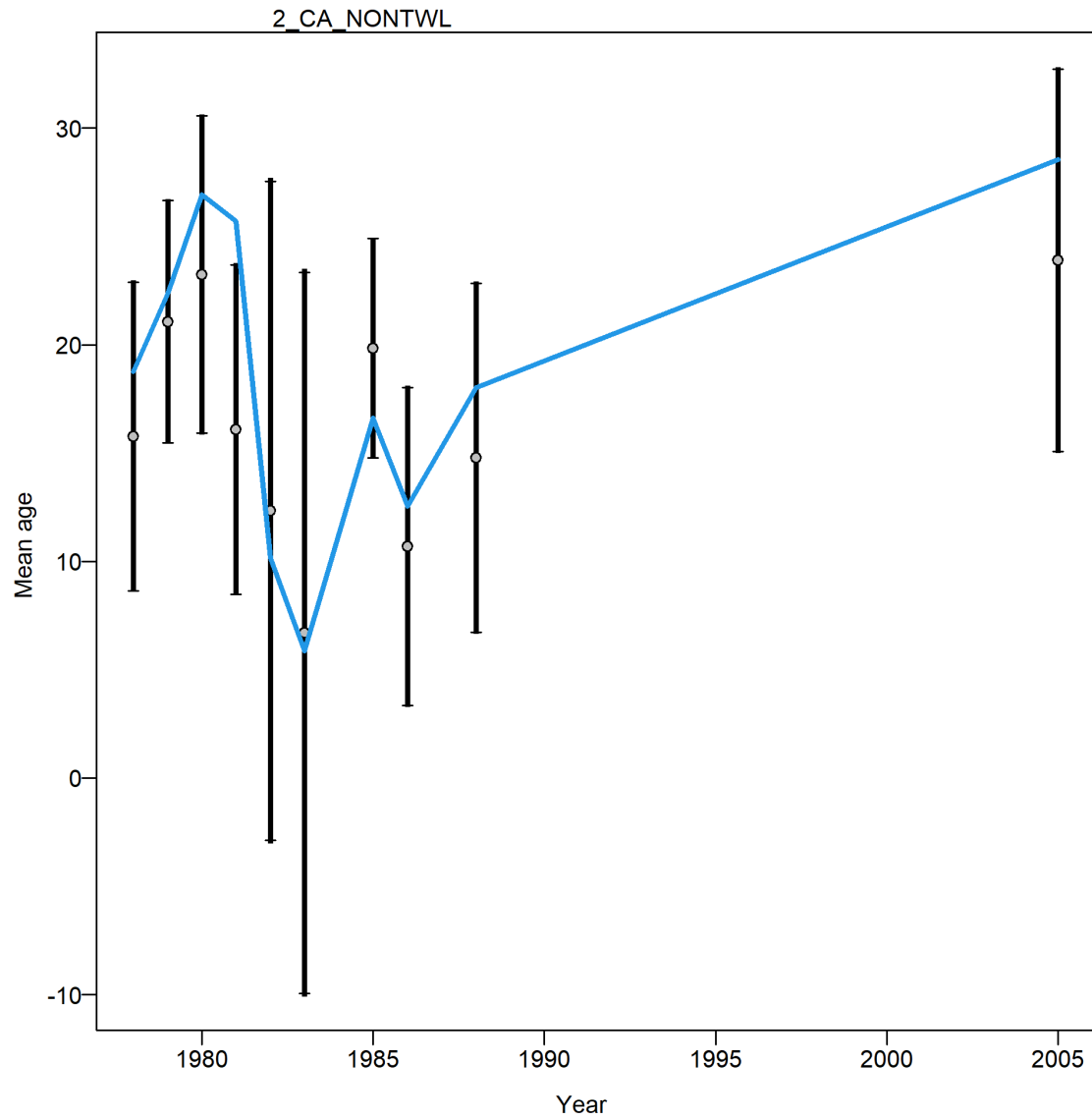


Figure 46: Mean age from conditional data (aggregated across length bins) for the CA NONTWL fleet with 95% confidence intervals based on input sample sizes. The blue line is the model expectation.

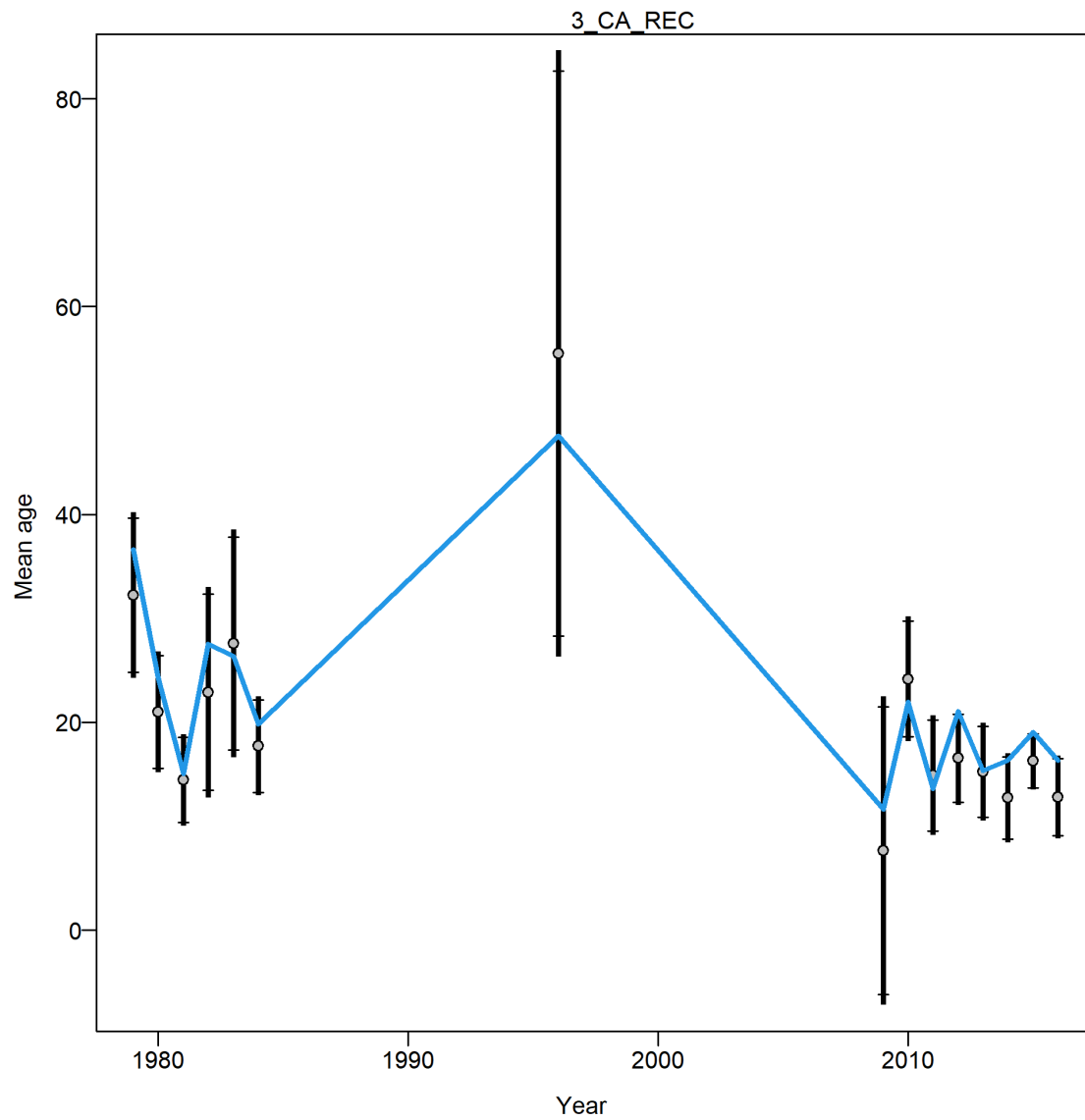


Figure 47: Mean age from conditional data (aggregated across length bins) for the CA REC fleet with 95% confidence intervals based on input sample sizes. The blue line is the model expectation.

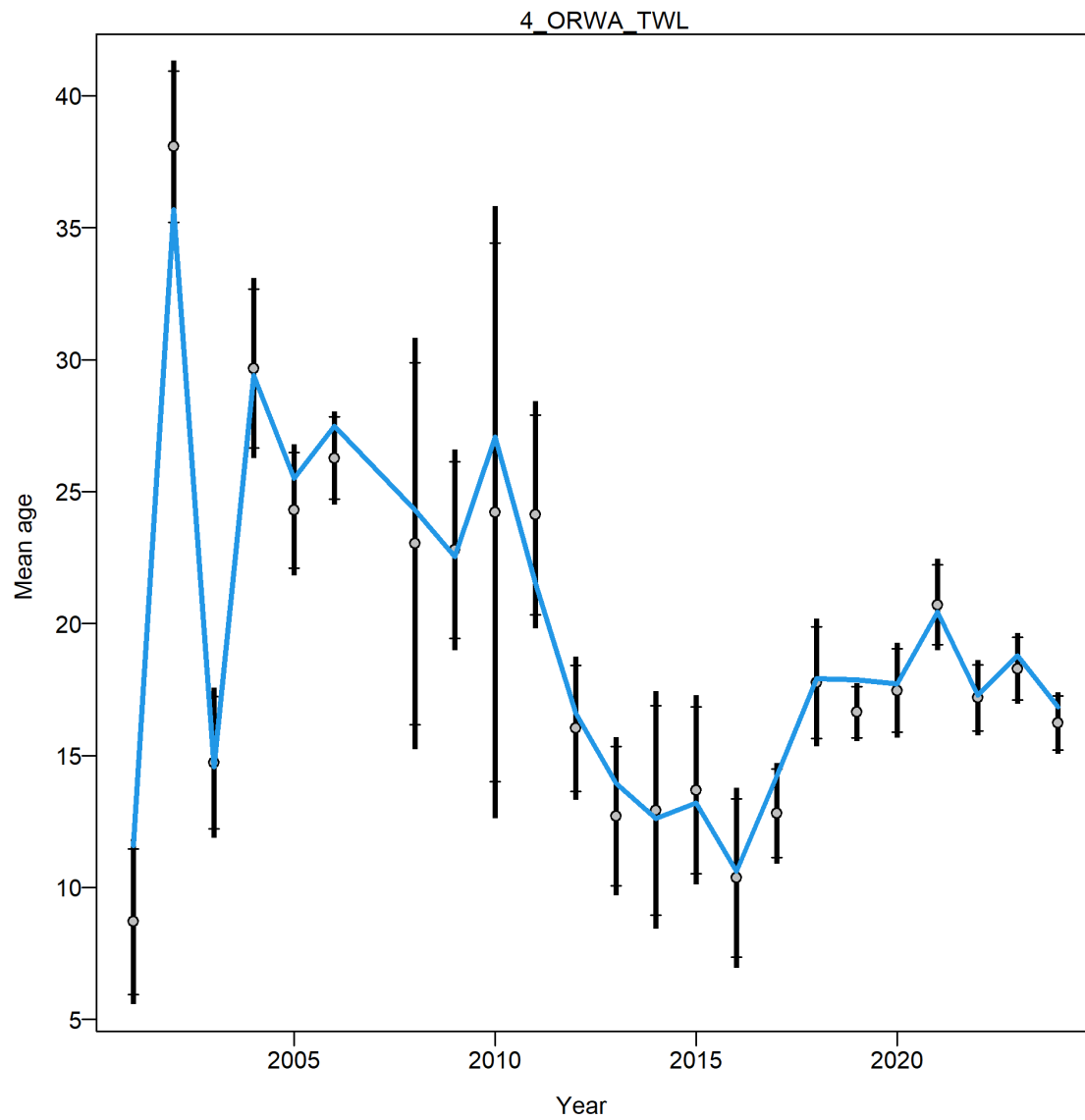


Figure 48: Mean age from conditional data (aggregated across length bins) for the ORWA TWL fleet with 95% confidence intervals based on input sample sizes. The blue line is the model expectation.

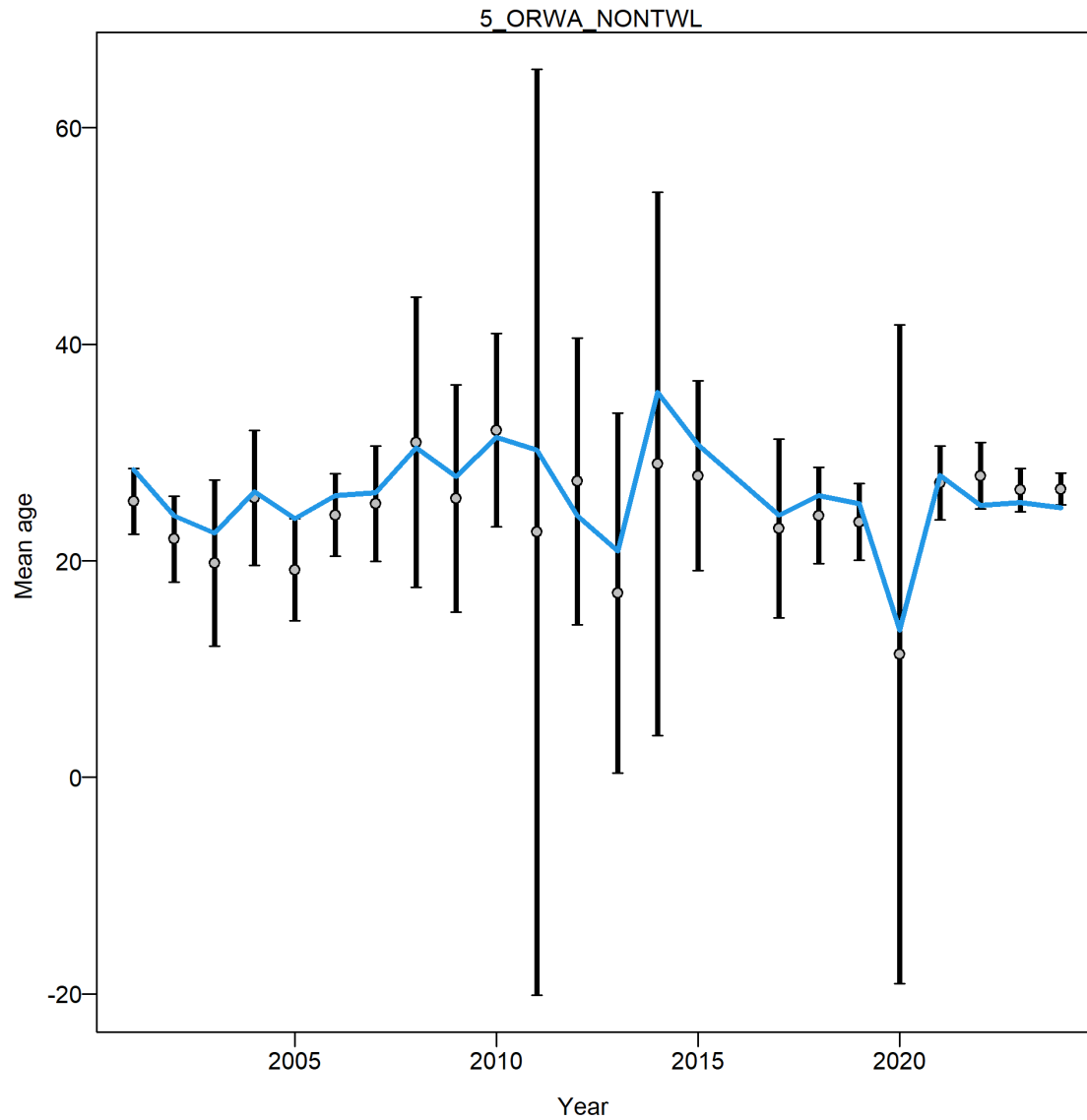


Figure 49: Mean age from conditional data (aggregated across length bins) for the ORWA NONTWL fleet with 95% confidence intervals based on input sample sizes. The blue line is the model expectation.

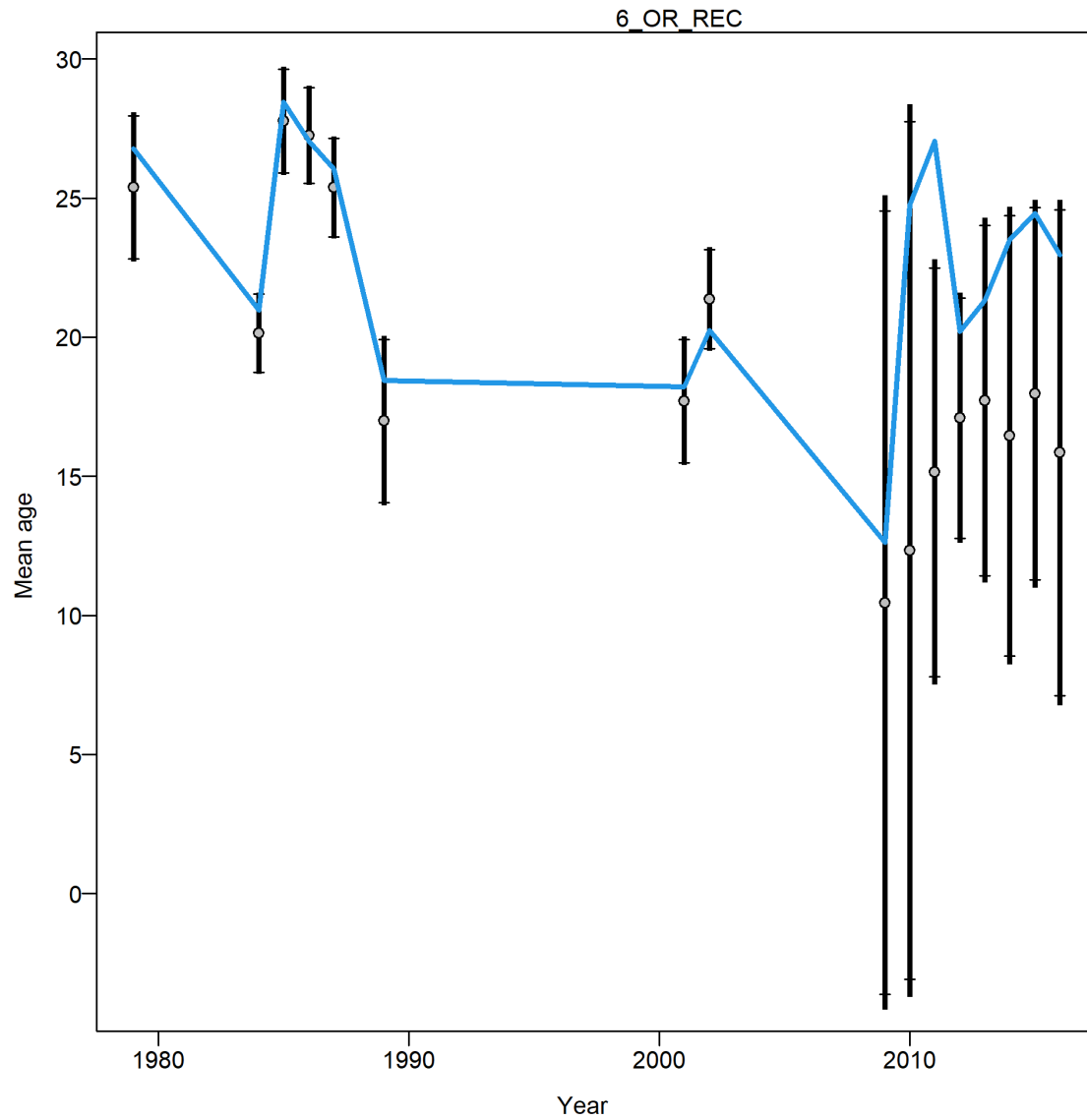


Figure 50: Mean age from conditional data (aggregated across length bins) for the ORWA REC fleet with 95% confidence intervals based on input sample sizes. The blue line is the model expectation.



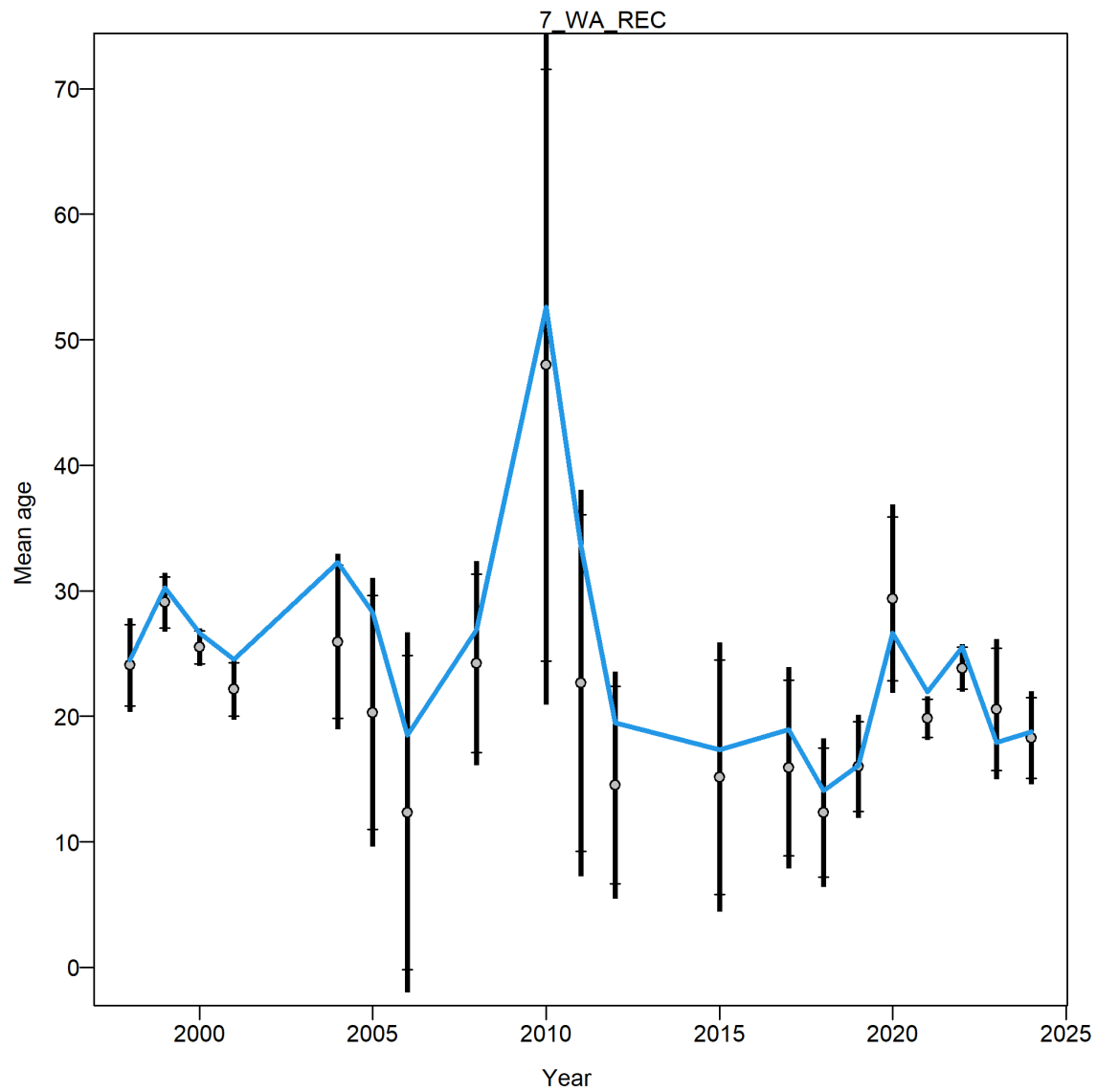


Figure 51: Mean age from conditional data (aggregated across length bins) for the WA REC fleet with 95% confidence intervals based on input sample sizes. The blue line is the model expectation.

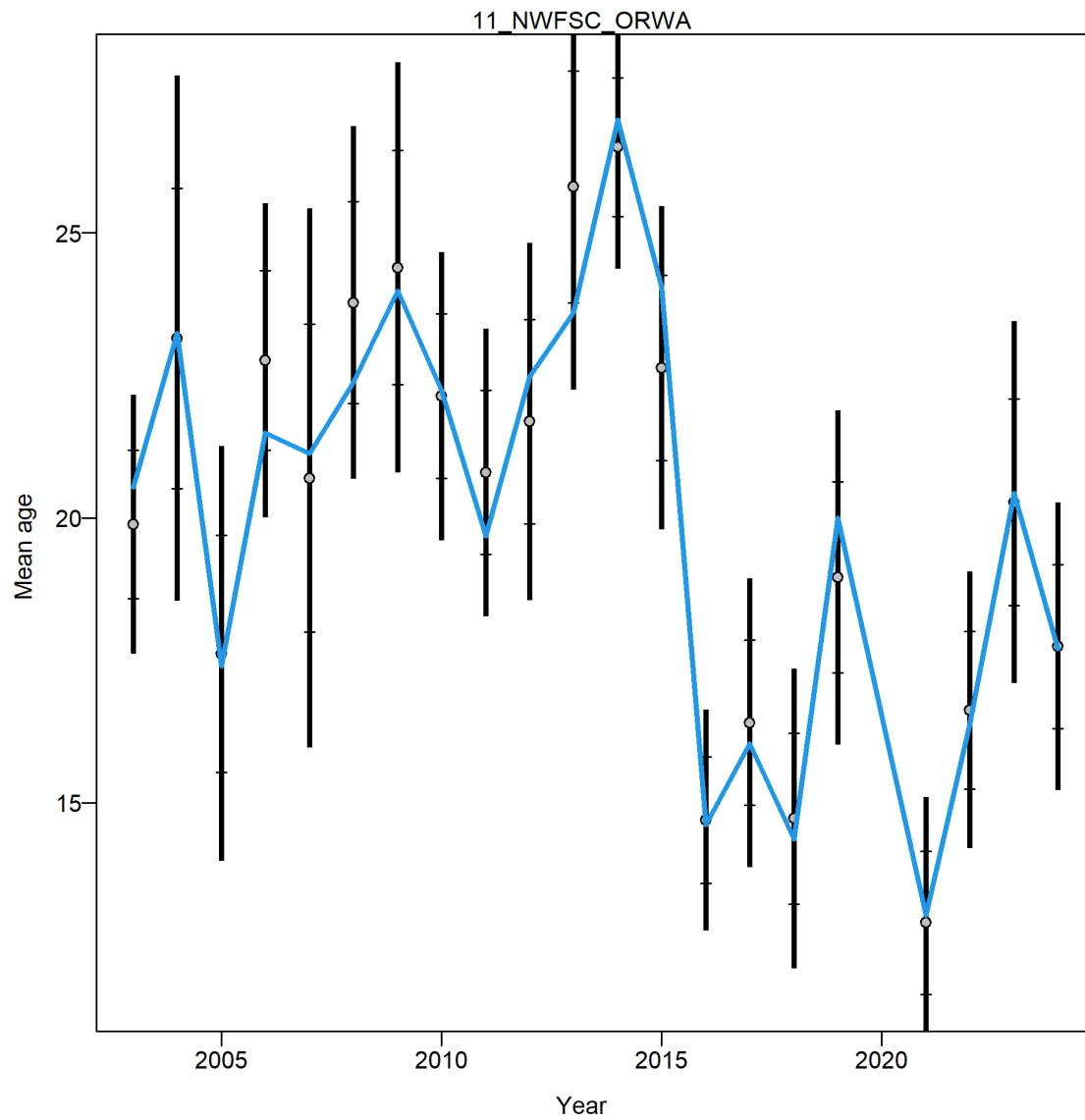


Figure 52: Mean age from conditional data (aggregated across length bins) for the WCGTBS with 95% confidence intervals based on input sample sizes. The blue line is the model expectation.

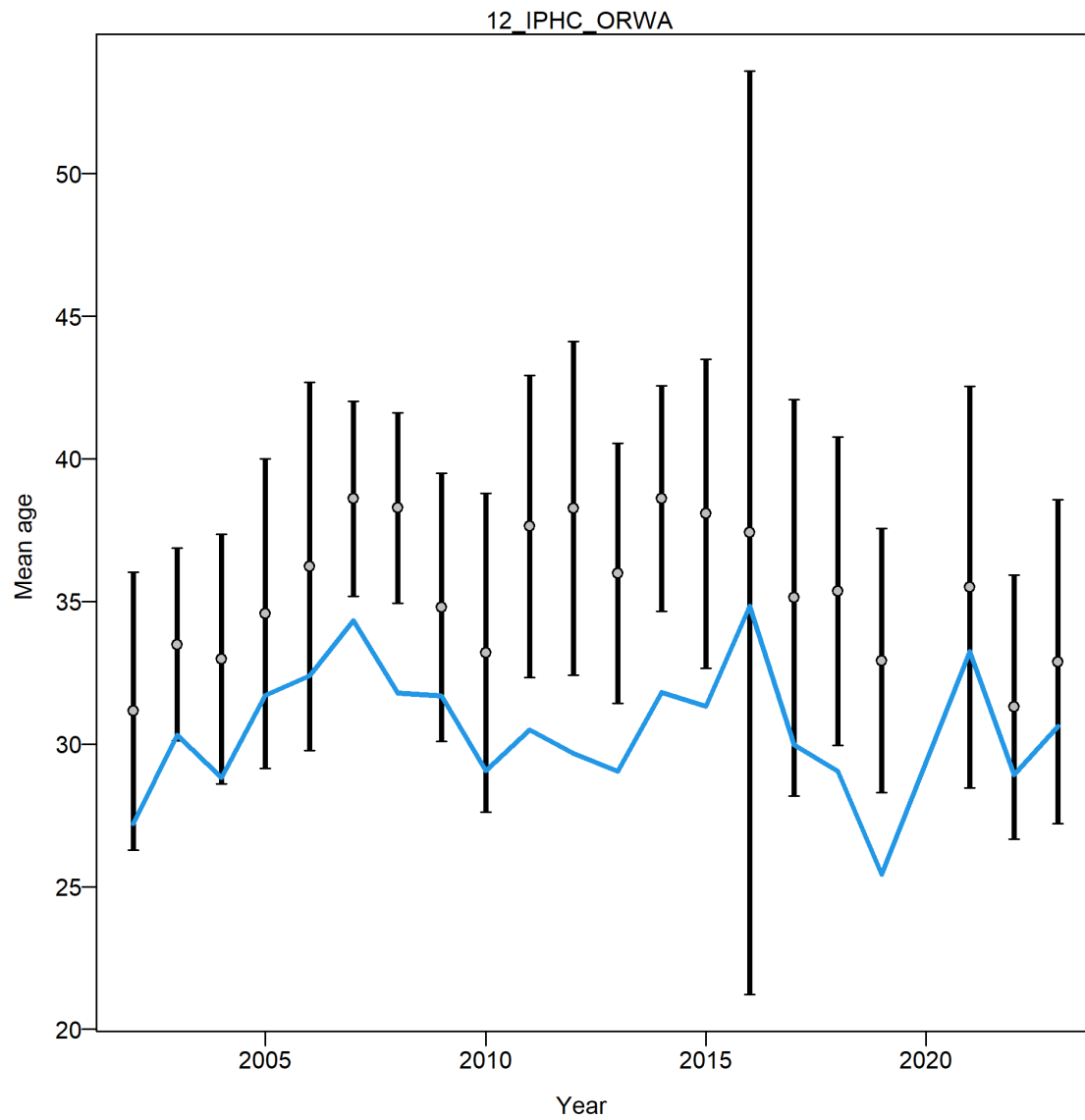


Figure 53: Mean age from conditional data (aggregated across length bins) for the IPHC survey with 95% confidence intervals based on input sample sizes. The blue line is the model expectation.

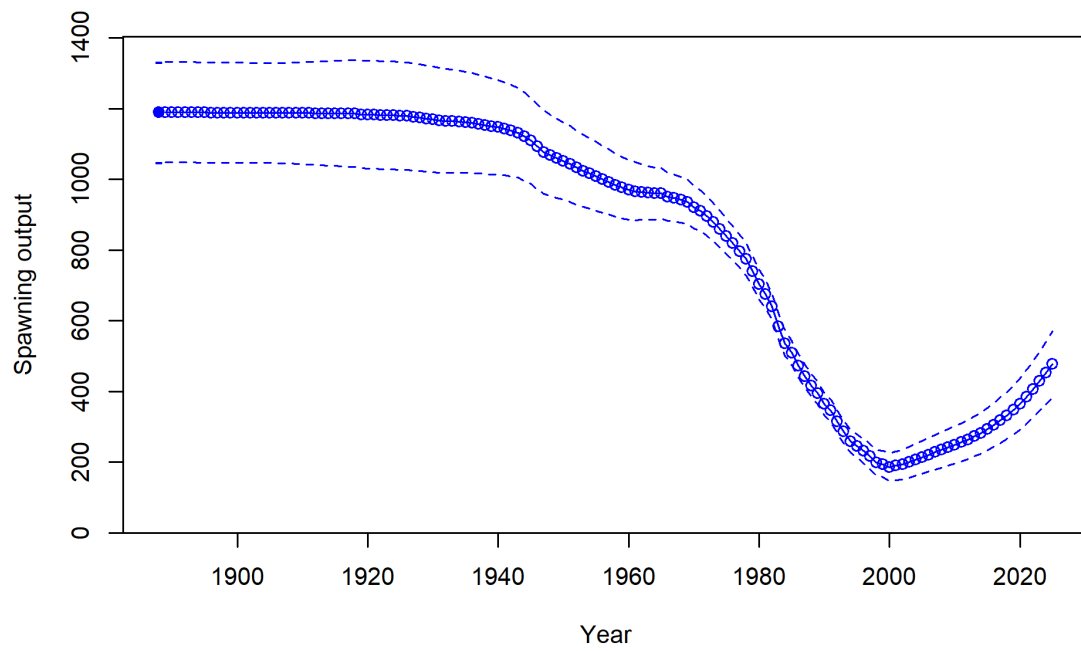


Figure 54: Estimated spawning output (billions of eggs) over time for both areas combined.

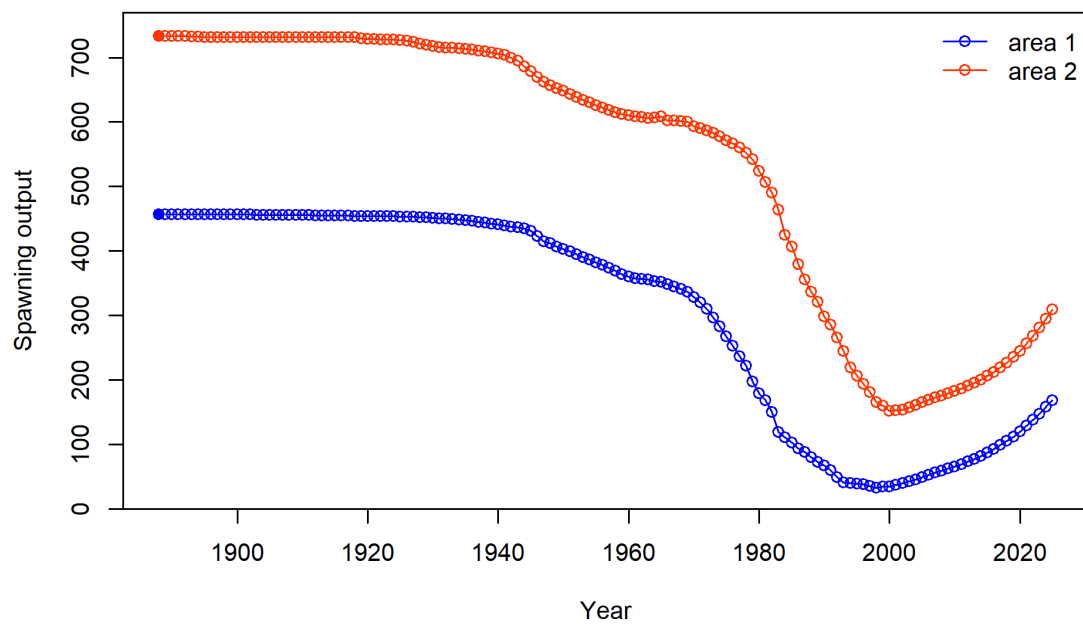


Figure 55: Estimated spawning output (billions of eggs) over time and by area (Area 1 is California, Area 2 is Oregon/Washington combined).

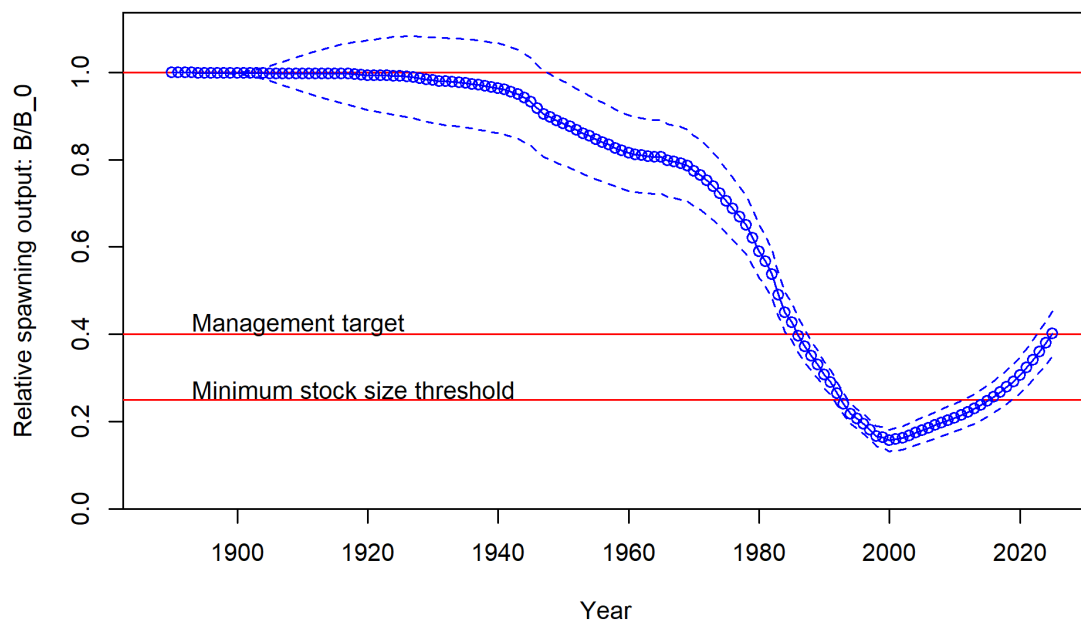


Figure 56: Time series of relative spawning output estimated in the assessment model (solid line) with ~95% interval (dashed lines).

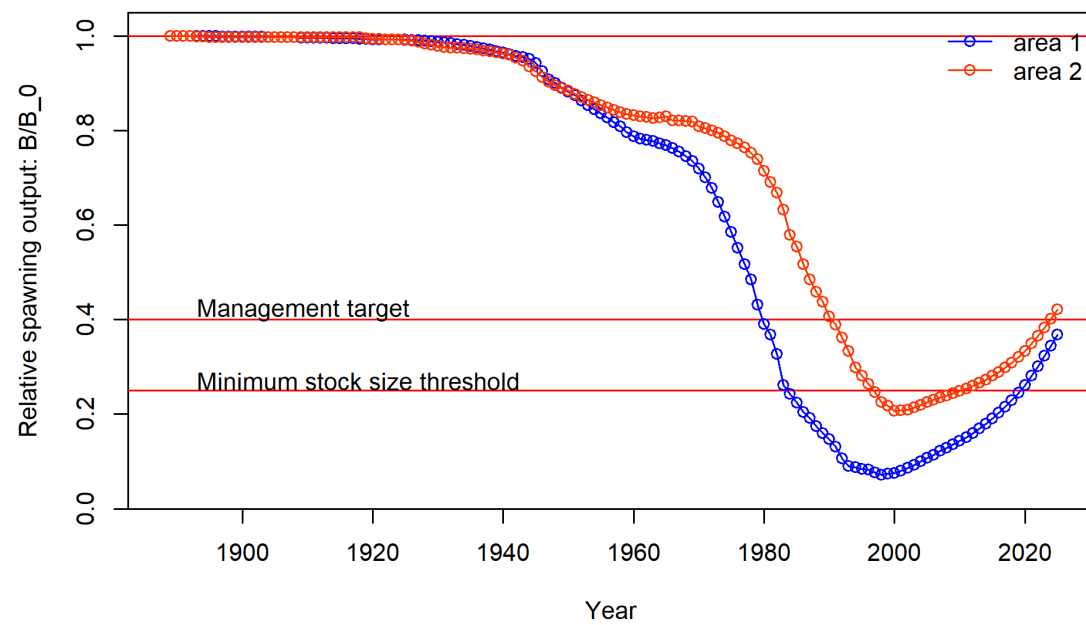


Figure 57: Time series of relative spawning output estimated by area (area 1= California, area 2 = Oregon and Washington).

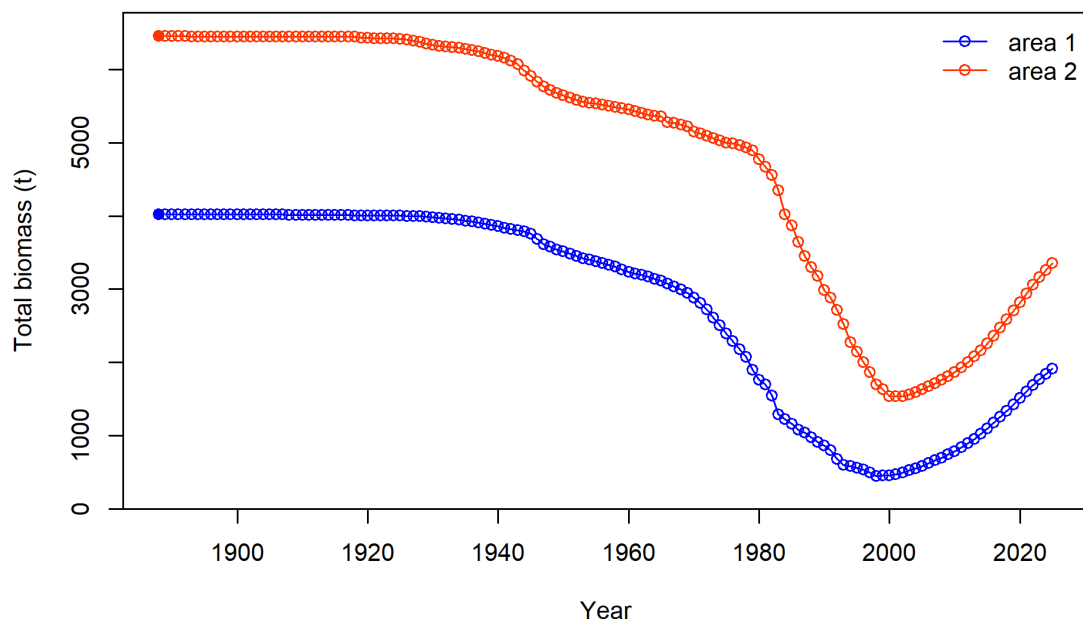


Figure 58: Total biomass (t) over time and by area (Area 1 is California, Area 2 is Oregon/Washington combined).



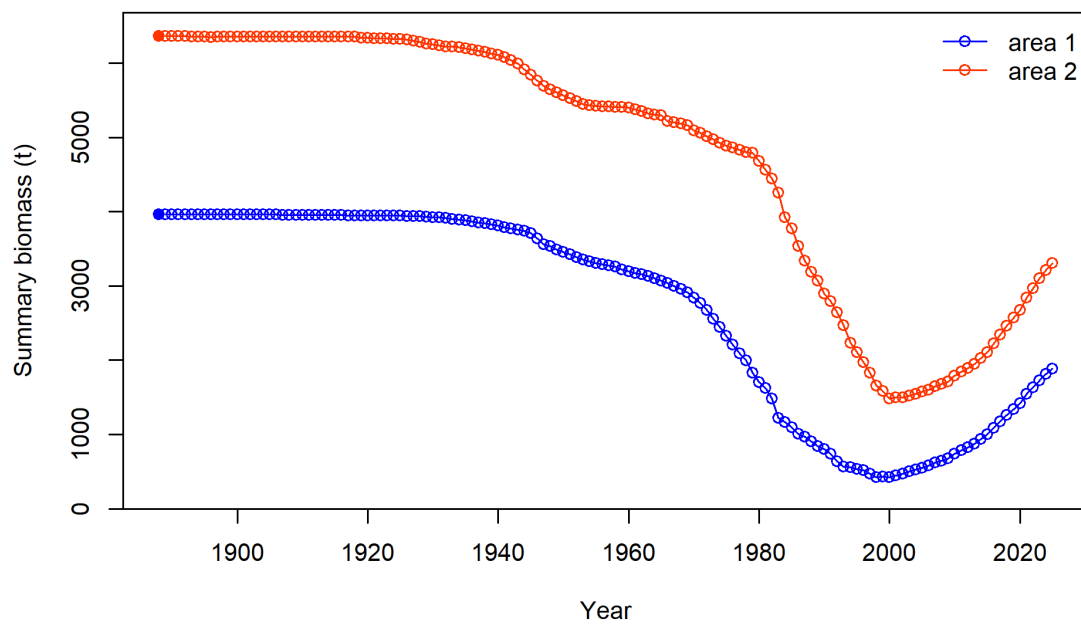


Figure 59: Summary biomass (t) of age 8+ fish over time and by area (Area 1 is California, Area 2 is Oregon/Washington combined).

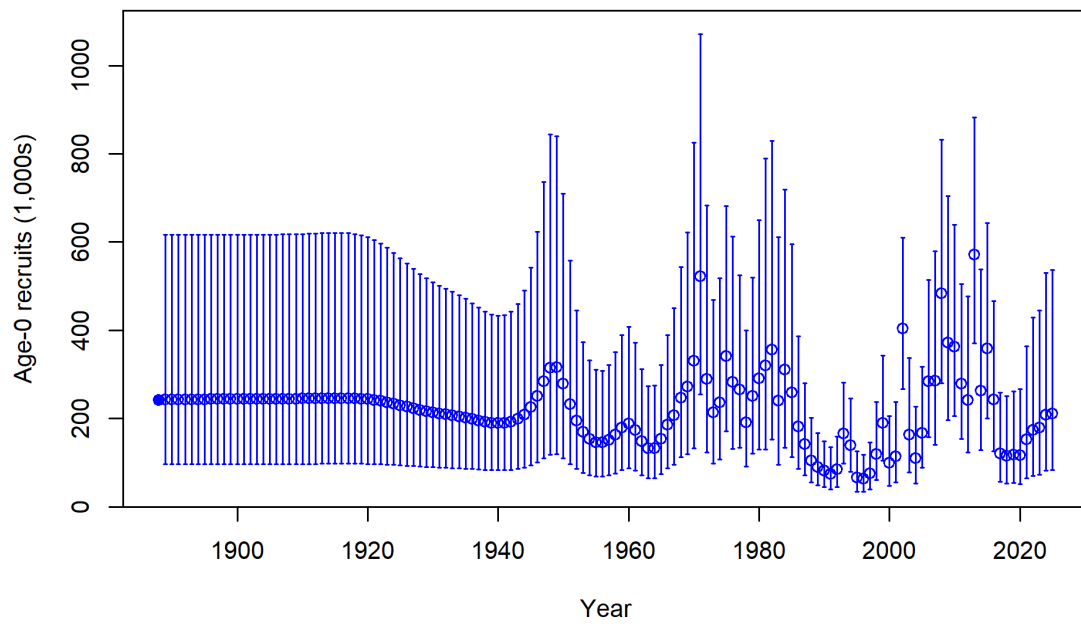


Figure 60: Time series of recruitment estimated in the assessment model with  $\sim 95\%$  interval.

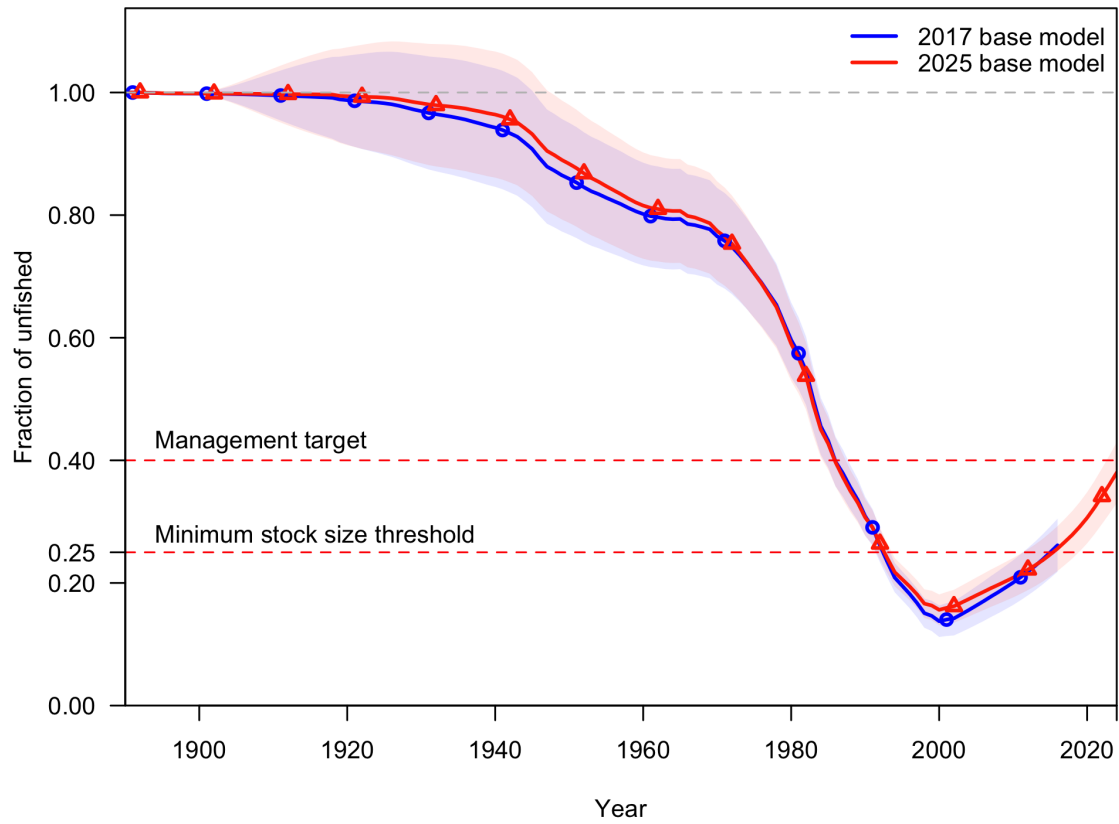


Figure 61: Time series of the fraction of unfished biomass with ~ 95% interval comparing the 2017 assessment trajectory with the 2025 base model.

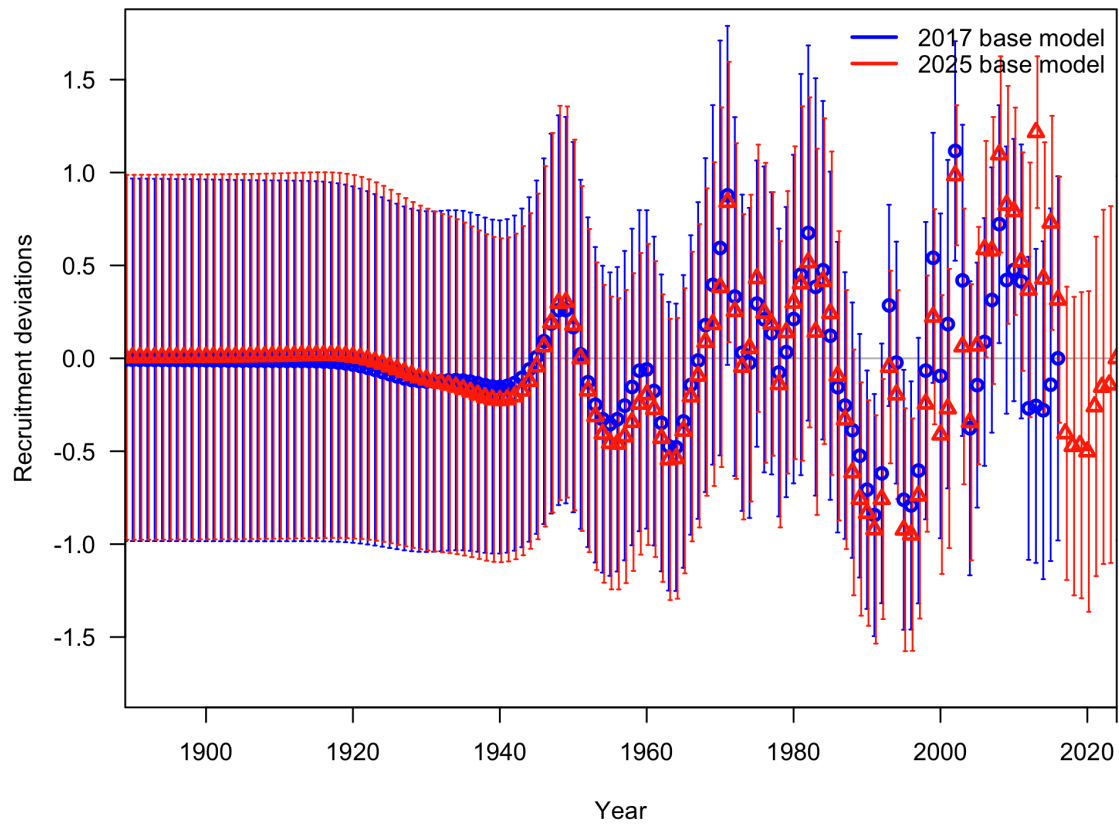


Figure 62: Time series of recruitment deviations estimated in the assessment model with  $\sim 95\%$  intervals comparing the 2017 recruitment deviations in blue, to the 2025 recruitment deviations in red.

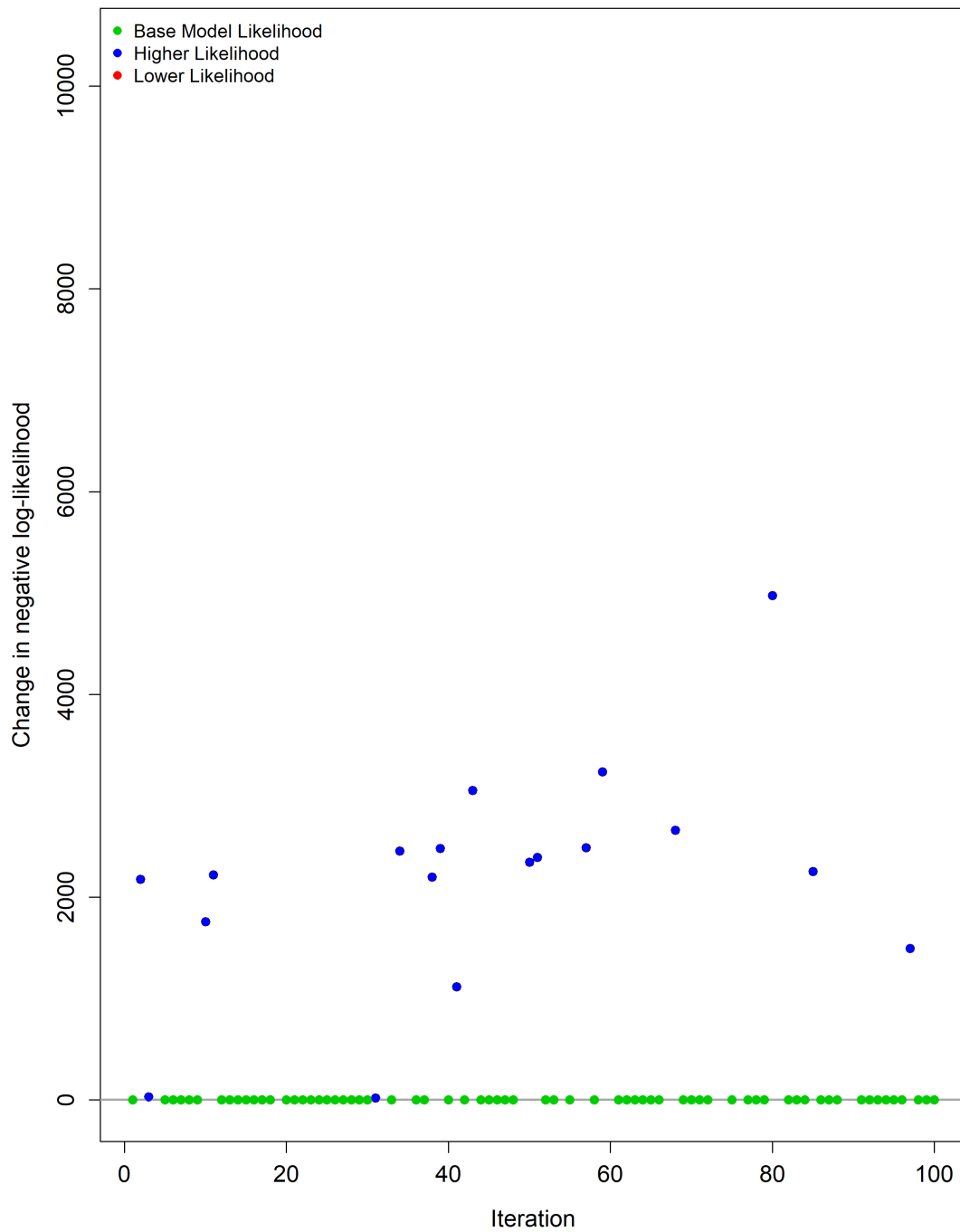


Figure 63: Results from 100 base model runs when starting parameters values are jittered by 0.1 units. Horizontal line indicates base model value.

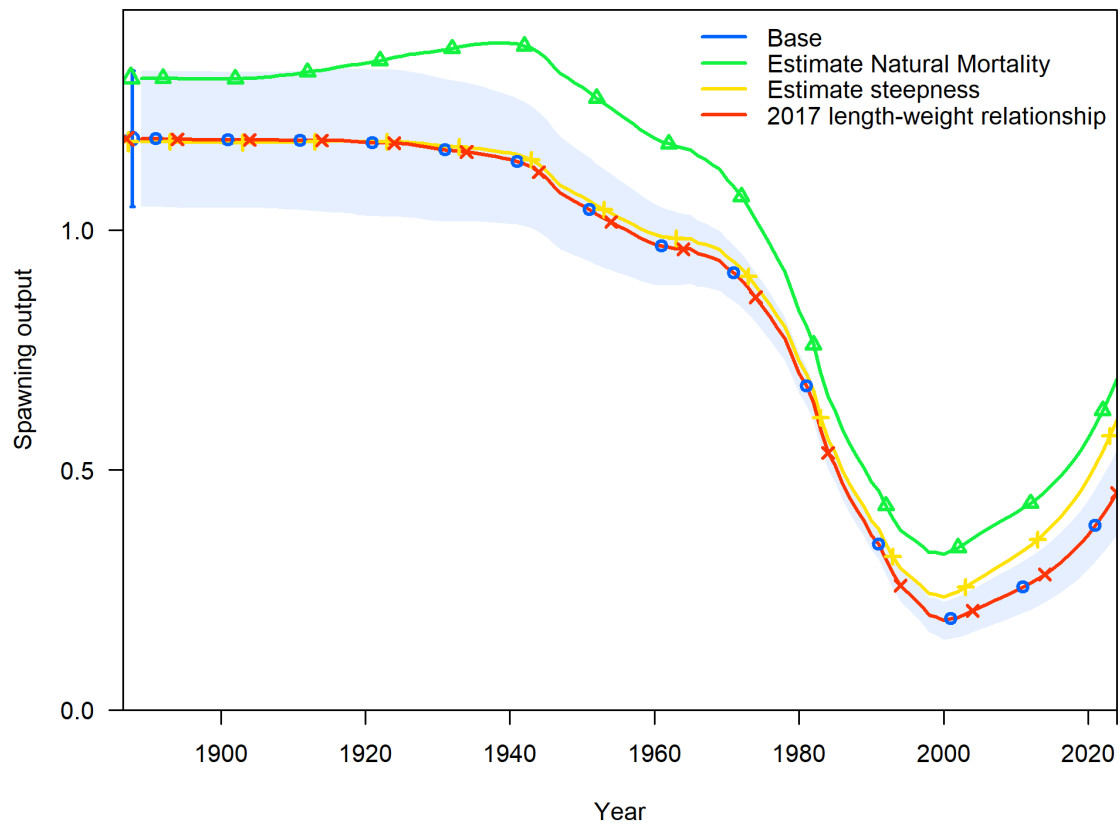


Figure 64: Spawning output (billions of eggs) across model structure sensitivities.

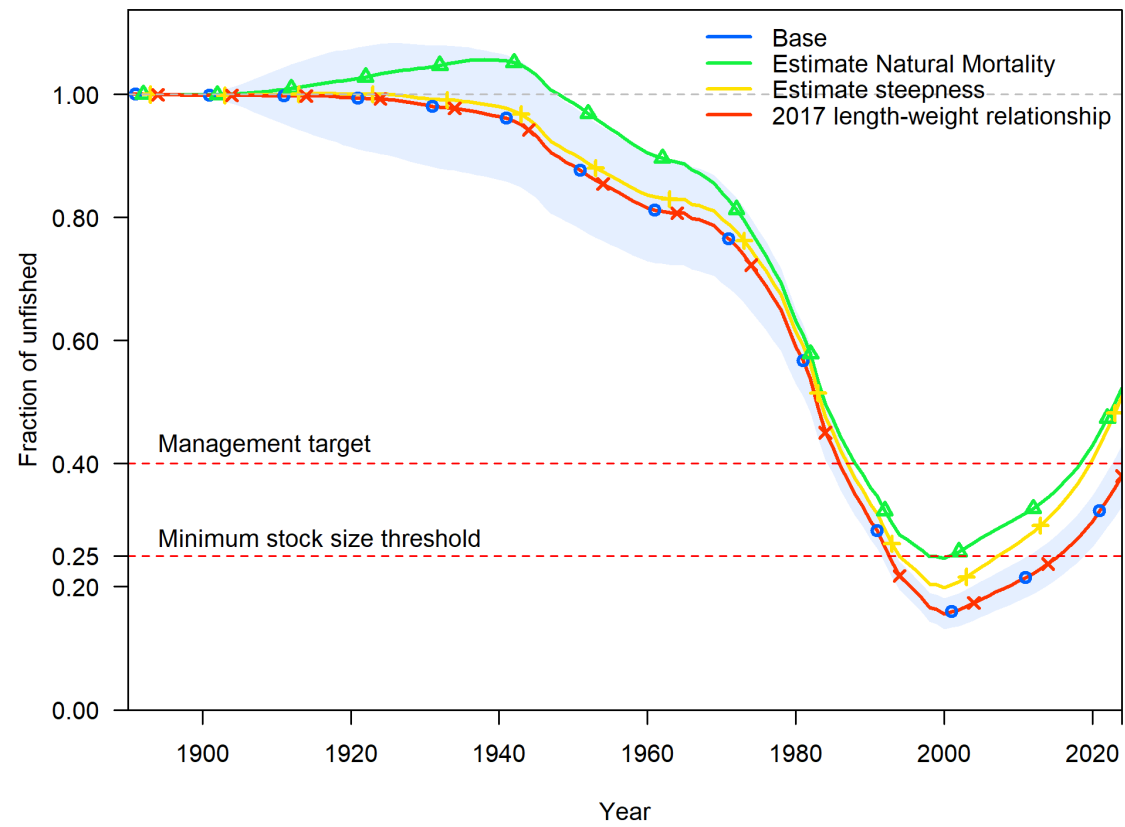


Figure 65: Relative spawning output across model structure sensitivities.

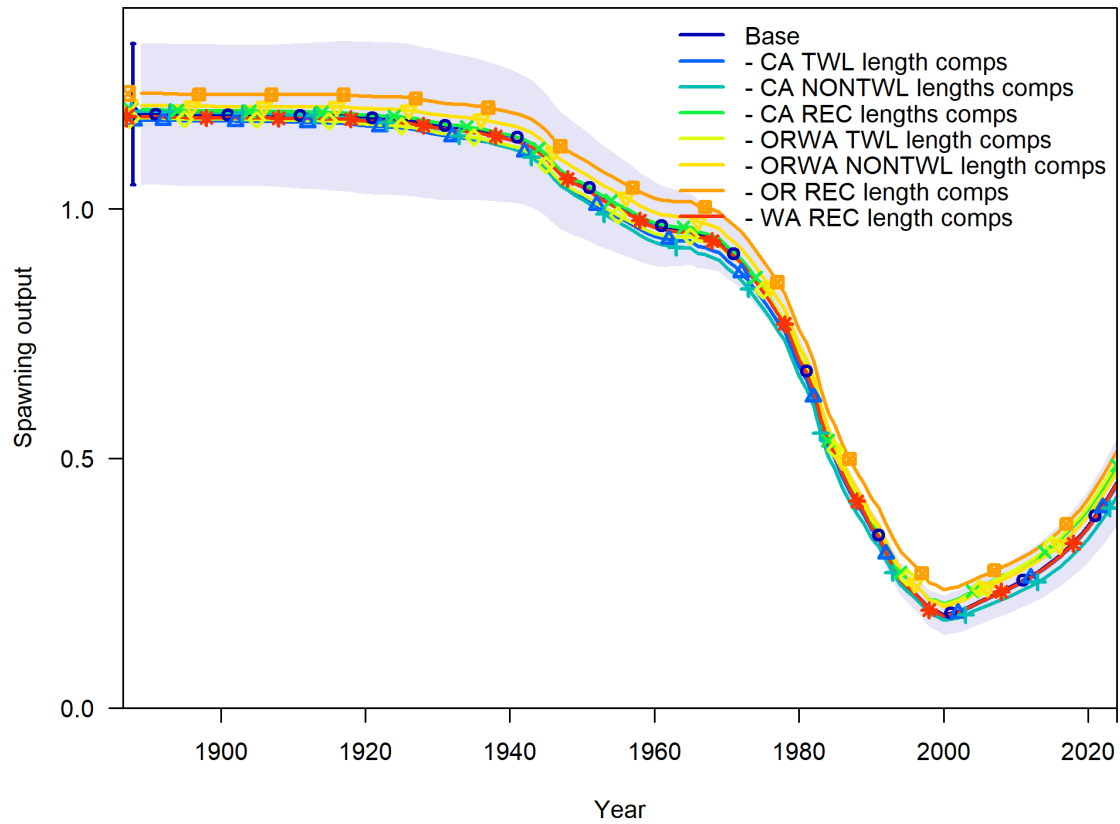


Figure 66: Spawning output (billions of eggs) across length composition inclusion sensitivities (1 of 2).



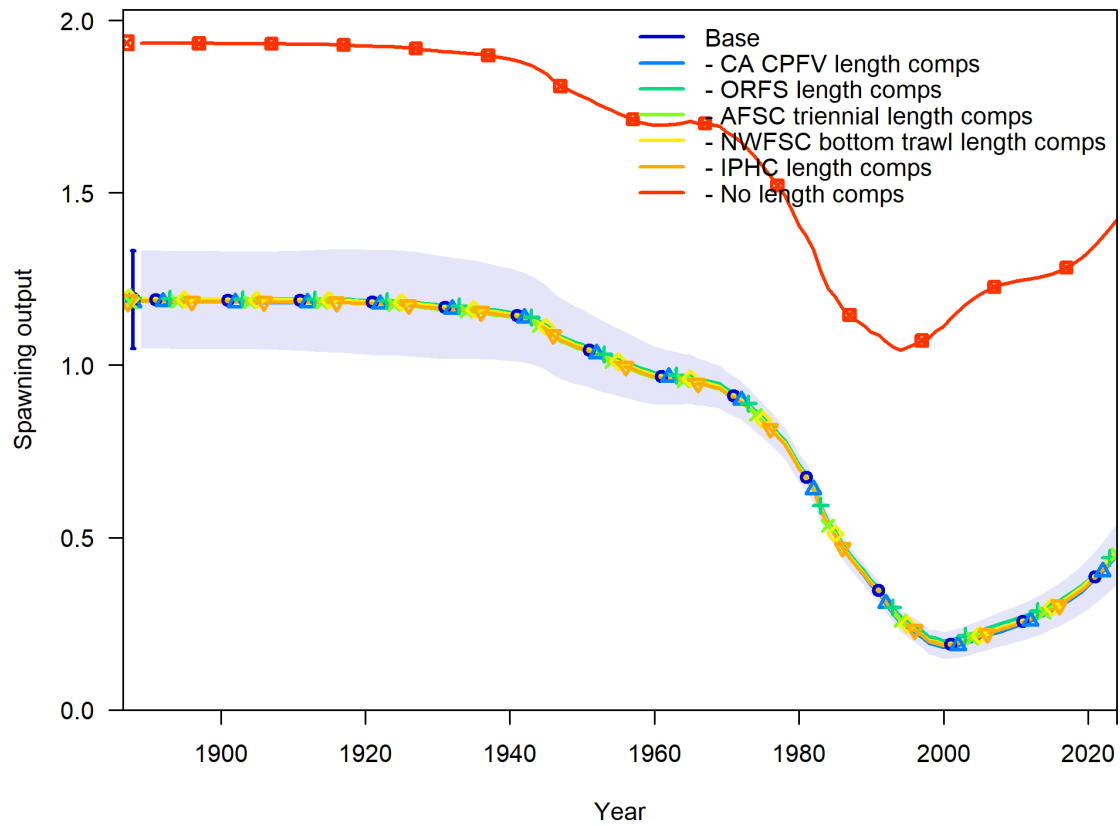


Figure 67: Spawning output (billions of eggs) across length composition inclusion sensitivities (2 of 2).

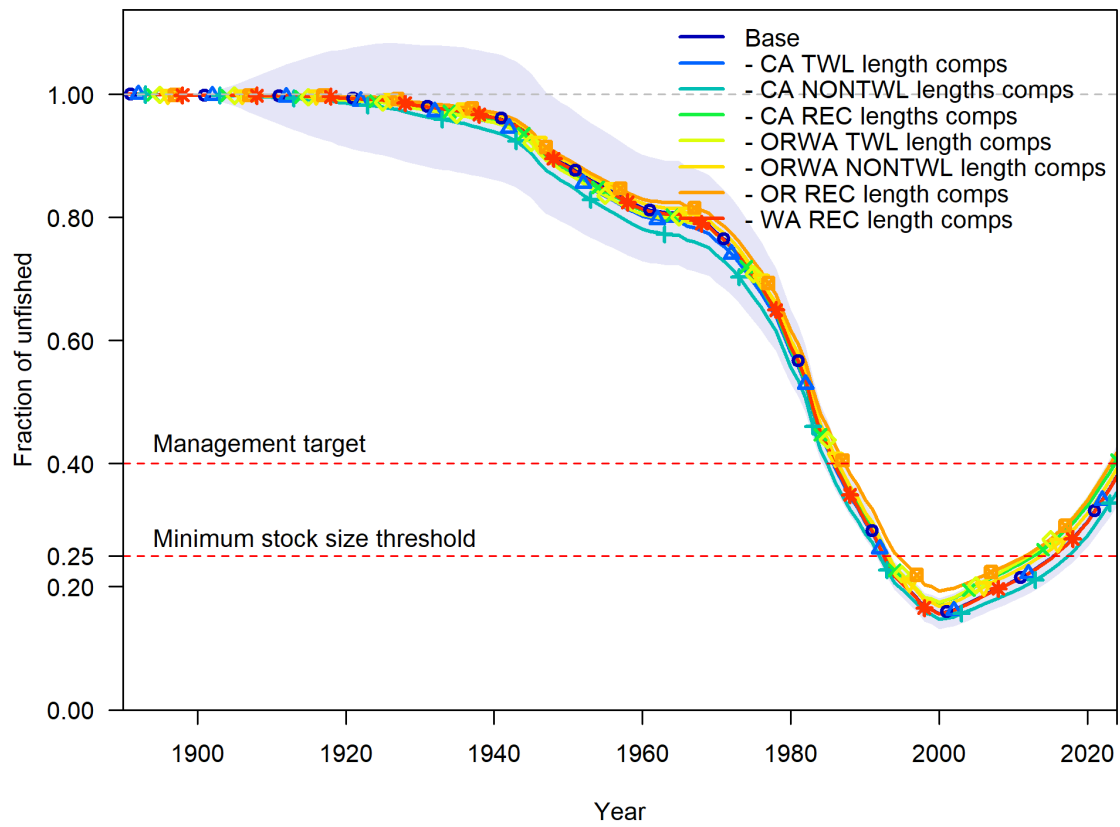


Figure 68: Relative spawning output across length composition inclusion sensitivities (1 of 2).

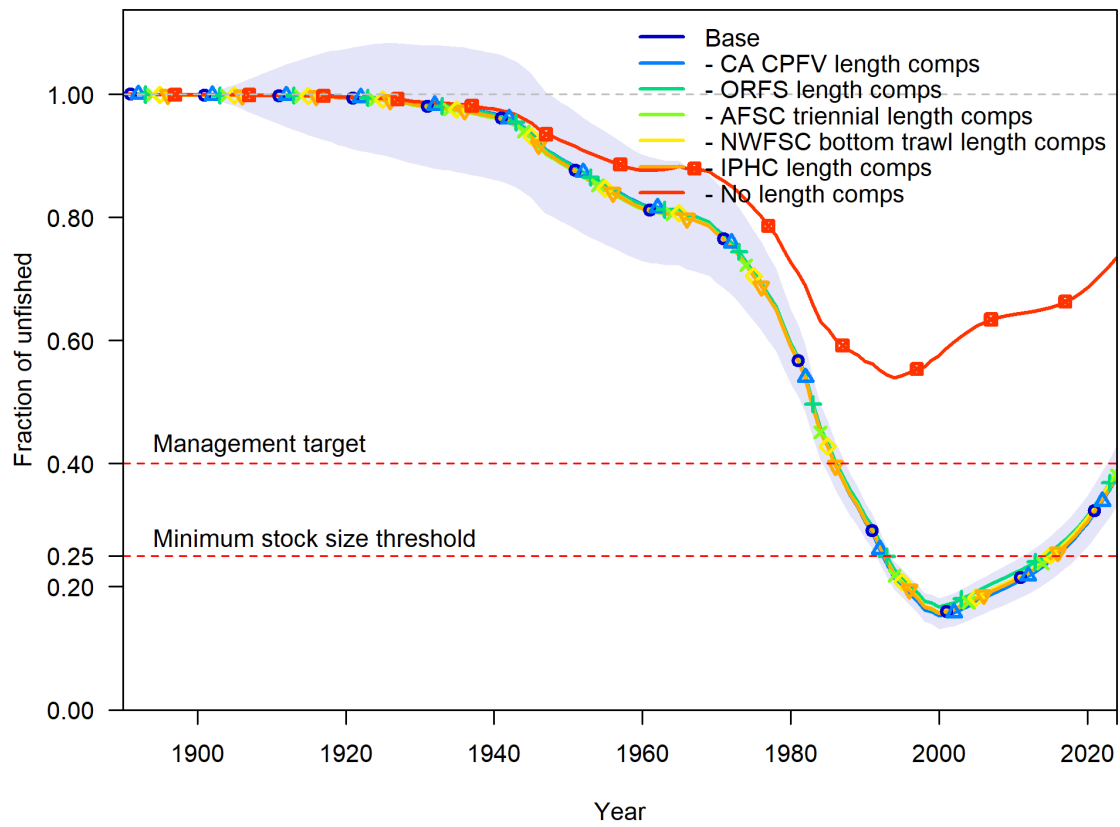


Figure 69: Relative spawning output across length composition inclusion sensitivities (2 of 2).

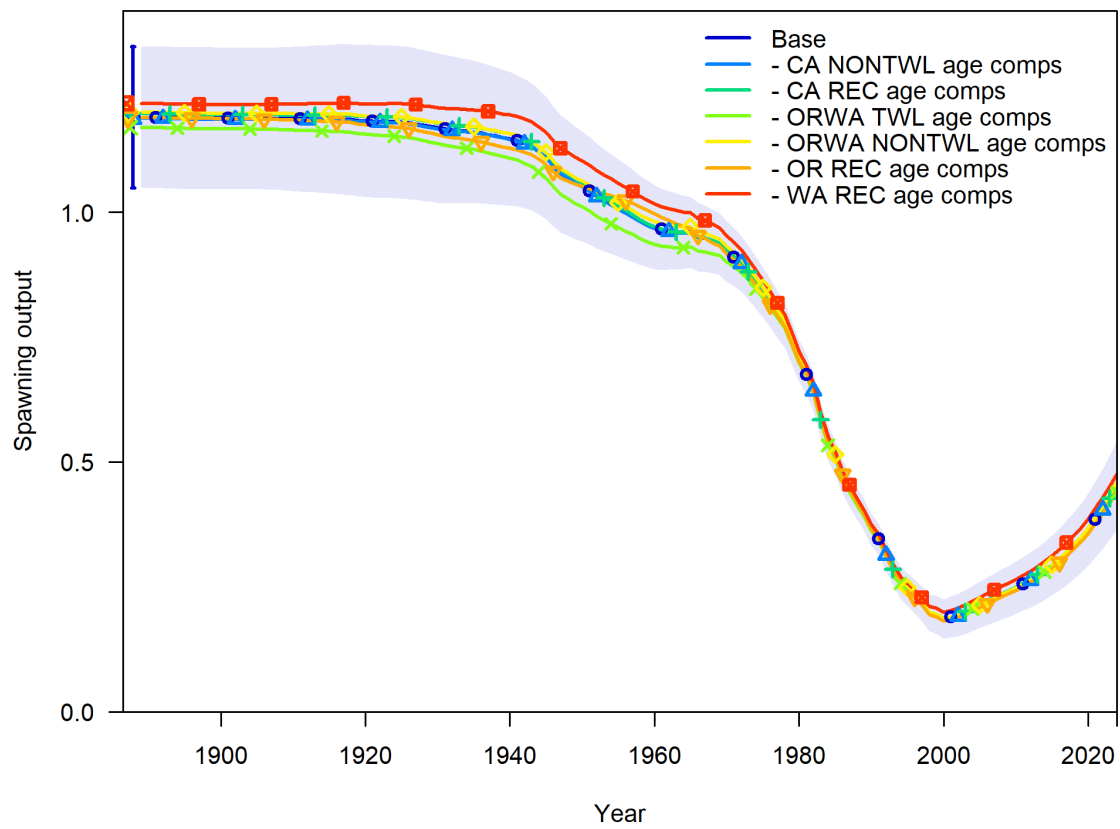


Figure 70: Spawning output (billions of eggs) across age composition inclusion sensitivities (1 of 2).

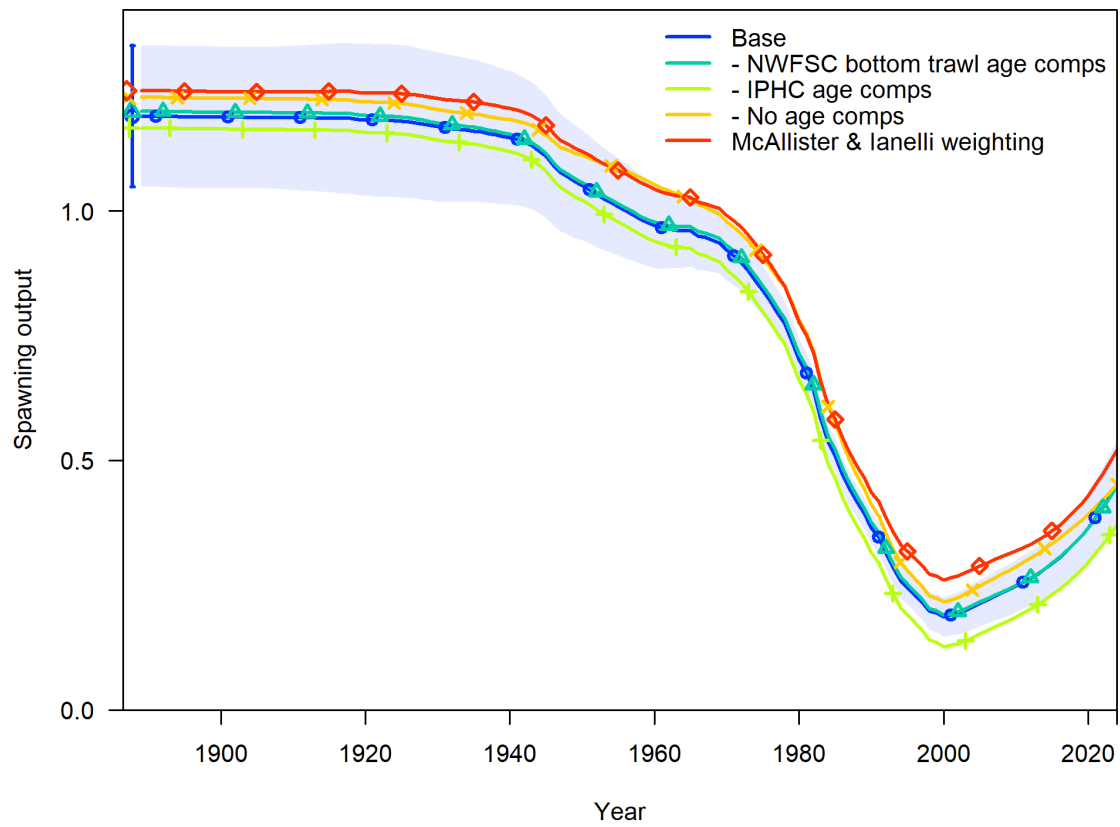


Figure 71: Spawning output (billions of eggs) across age composition inclusion sensitivities (2 of 2).

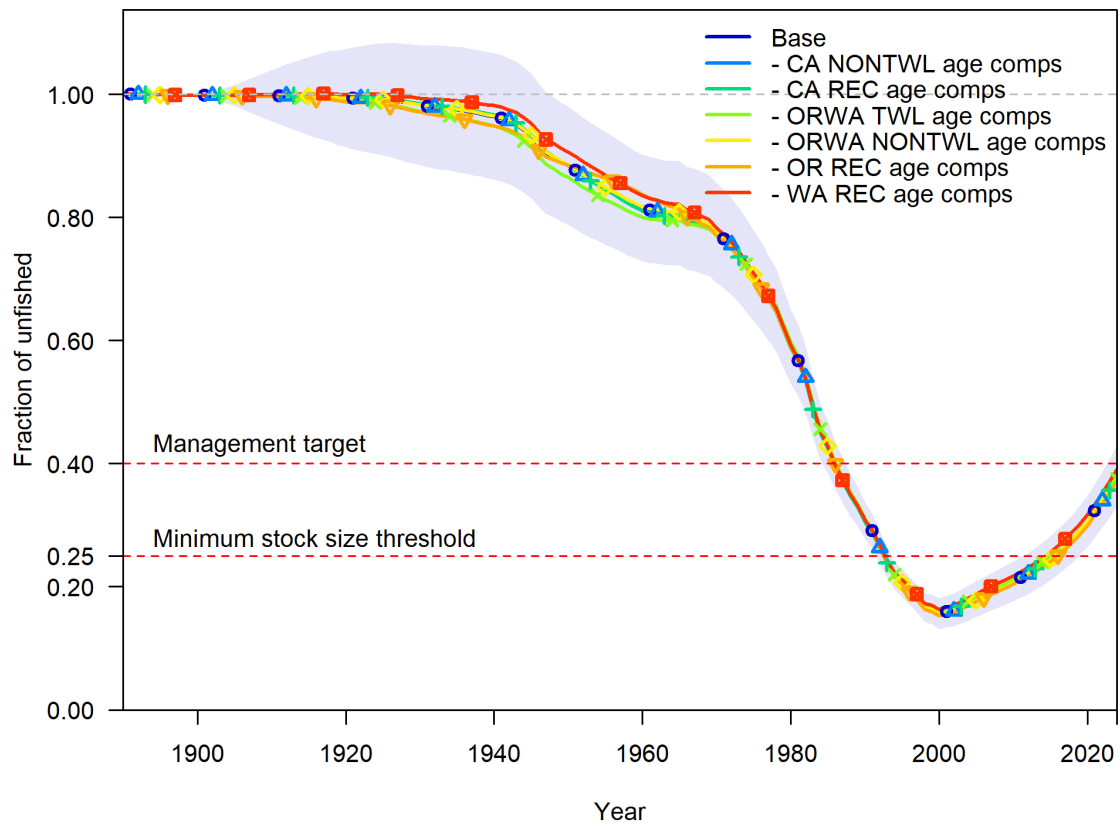


Figure 72: Relative spawning output across age composition inclusion sensitivities (1 of 2).

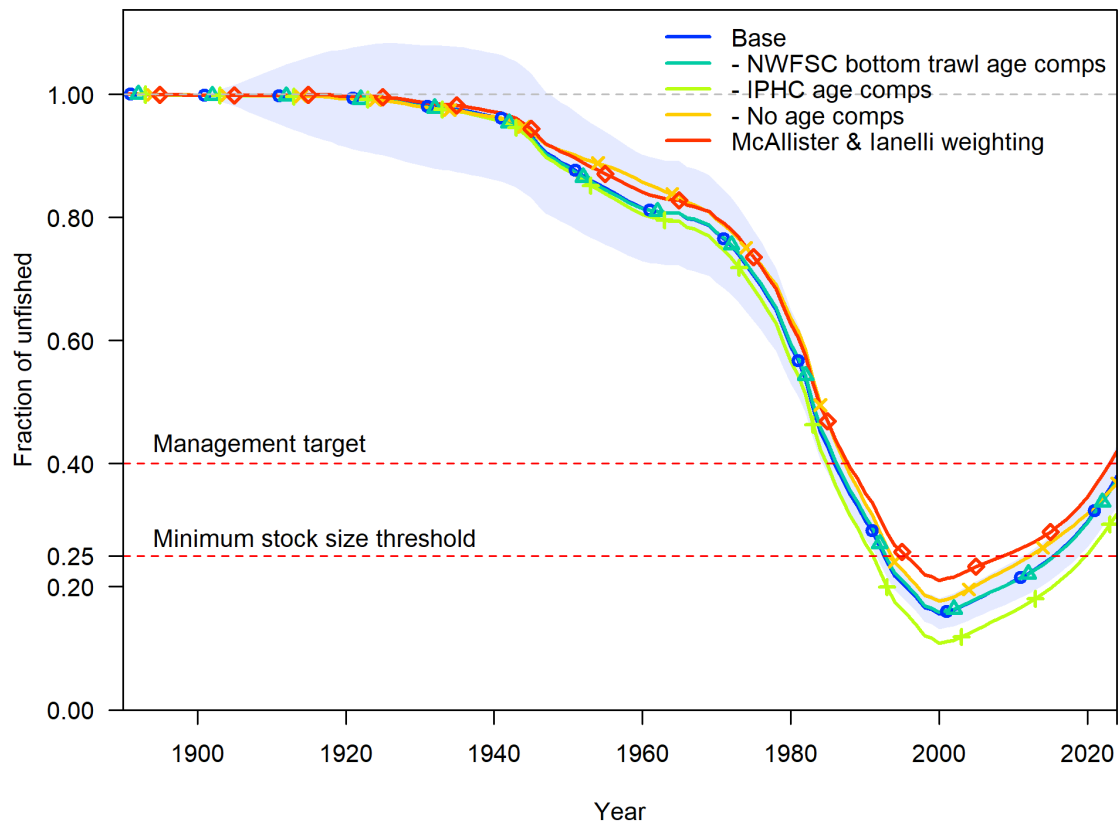


Figure 73: Relative spawning output across age composition inclusion sensitivities (2 of 2).

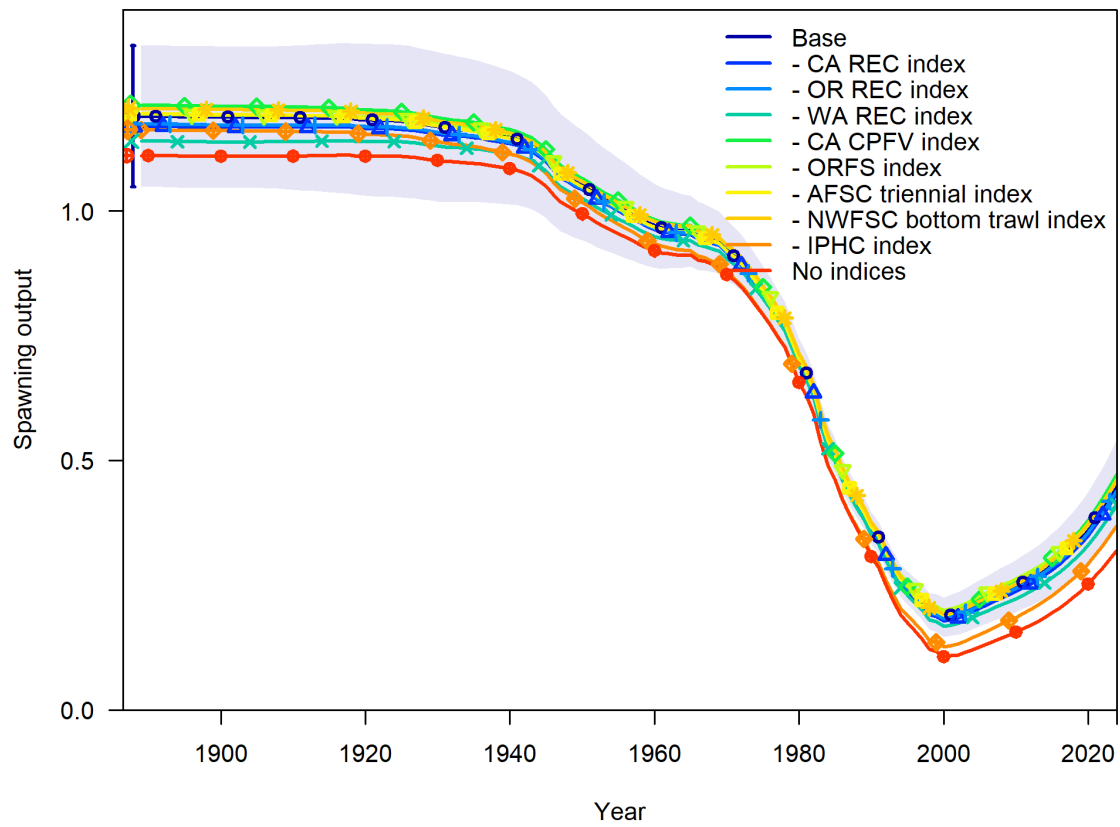


Figure 74: Spawning output (billions of eggs) across index inclusion sensitivities.



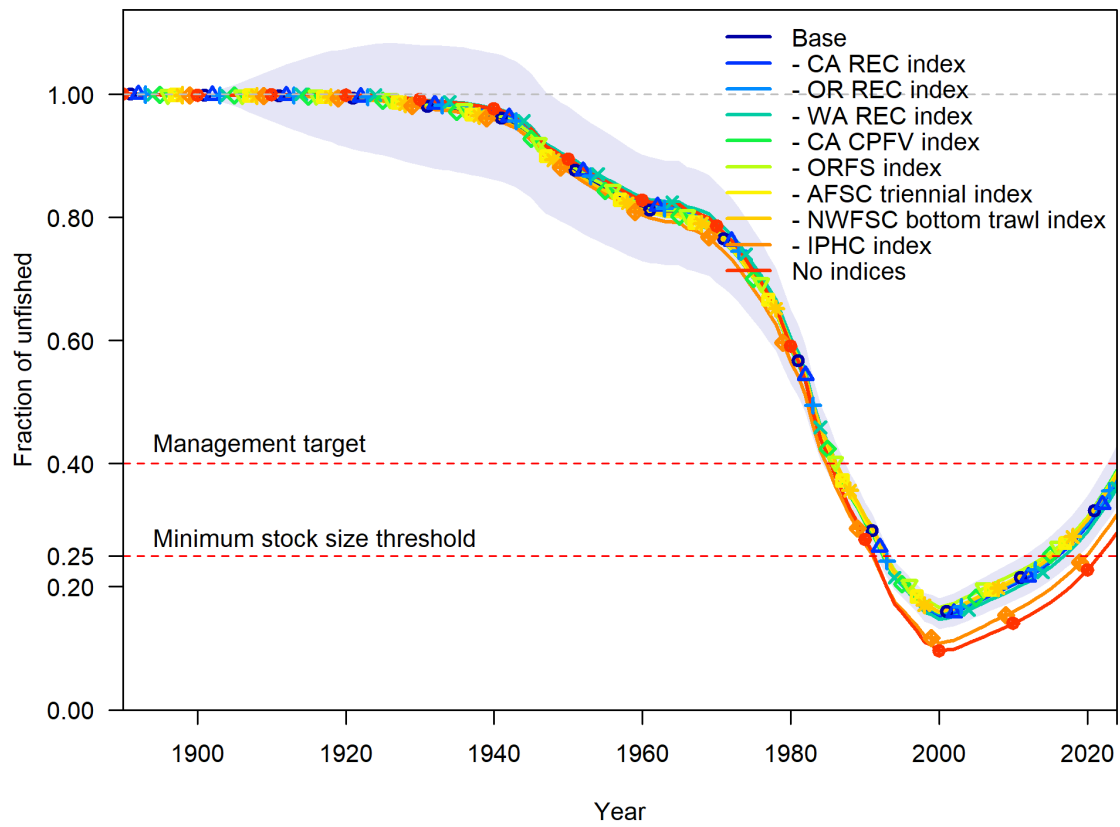


Figure 75: Relative spawning output across index inclusion sensitivities.

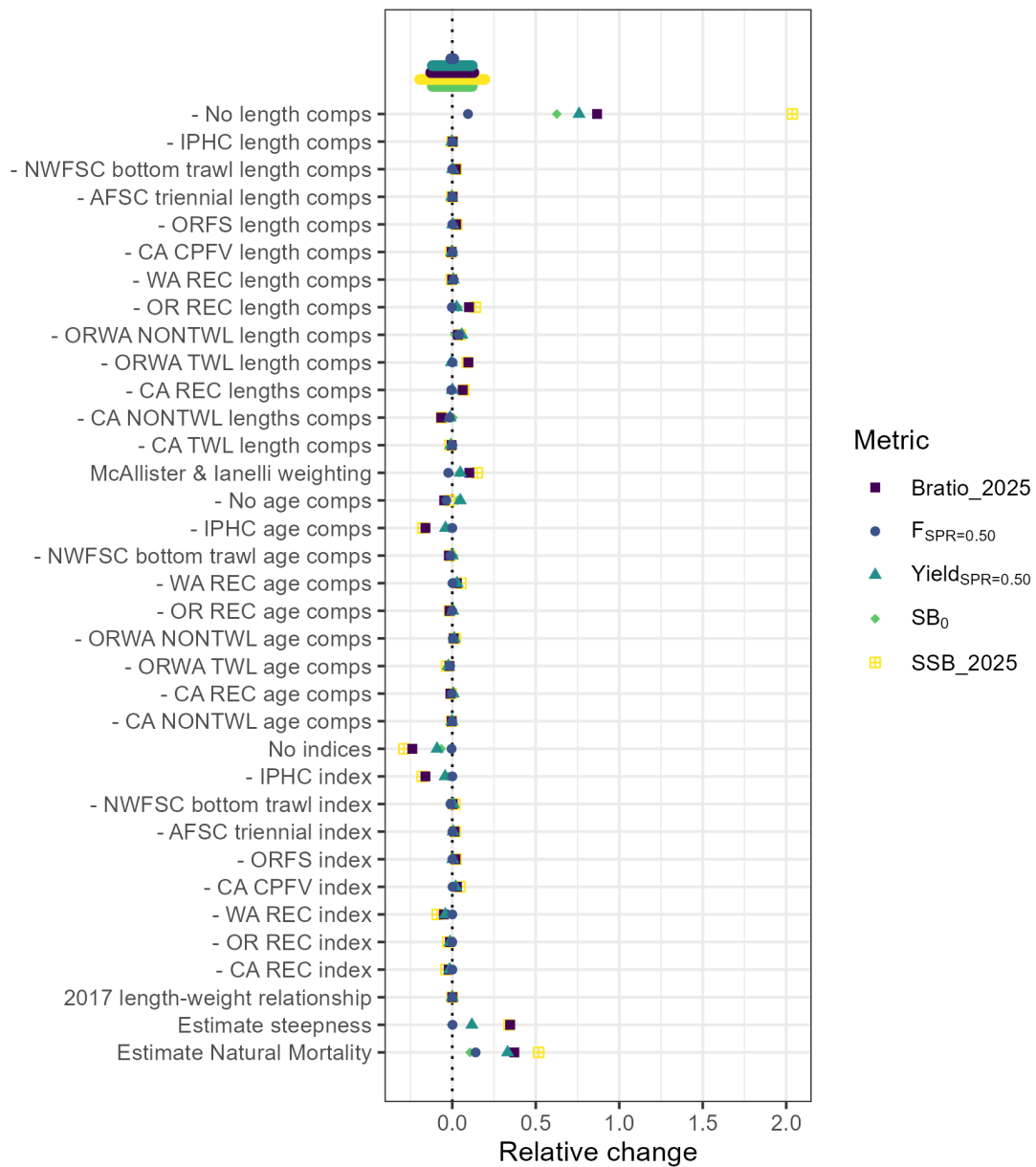


Figure 76: Relative change in management quantities across models conducted as sensitivities.

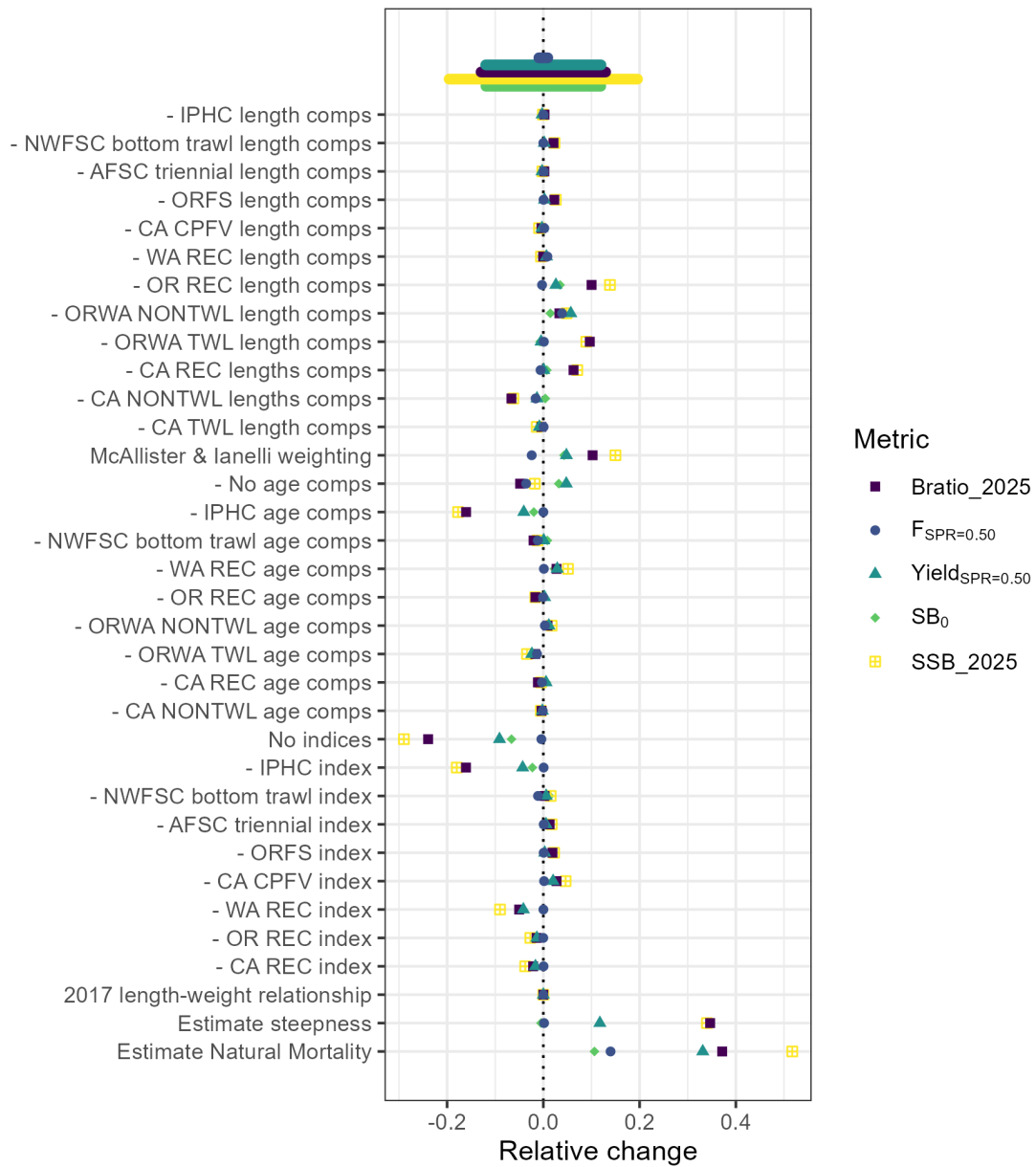


Figure 77: Relative change in management quantities across models conducted as sensitivities, without the removal of all length composition data.

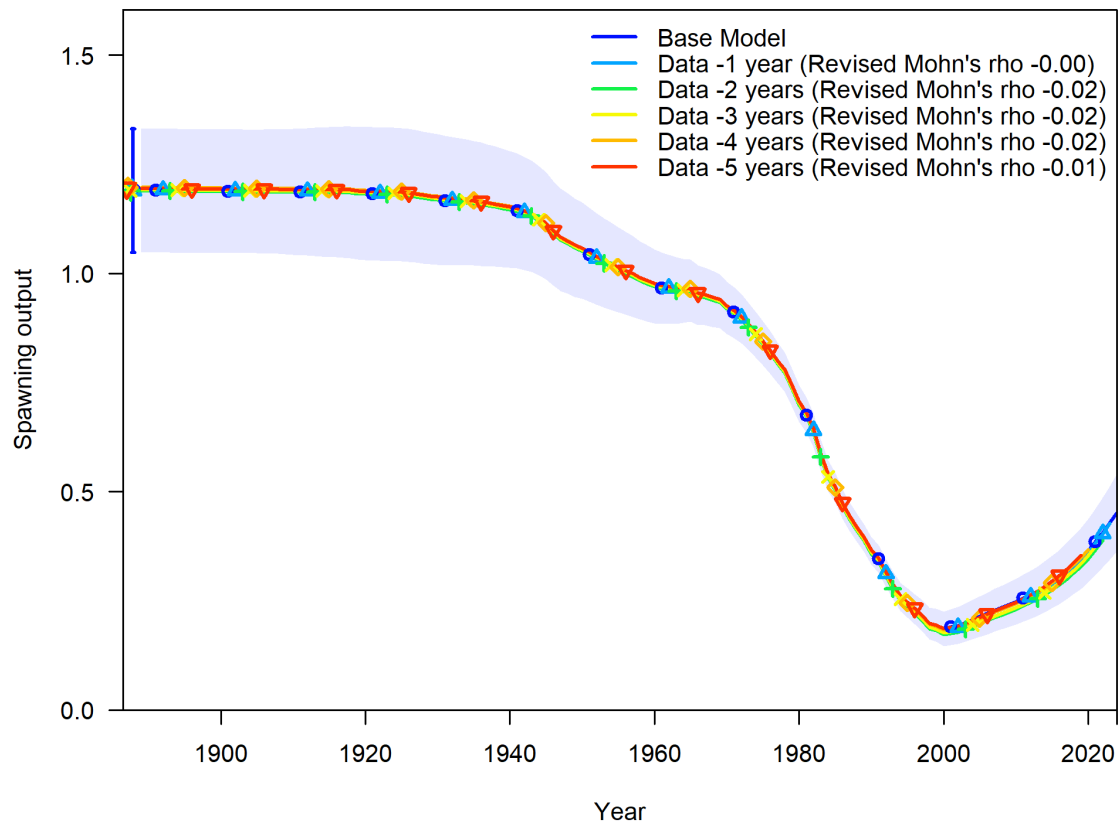


Figure 78: Results of retrospective analysis. Spawning output (billions of eggs) time series of this assesment base model are proved with ~95% interval.

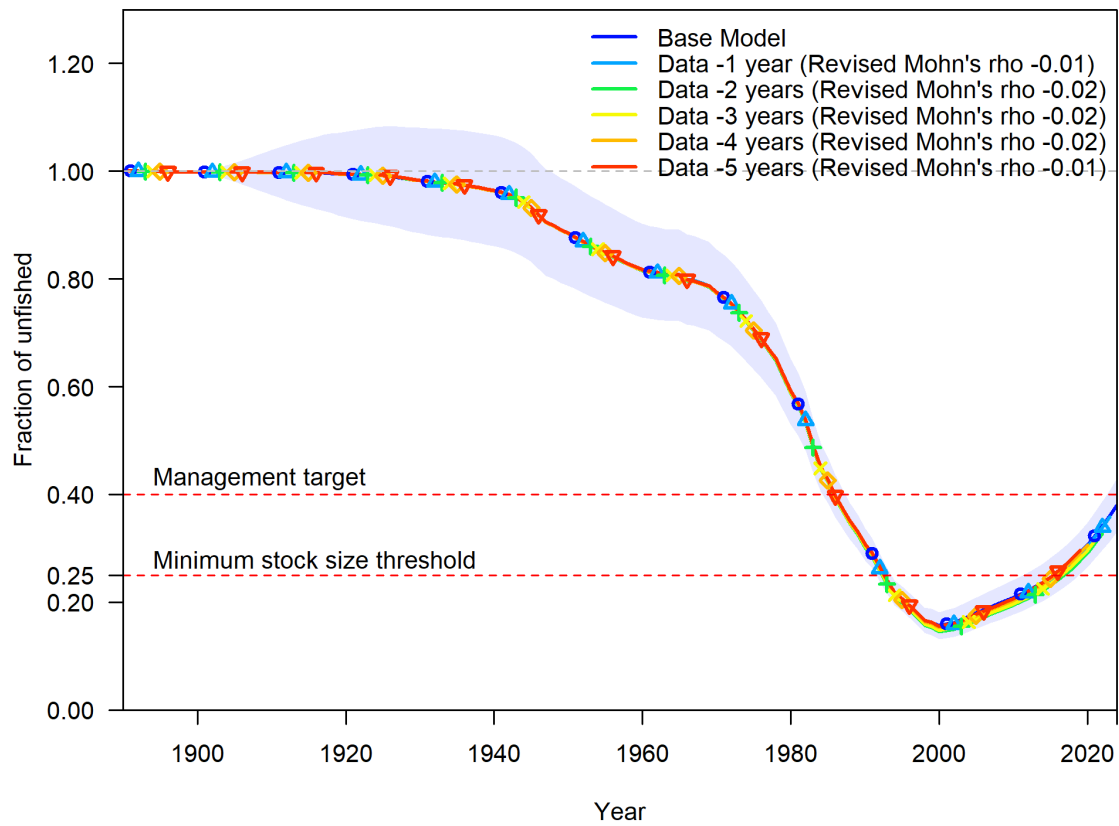


Figure 79: Results of retrospective analysis. Relative spawning output time series of this assessment base model are proved with ~95% interval.

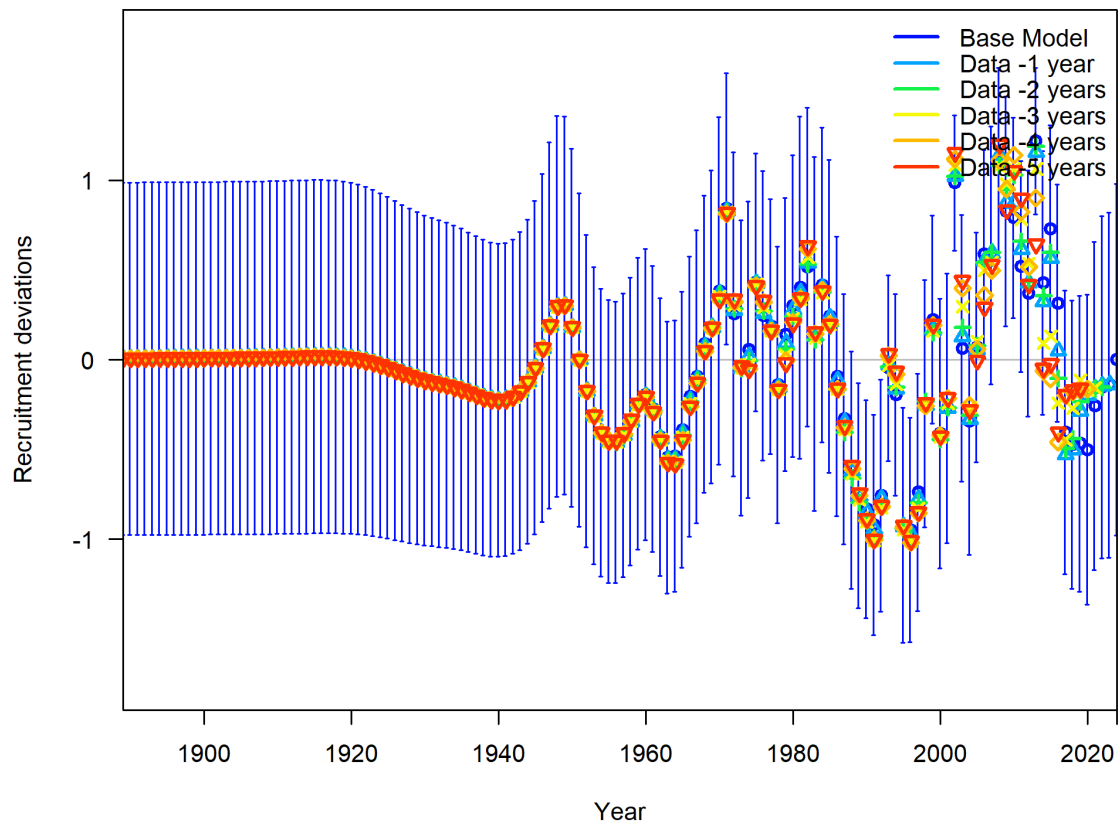


Figure 80: Recruitment deviation time series for each scenario of the retrospective analysis.

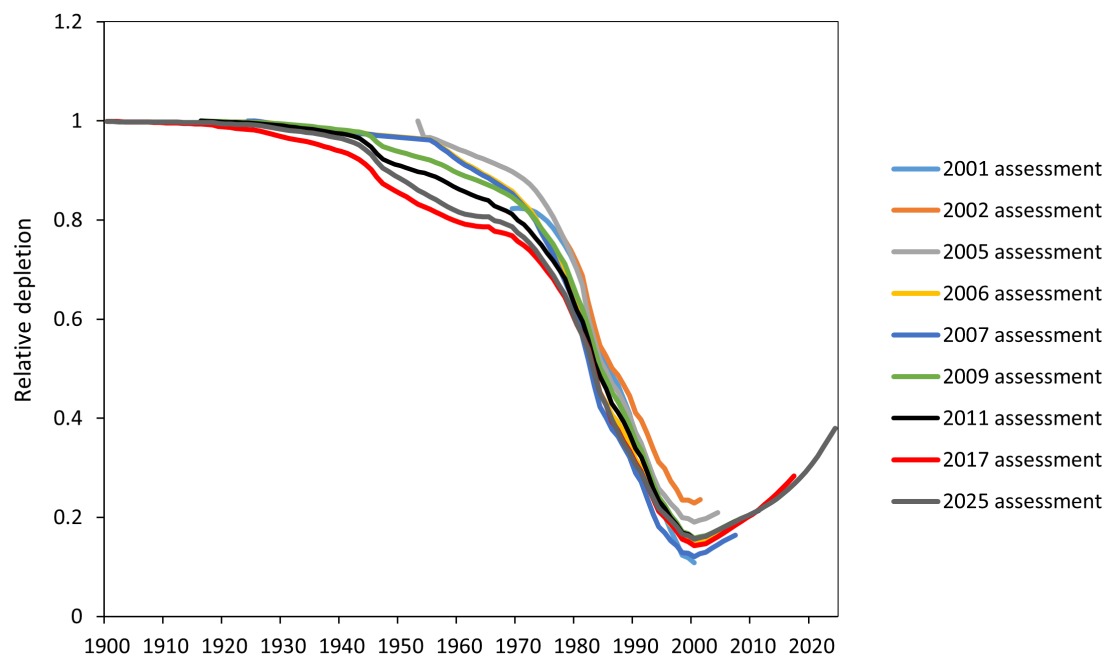


Figure 81: Relative depletion (spawning output) across Yelloweye Rockfish assessments over time.

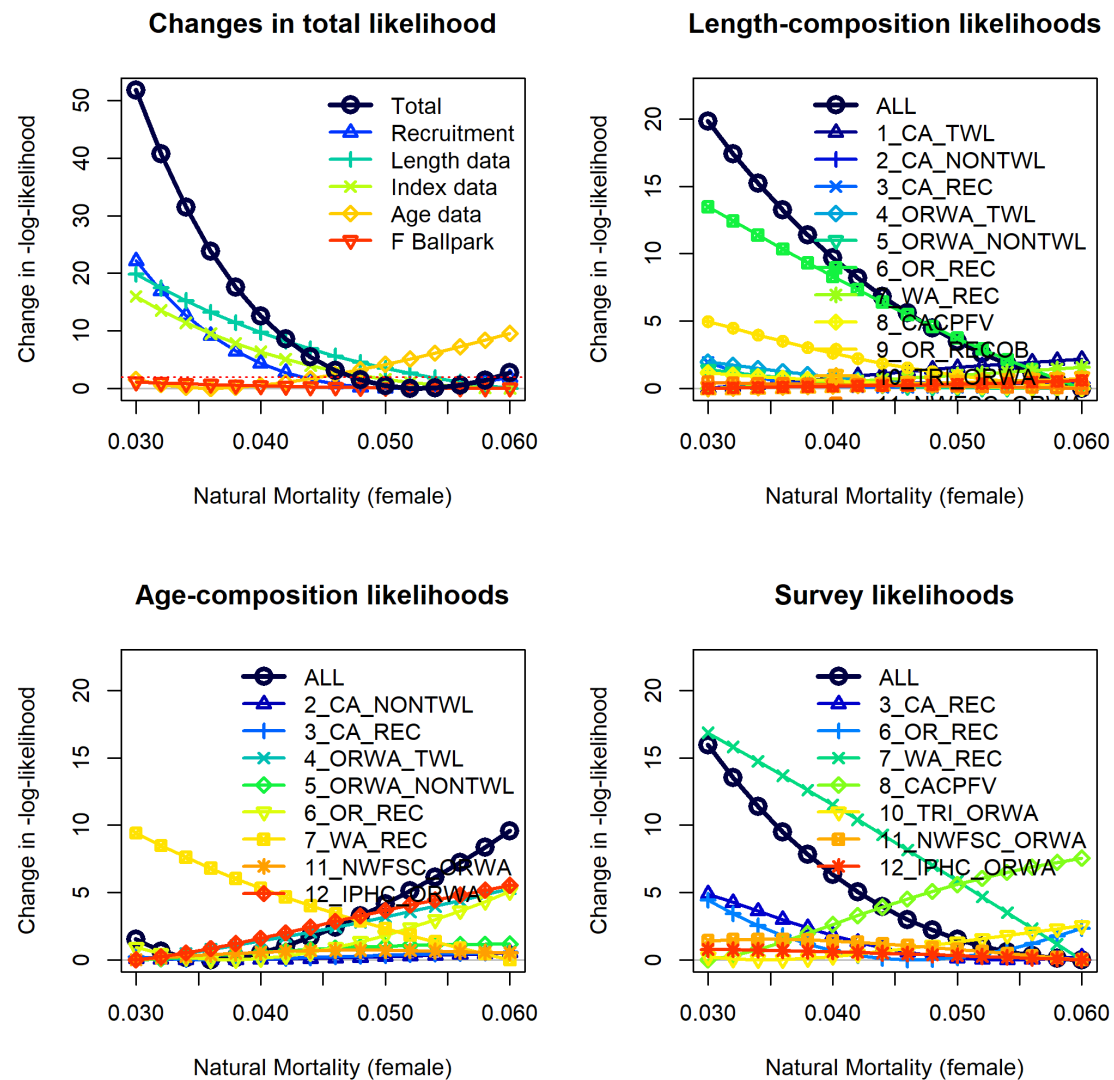


Figure 82: Negative log-likelihood profile for each data component and in total given different values of natural mortality ranging from 0.03 to 0.06 in increments of 0.002.



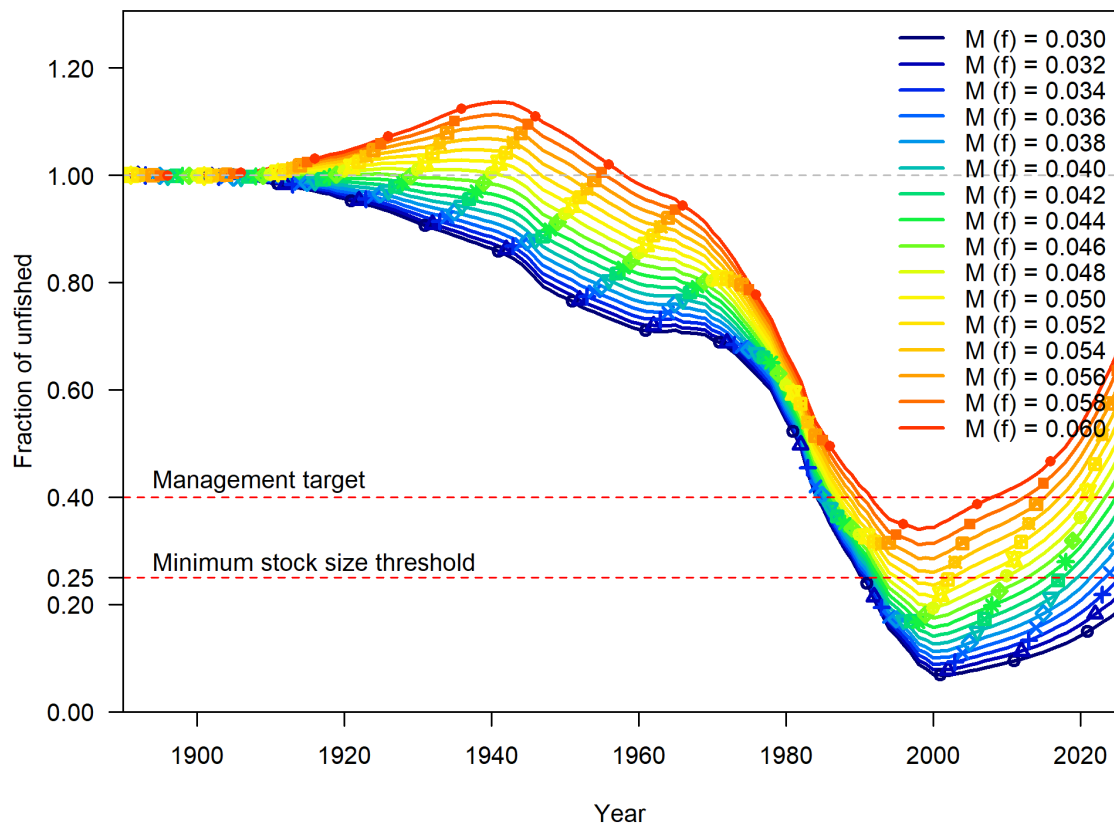


Figure 83: Time series of fraction of unfished biomass output associated with different values of natural mortality ranging from 0.03 to 0.06 in increments of 0.002.

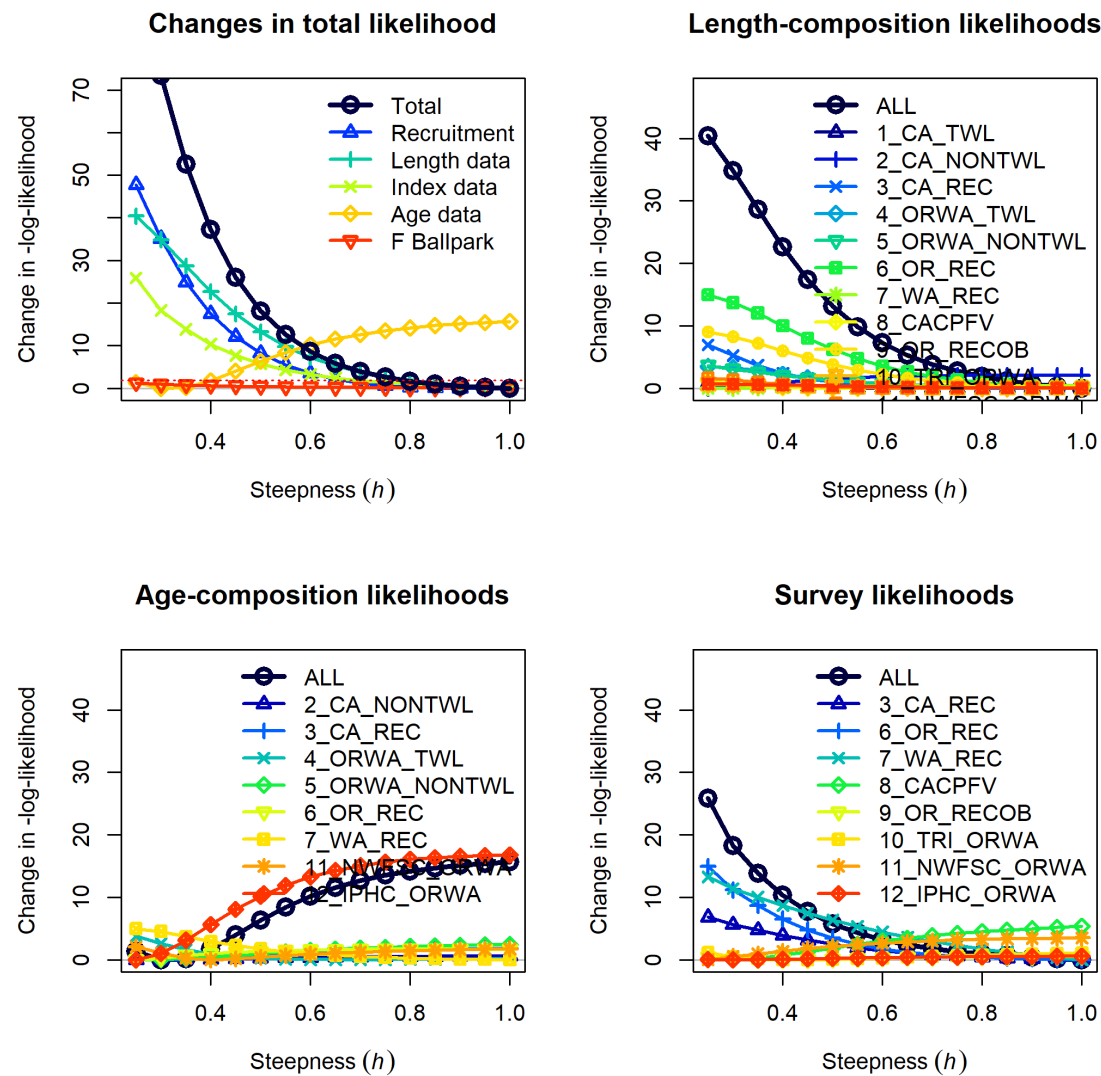


Figure 84: Negative log-likelihood profile for each data component and in total given different values of stock-recruit steepness ranging from 0.25 to 1.0 by increments of 0.05.

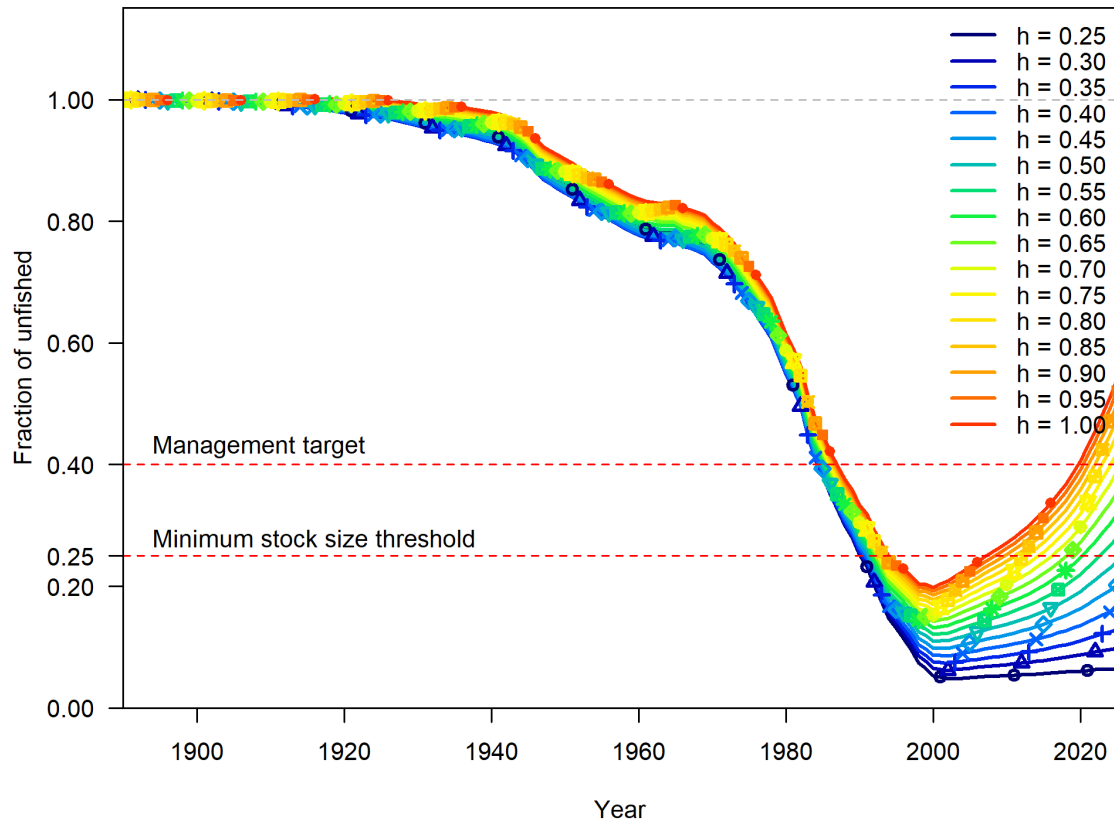


Figure 85: Time series of fraction of unfished biomass output associated with different values of steepness ranging from 0.25 to 1.0 in increments of 0.05.

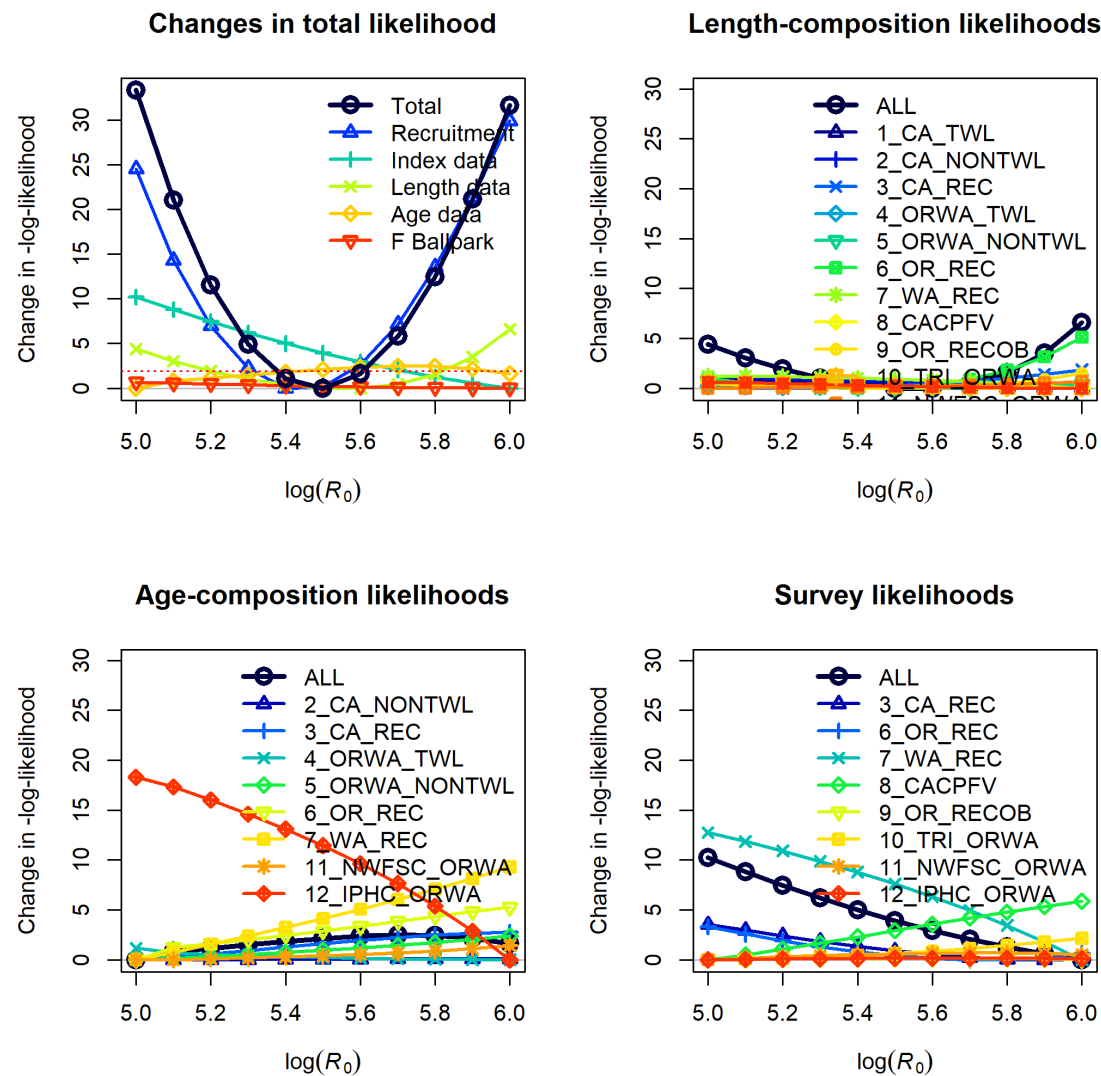


Figure 86: Negative log-likelihood profile for each data component and in total given different values of log initial recruitment ( $\ln R_0$ ) ranging from 4.5 to 6.0 by increments of 0.15.

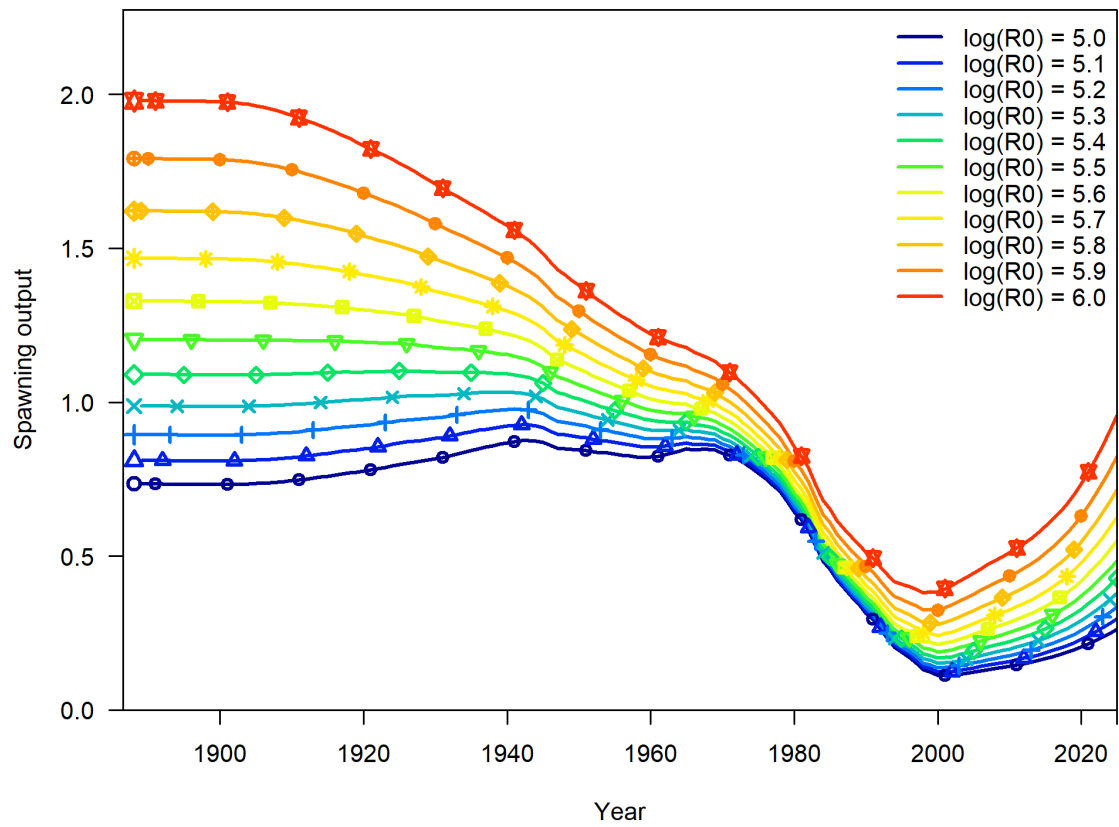


Figure 87: Spawning output (billions of eggs) as profiled over values of  $\ln R_0$ .

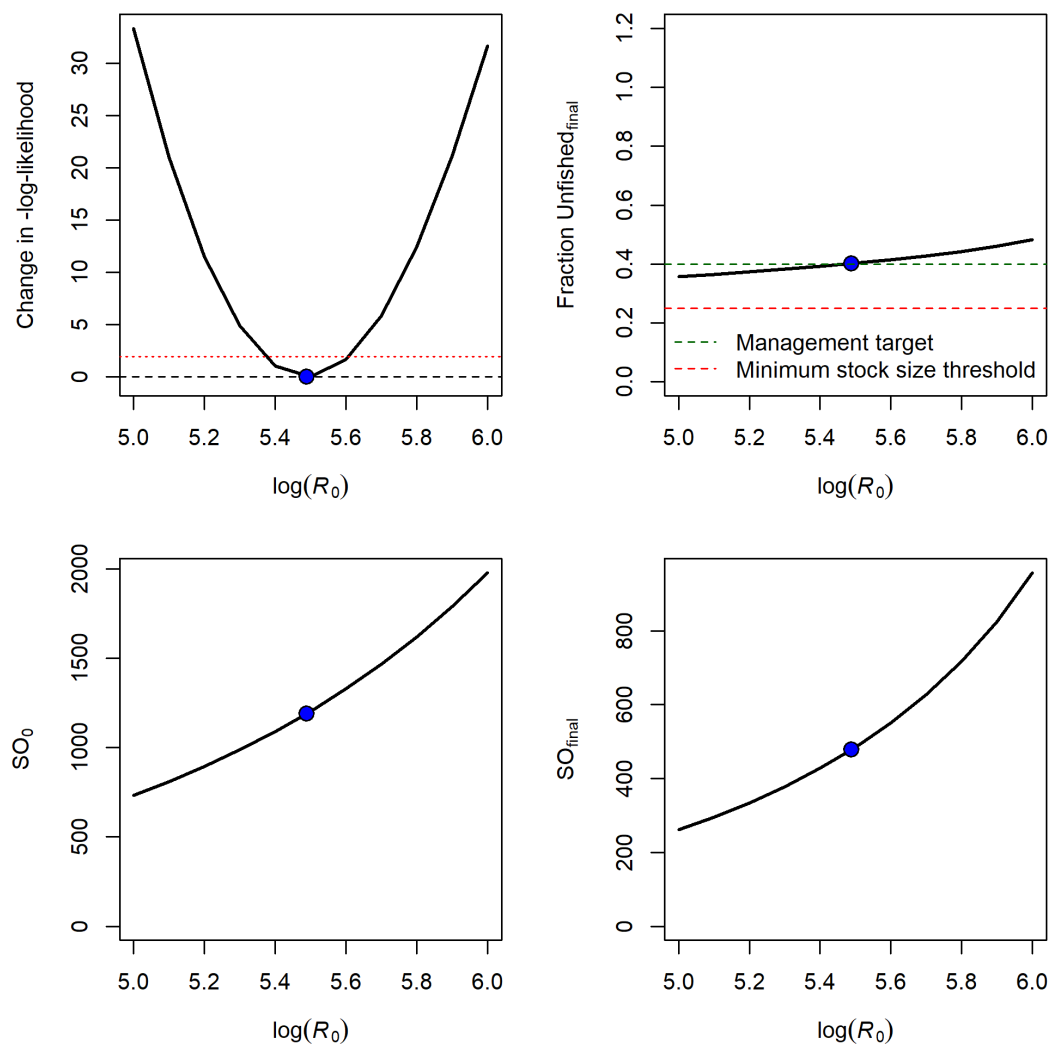


Figure 88: Likelihood profile (top left panel) for log initial recruitment ( $\ln R_0$ ), with associated changes in stock status in the current year ( $SB_{2025}/SB_0$ ; top right panel), initial spawning biomass ( $SB_0$ ; bottom left panel), and current year spawning biomass ( $SB_{2025}$ ; bottom right panel). Points indicate the base model MLE estimate.

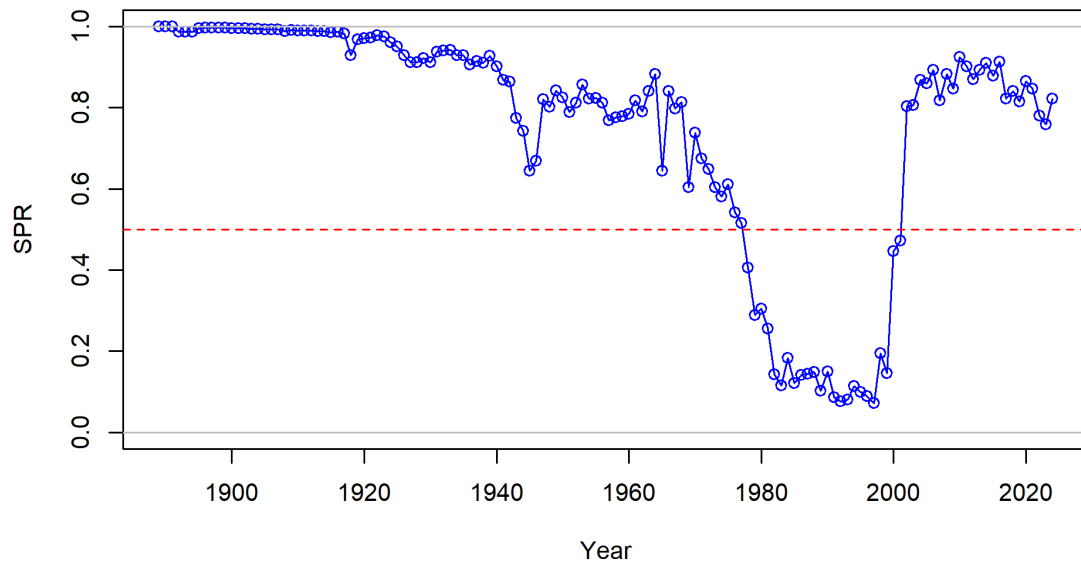


Figure 89: Time series of estimated SPR.

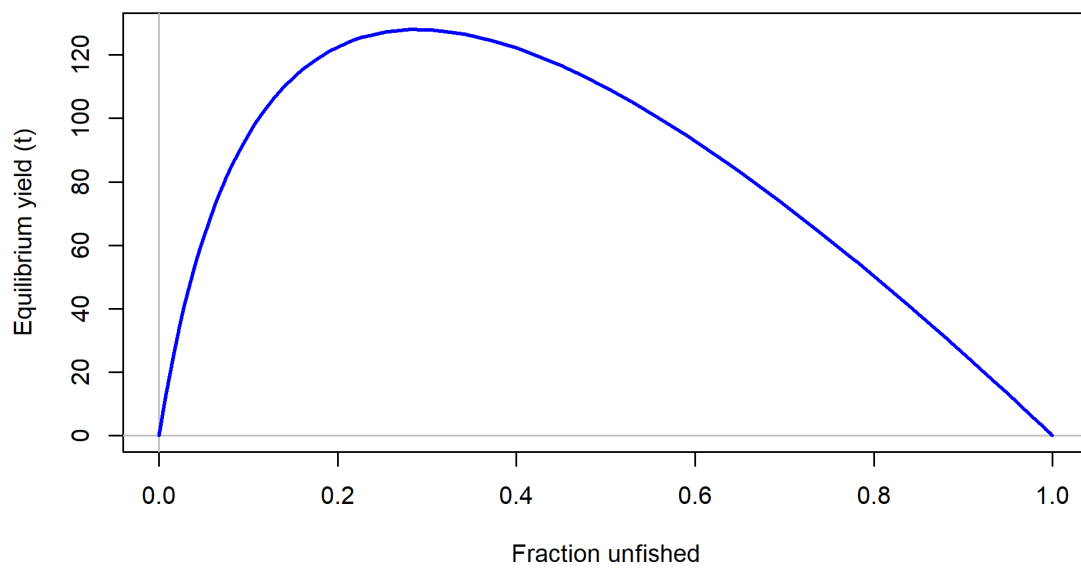


Figure 90: Equilibrium yield curve (derived from reference point values) for the base model. Values are based on 2024 fishery selectivity and distribution with steepness fixed at 0.72. The relative spawning output is relative to unfished spawning biomass.



